OECD Economic Surveys

Spain

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OECD ECONOMIC SURVEYS

2002-2003

Spain

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BASIC STATISTICS OF SPAIN (2002)

THE LAND

Area (1 000 km^2):	
Total	506.0
Cultivated (1998)	185.1

Major cities (thousand inhabitants):	
Madrid	3 017
Barcelona	1 527
Valencia	762
Seville	704

THE PEOPLE

In thousands:	
Population	41 838
Net natural increase (2001)	45
Net migration (2001)	247
Number of inhabitants per km^2	82.7

Employment (thousands)	16 258
Employment by sector (% of total):	
Agriculture	5.9
Industry	19.4
Construction	11.8
Services	62.9

PRODUCTION

Gross domestic product (GDP):	
Billion EUR	694
Per head (US$)	15 631

Gross fixed capital investment:	
% of GDP	25.4
Per head (US$)	3 973

THE GOVERNMENT

% of GDP:	
Consumption	17.6
Revenue	38.5
Deficit	–0.1
Fixed investment (% of gross fixed capital formation)	13.1

Composition of Parliament	
(No. of seats, March 2000):	350
Popular Party (PP)	182
Spanish Labour Socialist Party (PSOE)	125
Convergence and Union (CIU)	15
Izquierda Unida (IU)	8
Basque Nationalist Party (PNV)	7
Other	13

Next general elections: March 2004

FOREIGN TRADE

Exports of goods and services (% of GDP)	28.5
Exports as a % of total goods exports:	
Foodstuffs	12.5
Other consumer goods	28.8
Energy	2.0
Other intermediate goods	44.6
Capital goods	12.2

Imports of goods and services (% of GDP)	30.0
Imports as a % of total goods imports:	
Foodstuffs	6.6
Other consumer goods	20.9
Energy	10.7
Other intermediate goods	46.0
Capital goods	15.8

THE CURRENCY

Monetary unit: Euro

Currency units per US$, average of daily figures:	
Year 2002	1.0611
February 2003	0.9276

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Spain were reviewed by the Committee on 24 February 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 27 March 2003.

•

The Secretariat's draft report was prepared for the Committee by Claude Giorno, Miguel Jimenez, Douglas Sutherland and Ann Vourc'h under the supervision of Peter Hoeller.

•

The previous Survey of Spain was issued in June 2001.

Assessment and recommendations

Sustaining a strong performance over the medium term requires continuing the reform process

Spain's performance has remained remarkably strong thanks to the structural reforms implemented since the mid-1990s and the sound macroeconomic policy framework. Employment has expanded strongly, enabling the economy to grow more rapidly than the euro area average, even during the recent slowdown. To sustain this strong performance over the medium term, it is necessary to continue the reform process. F*irst*, the persistent inflation differential with the euro area is worrying as it erodes competitiveness progressively and could lead to a period of slower growth. This inflation differential, which is partly the consequence of exceptional factors, highlights the need for continuing labour market reforms, and for strengthening competition in certain sectors. Additional reforms in these areas would also help in bringing down unemployment, which is mainly structural and, in spite of its considerable reduction, remains among the highest in the OECD. S*econd*, productivity gains have remained small, partly as a consequence of positive effects of strong employment growth. Accelerating real convergence requires removing the obstacles to the development of higher value added activities, promoting the improvement of human capital, reducing the innovation gap and stepping up the slow diffusion of new technologies. T*hird*, a reform of the real estate market is desirable, to raise labour mobility, to improve resource allocation and to prevent the risk of a speculative bubble from developing. L*ast*, reforms are needed to prepare against the consequences of ageing. They should be phased in now to ensure the sustainability of the public finances, which will be under threat from sharply rising spending on pensions and medical and social services as from 2020-25.

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The slowdown was mild, but inflation remains high

Spain did not escape the economic slowdown that hit the OECD, but did perform better than most countries. Growth reached 2 per cent in 2002, the positive differential *vis-à-vis* the euro area average remaining at 1¼ percentage points. Housing investment has been the most dynamic demand component, while house prices have risen steeply. Job creation has been buoyant, increasing by nearly 1½ per cent, but productivity gains have remained small. Unemployment did, nonetheless, rise for the first time since 1994, reaching 11½ per cent in late 2002. Despite the easing of labour market pressures and a slightly negative output gap, inflation has not slowed – partly for exceptional reasons such as the euro changeover – and the differential with the euro area has not shrunk. This differential, which has been about 1¼ percentage points since 1999, tends to weaken international competitiveness.

The neutral stance of the 2003 Budget suits the economic situation and some slippage can be tolerated if downside risks materialise

Domestic liquidity conditions have become easier in recent years because the European Central Bank (ECB) has relaxed monetary policy. Monetary conditions remain accommodating despite the euro's appreciation since the second half of 2002. The expansionary effect has been limited, however, by fiscal tightening since 2000. Policy has, quite rightly, been more prudent than in most other European countries. Fiscal consolidation that began in the mid-1990s has led to the elimination of the general government deficit in 2001, despite the economic slowdown. The government account remained close to balance in 2002, even though activity was more sluggish than anticipated. The government aims again at a balanced budget in 2003 on the basis of an official forecast of 3 per cent, close to potential growth, implying a neutral fiscal stance following the tightening in 2001 and 2002. According to the OECD projections, gross domestic product (GDP) growth may be only marginally above 2 per cent in 2003 before reaching 3 per cent in 2004, maintaining the positive growth differential with the euro area, while inflation should progressively dip to 2½ per cent in 2004. These projections imply the persistence of a negative output gap and, given the relaxed monetary conditions, a neutral fiscal stance is appropriate. The stance also appears appropriate in view of the mainly negative risks surrounding the economic outlook, which largely

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concern international developments. A slight government deficit in 2003 cannot be ruled out, however, but should be tolerated, particularly if the outlook deteriorates further. Even though the inflation differential may decline, a permanent improvement in inflation performance requires tackling nominal wage and price rigidities rather than a tighter fiscal stance.

In the medium term, it will be necessary to steer a prudent fiscal course

Beyond the short term, the latest Stability Programme aims at maintaining the government accounts in balance or small surplus, which means adhering to a largely neutral policy stance through the period to 2006. The Programme includes the assumption that the inflation differential with the euro area will decline. If this were not the case, the effect on exports from the continuing loss of price competitiveness would be more than offset for a time by the impact on the domestic demand of lower real interest rates than in the rest of the euro area. This sort of situation would create an unbalanced policy mix. However, if the inflation gap is eradicated, the medium-term fiscal goals are appropriate and would allow a further reduction in debt or additional tax cuts. Fiscal policy faces, however, two main medium-term challenges.

In the context of decentralisation, the implementation of the new balanced budget rule should be flexible, as permitted by the law

The first medium-term challenge is to maintain fiscal prudence in the new, more decentralised way of organising the public finances. With health service management having been transferred to all the regions in early 2002, decentralisation of the largest spending areas is complete, making Spain one of the most decentralised OECD countries. In order to consolidate the fiscal progress made, the authorities have implemented two reforms. The first consists of increasing the regions' and municipalities' financial self-sufficiency and fiscal responsibility. The second is the adoption of a fiscal stability law that obliges all levels of government to balance their accounts as of the 2003 Budget. As regards the latter, it is important to ensure that the implementation of this fiscal rule, which takes no explicit account of the economic situation, does not engender pro-cyclical effects on activity. To avoid this danger, advantage should be taken of the flexibility built into the law. If deficits do develop, the law foresees that these need to be corrected

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only in the medium run, ensuring that fiscal discipline and automatic stabilisers are fully compatible. Moreover, the law explicitly allows the possibility of nominal deficits in "exceptional circumstances".

Improvements in the quality of the public finances need to be pursued

The second medium-term challenge concerns improving the quality of both taxation and spending. Decentralisation has the potential, in theory, to enhance the efficiency of public management. This, however, implies that the national authorities will have to re-think the way they act in a number of areas such as health, where their main role now is to co-ordinate the various regional systems. Co-ordination should stimulate the regions' efforts to be efficient. Regular publication of indicators for the communities would make it easier to assess their performance and encourage them to adopt best practice. In this connection, the recent decision to publish homogeneous information on regional hospital waiting time is welcome. Ensuring an equitable provision of public services throughout the country should also focus more on assessing services and results rather than only imposing minimum expenditure criteria, as is planned in the case of health. Lastly, co-ordination between the regions should take into account economies of scale or spillovers. With regard to taxation, the rationalisation of the system has moved forward. Distortions due to certain local taxes will be reduced. More importantly, the personal income tax has again been cut in 2003, and work incentives have been raised for older workers and mothers with young children, while the unemployed face better incentives to be geographically mobile.

Pension reform is more urgent than generally perceived

In the longer term, the biggest challenge is to ensure fiscal sustainability in the face of population ageing. Recent reform efforts have improved the incentives for older workers to remain active. Moreover, the pursuit of a prudent fiscal policy has allowed the social security reserve fund to be raised to above 1 per cent of GDP in 2003 – a year earlier than planned – and to reduce public debt. Solving the pension problem, however, implies adopting measures that go much further than those already approved. The strategy of the authorities separates the long-term financing problem posed by ageing from medium-term fiscal management,

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which makes a reform of the pension system essential. However, the official analyses tend to emphasise the pension system's present strong financial position, which should persist until 2015, whereas the uncertainty of the situation after 2015 is stressed. Nevertheless, even taking these uncertainties into account, the available studies point to the need for a prompt and ambitious reform. Further measures to promote a longer active life should be taken and, most importantly, a revision of the system's parameters is necessary, since currently pensioners receive, on a discounted basis, more in pensions than they contribute. If the sharp rise in pension spending as from 2015-20 is to be mitigated, a reform that makes the system actuarially neutral cannot be put off any longer, as many changes will have to be phased in. A significant first step would be to calculate pensions on the basis of earnings over a whole career instead of the last 15 years. The gradual introduction of this measure, which would not affect the pensions of current retirees or of those who are close to retirement but would lower future pensions of younger workers, should be discussed at the tripartite negotiations, which will start soon. Undertaking a comprehensive reform would reduce the need to raise fiscal revenues in the future to cope with ageing.

Labour market reform has progressed…

Structural reforms are needed to raise the employment rate further and to boost productivity growth. The remarkable labour market performance in recent years is attributable to the reduction in employment rigidities, highlighted by real wage moderation since 1995. Even so, there remain major distortions in the labour market. It is still very segmented, with temporary jobs in excess of 30 per cent. The flexibility of temporary jobs contrasts with the high level of protection enjoyed by workers on indefinite contracts, while the level and regional dispersion of unemployment also remain very high. A reform of unemployment insurance was adopted in 2002, enabling firms to slightly reduce their redundancy costs provided they agree to view dismissals as "unfair". It reforms the special compensation scheme for Andalusian and Extremaduran farmers which was being abused, and increases the incentive to look actively for work by clarifying the concept of suitable work, *i.e.* work that cannot be refused without incurring a penalty. The reform is a

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further step in the right direction, even if redundancy costs still need to be reduced more sharply. Also, the public employment service (INEM) will have to function better in providing job placement services, while active labour market policies, which are not very effective, need to be reviewed.

… but needs to be taken further, primarily as regards wage formation

Reforming the wage bargaining system is of crucial importance, even if difficult to achieve. The most urgent task is to reduce the strong nominal wage inertia caused by the widespread use of wage indexation clauses, which is one important reason for the inflation differential with the euro area. These arrangements provide workers with a guarantee against inflation surprises, but pose two major problems. First, they remove any responsibility for inflation developments from the social partners, whereas inflation in the short run is largely a consequence of their attitudes and the way they interact. The unions, in particular, are not concerned with nominal wage increases as long as they are guaranteed in real terms. Second, the indexation is based on overall rather than underlying inflation, reducing the economy's flexibility to cope with exogenous shocks – oil price shocks, for example. This rigidity, which penalises Spain, is harmful in the present economic situation. These shortcomings need to be remedied by reforming the wage bargaining framework. In view of the difficulty of reforming the system, a new attempt could have more success – as in 1997 – if it were to combine a number of mutually reinforcing measures. In addition to abolishing the indexation clauses and reforming the INEM, the level of protection enjoyed by established workers should be reduced while at the same time stricter controls against the abuse of temporary contracts should be implemented, so as to reduce employment segmentation without penalising overall labour market flexibility. Segmentation is likely to foster unskilled work, encourage investment in less productive sectors and is a heavy burden on younger workers, who are increasingly well trained. It also may reduce incentives to strengthen human capital and to incorporate new technologies in production, which could well be one of the causes for low productivity growth.

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The reform of education is welcome

Education policy is also one of the areas with a big potential impact on future productivity growth. The shortcomings of the current system are highlighted by the high failure rate in secondary education and, recently, the OECD's PISA report. Several reforms have been adopted with the objective of improving the quality of education. A new law broadens the range of free infant education services, which should raise female participation rates and help the integration of the children of immigrants. It also extends schools' independence in a number of areas, which is positive from the point of view of increasing experimentation and helping schools to adapt to their social environment. This independence could be further enhanced by basing teachers' pay and career development on their responsibilities and performance. Concerning the universities, changes have made the appraisal of university departments and the appointment of professors more independent. However, the system of university financing has not been touched by the reform, and should be rethought. The system is regressive, with low enrolment fees for university education, mainly benefiting students from middle and high income families who will later enjoy high salaries and low unemployment. An increase in fees, partly offset by a greater provision of grants or subsidised loans for disadvantaged students, would make the system less regressive. It would also help to cut the present length of courses, which is excessive. Higher university fees would also free up resources for improving quality of secondary and tertiary education and for occupational training which is not well developed.

Competition has been strengthened but needs to be rigorously applied in certain sheltered sectors

In the product market, competition policy plays a key role in fostering an efficient resource utilisation and reducing inflationary pressures. Noteworthy progress has been made in this area. The competition authorities' resources have been substantially increased and new regional courts have been set up, which should relieve the national authorities from having to investigate regional cases. Even so, co-ordination is needed to ensure that the competition rules are applied homogeneously across the regions and to prevent the development of a culture of indulgence *vis-à-vis* local firms. In this respect, continued efforts need to be

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made in the distributive sector, which is prone to inflationary pressures and where hypermarket locations are often subject to regional restrictions. In the network industries, competition has been strengthened thanks to an ambitious privatisation and liberalisation programme. Progress is ongoing, for example, in fixed telephony, thanks in particular to competition being actively promoted by the industry regulator. In the electricity and gas sector, liberalisation will soon be complete. However, the highly concentrated nature of production and distribution of energy products has resulted in some instances in anti-competitive behaviour and a significant increase in profits in part of the energy sector. The efforts already under way to improve linkages with other countries' networks are set to continue, while thought will have to be given to splitting up big companies with considerable market power.

The real estate market needs to be reformed

Reforms should also focus on correcting the distortions in the real estate market. They have underpinned the very steep rise in house prices and led to a shortage of rental housing, while the share of real estate assets in households' wealth is very high. Housing policy encourages house purchases by tax deductions while regional and local governments limit the supply of building land. The authorities have introduced new incentives to stimulate the supply of rental housing such as the reduction of the taxable base by 50 per cent of the net rental income, while the many unoccupied dwellings can now also be taxed more heavily by the municipalities. This would foster the development of the rental market and should lead to a better use of the housing stock. This reform would be even more effective if accompanied by the phasing out of the tax deductions for house purchases to establish a balanced mix of incentives. While scaling back generous fiscal support for house purchases is politically difficult, experience in some OECD countries shows that it is possible. Reduction of the incentives for future home buyers and introduction of rental subsidies for low-income households would be justified both from the standpoint of equity and by the now very low level of real interest rates. Such a reform would lessen the risk of a speculative bubble in the housing market. Homogeneous, transparent and more flexible management of the supply of building land by the communities and municipalities also appears

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necessary to damp house price increases. A reform of land management would require agreement among the different levels of government, that the central government has already started to promote. It would doubtless be easier to obtain an agreement if a reform of local finances were to reduce the heavy dependence of municipalities on resources derived from land zoning, which underlies their restrictive practices and could be offset, if necessary, by a more widespread recourse to user fees and charges.

Environmental policy should be strengthened, especially in response to climate change

As regards environmental policy, preference should be given to cost-efficient measures so as not to hamper growth. The delay in drafting a national climate change strategy and failure to implement cost-effective abatement strategies means Spain will have to rely on the flexible mechanisms allowed under the Kyoto Protocol to meet the EU Burden Sharing target. In addition, more should be done at the domestic level. Past policy has been of limited effectiveness and, therefore, needs to be reinforced. The climate change strategy should set more ambitious targets using measures that will equalise marginal abatement costs. The introduction of emission permits that can be integrated into an EU wide system should be the approach for achieving abatement targets. Where sector-specific targets are deemed necessary, care is required to prevent the emergence of undue abatement cost differentials. In this light, a green certificate market should be used to achieve the planned doubling of the share of renewable energy. Finally, the government should continue restructuring the coal industry and phase out coal subsidies.

More efficient water management requires better water pricing mechanisms

Pressure on water resources is serious, with agriculture representing the most important user. This pressure has led to the over-exploitation of ground water sources and environmental degradation in some areas. The water basin authorities should not let abstraction rates for ground water exceed those consistent with natural replenishment. The system of water allocation, including for agriculture, should, where appropriate, progressively move from an administered system to a marginal social cost system where supply and demand determine prices for each water basin, including a charge on the environmental externality of

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abstraction. Supporting legislation should be introduced to facilitate the operation of the recently allowed secondary market for water, and ensure that payment discipline is enforced vigorously, so giving all users incentives to use water sparingly. Getting the framework right for water management rapidly should complement the decision to invest in water transfer to ensure this resource is used efficiently.

The positive economic impact of immigration does not lessen the need for pension and labour market reforms

While many Spaniards emigrated in the past, immigration has surged in recent years. Despite the strong increase, the foreign population remains relatively small and has therefore had a rather limited macroeconomic impact so far. However, immigration is likely to continue, which makes it important to assess its economic effects in a number of areas. Immigration, which is the subject of the in-depth chapter of this *Survey*, has had a positive effect on growth, largely by boosting employment, but the overall effect on labour productivity is less clear. Non-EU foreigners, who account for most of the immigrants, often occupy low-paid jobs refused by natives and have contributed to sustain or develop the activity of some low value-added sectors. Immigrants have raised the employment rate both directly through their high participation rate, and indirectly by facilitating female participation through domestic services, boosting incomes for the natives. But immigration has probably not contributed to reducing regional differences in unemployment rates. Moreover, most non-EU foreigners work on temporary contracts, which contributes to sustaining the high share of such contracts, highlighting the need for reform in this area. Legally resident foreigners are making a positive contribution to public finance in the short run. In the longer run, however, while increased immigration would mitigate the expected increase in the dependency rate, the relief would be only transitory under current pension rules and further measures, especially reforms of the pension system, remain necessary.

Migration policy still needs to achieve an appropriate balance

The task of migration policy is difficult. It has to reconcile the strong demand for workers in low-wage sectors with the policy aim of preventing illegal immigration. The migration policy framework has been reformed, but at the moment inconsistencies persist. For example, a large number of

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places in the quota scheme go unfilled, while at the same time a large number of undocumented immigrants have regular but undeclared work. In agriculture, the demand for low-qualified labour partly reflects the heavy subsidisation of the sector. Migration policy needs to be sufficiently flexible to allow the undeclared immigrants, whose presence is *de facto* accepted, to obtain work permits. If an appropriate balance between legal inflows, border controls and labour inspections to reduce undeclared employment is not found, illegal immigration will continue to undermine formal immigration targets. Regarding enforcement, measures to reduce undeclared employment, if necessary by strengthening labour inspections, would be a cost-effective complement to border controls.

The integration of immigrants should be improved

The integration of immigrants, which is important both for economic and social reasons is another challenge for migration policy. The new Immigration Act clarifies the rights of immigrants in terms of access to health and education, thus contributing positively to their integration. However, the difficulty to find adequate housing, largely due to a very limited supply of rental housing is an obstacle to integration. Recent steps taken to favour rental housing may help in this respect. Integration policy should also take more into account the specificity of immigrants depending on their origin country. Experience from other countries suggests for example that efforts to raise proficiency in the local language facilitate integration. On these issues, the autonomous communities and municipalities have an important role to play.

Summing up

Output growth has remained strong for a number of years, with robust job creation, despite the recent slowdown. But inflation has been higher than the euro area average, raising competitiveness concerns while – partly as a result of strong job creation – productivity has been weak. The prudent fiscal policy of recent years has resulted in balanced public accounts. Any cyclical slippage can thus be tolerated if the economic upturn in 2003 is weaker than expected, though a structural deficit needs to be avoided. Prudence also needs to prevail in the medium term. In this regard, the recent reforms aimed at maintaining fiscal discipline in the context of public finance decentralisation are important. But the fiscal stability law, which requires all

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levels of governments to balance their budget, has to be applied with the flexibility embodied in the law. In the current situation the inflation differential with the euro area is worrying. However, it should be tackled by structural reforms, rather than a tighter fiscal policy. This implies elimination of the wage indexation clauses and stronger competition in certain sectors. In addition, labour market reform, which moved ahead in 2002, should aim to further reduce labour market segmentation and facilitate a pickup in productivity gains. While growth based on strong job creation can continue for a few years if wage moderation prevails, strengthening productivity over the longer term will be essential to ensure real convergence. In this respect, the recent reform of education will help, but it should be reinforced by an overhaul of the university funding system. Further efforts are also needed to reduce the innovation and research and development gap *vis-à-vis* the OECD average. Changing housing policy is essential for a better functioning of the labour market, a more efficient allocation of resources, and to reduce the pace of house price increases. In addition to a reform of land management, phasing out the fiscal incentives for home purchases would be a first-best approach. The authorities' choice of granting new tax deductions to increase the supply of rental housing is, in this perspective, a second-best solution. Many structural reforms necessitate active participation of the communities, whose responsibilities now extend to such crucial areas as land use and competition policy. They also have an important role to play in environmental policy, for example in water management, where efforts should focus on limiting water wastage through the introduction of adequate pricing mechanisms. In recent years, a notable development has been the steep rise in immigration, which has had positive effects on the economy. Nevertheless, reducing the number of illegal entries while responding to the needs of the labour market is a policy challenge, as is the integration of foreigners. The positive short-term impact of immigration on public finances will be temporary and does not lessen the need for pension reform to cope with population ageing. Guaranteeing the long-term viability of the public finances and the continuation of structural reforms are key to keeping Spain's growth and employment performance among the best in the OECD area.

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I. Economic developments, the fiscal stance and prospects

Overview

Spain has not escaped the international economic slowdown since 2000 but has fared better than most countries. Output growth for 2002 reached 2 per cent, which means that the positive growth differential with the euro area remained at 1¼ percentage points, its average level since 1996. This strong growth performance, which is similar to that of the United States since the mid-1990s (Figure 1), is due mainly to strong job creation which has boosted domestic demand, especially private consumption and residential investment. However, robust growth of employment did not prevent a cyclical rise in the unemployment rate to 11½ per cent in late 2002, the first increase since 1994, which is also due to the low sensitivity of the labour force to economic conditions and the steep rise in immigration. Despite an easing of labour market pressures and a slightly negative output gap, inflation has not slowed, partly because of exceptional factors such as the changeover to the euro. In February 2003, consumer prices were 3¾ per cent higher than a year earlier. The inflation differential with the euro area of about 1¼ percentage points since 1999 tends to erode the economy's international competitiveness.

Recent developments, while favourable overall, have revealed three main sources of strain that are essentially structural in nature. First, the inflation differential with the euro area constitutes a major threat to the continuance of good performance in the medium term. This problem is largely due to nominal wage rigidity and insufficient moderation of profit margins in the sheltered sector. Second, productivity gains have remained weak, even though uncertainties surround productivity estimates. Continuing sluggish productivity growth during the next few years will not pose a problem if real and nominal wage growth moderates. But if the present trend continues because of labour market rigidities, Spain's real convergence with the most developed countries will be compromised. A third notable characteristic has been the steep rise in house prices, which have amounted to some 40 per cent since 1999 and shows little sign of slowing. This has led some analysts to express concern about the emergence of a speculative

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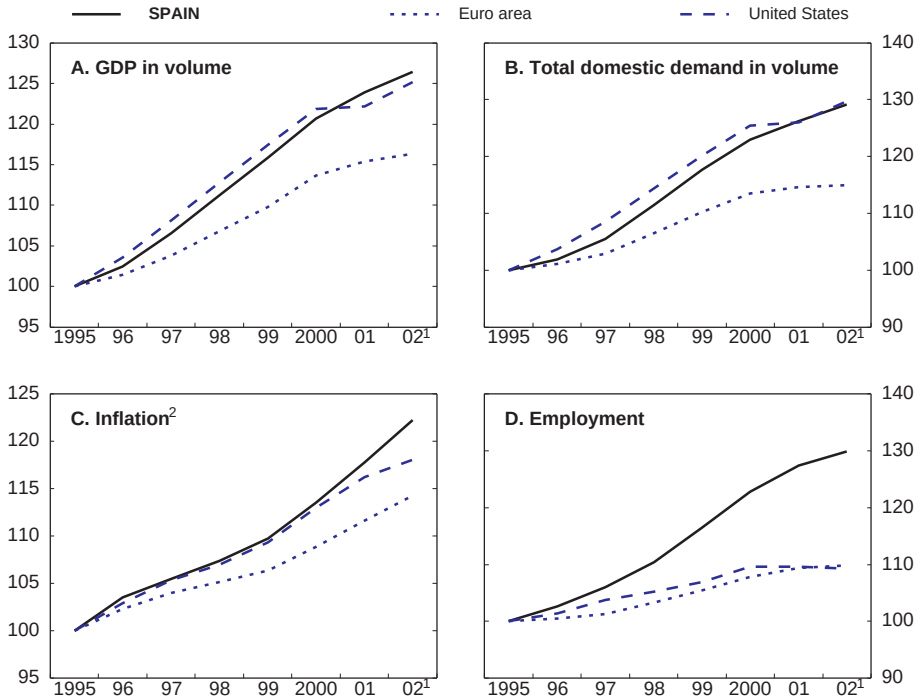
Figure 1. **Key indicators in international perspective**
Index 1995 = 100



1. OECD estimates.
2. CPI for the United States and harmonised CPI for Spain and the euro area. The break of the Spanish harmonised CPI in 2001 has been adjusted using the national CPI as benchmark.

Source: INE, EUROSTAT and OECD.

bubble. Although a bubble is unlikely to exist at present, it is important not to underestimate the risks of a speculative spiral developing, given the large distortions in the real estate market.

Domestic liquidity conditions have become easier in recent years because the European Central Bank (ECB) has relaxed monetary policy. Monetary conditions remain accommodating in Spain despite the euro's appreciation since the second half of 2002. The expansionary effect has been limited, however, by fiscal tightening since 2000. This policy has continued the fiscal consolidation begun in the mid-1990s and eliminated the general government deficit. In 2003 the government aims again at a balanced budget on the basis of an official growth forecast of 3 per cent, close to potential, implying a neutral fiscal stance. According to the

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OECD projections, gross domestic product (GDP) growth may be only marginally above 2 per cent in 2003 before reaching more than 3 per cent in 2004, maintaining the positive growth differential with the euro area. These projections imply the persistence of a negative output gap, and given the relaxed monetary conditions, a neutral fiscal stance is appropriate. This stance also seems appropriate in light of the risks to the economic outlook, which are mainly on the downside. In the short term, some fiscal slippage could be tolerated, notably if prospects deteriorate.

Recent trends

Household consumption has remained relatively buoyant

Private consumption slowed further in 2002, spending rising by less than 2 per cent, as against over 4½ per cent in 1998-99 (Table 1). With the economy more sluggish and consumer confidence turning down, there was a slight rise in the saving ratio, while wages increased less because of the slowdown in job creation. Households, whose lending capacity has fallen sharply since the mid-90s (Table 2), have adopted a more cautious attitude as a result of mounting uncertainty about the international situation and the increasingly gloomy outlook regarding their incomes. Furthermore, the euro changeover in early 2002 seems to have triggered the use of old notes to advance purchases of consumer durables, particularly cars, which underpinned consumption in late 2001

Table 1. **Demand and output**
Year-on-year percentage changes (1995 prices)

	Average 1991-2000	1998	1999	2000	2001	2002	2002		
							Q2	Q3	Q4
Private consumption	2.4	4.4	4.7	3.9	2.5	1.9	1.8	1.5	1.8
Government consumption	3.2	3.7	4.2	5.0	3.1	3.8	3.8	3.7	4.0
Gross fixed investment	3.0	10.0	8.7	5.7	3.2	1.4	1.0	1.6	2.5
Construction	2.5	7.8	9.0	6.1	5.8	4.5	4.5	5.0	4.6
Machinery and equipment	3.6	12.6	8.4	5.1	0.3	–2.2	–3.0	–2.4	–0.2
Final domestic demand	2.7	5.5	5.6	4.5	2.8	2.1	1.9	1.9	2.3
Total domestic demand	2.6	5.7	5.6	4.4	2.7	2.2	1.9	1.9	2.7
Exports of goods and services	10.1	8.2	7.7	10.1	3.4	1.4	–1.6	4.2	6.0
Imports of goods and services	9.2	13.2	12.7	10.6	3.5	2.2	–1.7	4.2	7.4
Gross domestic product	2.7	4.3	4.2	4.2	2.7	2.0	2.0	1.8	2.1
GDP price deflator	4.1	2.4	2.7	3.5	4.2	4.4	4.2	4.5	4.8
Private consumption deflator	4.2	2.2	2.4	3.2	3.3	3.6	3.6	3.6	4.0
Industrial production	2.2	5.5	2.6	4.4	–1.4	0.3	–0.9	0.6	2.0
Household saving ratio (level)	12.7	12.2	11.1	10.6	10.2	10.4			

Source: INE.

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Table 2. **Financial position of the economic sectors**

Per cent of GDP

	1996	1997	1998	1999	2000	2001
Gross saving						
Total economy	22.1	22.6	22.6	22.3	22.4	22.9
Financial corporations	1.6	1.3	1.4	1.0	1.4	1.8
General government	–1.2	0.4	1.2	2.9	3.3	3.9
Non-financial firms	11.8	11.7	11.8	11.3	11.2	10.7
Households[1]	9.9	9.1	8.2	7.2	6.5	6.5
Capital transfers and investment						
Total economy	–20.8	–21.0	–22.1	–23.4	–24.9	–24.9
Financial corporations	–0.6	–0.6	–0.5	–0.5	–0.7	–0.6
General government	–3.7	–3.6	–3.9	–4.0	–3.9	–4.0
Non-financial firms	–11.7	–12.0	–12.7	–13.3	–14.4	–14.3
Households[1]	–4.9	–4.8	–5.1	–5.5	–5.9	–6.0
Net lending						
Total economy	1.2	1.6	0.5	–1.1	–2.5	–2.0
Financial corporations	1.0	0.7	1.0	0.5	0.7	1.2
General government	–4.9	–3.2	–2.7	–1.1	–0.6	–0.1
Non-financial firms	0.2	–0.3	–0.9	–2.1	–3.2	–3.6
Households[1]	5.0	4.3	3.1	1.6	0.6	0.5

	2001			2002		
	Q2	Q3	Q4	Q1	Q2	Q3
Net lending						
Total economy	–1.9	–1.9	–1.8	–2.0	–1.7	–1.1
Financial corporations	1.6	1.2	1.0	1.8	1.3	1.6
General government	–0.4	0.6	–3.0	4.7	–2.2	2.5
Non-financial firms and households[1]	–3.1	–3.7	0.2	–8.4	–0.7	–5.2

1. Including non-profit institutions serving households.

Source: Bank of Spain, *Financial accounts of the Spanish economy*.

(Figure 2),[1] but depressed consumption thereafter. On the other hand, the influence of the decline in financial wealth on the saving ratio following the stock market correction is likely to have been largely offset by rising house prices, which account for some 70 per cent of households' total assets.[2] Despite the deceleration, private consumption has remained more buoyant in Spain than in the rest of the euro area, with job creation continuing to be more robust and the rise in the saving ratio more limited.

Housing investment remains strong, while property prices rise briskly

With growth of 4½ per cent in 2002, construction investment is the most dynamic demand component and confidence indicators for this sector have

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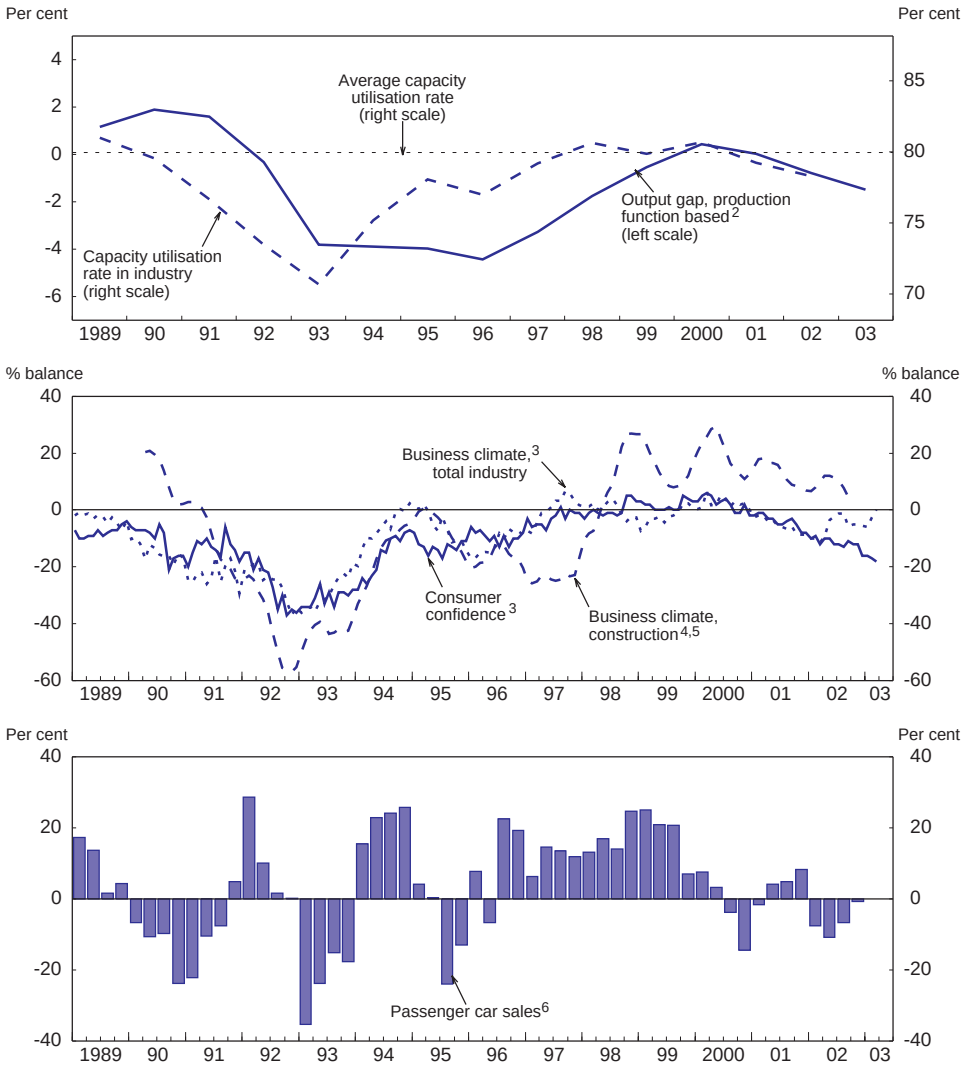
Figure 2. **Output gap[1] and economic activity indicators**



1. Deviation of GDP from potential. Data for 2002-03 are OECD estimates and projections.
2. Estimate based on OECD method.
3. Balance of positive and negative responses.
4. Simple average of the level of the orders and the employment tendency in the construction industry.
5. Six-month moving average.
6. Year-on-year percentage change.

Source: INE, Ministry of Science and Technology and OECD, *Main Economic Indicators.*

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remained higher than those for industry and consumption (Figure 2). Activity in the civil engineering and residential construction sectors in particular has remained very buoyant, though it has weakened in the non-residential sector. Civil engineering, which accounts for nearly one third of total construction expenditure, increased by over 10 per cent in 2001, partly as a result of the implementation of the 2000-07 infrastructure plan. Growth in this sector has also benefited from increased local authority spending in the run-up to the 2003 municipal elections.

In contrast with private consumption, residential investment has lost little steam despite the slowdown in activity. Housing demand has remained well in excess of its long-term trend based on population factors, while at the same time the number of dwellings has increased strongly.[3] This is partly due to the continuing adjustment to the fall in real interest rates (Figure 3). Lower rates, coupled with the more extensive and flexible supply of mortgage loans, have enabled households to increase their debt to a level close to the euro area average (Figure 4). Housing expenditure also appears to have been stimulated by the euro changeover and by investment shifting from the financial sector to the real estate sector because of the stock market fall.[4] As a result, house price increases have outstripped rises in most OECD countries, prompting some analysts to worry about the risk of a speculative bubble. However, it is unlikely that such a bubble has already developed. Although real estate prices have reached historically high levels, a sizeable portion of the increase reflects the improvement in households' average incomes, due in particular to the increase in female employment. House prices relative to average household income are currently not higher than in 1991 (BBVA, 2002b).[5] Yet it is important to bear in mind the risk of triggering a speculative spiral in the present economic situation. This risk is accentuated by the distortions in the real estate market, which are restricting supply and stimulating housing demand (Chapter III). The distortions include, in particular, tax concessions for the purchase of the main residence, the scale of which (½ per cent of GDP per year) corresponds to a reduction in mortgage rates of some 1½ percentage points, making them negative in real terms.

The downturn in equipment spending was fairly moderate

With an international environment that had been deteriorating since 2000 amid mounting uncertainty, firms, unlike households, cut back investment appreciably. Particularly hard hit were purchases of machinery and equipment. Order books in industry fell to their lowest level since 1996, as did capacity utilisation (Figure 2). The negative impact of unfavourable demand prospects and the low capacity utilisation rate was accentuated by the increase in corporate debt since 1995. This increase in indebtedness partly reflects, as in

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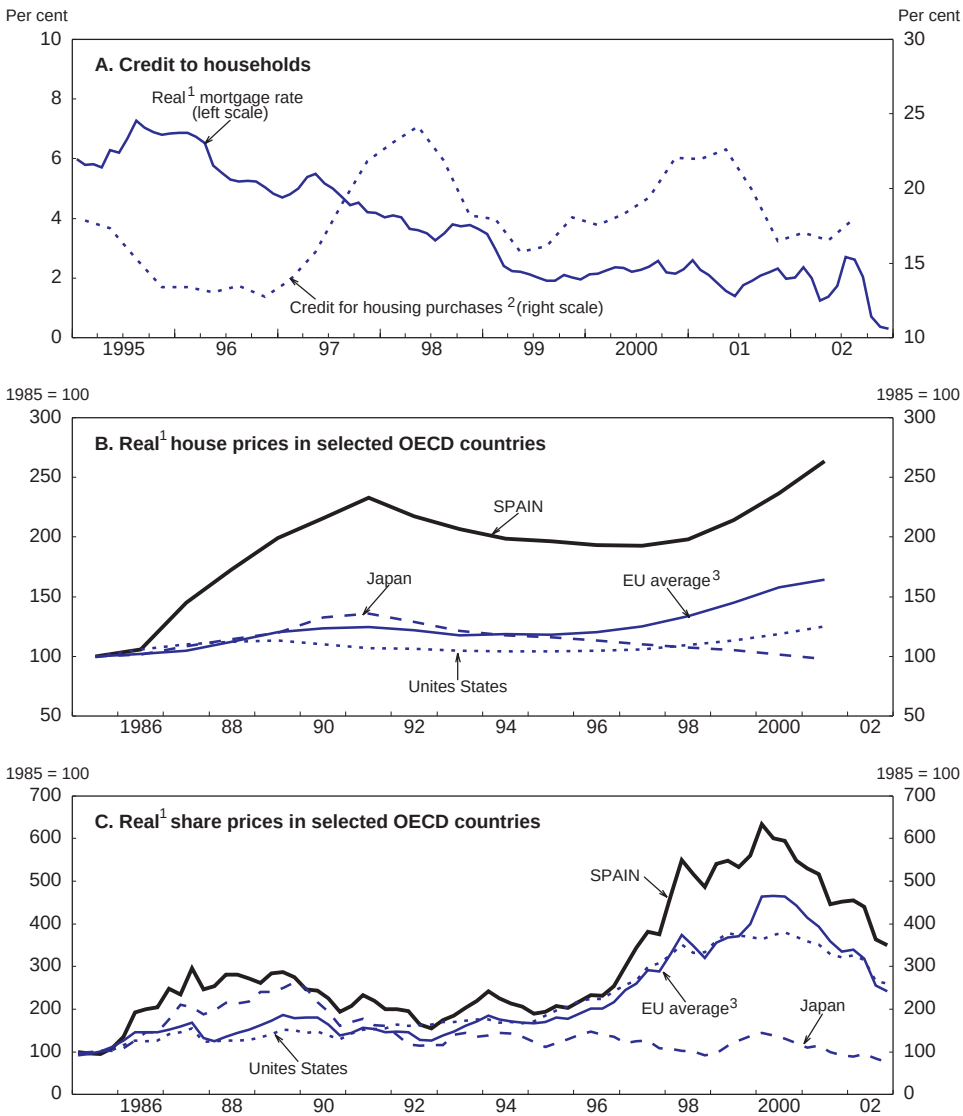
Figure 3. **Credit to households and asset prices**



1. Deflated by the consumer price index.
2. Year-on-year percentage changes.
3. Non-weighted average of Denmark, France, Germany, Ireland, Italy, Netherlands, Sweden and the United Kingdom.

Source: Ministry of Economy and OECD, *Main Economic Indicators.*

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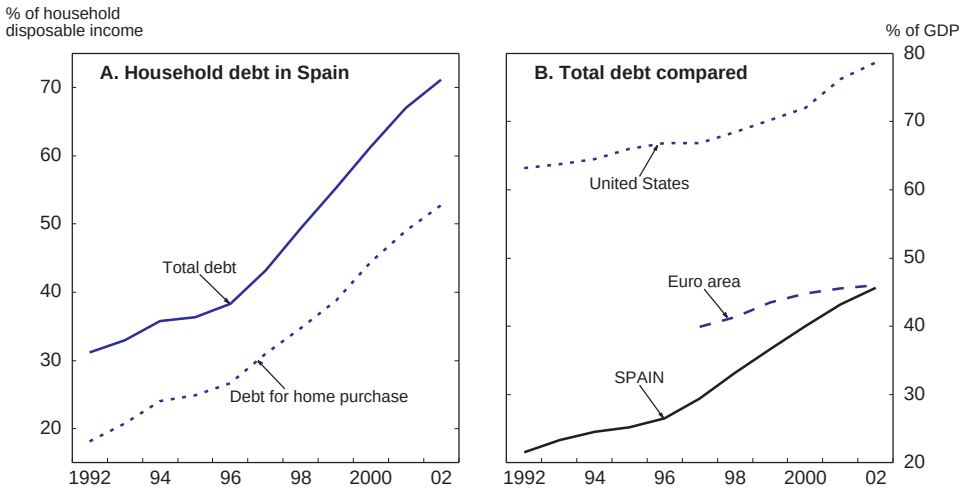
Figure 4. **Household indebtedness**[1]



1. End-of-period data. For 2002, these are Q3 2002.
Source: Bank of Spain, Board of Governors of the Federal Reserve System and European Central Bank.

the case of households, a structural adjustment to lower interest rates, and served in some cases to finance investment in Latin America or in the telecommunications sector, which was not always profitable and generated substantial losses for a few big groups. The Bank of Spain survey shows that these losses resulted in a downturn in company results which became appreciably more marked in 2002 (Table 3). However, corporate profitability excluding exceptional provisions has remained relatively high, because of the steep fall in financial costs due to lower interest rates, which more than offset the debt increase. Labour costs also eased as wage payments grew more slowly. These developments had the effect of limiting the slowdown in machinery and equipment investment, which has been less pronounced since end-2000 than during previous episodes of soft demand (Figure A1, Annex I).

Exports falter due to weakening markets and competitiveness

The marked weakening in exports is one of the main reasons for the cyclical slowdown. Foreign sales of both goods and services were affected by the deceleration. Tourism receipts fell in nominal terms in 2002, whereas they had risen uninterruptedly by over 10 per cent per year between 1994 and 2000. This fall reflects not only the decline in international tourism following the 11 September terrorist attacks, but is also attributable to certain traditional,

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Table 3. **Financial performance of non-financial firms**[1]

Per cent

	Annual sample				Quarterly sample[2]			
					2001		2002	
	Average 1991-98	1999	2000	2001	S1	S2	S1	Q3
Growth, annual rate								
Gross value added	5.2	4.7	7.5	3.9	4.2	3.9	2.2	3.0
Labour costs	4.1	6.4	7.5	5.4	5.5	4.5	3.7	3.8
Financial costs	–3.3	–2.2	26.3	12.0	29.6	9.4	–9.9	–9.5
Ordinary net result[3]	22.8	0.1	8.3	10.2	7.2	7.6	4.2	6.9
Net result[4]	17.4	–5.0	15.3	–3.1	–6.0	–11.5	–47.6	–34.4
Profitability ratios								
Return on net assets	7.1	7.5	7.4	7.6	7.7	8.2	8.3	8.5
Financial costs on total liabilities	9.2	5.0	5.0	5.0	5.5	5.3	4.6	4.6
Leverage ratio[5]	–2.1	2.5	2.3	2.5	2.2	2.9	3.7	3.9
Debt ratio	45.0	41.1	46.0	48.7	46.7	48.8	50.0	49.3

1. Data coverage is biased towards large, public and industrial enterprises.
2. For each period, growth rates are computed as the cumulative quarterly data since the beginning of the specified year over the same cumulative data of the previous year.
3. Gross economic results plus financial income less financial costs and amortisation allowances.
4. Resources less depreciation allowances.
5. Rate of return on assets less financial costs on total liabilities (in percentage points).

Source: Bank of Spain, *Statistical Bulletin*.

mainly German customers travelling less, no doubt because of the depressed state of their economy.[6] However, the competitiveness of Spanish tourism is also being eroded, with prices rising sharply in the hotel and catering trade[7] and, perhaps, too little use being made of new technologies: only 3 per cent of hotel reservations are made via the Internet, compared with nearly 40 per cent in the Nordic countries and 25 to 35 per cent in the United States and Canada (Canadian Tourism Commission, 2002). As with tourism, foreign sales of goods have also lost competitiveness, which has accentuated the effect of the slower growth of foreign markets on exports. The loss of competitiveness partly reflects the euro's appreciation against the dollar, which has reached some 20 per cent since the second half of 2000, but it is also due to prices and unit labour costs continuing to rise faster than in the rest of the euro area. Measured on the basis of prices, the loss of competitiveness has however only been perceptible since 2000, whereas in terms of labour costs it has been ongoing since the mid-1990s (Figure 5). The underlying decline in firms' financial costs as a result of lower interest rates, which has been larger than in other countries, has doubtless offset the effect of increasing labour costs on corporate profit margins. However, Spanish exporters, who steadily increased their market shares following the 1992 devaluation, are now finding it difficult to hold their position, particularly as they

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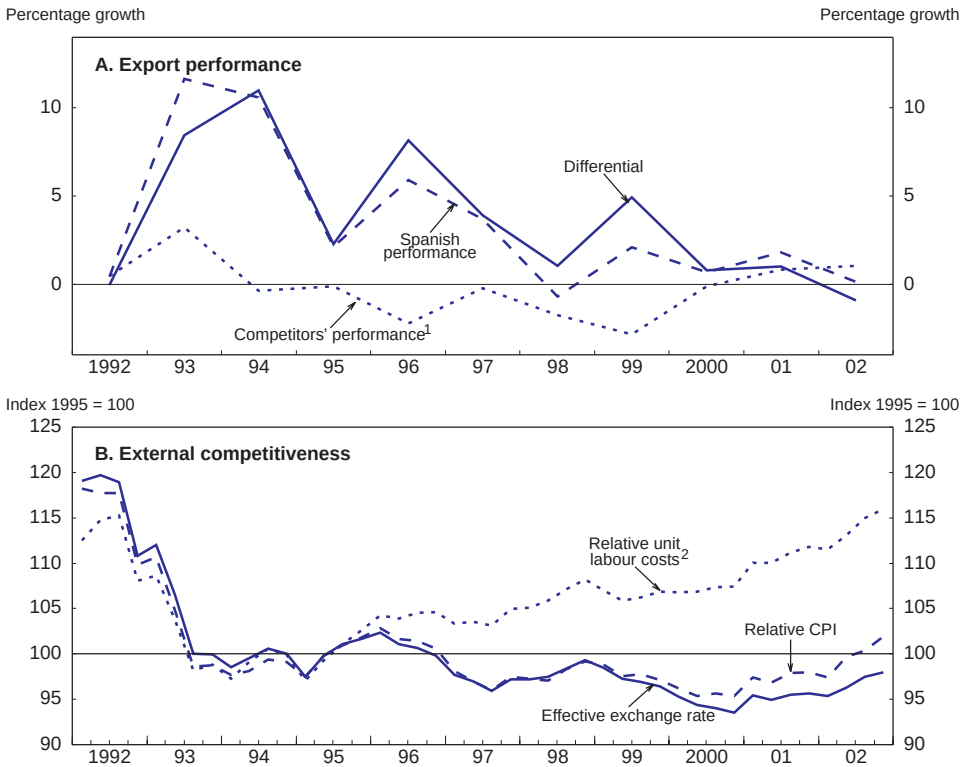
Figure 5. **Competitiveness indicators and export performance**



1. Represented by Belgium, France, Germany, Italy and Portugal.
2. In manufacturing.
Source: OECD.

are also having to contend with increased competition from the Central European countries that are in the process of joining the Union (Figure 5).[8]

In 2002, weaker exports were partly offset by a similar trend in imports, so that the contribution of the foreign balance to activity was slightly negative (Figure 6). The improvement in the terms of trade, resulting from the oil price fall (expressed in United States dollars (USD)) of some 12 per cent since the second half of 2000, plus the 5 per cent nominal effective appreciation in the exchange rate over the same period, helped to reduce the negative current account balance from 3.3 to 2.7 per cent of GDP between 2000 and 2002, despite the slight increase of the investment income deficit (Table A1, Annex I).

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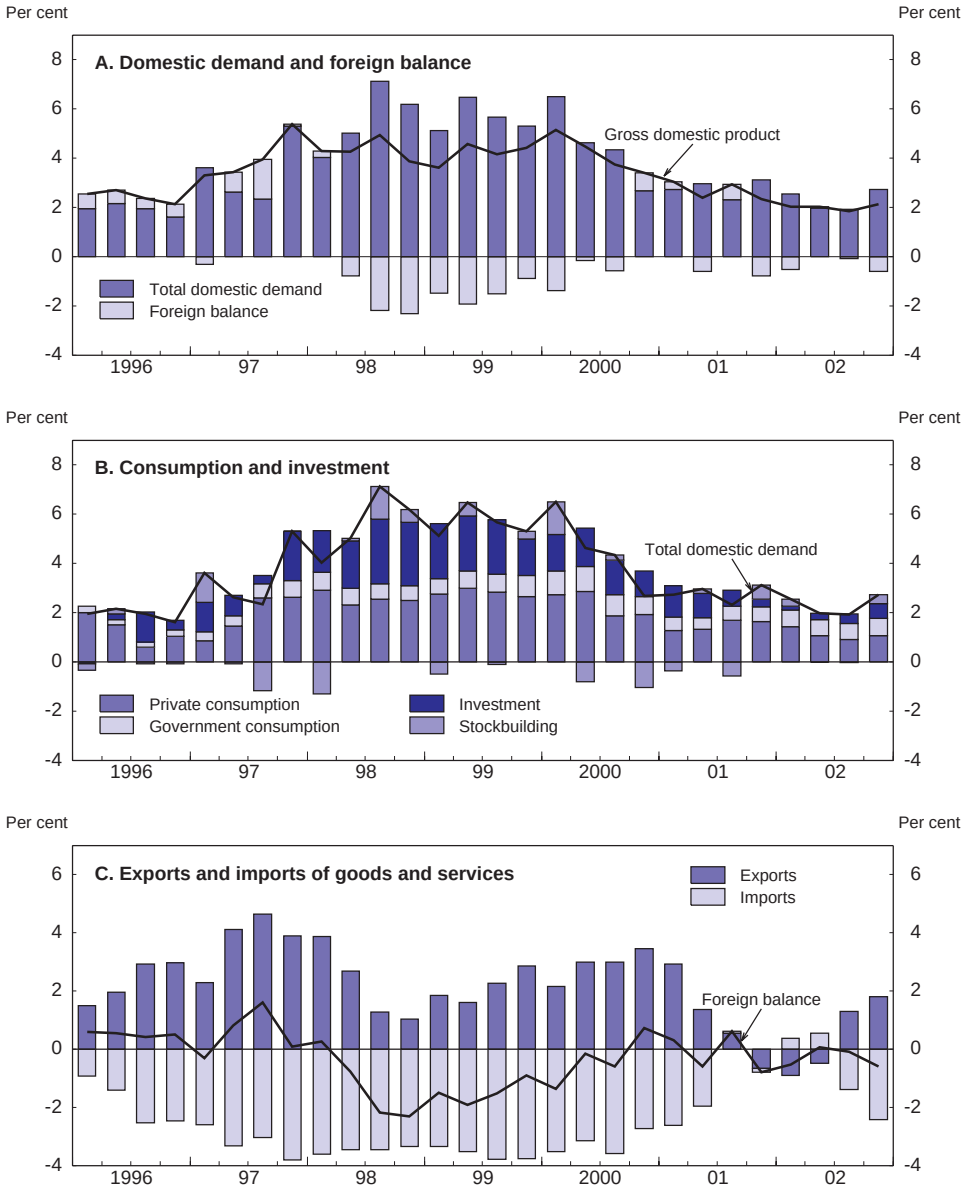
Figure 6. **Contributions to GDP growth**[1]



1. Seasonally adjusted data.
Source: OECD, *Quarterly National Accounts.*

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Job creation slows

Employment growth has slowed with the slackening of activity since early 2001, but the scale of the slowdown is uncertain. According to the labour force survey, recruitment grew by some 2 per cent in 2002, which was about half the 2000 rate (Figure 7). Though fairly similar to that shown in the national accounts, this deceleration is less marked according to the social security figures. The latter show employment growth of about 3 per cent since the beginning of 2002, reflecting an especially sharp increase in the recruitment of foreign workers. A significant proportion of this increase could, however, be due to the regularisation of immigrants, who had already been recorded in the labour force and national accounts figures (Chapter IV). All indicators agree, however, that the slowdown in job creation has eased since spring 2002. From a sectoral standpoint, the downturn in recruitment in industry seems to be over, but meanwhile job creation in construction and in market services has also lost momentum. The employment figures also point to a pick-up in the number of indefinite contracts. This is a result of the March 2001 reform which extended and indefinitely prolonged the system of subsidies for workers hired on permanent contracts adopted in 1997 (OECD, 2001a). On the other hand, the introduction under this reform of redundancy costs, albeit at a reduce rate of eight days per year of work for the termination of temporary contracts has not resulted in a marked fall in the proportion of the latter in total employment. It stood at 30.7 per cent in the fourth quarter of 2002, *i.e.* 1 percentage point less than in 2000 and in 2001, and remained twice as high as in the other EU countries.[9] Nor did the measures in favour of part-time work, adopted in March 2001, stimulate this type of recruitment, whose share (7.9 per cent of total employment) is still smaller than in the other OECD countries, especially the European ones (where it accounts for 19 per cent of total employment).

The persisting weakness of productivity gains partly reflects increased employment flexibility…

Job creation decelerated less than activity. Labour productivity growth, which was already low, thus slowed to an annual rate of only ¼ per cent since end-2000. This further slowdown is unusual, as productivity has usually accelerated during previous episodes of economic weakness (Table 4, Part A). The slowdown is probably little affected by labour hoarding because of the large share of temporary labour, for which the redundancy costs are low. Conversely, the persistence of low productivity growth could indicate greater flexibility than in the past – particularly as regards unskilled jobs – following the reforms of the mid-1990s. The economy has enjoyed an abundant and flexible supply of unskilled labour thanks to strong immigration in recent years, and this has contributed to the persisting buoyancy of employment growth in sectors such as retailing and construction, without generating excessive wage pressures. More generally, the moderation of real wages

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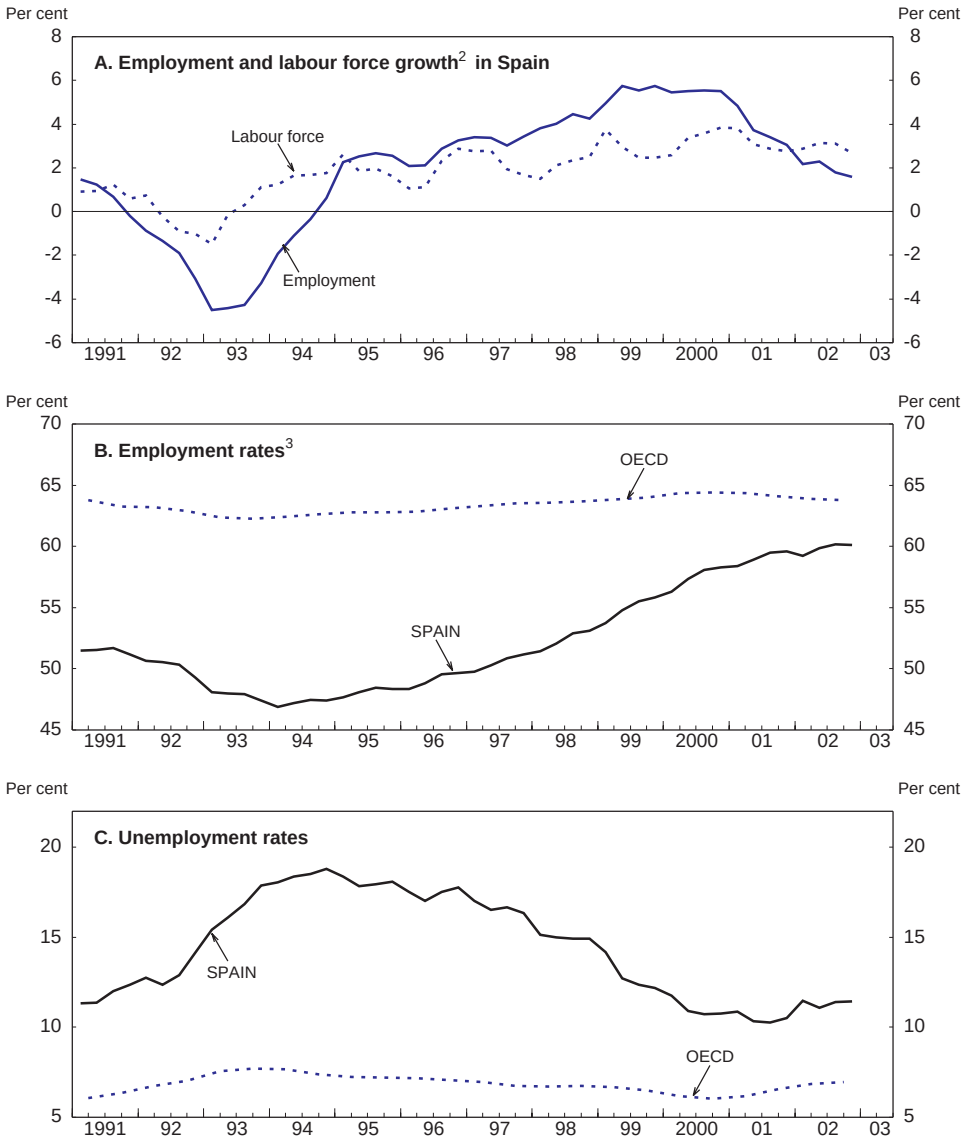
Figure 7. **Employment and unemployment:[1] key features**



1. Labour force and unemployment data before 2001 have been estimated by the Bank of Spain.
2. Year-on-year percentage changes.
3. Employment as a per cent of population aged 16-64.

Source: Bank of Spain, INE and Ministry of Economy.

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Table 4. **Productivity and real wage growth by sector**
Per cent, average growth rate over the period

		Total	Agriculture, hunting, forestry and fishing	Industry and energy	Construction	Market services	Non-market services
A. Productivity[1] growth							
1981Q1-2002Q4		1.5	5.1	2.2	1.1	0.6	0.8
1981Q1-1994Q4	(1)	2.0	6.6	2.8	1.8	0.8	0.8
1982Q4-1985Q2	(1a)	3.4	8.3	3.6	6.6	3.1	0.2
1991Q4-1994Q4	(1b)	2.4	5.5	3.0	2.2	1.8	1.7
	(1a)-(1)	1.4	1.7	0.8	4.8	2.4	–0.6
	(1b)-(1)	0.5	–1.1	0.2	0.3	1.0	1.0
1995Q1-2002Q4	(2)	0.7	2.6	1.1	–0.2	0.3	0.7
2000Q3-2002Q4	(2a)	0.4	1.0	0.6	1.1	0.1	0.5
	(2a)-(2)	–0.3	–1.6	–0.5	1.3	–0.2	–0.2
	(2)-(1)	–1.3	–4.0	–1.7	–2.0	–0.5	0.0
B. Real wage[2] growth							
1981Q1-2002Q4		0.9	1.3	1.0	0.8	0.7	0.4
1981Q1-1994Q4	(1)	1.4	2.1	1.7	1.4	1.1	0.6
1982Q4-1985Q2	(1a)	0.5	2.2	–0.4	0.2	1.3	–0.1
1991Q4-1994Q4	(1b)	2.3	1.4	3.2	3.7	1.6	1.2
	(1a)-(1)	–0.8	0.0	–2.1	–1.2	0.2	–0.7
	(1b)-(1)	1.0	–0.7	1.5	2.3	0.5	0.6
1995Q1-2002Q4	(2)	0.0	–0.2	–0.2	–0.2	0.0	0.0
2000Q3-2002Q4	(2a)	–0.2	–0.7	0.1	–0.2	–0.2	–0.9
	(2a)-(2)	–0.2	–0.5	0.3	0.1	–0.2	–0.9
	(2)-(1)	–1.4	–2.3	–1.9	–1.6	–1.0	–0.6

1. Real GDP of the sector divided by the number of full-time equivalent employment.
2. Nominal wage per full-time equivalent employee deflated by the GDP deflator.
Source: INE.

since 1995 (Table 4, Part B), which compared to expectations partly reflects negative inflation surprises, is evidence of the reduced real rigidities in the labour market. This has resulted in a somewhat higher labour intensity of production, as the slowdown in labour productivity in all branches of activity since 1995 suggests.[10]

… but it could be problematic if this trend were to persist

The deceleration in productivity gains is partly a natural consequence of the positive effects of past reforms, which have boosted job creation. The

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scale of the slowdown is moreover uncertain since both the level of GDP and its growth have possibly been underestimated in recent years.[11] The slowdown has been accentuated by the concentration of job creation in less productive sectors.[12] The sectoral shift in employment towards low value-added branches is estimated to have reduced the growth of apparent labour productivity by an average of 0.2 percentage point per year between 1995 and 2002 (Chapter IV). Timid productivity growth for a few further years will not create difficulties as long as there is real and nominal wage moderation. If, however, the trend were to persist for longer, it would jeopardise real convergence with the other European countries. Moreover, even if productivity were to pick up again in the future spontaneously in the low value-added sectors, this sort of specialisation would perpetuate a standard of living differential with the other countries. As the Bank of Spain (2001) shows, the country's productivity lag compared with its European partners is attributable mainly to the production structure, which is concentrated in low value-added sectors.[13] This specialisation, which seems to have increased in recent years, could be related to the inadequate up-take of new technologies, notably because of the distortions in the way the labour and goods markets function (Chapter III).

The recent increase in unemployment is mainly cyclical

Labour force growth, which has accelerated since 1999 to an annual rate of 3 per cent, barely slowed with the deceleration in employment, with the result that the unemployment rate has risen for the first time since 1994. It was up to 11.5 per cent in the fourth quarter of 2002 (Figure 7, Part C). The labour force has in fact continued to be fairly insensitive to economic conditions. Its growth has been dominated by the increase in immigration and the continuous rise in female participation in the labour market, which mainly reflects the trend improvement in women's level of education (Álvarez-Llorente, 2002). The slight upturn in unemployment in 2002, which affected women more than men, and also young people seeking their first jobs, would appear to be mainly cyclical. Long-term unemployment, which accounted for 38.1 per cent of total unemployment in the fourth quarter of 2002, increased slightly recently, but is more than 13 percentage points lower than in 1998. Scrutiny of the way the employment and unemployment rates have moved over the long term reveals, moreover, the remarkable structural improvement in the labour market since the mid-1990s (Figure 7, Part B). Although there is still considerable progress to be made before Spain's performance will match the OECD average, the new INE statistics[14] show that the employment rate has risen by more than 10 percentage points since 1995 and the unemployment rate has fallen by 11 percentage points.

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Nominal wage rigidity partly explains the persisting inflation differential with the euro area

Real wage moderation since the mid-1990s was accompanied by an acceleration in nominal wages as of 1999. According to national accounts data, the rise in wages per employee has reached 4 per cent on average since 2000, whereas it was limited to 2.6 per cent between 1997 and 1999. This acceleration was much more marked than in the other euro area countries (Figure 8). This was because the wage negotiation system in Spain is based on collective agreements often containing safeguard clauses that guarantee workers' purchasing power in the event of inflation surprises. Moreover, the proportion of workers covered by agreements incorporating such clauses rose from about 50 to nearly 75 per cent between 1998 and 2002. The existence of the clauses, which provide workers with an ex-post real income insurance, contributed to the adoption in 2002 of a national agreement on wage moderation negotiated on the basis of a 2 per cent inflation target. In fact, there has been a slight easing of nominal wage increases since the beginning of 2002, with collective wages rising by only 3 per cent, *i.e.* ½ percentage point less than in 2001. This moderation is likely, however, to be offset by the year-end activation of the catch-up clauses which, according to a first official estimate of the Ministry of Labour and Social Affairs, have increased wages by an extra 0,7 percentage points in 2002, instead of 0.4 per cent on average in recent years (Bank of Spain, 2002b). According to the national account data, compensation per employee growth decelerated by only 0.1 percentage points in 2002. The present system of negotiating compensation accentuates the nominal rigidity of wages, causing labour costs per worker to increase more than in the other euro area countries.[15] According to OECD estimates, this rise in per capita wage costs, coupled with slightly lower productivity growth, accounts for almost 60 per cent of the nearly 1½ percentage point inflation differential with the rest of the euro area between 1994 and 2002, as measured by the GDP deflator. The national wage moderation agreement has been renewed for 2003. It includes the same main features as the 2002 agreement. It contains again certain guarantees with regard to keeping permanent jobs in exchange for restraint on wages, which were set in accordance with Spain's official inflation target (2 per cent) and increases in productivity accruing either at the sectoral or firm level.

The indexation clauses frequently included in the collective agreements were in many cases retroactive so as to offset the impact of an inflation surprise. Such agreements pose two main problems. First, the social partners feel less responsible for inflation, whereas inflation (excluding energy and fresh food prices) is in the short term precisely the result of the interplay of their behaviour. While the authorities have stopped setting an official inflation forecast since the creation of European Monetary Union (EMU) and now refer only to an inflation target (2 per cent), the way the negotiations are organised results in the unions being

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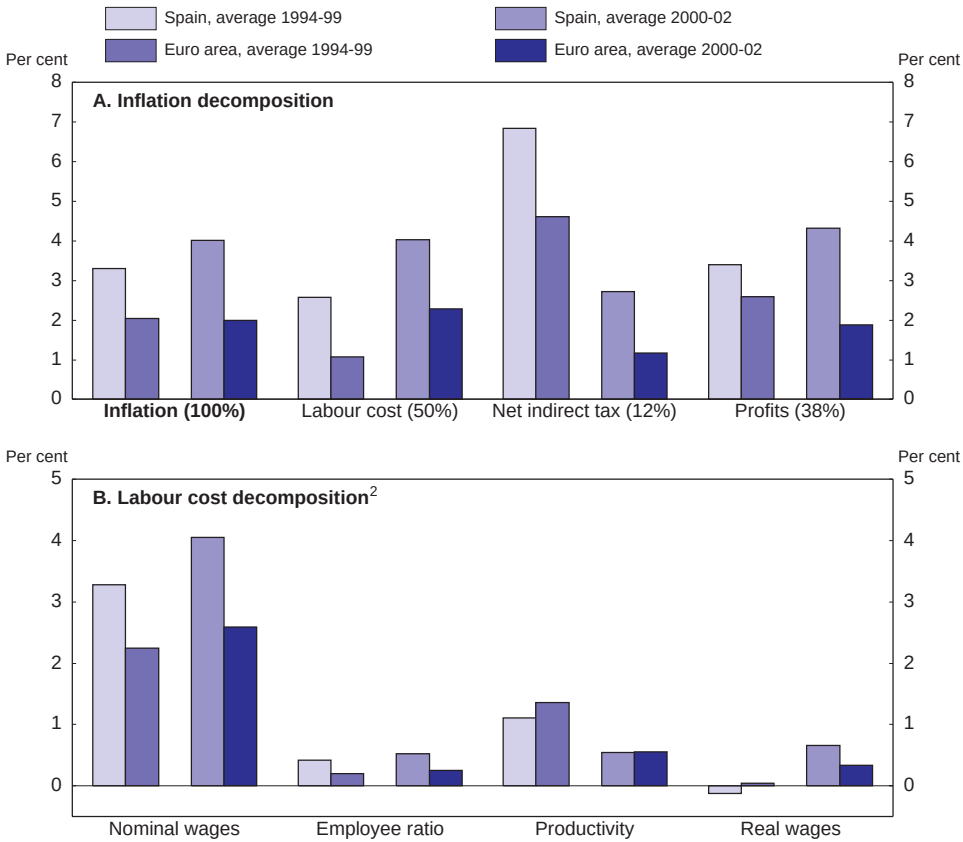
Figure 8. **Inflation decomposition for Spain and the euro area**[1]



1. The breakdown of inflation (measured by the GDP deflator) is obtained using the following accounting identity expressed in logarithm:

$$\frac{\text{GDP}}{\text{GDP volume}} = \frac{\text{Total compensation of employees}}{\text{GDP volume}} + \frac{\text{Net indirect taxes}}{\text{GDP volume}} + \frac{\text{Profits}}{\text{GDP volume}}$$

The first term represents unit labour cost of the whole economy, the second term represents the costs linked to net indirect taxes and the final term (defined as a residual of the identity) represents the contributions of profits and other revenues. Their weight in total inflation for the whole period is indicated in parentheses.

2. Labour cost (shown in panel A) is decomposed in panel B using the following identity expressed in logarithm:

$$\frac{\text{Total compensation of employees}}{\text{GDP volume}} = \frac{\text{Total compensation of employees}}{\text{Total employees}} + \frac{\text{Total employees}}{\text{Total employment}} + \frac{\text{GDP volume}}{\text{Total employment}}$$

The first term represents nominal wages, the second the employee ratio and the third productivity. Real wages are provided for information.

Source: OECD calculations based on national accounts data.

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less concerned with nominal wage increases as long as they are guaranteed in real terms. Insofar as they are able to protect their profit margins – at least in the short term – by increasing prices, firms also seem to find such agreements satisfactory, even if in the longer term the impact on international competition, coupled with the lack of any possibility to adjust the exchange rate, will no doubt be prejudicial to them, especially in the exposed sector. Second, the present system of wage indexation depends on total rather than underlying inflation, which reduces the economy's ability to adjust to exogenous shocks such as an oil price shock. This form of indexation is moreover asymmetric, compensation not being adjusted in the event of price surprises that are favourable to wage earners. This lack of flexibility penalises Spain in relation to its main partners. In the current situation, a reform of the wage formation system is vital (Chapter III).

Profit behaviour in certain sectors also keeps inflation high…

Apart from wages, profit developments since the mid-1990s also partly explain the inflation differential between Spain and the euro area average. Profit margins, calculated on the basis of the gap between the GDP deflator and unit labour costs, account for approximately one-third of the differential. Although corporate profits have tended to decrease since 1995, that trend has been less marked than in the other euro area countries. Since 1995, in fact, profit margins have fallen markedly only in manufacturing, remaining stable in market services and picking up spectacularly in the construction and energy sector since 1999 (Figure 9).

A number of possible reasons have been put forward for the differences in profit margin behaviour across branches of activity. First, demand pressures differ between sectors, this having no doubt played a part in the construction industry and tourism in recent years. Second, several analysts have pointed to price convergence with the euro area countries, even though the empirical foundation for this hypothesis is fragile.[16] There can be a number of reasons for the existence of convergence, but some of them are improbable.[17] Amongst the possible causes, mention is sometimes made of the existence of a price adjustment phenomenon caused by Spain's entry into the euro area at an exchange rate that was undervalued in relation to its equilibrium value. However, while this sort of adjustment appears consistent with the fact that profit margins have held up better than in the other countries, it ought to happen mainly in sectors exposed to international competition, a low exchange rate enabling enterprises to raise their prices without running the risk of losing markets: this has happened in tourism. In the industrial sector, which is more exposed to foreign competition, Spanish margins developed as in the other euro area countries. All in all, the most likely explanation for profits increasing more steeply than in the other countries is the differential of demand while some analysts also have suggested in the past the possible inadequacy of competitive pressures in certain sectors of the economy (Chapter III). This lack of

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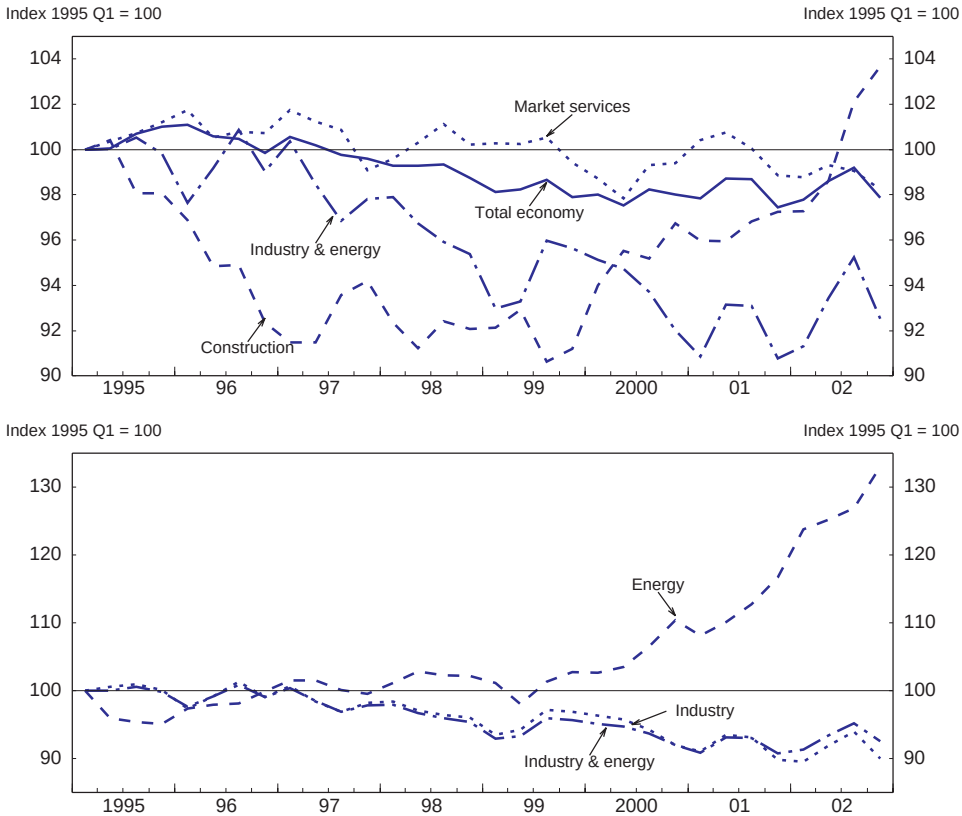
Figure 9. **Profit margins by sector**



Source: INE.

competition, according to Estrada and López-Salido (2002) may have been a major cause for Spain's dual inflation problem in the 1990s. However, the increasing inflation differential in 2002 is mainly due to exceptional reasons such as the stronger than expected impact of the euro changeover, which may have added about ½ percentage point to consumer price inflation,[18] as well as other one-off factors.

... while certain specific factors have prevented greater price moderation

The rise in the consumer price index reached 3¾ per cent in February 2003 (year-on-year), which is twice the official inflation target, while underlying inflation was 3¼ per cent (Figure 10). The lack of any disinflation

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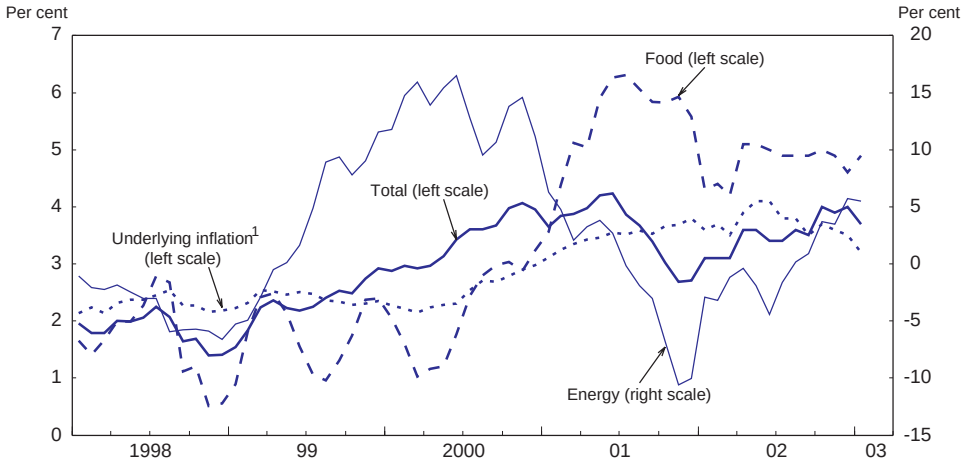
Figure 10. **Consumer prices**
Year-on-year percentage changes



1. Excluding non-processed food and energy.
Source: INE and EUROSTAT.

compared with 2001, despite lower import prices and the existence of a slightly negative output gap, was no doubt also affected by the euro changeover, the negative impact of weather on food prices and the rise in indirect taxes in early 2002, the impact of which is put at about ½ percentage point. Obtaining an accurate measurement of these specific effects, which are transitory in nature, is made more difficult by the change in method affecting the price index since 2002 (OECD, 2001a). In some sectors, the change has contributed to affect inflation measurement. With these particular factors disappearing in 2003 a more moderate increase in prices over the coming months is likely plus, perhaps, some narrowing of the inflation differential with the euro area, which has been some 1¼ to 1½ percentage points since 1999 and a cumulative 6 to 7 percentage points since 1997 (Table 5).

Fiscal policy in the short run

Prudent fiscal policy has limited the expansionary effect of monetary conditions

Monetary conditions have remained accommodating (Figure 11). With the easing of the ECB's monetary policy since end-2000 and inflation remaining higher than in the other euro area countries, the already low level of real interest

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Table 5. **Inflation differential between Spain[1] and the euro area**

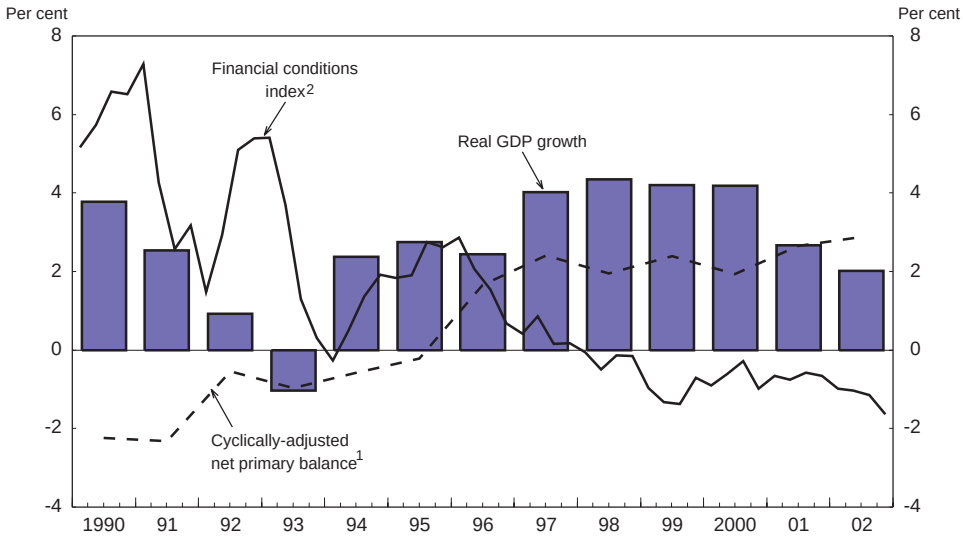
Percentage points

	Weight	1997	1998	1999	2000	2001	2002
Total	100.0	0.3	0.7	1.1	1.2	1.1	1.4
Energy	9.4	–0.3	–1.2	0.9	0.4	–3.3	0.5
Total excluding energy	90.6	0.4	0.7	1.2	1.7	1.7	1.5
Food	25.1	–1.5	–0.2	1.1	1.2	–0.2	1.9
Industrial goods	40.7	0.7	0.3	0.8	1.1	0.4	1.0
of which:							
Industrial goods excl. energy	38.1	1.0	0.5	0.9	1.5	1.7	1.2
Services	34.3	1.2	1.6	1.7	2.0	1.6	1.5
Memorandum items:							
Core inlfation[2]	82.3	0.5	0.8	1.3	1.4	1.5	1.4
Producer prices		–0.1	0.1	1.1	–0.1	–0.5	0.8

1. The break of the Spanish harmonised CPI in 2001 has been adjusted using the national CPI as benchmark.
2. Total excluding unprocessed food and energy.

Source: INE and EUROSTAT.

Figure 11. **The macroeconomic policy stance**



1. As a per cent of potential output.
2. The financial conditions indicator is computed as the weighted average of the real short-term and long-term interest rate and the real effective exchange rate (based on unit labour costs in manufacturing). The interest rate variable has a unitary coefficient while the exchange rate coefficient is the ratio of exports to GDP.

Source: OECD.

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rates dipped further, becoming negative for short rates. While the impact of this easing of conditions was to some extent offset by the appreciation of the euro against the dollar since the second half of 2002, domestic lending nevertheless continued to grow rapidly, reaching some 14 per cent during 2002. This indicates the ease of monetary conditions. Another indicator is provided by a Taylor rule, which suggests that three-month interest rates should currently be 1¾-2¼ points higher.[19]

The easing of monetary policy throughout the euro area, which reflected the general weakening of the economic situation, was accompanied by a tightening of fiscal policy in Spain. The Spanish authorities quite rightly showed greater prudence than many other EU countries, even if the restrictive stance of fiscal policy has only partly offset the expansionary effect of monetary conditions on activity. It was also possible, thanks to this policy, to continue the consolidation of the government accounts that had begun in the mid-1990s by eliminating the public deficits.

Fiscal consolidation has been structural in recent years

The fiscal consolidation begun in the mid-1990s has continued. The general government deficit, which stood at 6½ per cent in 1995, was eliminated in 2001 and a balanced budget was broadly maintained in 2002 (Figure 12). This performance, which is consistent with the official targets, was achieved despite the economic situation being less favourable than expected in 2001 and 2002. The structural improvement in the budget balance since 2000 has more than offset the negative effect of the cycle on public finances, in contrast with the euro area average. The consolidation process, which resulted in the general government primary structural balance improving by 1 per cent of GDP in the last two years, was based largely on better control of current expenditure, while the share of public investment in GDP has continued to increase since 1997 (Table 6). As a result, the reduction in the public debt/GDP ratio exceeded 12 percentage points between 1996 and 2002, falling to less than 56 per cent (Maastricht definition). Moody's has hence raised Spain's public debt rating and the long-term interest rate differential with Germany has practically disappeared.

As in 2002, the 2003 Budget is designed to balance the accounts

According to official estimates, the general government account recorded a slight deficit of 0.1 per cent of GDP in 2002. There are three main reasons for these favourable developments, despite an economic situation that is gloomier than initially forecast.[20] First, the budget was again prudent, the authorities indicating in particular that revenue projections had been based on historic elasticities (latest ten years) with respect to GDP that are notably lower with respect to

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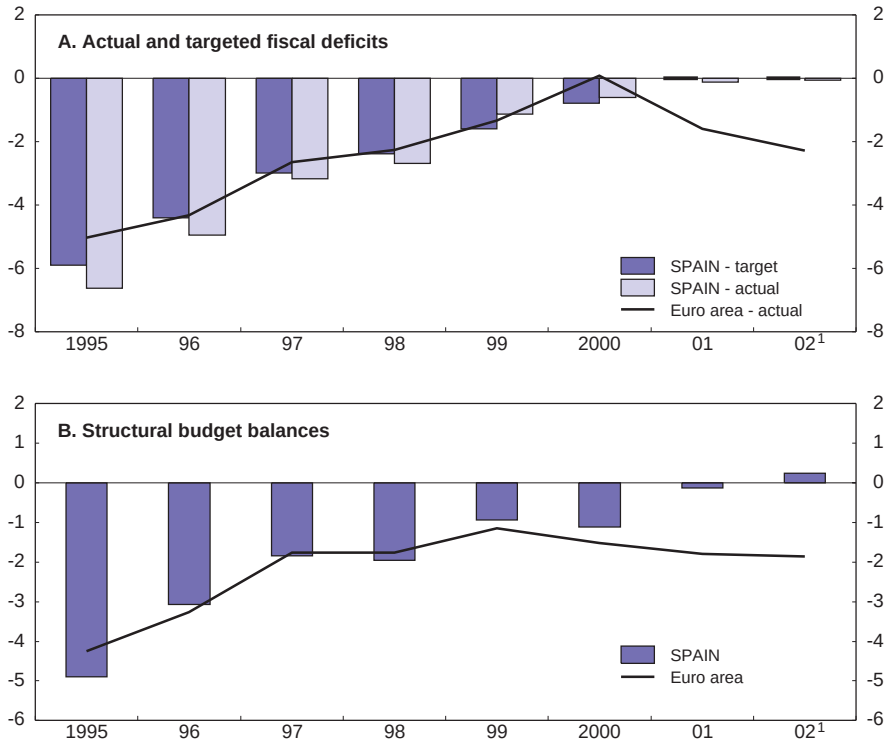
Figure 12. **Budget balances**
As a percentage of GDP



1. Preliminary estimates.
Source: OECD.

the elasticities of the latest five years. This surprise partly reflected past reforms aimed at simplifying tax collection and combating tax evasion. Second, despite the slowdown in activity, employment grew more than expected. As a result of the increase in contributions, the social security surplus and that of the public employment services (INEM) have systematically exceeded projections in recent years, despite unforeseen increases in retirement expenditure linked to adjustments in pensions, the purchasing power of which is guaranteed in the event of a surge in inflation.[21] The unexpected size of the surplus counterbalanced at least partly the overshoot on other government accounts – notably regional –

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Table 6. **The fiscal consolidation process**

	1995	1997	2000	2002[1]	2003[2]	Change over period		
						1995-97	1997-2000	2000-02[1]
Per cent of GDP								
Net lending	–6.6	–3.2	–0.6	–0.1	–0.4	3.5	2.6	0.5
Cyclical effects	–1.2	–0.8	0.5	0.1	–0.3	0.4	1.3	–0.4
Interest payments[3]	–5.2	–4.8	–3.3	–3.0	–2.9	0.5	1.5	0.3
Per cent of potential GDP								
Structural primary balance	–0.2	2.4	2.2	2.8	2.8	2.6	–0.2	0.6
Structural revenues	35.4	36.1	37.1	36.9	36.6	0.7	1.0	–0.2
Direct taxes	10.2	10.5	10.5	10.5	10.2	0.3	0.0	0.0
Indirect taxes	10.2	10.5	11.7	11.1	11.1	0.3	1.2	–0.6
Social security contributions	12.9	13.0	13.3	13.5	13.5	0.1	0.3	0.2
Other revenues	2.1	2.1	1.6	1.8	1.8	0.0	–0.5	0.2
Structural expenditure	35.8	33.8	35.1	34.2	33.9	–2.0	1.3	–0.9
Consumption	17.4	17.0	17.8	17.3	17.2	–0.4	0.8	–0.5
Investment	3.6	3.0	3.2	3.3	3.3	–0.6	0.2	0.1
Social security expenditure	13.3	12.8	12.5	12.1	12.0	–0.5	–0.3	–0.4
Other expenditure	1.5	1.0	1.6	1.5	1.4	–0.5	0.6	–0.1

1. Preliminary estimates.
2. OECD projections.
3. The fall in interest payments can be included among the structural factors as it reflects the permanent elimination of risk premiums on interest rates due to the sounder economic conditions achieved prior to joining the euro area.

Source: OECD.

in 2001 and 2002 (Table 7). Third, the rise in inflation helped to maintain brisk growth of nominal GDP and partially offset the effect of real growth falling short of the projection. In 2002, despite the increase in volume GDP perhaps falling 1 percentage point short of expectations, in nominal terms it will no doubt exceed the budget estimate.

A number of innovations affected the budget process in 2003. Drawn up on the basis of real GDP growth of 3 per cent, the aim of the budget is to keep the government accounts in balance, in accordance with the fiscal stability law which is being applied for the first time (Chapter II). This budget is also the first to be presented under the new process enacted by this law. A ceiling was thus placed on central government expenditure as early as March 2002 (with an increase restricted to 4.1 per cent above the 2002 spending projection). This ceiling includes a contingency fund equal to 2 per cent of expenditure, which has been established for unscheduled but non-discretionary expenditure inasmuch as it will no longer be possible to use end-of-year budget packages to finance unanticipated overruns. For instance, it will be used to cope with the aftermath of the

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Table 7. **General government accounts**
As a percentage of GDP

	1999	2000	2001	2002[1]	2003[2]
Current receipts	38.3	38.4	38.5	38.5	38.3
Direct taxes	10.2	10.5	10.5	10.9	10.5
Household	7.1	7.2	7.4	7.4	7.1
Business	3.1	3.3	3.1	3.5	3.5
Indirect taxes	11.7	11.7	11.4	11.2	11.2
Social security contributions	13.1	13.3	13.6	13.4	13.5
Other	3.3	2.9	3.1	3.1	3.1
Current expenditure	36.9	36.7	36.2	36.1	36.2
Government consumption	17.4	17.6	17.5	17.6	17.8
Subsidies	1.2	1.2	1.1	1.1	1.1
Social security outlays	12.4	12.3	12.2	12.2	12.2
Property income paid	3.5	3.3	3.1	3.0	3.0
Other	2.3	2.3	2.3	2.2	2.2
Net saving	1.4	1.7	2.3	2.4	2.1
Net capital outlays	2.5	2.3	2.5	2.5	2.5
Net lending	–1.1	–0.6	–0.1	–0.1	–0.4
Memorandum items:					
Net primary balance[3]	2.2	2.4	2.7	2.6	2.2
Net borrowing of:					
Central government	–1.1	–0.7	–0.5	–0.6	–0.5
Social security	0.2	0.5	0.8	0.8	0.5
Local and regional government	–0.2	–0.4	–0.5	–0.4	0.0

1. Preliminary estimates.
2. OECD projections.
3. Difference between the actual balance and net interest payments.
Source: Ministry of Finance and OECD.

Prestige oil tanker disaster, with budget costs initially estimated at around EUR 250 million. The 2003 Budget is, moreover, the first to be drawn up since the full implementation of the new financing agreement with the autonomous communities and the transfer to these regions of health spending, which changes the contours and responsibility of the main levels of government (Chapter II). However, these institutional changes make detailed comparisons of the government accounts with previous years difficult.[22] In the official documents, moreover, detailed accounts are provided only for central government (*i.e.* excluding regional and local government), which accounts for less than 55 per cent of the total budget.

On the revenue side, the main measure in the 2003 Budget concerns personal income tax cuts, which are officially estimated to amount *ex ante* to

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EUR 3 billion in 2003 and EUR 0.6 billion in 2004 (making a total of ½ per cent of GDP). Coupled with the tax cut is a freeze on excise taxes on fuels and the abolition, for about 95 per cent of taxpayers, of the local tax on economic activities, which affects companies. Taken together, these measures explain why the 4.3 per cent increase in receipts is lower than the projection for nominal GDP growth (5.8 per cent). As already noted, the increase in State spending is capped at 4.1 per cent, with investment, spending on justice and security and social expenditure being given priority. When preparing the budget, an agreement was signed with public sector employees that provides for a 3.3 per cent increase in per capita wages in 2003, which includes a uniform 2.7 per cent adjustment and a productivity-linked increase of 0.6 per cent.[23] Against this background, it is planned to extend the opening hours of certain public services and to cut absenteeism by 20 per cent. The agreement also aims to put an end to the requirement that only one retiring civil servant in four be replaced, which has led to a sharp rise in temporary employment in the public sector. The intention in fact is to reduce the proportion of temporary jobs in the civil service from 22 to 8 per cent over the next few years.[24]

In the present context, the neutral stance of fiscal policy is appropriate

A number of criticisms have been levelled at the budget, the main one concerning the lack of transparency of some of the statistical information. This is probably a temporary problem deriving in particular from the ongoing process of decentralisation, but it is also due to sometimes not very homogeneous presentation of the government accounts.[25] Also, the macroeconomic projections look somewhat optimistic. The 3 per cent growth assumption, close to potential growth, which was made by the authorities in March 2002, when the central government spending ceiling was decided, is in fact ½ percentage point higher than the consensus forecast of March 2003. Moreover, consumers may save more of the tax cut than projected. According to the authorities, the net cost of the tax reform will be EUR 2 450 million in 2003, *i.e.* 80 per cent of the tax reduction estimated *ex ante*, because of the positive effect of the increase in activity on receipts. Since the tax cut takes place in a situation where government expenditure growth is subject to strict control, agents could however perceive it as permanent and therefore spend a sizeable part of the cut. Moreover, the social security budget appears to be based on prudent assumptions, bearing in mind the still brisk growth in employment and the continuing strong rise in nominal wages. A surplus in excess of the ½ per cent of GDP forecast for social security would again make it possible to offset a possible overrun on the regional accounts which are supposed to balance their budgets. All told, the budget could record a slight cyclical deficit in 2003 as the economy could be a little less robust than expected in the official projections. The neutral stance of fiscal policy

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in 2003, following the restrictive policy pursued in 2001 and 2002, suits economic conditions, with growth remaining slightly short of potential and the output gap negative. Even though inflation developments remain worrying, the solution to this problem should come from structural reforms rather than fiscal policy.

Prospects

According to the OECD's projections, presented in Table 8, economic activity should accelerate during the second half of 2003.[26] Conjunctural indicators at the beginning of the year do not suggest an imminent pick-up of activity. Industrial production has improved during the last months of 2002 as well as business confidence in January 2003, but consumer confidence weakened and car sales continued to fall. An export-led recovery in the second half of 2003 should be moderate due to the recent appreciation of the exchange rate and should pull equipment investment, while growth in the construction sector should ease slightly but still remain buoyant. In a context of a neutral fiscal policy overall, private consumption should strengthen, incomes being bolstered by income tax cuts, sustained employment growth and the wage catch-up pursuant to the indexation clauses activated following the inflation spurt in 2002. All told, GDP should grow marginally above 2 per cent in 2003, as in 2002, before accelerating to over 3 per cent in 2004, a rate higher than the euro area average. These figures are slightly below the March 2003 consensus projections and about 1 percentage point below the official forecast.

With the soft recovery and the continued strength of the labour force, unemployment rate might increase slightly in 2003 before edging down in 2004. Given a negative cyclical gap over the period, import prices remaining moderate with the exchange rate appreciation, and the absence of special factors like the euro changeover, inflation could fall back to less than 3 per cent in 2003 and below 2½ per cent in 2004. The decline in the inflation differential with the euro area is likely to be limited.

There are two major risks to this central scenario. The first relates to the strength of external demand which supports the recovery as from the second half of 2003. Although a sharper than expected acceleration in world economic activity cannot be ruled out, recent developments in the external environment point to the existence of downside risks, especially in some large European countries. A slower than expected international recovery would undoubtedly worsen the outlook. Business investment would also be lower if business confidence is not fully restored because of continuing international uncertainties or if new disturbances in Latin America further erode the profits of Spanish firms that have invested in that region. In such a context, a further loss of confidence on the part of consumers might cause them to reduce their spending and instead save most of the tax cut

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Table 8. **Short-term prospects**[1]
Percentage changes from previous period[2]

	Current prices EUR billion 1999	2000	2001	2002	2003	2004
Private consumption	335.2	3.9	2.5	1.9	2.3	3.2
Government consumption	98.6	5.0	3.1	3.8	3.5	3.0
Gross fixed investment	136.1	5.7	3.2	1.4	2.6	4.1
Construction	72.4	6.1	5.8	4.5	3.3	3.3
Machinery and equipment	63.8	5.1	0.3	–2.2	1.8	5.0
Stockbuilding[3]	2.6	–0.1	–0.0	0.1	0.0	0.0
Total domestic demand	572.6	4.4	2.7	2.2	2.6	3.4
Exports of goods and services	155.5	10.1	3.4	1.4	3.9	5.8
Imports of goods and services	162.8	10.6	3.5	2.2	5.3	6.6
Foreign balance[3]	–7.4	–0.3	–0.1	–0.3	–0.5	–0.4
Gross domestic product	565.2	4.2	2.7	2.0	2.1	3.1
Household saving ratio (%)[4]		10.0	10.2	10.3	10.6	10.4
Employment[5]		5.5	3.7	2.0	1.4	2.1
Unemployment rate (%)		11.0	10.5	11.4	12.0	11.7
Private consumption deflator		3.2	3.3	3.6	2.9	2.4
GDP price deflator		3.5	4.2	4.4	3.0	2.4
Wage rate, total economy		3.7	4.1	4.3	3.7	3.5
Unit labour cost, total economy		3.7	4.2	4.3	3.2	2.6
Output gap (%)[6]		0.4	0.0	–0.8	–1.5	–1.2
As a % of GDP:						
Current account balance		–3.5	–2.6	–2.3	–2.4	–2.4
Net government lending		–0.6	–0.1	–0.1	–0.4	–0.2
Primary government balance		2.4	2.7	2.6	2.2	2.2

1. The projections are based on the following assumptions: oil prices are assumed to stay at USD 31 per barrel in the first half of 2003 before falling to USD 25 in the second half and remain there until the end of the projection period; exchange rates are fixed at their level of 7 February 2003 (1 EUR = 1.08 USD), implying an appreciation of the euro *vis-à-vis* the US dollar of 20 per cent since the first half of 2002.
2. At constant 1995 prices for GDP and its components.
3. Contribution to GDP growth.
4. Household saving as a percentage of household disposable income.
5. Labour force survey definition.
6. Deviations of actual GDP from potential GDP as a percentage of potential GDP.

Source: OECD.

granted in 2003. The materialisation of this risk, a quantification of which is provided in Box 1, would result in a cyclical deterioration of the budget balance, which could be accommodated, and a more marked reduction of inflation, although this would have little effect on the differential with the euro area. The second major risk concerns the evolution of the price of oil, which could pick up in case of a prolonged conflict in the middle-East, but could also be weaker in particular if the

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Box 1. Risks to the outlook

In order to evaluate the main risks surrounding the projections for 2003 and 2004, an alternative scenario has been drawn up with the OECD's Interlink model. It chiefly concerns possible developments in the international environment, but also the implications for households' behaviour.

The scenario quantifies the impact of a weaker than projected pickup of output (½ percentage point) in 2003 and 2004 in the major European countries and their close neighbours, because of a less dynamic domestic demand.* Activity would then also weaken in Spain by ¼ percentage point this year and next relative to the baseline scenario, so that growth would amount to 1¾ and below 3 per cent respectively in 2003 and 2004 (Figure 13). Inflation, which would show little change in 2003, would slow by almost ½ percentage point to around 2 per cent in 2004. In the alternative scenario, where the household saving ratio is virtually the same as in the baseline scenario, the slowdown in activity induces a rise in the unemployment rate, which does not fall below 12 per cent until 2004. A variant of this scenario evaluates the impact of a more prudent saving behaviour by households. It is assumed that in the event of a less favourable macroeconomic performance, households would increase their saving by an amount equivalent to half of the tax reduction granted to them *ex ante* in 2003 and 2004 (equivalent to ¼ per cent of GDP bearing principally on 2003). In that case, GDP growth would amount to 1.6 and 2.9 per cent respectively in 2003 and 2004. With this variant, the budget would show a deficit of 0.6 per cent of GDP in 2003.

* In the alternative scenario, as in the variant of it discussed a little later, nominal interest and exchange rates are the same as in the baseline scenario. The fiscal stance as measured by the structural primary balance is also unchanged. Given the linearity of the OECD's Interlink model, the simulations made can also be used (changing the sign of the impact) to evaluate the consequences of a stronger international upturn.

current crisis is resolved quickly. An oil-price shock would probably have much the same effect on short-term growth in Spain as in other European countries. According to the OECD's standard simulations, an oil-price hike (fall) of USD 10 per barrel in early 2003 would cut (raise) output growth by ¼ percentage point in 2003. The rise (fall) in inflation might be roughly the same as in other European countries in 2003 (½ percentage point) but steeper in 2004 owing to wage indexation. The result would be a widening (reduction) of the inflation differential. The gap, were it to persist, would erode the economy's competitiveness and in the medium term make it difficult to sustain the good performance recorded in recent years (Chapter III).

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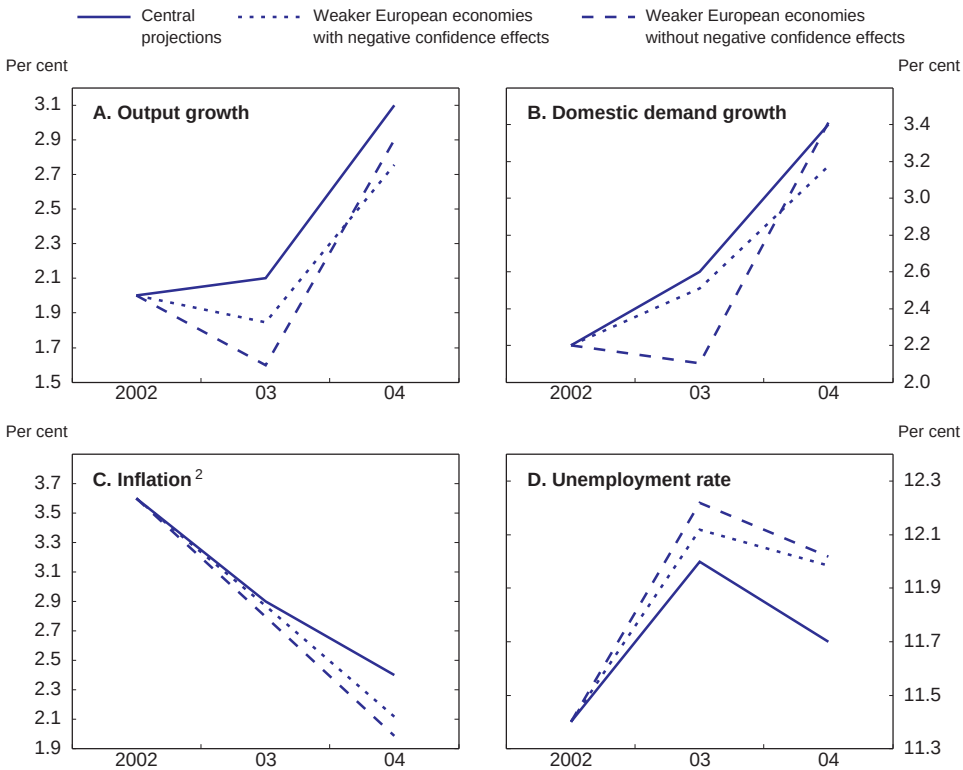
Figure 13. **Risks of a weaker European recovery**[1]



1. Represented by Belgium, Austria, France, Germany, Italy, Netherlands, Portugal and United Kingdom.
2. Change of the private consumption deflator over the previous year.

Source: OECD.

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II. Medium and long-term fiscal policy challenges

Beyond the short term, fiscal policy will face three major challenges. The first challenge will be to preserve the progress made with fiscal consolidation, while at the same time maintaining sufficient flexibility in the new, more decentralised fiscal framework that is in force since 2002-03. The second medium-term challenge relates to raising public sector efficiency further in this new framework. Thirdly, and over the longer term, reforms are needed to prepare Spain for the consequences of population ageing as the viability of the public finances will be threatened by the steep rise in pensions and medical and social expenditure as of 2020-25.

Fiscal policy: being prudent, while ensuring flexibility

A *prudent fiscal policy and further structural reforms are necessary in certain areas*

A prudent fiscal policy over the medium term is desirable as it would underpin the gains from the structural reforms that are still necessary to achieve real convergence with the most developed countries. To this end, the latest Stability Programme (2002-06) is designed to keep the general government account balanced or to generate a small surplus (Table 9). The Programme is based on projected real GDP growth of 3 per cent per year between 2004 and 2006, which seems realistic and implies a neutral fiscal policy up until 2006. Revenue will remain approximately stable as a proportion of GDP, following the 2003 tax cut, while the share of expenditure will dip marginally as a result of a further reduction in current spending, which will be only partly offset by higher public investment. The Programme, however, rests on the assumption that the inflation differential with the euro area will progressively decline (inflation, measured on the basis of the private consumption deflator, is projected at 2½ per cent between 2004 and 2006). If the differential persists, however, Spain would probably have to contend with lower real interest rates and more relaxed monetary conditions than the rest of the euro area. This could create an unbalanced policy mix. Therefore,

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Table 9. **Stability Programme 2002-06**

	Annual average percentage growth		
	2001	2002	Average 2003-06
Macroeconomic scenario for Spain			
Real GDP	2.7	2.2	3.0
Private consumption	2.5	1.9	2.9
Government consumption	3.1	3.0	3.0
Gross fixed capital formation	3.2	1.7	3.7
External sector[1]	–0.1	0.0	–0.2
Private consumption deflator	3.3	3.5	2.7
Employment	2.4	1.1	1.8

	Per cent of GDP					
	2001	2002	2003	2004	2005	2006
Government net lending[2]						
Spain	**–0.1**	**–0.2**	**0.0**	**0.0**	**0.1**	**0.2**
Austria	0.0	0.0	0.0	0.2	0.5	..
Belgium	0.0	0.0	0.0	0.3	0.5	..
Denmark	2.8	2.1	2.2	2.5	2.4	2.2
Finland	4.9	3.8	2.7	2.1	2.6	2.8
France	–1.4	–2.8	–2.6	–2.0	–1.3	–0.5
Germany	–2.8	–3.8	–2.8	–1.5	–1.0	0.0
Greece	–1.2	–1.1	–0.9	–0.4	0.2	0.6
Ireland	1.6	–0.3	–0.7	–1.2	–1.2	–1.2
Italy	–2.2	–2.1	–1.5	–0.6	–0.2	0.1
Luxembourg	4.1	–0.3	–0.3	–0.7	–0.1	..
Netherlands	0.1	0.7	–1.0	–0.7	–0.4	0.1
Portugal	–2.2	–2.8	–2.4	–1.9	–1.1	–0.5
Sweden	4.8	1.7	1.5	1.6	2.0	2.0
United Kingdom	–0.2	–1.8	–2.2	–1.7	–1.6	–1.6
Euro area[3]	–1.5	–2.2	–1.7	–1.0	–0.6	0.0

1. Contribution to GDP growth.
2. According to the 2001-05 Stability and Convergence Programmes (SCP) for Austria; 2002-06 SCP for the other countries. Government balances in 2001 exclude one-off proceeds from the sale of UMTS licenses.
3. OECD estimates.

Source: Ministry of Economy, *Stability Programme Update Kingdom of Spain* 2002-2006; European Commission, "Public Finances in EMU – 2002", *European Economy*, No. 3/2002 and Ministry of Finance for the countries publishing their 2002-06 SCP.

further structural reforms in certain areas would be required to reduce nominal wage and price rigidities and remove the inflation differential permanently.

The reform of autonomous communities' finances will increase the regions' responsibilities…

Fiscal consolidation has prompted a change in attitudes, with a large majority of the Spanish population endorsing the goal of balancing the accounts.

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Maintaining a balanced budget in the medium term could be, however, made more complicated by decentralisation. The transfer of health service management to regions ended in 2002, so that the decentralisation of the biggest spending items is complete, making Spain one of the most decentralised OECD countries (Table 10). In order to preserve the progress made with fiscal consolidation in the context of a more decentralised state, the authorities recently introduced two major reforms. The first establishes a closer link between responsibility for spending and taxation among the different levels of government. The decentralisation of health spending was accompanied by a reform of the method of financing the regions, which had been in force since 1997. The new system, described in Annex II, increases the communities' financial self-sufficiency and fiscal responsibility. Regional tax receipts are expanded and made more stable, covering 65 per cent of these authorities' resources, as opposed to 29 per cent under the previous system, and are supplemented by unconditional transfers. The tax revenue at the regions' disposal consists of a broad basket of taxes comprising a third of personal income tax, 35 per cent of the value added tax (VAT), and 40 per cent of excise tax

Table 10. **Expenditure by level of government**
In per cent of total consolidated public expenditure

	Consolidated central government	State, regional or provincial governments	Local governments
Spain (2001)[1]	59.0	27.5	13.5
Spain (2002 estimates)[1, 2]	**52.6**	**33.9**	**13.5**
Australia (1999)	55.6	38.8	5.6
Austria (1999)	68.9	15.7	15.4
Canada (2000)	41.4	43.0	15.6
Germany (1998)	62.5	23.7	13.9
Switzerland (1999)	52.6	28.0	19.4
United States (1999)	48.5	29.7	21.8
Federal countries	54.9	29.8	15.3
Denmark (2000)	53.6		46.4
Finland (1998)	64.0		36.0
France (1997)	82.5		17.5
Italy (1999)	75.6		24.4
Netherlands[1] (1997)	78.3		21.7
Norway (1998)	66.2		33.8
Sweden (1999)	64.6		35.4
United Kingdom (1998)	77.9		22.1
Unitarian countries	70.3		29.7

1. Local governments also include provincial ones.
2. The structure of expenditure has been estimated assuming that all health spending managed by the central government, which represented 1.7 per cent of GDP (or about 4¾ per cent of total government expenditure), was transferred to the regions.

Source: IMF, *Government Finance Statistics Yearbook*, 2001 and information provided by the Spanish authorities.

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on alcohol, tobacco and fuels, as well as the totality of a number of other taxes collected in the region.[27] Moreover, the reform extends appreciably the communities' powers with regard to taxation, particularly income tax, and it abolishes the state guarantee ensuring that growth of their main source of tax revenue is at least equal to that of GDP. In conjunction with these changes, a reform of the system of local authority financing has also been launched with the objective, in particular, of increasing the municipalities' budgetary responsibilities. The former will also have increased powers over their taxation, while large towns will, like the regions, have a proportion of central government taxes transferred to them (Annex II).

... as will the fiscal stability law...

The second important reform is enshrined in the fiscal stability law, which was reviewed in the previous *Survey* (OECD, 2001a) and which came into effect with the 2003 Budget. The law is based on four main principles:

- *Transparency*: comprehensive information will be made available rapidly so as to monitor the financial situation of the different levels of government.
- *Joint financial responsibility*: each level of government will be required to keep its accounts in balance or in surplus. In the event of a deficit, a plan of action aimed at correcting that situation will be presented.
- *Central government expenditure* will be limited by setting a spending ceiling consistent with the medium-term projections and a contingency fund will be established, as was already the case with the 2003 Budget.
- *Fiscal discipline* will be mostly underpinned by monitoring and peer pressure. It will also be guaranteed by the central government's power to impose debt limits on both regional and local authorities should these entities fail to comply with their budgetary stability commitment. It is not, however, planned to impose penalties on authorities that do not comply with fiscal discipline, unless this results in the overall government accounts overshooting the Maastricht deficit limit of 3 per cent of GDP.

... which needs to be implemented flexibly

Coupled with the increased fiscal responsibility of the regions and local authorities, the law should help to maintain fiscal discipline over the medium term. To the extent, however, that the new framework does not take the effect of the economic situation on the budget explicitly into account, it is likely to lack flexibility, especially where regional and local government is concerned. The resources of the communities (and of the big municipalities), which are more dependent on taxes, are now more influenced by the economic situation than they were in the past. However, a large proportion of their expenditure, such as that on health, is not sensitive to the cycle. Strict application of the stability law

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would have pro-cyclical effects on activity. Furthermore, it would oblige all the regions to keep their accounts in significant surplus so as to comply with the balanced budget rule in a downturn. To avoid these drawbacks, advantage should be taken of the flexibility built into the law. The deficits need to be corrected only in the medium run, as foreseen in the law, ensuring that fiscal discipline and automatic stabilisers are fully compatible. Moreover, the law explicitly allows the possibility of deficits in "exceptional circumstances". Alternatively, a change in the rule could be envisaged with the objective of ensuring structural rather than nominal budgetary balance, as was done in Switzerland for example. Nevertheless, the practical difficulty of calculating structural budgets for the different levels of government, including the local level and public companies, should not be underestimated.

Increasing public sector efficiency has to remain a priority

In addition to continuing prudence, another major fiscal policy concern over the medium term will be to improve the quality of public finances from the standpoint of both taxes and expenditure.

Another reform of the personal income tax in 2003

A better tax system would underpin the medium-term strategy of converging with the EU average per capita GDP. An important reform of the personal income tax was introduced in 1999, consisting in a reduction of the tax burden, an extension of the tax base to include all forms of capital revenues and a re-organisation of tax deductions, which were simplified and mostly applied to the tax base rather than as tax rebates. The reform was endorsed in a special chapter of a previous survey (OECD, 2000), although it was also signalled that some areas remained where further improvement is desirable, especially concerning the neutrality of taxation with respect to long-term savings. At the end of 2002, another reform of the personal income tax was approved to be implemented in 2003, including average tax cuts of 11.1 per cent with a budgetary cost of EUR 3.6 billion (0.5 per cent of GDP). In particular, the number of tax brackets was reduced from 6 to 5, and the top and bottom marginal rates have been brought down from 48 to 45 per cent and from 18 to 15 per cent, respectively.[28] The reform also includes increased family allowances and their extension to new family circumstances, while new work-related deductions aim at improving the supply of labour of older workers, working mothers of young children and unemployed workers, who accept a job in a different province.

Concerning the taxation of income from savings, the basic structure of tax allowances has not been changed although some aspects have been modified. The fixed rate for capital gains has been reduced (from 18 to 15 per cent), while the deduction for income on capital held for more than two years has been raised

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(from 30 to 40 per cent). Capital gains generated in mutual funds will be exempted from the tax when they are re-invested in other funds, in order to raise capital mobility across funds and thus improve competition in the sector.[29] The ceiling of tax-exempted contributions to private pension funds has been increased, and will be extended to new types of schemes. To favour the market for rental housing, the reform introduces a reduction of 50 per cent of net rental income for owners who offer residential dwellings for rent, plus deductions of all types of expenses, which in practice will imply a tax rate close to zero for rental income.[30] The reform also modifies the special regime for small self-employed and unincorporated businesses, which is based on special parameters for these businesses instead of income measured by proper accounting of their profits. A new threshold defined in terms of purchases (fixed to a maximum of EUR 300 000), which should be easier to control, has been set up for the small enterprises to benefit from this tax regime. Also, it is now required that activities are carried out in the territorial scope of the personal income tax.[31]

The reform will result in a further reduction of the tax burden, which will have a positive impact on the supply of labour and investment (Annex III). Incentives to work will be further enhanced by the specific deductions for older workers and mothers of young children, which should improve participation in the labour market of these two groups with lower than average employment ratios. The mobility incentive for unemployed workers should also help to reduce structural unemployment marginally. Tax brackets and many tax allowances and limits are not indexed, but the reform more than compensates for the lack of inflation adjustment in recent years.[32] In order to make the tax system more transparent, a more systematic inflation adjustment of tax brackets and allowances could be advisable, though this should be done in the context of comprehensive tax structure decisions. The new conditions on the application of the special regime for self-employed income is welcome, as it will help to limit tax fraud, but it would be better to completely phase out the special regime, which would improve horizontal equity.

With respect to taxation of savings, progress towards greater neutrality is limited, and the tax system continues to favour savings in the form of pension funds, insurance contracts and housing purchases (OECD, 2000). Though the first may raise incentives to contribute to complementary private pension funds, the latter two are more difficult to justify. In particular, tax credits (which amounted to 0.5 per cent of GDP over the last years) benefit all tax payers when buying their main residence independently from their income level, with uncertain distributional outcomes even though the ceiling of this deduction has been kept unchanged since 1998. Neutral taxation of housing relative to other forms of savings would require imputing rents for home ownership and taxing capital gains from home sales in line with other forms of savings after deducting mortgage interest payments (OECD, 2001b). In 1999, taxation of imputed income of the main

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home was suppressed for the sake of simplicity, since such income was difficult to estimate, but interest deductibility was maintained. When imputed rents are not taxed, suppressing mortgage interest deductibility would go in the direction of more neutrality, and would also help to re-direct housing demand to the rental market. This would be less costly than the granting of the new deduction for rental housing included in the recent reform; if problems concerning the supply of rental housing are detected (such as slow procedures to recover unpaid rents), the root of this problem should be addressed.

The rationalisation of the local tax system has also progressed

A rationalisation of certain aspects of local taxation has also been recently adopted. The reform includes the abolition of the tax on business activities (IAE) for about 95 per cent of taxpayers. This tax affects now only companies with a net turnover over EUR 1million without taking account of their income or wealth and undermines horizontal equity and incentives to recruit and invest. The reform will ensure that local authorities continue to have sufficient resources, while at the same time removing some major distortions (Annex II). However, it is unlikely to reduce the dependency of municipal resources on real estate taxes, which appears excessive in the case of Spain. Two-thirds of local taxes (or a quarter of all non-financial resources) are property taxes which municipalities tend to increase with rising house prices using restrictive practices as regards the supply of building land (Chapter III). User fees or charges are, on the other hand, less utilised than in other countries (Table 11). More frequent recourse to this type of revenue

Table 11. **Local government revenue in OECD countries with federal structure**
In per cent of total revenue

	Tax revenue	Non tax revenue		Transfers
		Total	*of which*: Fees, sales, fines	
Spain (1997)	**50.4**	**16.2**	**7.7**	**33.4**
Australia (1998)	45.2	38.5	24.2	16.3
Austria (1999)	56.7	25.0	21.0	18.3
Belgium (1998)	37.9	9.1	..	53.0
Canada (2000)	41.7	17.3	13.7	41.0
Germany (1999)	41.0	24.7	14.2	34.3
Mexico (1998)	68.0	19.9	9.2	12.1
Switzerland (1999)	47.9	34.7	26.6	17.4
United States (1999)	38.7	22.9	11.8	38.4
Non-weighted average (excl. Spain)	47.1	24.0	17.2	28.9

Source: OECD, *Revenue Statistics* 1965-2001, 2002 ed.

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would be a good way of financing local services, including those relating to environmental concerns (for instance, for water or waste management), a source of revenue that is likely to increase in the future. Greater use might be made of this type of revenue[33] if there was a better understanding of the costs of the services and of the externalities to be taken into account. This could be fostered by technical assistance of the central government to the many very small municipalities.

The tax to GDP *ratio is well below the* EU *average*

Following the past reforms, in particular the 1999 reform of the personal income tax, the tax to GDP ratio is considerably below the EU average (Table 12), and it will be reduced further by the recent reform. As a proportion of GDP, Spain has lower income tax revenue than most OECD countries, but indirect taxes and social security contributions are above the OECD (but below the EU) average. Future tax cuts should contribute to reduce distortions concerning labour market and savings behaviour, but must be balanced against the need for improving public infrastructure, R&D investment and human capital formation, where Spain lags

Table 12. **Tax revenues in OECD countries**
In per cent of GDP, 2001

	Indirect taxes	Taxes on individuals	Corporate taxes	Social security contributions	Total taxes
Spain	**11.4**	**7.4**	**3.1**	**13.6**	**35.4**
Austria	14.7	11.8	3.3	17.0	46.8
Belgium	12.6	14.1	3.5	15.3	45.5
Denmark	17.3	27.4	2.5	3.2	50.4
Finland	13.2	14.9	4.6	12.5	45.2
France	15.1	9.3	3.3	18.3	46.0
Germany	11.9	10.4	0.6	18.5	41.5
Greece	14.4	5.8	4.3	13.5	37.9
Ireland	12.2	9.1	3.8	5.7	30.8
Italy	14.6	11.9	3.3	12.7	42.4
Netherlands	12.6	7.6	4.2	15.3	39.9
Portugal	14.4	5.4	4.4	12.0	36.2
Sweden	16.4	18.2	4.1	15.5	54.1
United Kingdom	13.6	13.2	3.6	7.6	38.0
European Union[1]	13.5	11.0	2.8	14.5	41.8
Australia	13.7	12.1	4.4	0.0	30.2
Canada	12.8	13.2	3.9	5.0	34.9
Japan	8.5	5.8	3.4	10.7	28.3
United States	7.7	12.8	2.0	7.2	29.7
OECD[1]	10.6	10.4	2.9	9.9	33.8

1. Weighted average using 1995 GDP weights expressed in 1995 purchasing power parities.
Source: OECD.

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behind other countries (Chapter III). Changing the tax mix is another option, but trade-offs need to be assessed. For instance, lowering social security contributions in a revenue-neutral manner would benefit employment creation, but would further damage the long-term sustainability of the pension system. Raising indirect, while lowering corporate taxes, could have positive effects on business and the environment, but would also have negative short-term inflationary consequences.

Following decentralisation, the authority should reassess their mode of action in several areas

Efforts aimed at improving expenditure efficiency need to continue. The authorities, aware of this need, have increased capital spending over the last years in a context of fiscal consolidation mainly based on expenditure restraint. The decentralisation to the regions also has the potential to improve the management of public finances. The communities' increased financial dependence on the taxes raised on their territory, plus their increased powers, should prompt the autonomous communities to look for tax and spending levels and structures that underpin their region's activity and respect local preferences. Being less dependent on transfers will no doubt also reduce the risk of fiscal illusion and resource wastage.

In general however, this decentralisation also implies that the authorities should reassess their mode of action in several areas. This is the case, for example, in the health sector. The main role of central government now is to co-ordinate the different regional health systems in such a way as to guarantee equality of access to good basic medical care throughout the country. It is important, however, that this co-ordination, the details of which were spelled out in a draft bill (*Ley de Cohesión y Calidad Sanitaria*, LCC) at end-2002, is organised in such a way as to exploit the opportunities that decentralisation provides to stimulate the regions' efforts to improve efficiency.

The effectiveness of spending would benefit from a greater use of benchmarking…

Efficiency gains can be encouraged by regularly publishing performance indicators for the different regions. The existence of reliable and homogeneous indicators comparing the quality of financial management and the provision of regional public services would make it easier to assess the communities' performance and act as an incentive to adopt better practices.[34] For this to happen, decentralisation must be accompanied by improvements in regional and national statistical information, not the loss or segmentation thereof. In this context, the Health Ministry decision to set up a Health Information Institute enabling all the communities to access information on the National Health System, such as on hospital waiting lists, is very welcome.

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… and management based on output and from enhanced co-ordination between regions

The reform of autonomous community financing adopted with the decentralisation of health spending, which is fairly complex, is designed to achieve a balance between reinforcing budgetary responsibilities and maintaining solidarity between communities. Greater co-ordination between central government and the regions is however needed to reduce duplication. Such a duplication is no doubt the cause of the increase in total public employment despite the rule (just terminated) stipulating that, at every level of government, only one in every four civil servants retiring should be replaced. Also, some of the characteristics of the reform could have the effect of restricting the expected benefits. The minimum expenditure obligation concerning health limited to the first three years, for example, is liable to curb the regions' efforts to be efficient. It would be preferable to monitor the homogeneity of regional service supply on the basis of indicators concerning outputs, or even outcomes, rather than input criteria. Also, renewed efforts are probably needed to step up co-ordination between the communities in a number of sectors such as health, education and taxation. The experience gained by other federal countries, such as Switzerland, demonstrates the importance of this sort of co-ordination in taking more fully into account the effects of economies of scale and spillovers that a decentralised supply of services such as hospitals, universities or employment services does not provide (OECD, 2002a). Taking insufficient account of such spillovers tends to result in sub-optimal supply, which can, for example, generate an excessive number of universities (Chapter III), an inadequate matching between labour supply and demand and long waiting lists in certain hospitals. In this regard, the introduction of compensatory financial transfers in the event of one region's hospital services being used by the inhabitants of another region in specific services, as well as the Health Cohesion Fund, will contribute to a more efficient utilisation of this infrastructure.

Provision of health care and control over pharmaceutical spending should be more efficient

Overall, public spending on health, which is financed by taxation and amounted to 5½ per cent of GDP in 2000, has been effectively contained over the past few years (Table 13). However, substantial upward pressures persist on demand for health care. Evidence is provided by the fact that public consumption of drugs has been rising steeply for more than ten years and accounts for a larger proportion of the health budget than in other countries. This restricts the resources available for ambulatory, hospital and preventive care. Different measures have been adopted over the last years to curb the consumption of pharmaceuticals. It remains however vital to increase the efficiency of public health care provision and to exercise tighter control over consumption of

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Table 13. **Evolution of health expenditure**

	Spain			United States			Japan			EU average		
	Change over period[1]		Level 2000[2]	Change over period[1]		Level 2000[2]	Change over period[1]		Level 2000[2]	Change over period[1]		Level 2000[2]
	1990-95	1995-00		1990-95	1995-00		1990-95	1995-00		1990-95	1995-00	
In % of GDP	1.1	–0.1	7.7	1.4	–0.2	13.0	1.1	0.8	7.7	0.7	0.0	8.1
Private expenditure on health	0.8	0.1	2.3	0.1	0.0	7.2	0.2	0.3	1.8	0.4	0.1	2.2
Public expenditure on health	0.2	–0.1	5.4	1.3	–0.2	5.7	0.9	0.5	5.9	0.3	–0.1	5.9
of which:												
In % of public expenditure on health												
Investment	–1.6	0.2	3.2	–0.1	–0.2	0.8	1.0	–2.4	3.0	–0.7	–0.2	3.1
Current expenditure	1.6	–0.2	96.8	0.1	0.2	99.2	–1.0	2.4	97.0	1.6	0.3	96.9
of which:												
Pharmaceutical and other medical goods[3]	2.5	2.7	22.3	0.5	1.9	5.9	1.2	–4.6	13.5	0.8	0.3	13.9
In-patient care[4]	2.2	–1.6	55.4	–2.6	–3.9	53.0	–4.5	9.1	44.1	0.0	3.9	55.3
Out-patient care[5]	–1.8	–1.0	15.6	2.0	1.0	27.1	–2.7	–9.5	35.5	–1.1	1.3	25.4
Prevention and public health	–0.3	0.2	1.0	–0.5	0.7	8.5	. .	. .	1.8	–0.4	0.5	2.1
Other	–1.0	–3.0	2.3	0.7	0.4	4.7	. .	. .	2.1	2.3	–5.7	0.2

1. Changes expressed in percentage points.
2. Or latest available data.
3. Including therapeutic appliances.
4. Including expenditure on ancillary services (clinical laboratory, diagnostic imaging, transport and emergency rescue).
5. Including expenditure on home health care services.

Source: OECD *Health Data*, 2002.

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pharmaceuticals in order to avoid pressures on demand for medical spending to weigh on the communities' budgets or to lead to a deterioration in the quality of health care provision.

Improvements are still needed in hospital management. Few hospitals have taken advantage of the possibility they have had since 1999 of opting for public foundation status, with greater administrative autonomy and the possibility of recruiting staff on more flexible contracts. Only newly created establishments have opted for this more flexible legal framework, while converting existing hospitals has proved difficult, principally because of the obstacles deriving from the status of serving staff.[35] Yet the lack of any link between the remuneration and the performance of such staff, who enjoy job security thanks to the civil servant status, does undermine hospitals' efficiency. Moreover, the latter also suffer from a lack of flexibility in the management of their resources and insufficient information about their costs (Monge, 2002). This leads to wastage, such as the overbilling of hospitals' spending on drugs, which was uncovered by the Court of Auditors in 2001.[36] Steps also need to be taken to ensure better co-ordination between hospital and ambulatory care. An example of this lack of co-ordination is the handling of hospital emergencies, a very large proportion of which requires care that is neither really urgent nor specialised. Apparently, patients prefer to visit a hospital, and sometimes wait for a long time, rather than consulting their own doctor because in this way they can choose when to go and, above all, they avoid having to come back for more analyses. This situation shows that primary care provision is not properly tailored to user needs. If doctors in primary care centres were properly organised, they would be better placed than hospitals to meet these needs, bearing in mind their comparative advantage in keeping medical track of patients. What happens in practice is that doctors providing basic care, whose resources have been hardest hit by the budgetary restrictions – perhaps undermining their motivation[37] – tend to focus primarily on monitoring patients suffering from chronic complaints and on prescriptions, the amount of which perhaps makes up for shorter consultations.

Public spending on prescription drugs is continuing to increase rapidly, despite efforts to restrain growth. The rise in spending, which exceeded 10 per cent per year between 1991 and 1999, did however fall to some 7¾ per cent in 2000-01 as a result of a series of measures which included incentives to prescribe generic medicines, the introduction of a system of reference prices, cuts in the prices of certain products and a squeeze on distributors' profit margins on reimbursable drugs. Though these results were encouraging and did result in more widespread use of generic products, they may not last because of the temporary effect of some of the measures adopted.[38] Indeed, growth of spending on prescription drugs accelerated in 2002 and reached almost 10 per cent despite the implementation of an agreement for the period 2002-04 between the authorities and the various players that influence drug consumption. This agreement sharply

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increases the part that the pharmaceutical industry and distributors play in financing public expenditure on research, and it includes a mechanism for increasing the level of financing if consumption of medication grows at an excessive rate. This mechanism will raise the contribution of the pharmaceutical industry to public research by EUR 112 million, equivalent to 1.4 per cent of public drug consumption in 2002. Such an increase does not appear to be much of a deterrent, however, for a sector accustomed to spending large amounts on research, and the authorities have announced that they intend to modify this agreement. The agreement with the pharmaceutical industry also encourages the use of reference prices and generic medicines, which are not as yet very widespread, compared to other OECD countries. In addition, the authorities have recently decided to review antibiotic medicine packaging in order to reduce waste and to limit the pressure of the pharmaceutical industry on doctors' prescriptions. These positive changes could be supplemented by additional measures. One interesting route, for example, would be to give pharmacists freedom of substitution between generic and non-generic products and to encourage doctors to prescribe on the basis of active ingredients instead of brand names. In addition, pharmacists' remuneration could also depend partly on services rendered to customers on a fixed basis, instead of always being proportionate to expenditure, as in other OECD countries. Lastly, as previous surveys have suggested, cost/benefit analysis of new medicines should be improved before they are made refundable. Moreover, the practice of providing drugs free for retired people, which has often been abused, could be replaced by a means-tested system of patients' co-payments, which would also help to curb consumption of drugs.

The consequences of ageing: pension reform is more urgent than generally perceived

The previous S*urvey* showed that population ageing was posing a serious problem as regards the long-term viability of public finances. The recent growth of immigration, as revealed by the latest population census, will, even if it continues, have only a limited and doubtless temporary effect on the budgetary consequences of ageing (Chapter IV). In the light of the latest demographic projections published by the National Institute of Statistics (INE), the increase in public expenditure on pensions might amount to about 6 per cent of GDP by 2050, compared with 3 to 4 per cent in other OECD countries. Furthermore, labour force growth due to increased immigration might aggravate the problem of financial viability beyond the 2050 horizon. The present pension system is not actuarially neutral, with the result that pensioners on average receive more than they have contributed on a discounted basis.

The recent efforts of the authorities with regard to pension reform have focused mainly on improving incentives for older persons to continue working, in

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line with the agreement signed by the social partners in March 2001. Moreover, since January 2002, the possibilities of phased and flexible retirement have been expanded. Older workers are now entitled to draw a (smaller) pension while remaining in employment on a part-time basis, and persons continuing to work after reaching the age of 65 are exempted from social security contributions and are able to build up their pension entitlement even if they have already contributed for 35 years. Partial exemption of social contributions is also granted to workers over 60 years old with an indefinite contract and a minimum experience of five years within the firm. The possibilities of combining severance pay with unemployment benefit, often used by firms proposing early retirement schemes, have been reduced. These measures, which *a priori* increase the financial sustainability of the pension system, have been accompanied by appreciable increases in minimum and survivors' pensions[39] and new fiscal incentives for private pension plans. On the other hand, maintenance of a prudent fiscal policy over the recent period has made it possible to increase the social security reserve fund to 1 per cent of GDP as of 2003, a year earlier than expected, and to reduce the public debt.

To ensure the long-term sustainability of the pension system, more far-reaching measures are needed. The authorities have accordingly drawn up a strategy for meeting the challenges posed by the ageing process. They propose to separate medium-term budget policy from the long-term financial problem of ageing. Although the importance of a prudent fiscal policy is recognised, a wide range of measures in the context of a multidimensional strategy is judged to be necessary (Ministry of Economy, 2001). Apart from balancing the budget that will allow a progressive reduction of the debt ratio, this strategy aims at improving the labour market performance by fostering participation rates and reducing early retirement incentives. In addition, a progressive consensual approach to pension reform is promoted through the process initiated with the Toledo Pact, which provides for regular reviews of the situation. In that context, discussions should recommence in 2003. Generally, however, the official analysis by the Ministry for Social Affairs stresses the present financial ease of the pension system and the absence of upward pressure on expenditure until 2015. The uncertainties inherent in the projections beyond 2015 are also stressed, thus downplaying the urgency of reform (Ministry of Labour and Social Affairs, 2002).[40]

The authorities' aim of building a consensus on pension reform is commendable. But it necessitates the issuing of adequate information on the consequences of ageing. In this regard, analyses bearing on the period beyond 2015-20, *i.e.* when the impact of ageing will be felt, is useful. The findings of the numerous studies already carried out are clear and convergent, even though uncertainties exist and there is a need for regular updates.[41] Spain will experience a demographic shock later but more severely than most other OECD countries. And the key parameters of the national pension system are more generous in Spain than in most OECD countries. The projections on which this assessment is based

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necessarily extend quite far into the future and are therefore uncertain, like those for all other countries.[42]

In the light of the analyses available, calls for an ambitious and speedy reform of the pension system appear justified. Essentially this involves a revision of the pension system's parameters so as to make it actuarially neutral and to slow the growth of pension expenditure as from 2015-20.[43] A reform of this kind, needs to be anticipated and phased in.[44] A first significant step towards bringing the method of pension calculation into line with the one applying in most other OECD countries would be to base the pension on the worker's entire career instead of the last 15 years, as is the case at present. The possibility of phasing in such a measure could be discussed during the negotiations scheduled for 2003. The reform, which would reduce the pension income disparities generated by the present system (Jimeno, 2002a), would not affect current pension levels but would slow the growth of future pensioners' incomes. This type of reform is not incompatible with measures to protect the elderly against the risks of poverty to which some of them are still exposed, notably elderly women because of their low labour market participation.

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III. Structural reforms to sustain a strong performance

In recent years, strong growth of output and employment were underpinned by a wide-ranging structural reform programme and a sound macroeconomic policy framework, while real interest rates have come down to low levels. Privatisation of large public companies was almost complete by 2000, and was accompanied by several liberalisation "packages" that included an early introduction of competition in network industries, among other measures. Labour market reforms that lowered severance payments for most new permanent contracts, together with real wage moderation and the use of temporary contracts have boosted the employment ratio from 50.7 per cent in 1997 to 58.8 in 2002. At the same time, the unemployment rate fell by 5.5 percentage points, even though it is still – at over 11 per cent in late 2002 – one of the highest in the OECD.

Strong job creation was, however, accompanied by sluggish productivity, which has hampered the catch-up process with the leading countries from being even faster. Other countries have also experienced strong employment growth during the upswing of the late 1990s while maintaining significant productivity gains (Figure 14). In Spain, productivity growth was held back by a range of factors, whose relative contribution is difficult to quantify. Part of the explanation lies in the labour market integration of a large number of workers, such as long-term unemployed or young workers, who are initially less productive; but this is likely to be a temporary phenomenon. More important is probably the influence of the dual structure of the labour market, with a large share of temporary workers. This not only raises equity concerns, but may also reduce incentives to strengthen human capital, since temporary workers may not receive much specific training if they have a short career perspective in the firm (Dolado *et al.*, 2001), while permanent workers may lack strong incentives to improve their human capital because their job is secure. Breaking this duality would be positive for productivity growth. Another reason for low productivity is the bias of growth towards labour-intensive sectors, like construction, and some service sectors (Figure 15). This bias has

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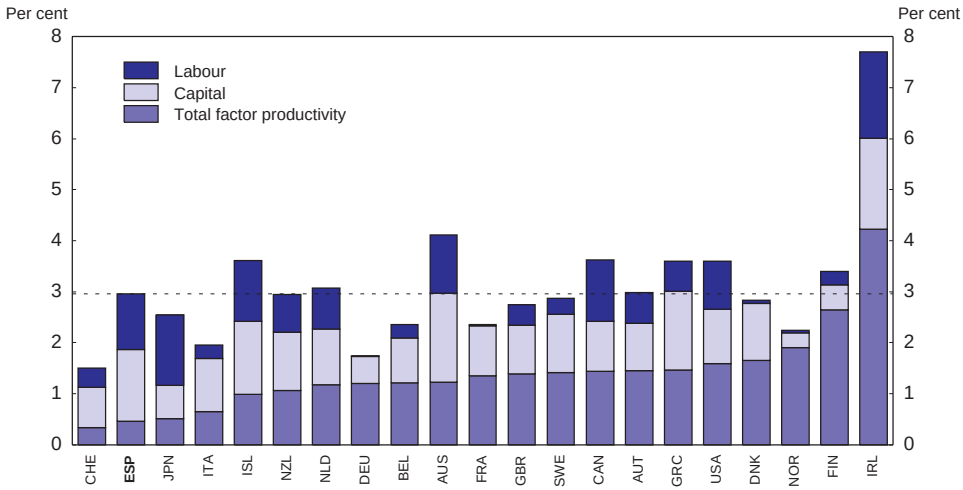
Figure 14. **Decomposition of the business sector potential growth**
1998-2001



Source: OECD.

helped job creation and is to some extent the result of demand pressures, like buoyant housing demand, which benefits also from tax deductions.

Slow productivity growth might be explained to an unknown extent by statistical factors, since GDP growth could be higher than measured by the National Statistics Institute (INE).[45] The INE will present at the end of 2003 revised National Accounts. Apart from this statistical question, there are other important reasons that explain slow productivity growth relating to the formation of physical and human capital. The education system, apart from pursuing important social goals, has a clear impact on productivity, and in several aspects Spain lags other OECD countries in education outcomes. Recent reforms in different areas show that the authorities are aware of shortcomings and ready to solve them. Investment in physical capital (excluding construction) has diminished in the last two years mainly for conjunctural reasons, but the stock of public capital per head is still only 81 per cent of the EU average and additional investment is required in the coming years to close the gap.[46] Finally, the production and diffusion of R&D and telecommunication technologies – in particular the use of the internet – are two areas which have an obvious impact on productivity and where the gap with the rest of the European Union is large.

An important disequilibrium that poses risks to growth in the coming years – the inflation differential with the euro area – has to a large extent structural

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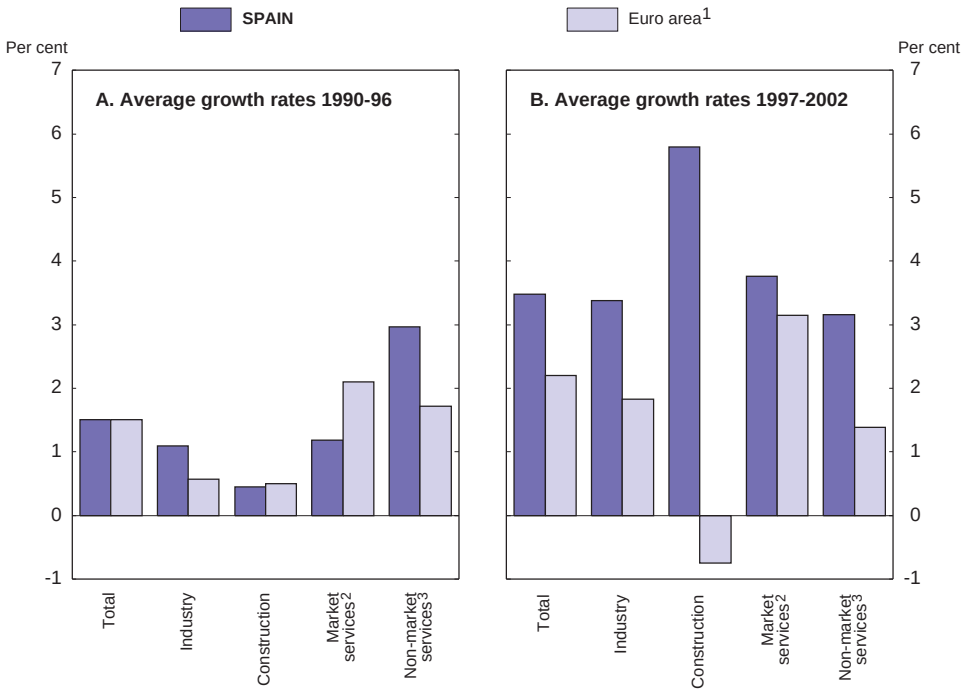
Figure 15. **Real GDP growth by sector**



1. Period 1990-96: Austria, Belgium, Finland, France, Italy, Netherlands, and Germany from 1992 only. Period 1997-2002: Austria, Belgium, Finland, France, Germany, Italy, Netherlands and Portugal.
2. Whole and retail trade; repairs; hotels and restaurants; transport, storage and communication; financial intermediation; real estate, renting and business activities.
3. Other services not included above.

Source: OECD, *Annual National Accounts*.

roots. As noted in Chapter I, nominal wage rigidity explains about two thirds of this differential. In this respect, the wage bargaining system, which dates from a period when the social and political environment was very different, is in need of reform. But the lack of success of recent reform efforts in this area suggests that there is strong opposition by at least part of the trade unions and employer federations. Part of the inflation differential can also be attributed to rising profit margins in some sectors. Despite progress in liberalisation and the positive role of the competition authorities, there is some evidence that the degree of market power in some network industries is still high, and that entry barriers in distribution channels need to be removed, especially by the regional authorities.

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This chapter deals with these structural policy issues, highlighting in particular the recent reform of unemployment benefits and the reform of the education system and its financing. It highlights the need to reform the wage bargaining framework and the housing market. It also reviews recent policy initiatives concerning goods and services markets and the financial sector, where a new law has modified several aspects of financial markets and corporate governance. A synopsis of the recommendations for structural reform is provided in Table 14.

Labour markets

The wage bargaining system is in need of a major reform

The design of the wage bargaining system is key for the performance of the labour market. A good system should allow for an efficient allocation of labour while ensuring that wage growth moderates when unemployment is high. A *decentralised* bargaining set-up is ideal in this respect, since it allows wages to follow productivity at the firm or individual level and to respond with moderate wage growth when there is excess labour supply. However, in some circumstances *centralised* wage bargaining may help to take into account externalities that are not considered by individual workers and employers.[47] The current Spanish system, based on negotiations at the intermediate level, is usually considered as the least favourable option, as it lacks the advantages of either type of negotiation (Calmfors and Driffill, 1988).[48] The need for a well-functioning system has become more imperative with monetary union, as the loss of the monetary and exchange rate tools have reduced policy options. Moreover, increasing demand for skilled labour linked to growth of the information and communication technology (ICT) sector and diffusion and increasing competition from low-wage countries also require an appropriate wage dispersion that provides job opportunities for the low-skilled, while improving the incentives for human capital investment (Bentolila and Jimeno, 2002).

Despite its important drawbacks, the current bargaining system, which dates from the end-1970s, has resisted serious reform attempts.[49] Wage agreements are mostly sectoral, be it at the national, autonomous community and provincial level. Enterprise agreements only exist in 11 per cent of firms, mainly in large firms and public sector enterprises. Sometimes conditions set at higher levels are re-negotiated at lower levels, but only to raise wages further. Since the 1994 reform, all agreements above the enterprise level must have special provisions that allow individual firms to adopt lower wage increases than those fixed in the agreement (*opt-out* clause), but in practice these clauses have been too restrictive and have not been used by firms. Recent reform attempts have not changed the situation much. In 1997, a voluntary pact among trade unions and entrepreneurs' associations to develop new forms of collective bargaining was

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Table 14. **Recommendations for further structural reforms**

Based on previous and current *Surveys* and actions taken

New (in *italics*) and previous recommendations	Actions taken since previous *Survey*
LABOUR MARKET AND EDUCATION SYSTEM	
Reduce the gap in protection between permanent and temporary workers by lowering further severance payments for permanent contracts and *improving controls on the abuse of temporary contracts*.	The reform that reduced severance payments for many types of new open-ended contracts has been made permanent. Wage payments during litigation over firing cases were suppressed if the firm accepts that the firing is "unfair". The administration can petition the labour court if an abusive pattern of temporary contracts is detected.
Consider taking into account severance payments when compensating the unemployed for loss of a job.	In general, severance payments exceeding a given amount are taken into account for access to unemployment assistance.
Restrict eligibility conditions of the unemployment subsidy programme for the rural sector to enhance regional labour mobility.	The special unemployment benefits regime for agricultural workers in Andalusia and Extremadura will be reformed for new entrants, with more conditionality and benefits provided only up to a maximum of six years.
Unemployment benefits should be linked to training and/or workfare. Enhance monitoring of the effectiveness of active labour market programmes.	Reform of the unemployment insurance system that clarifies the obligation to participate in active measures and the conditions under which the unemployed is obliged to accept a new job.
Improve efficiency and resources of the INEM on active labour market policies.	The resources for personnel of the INEM have been increased by 14 per cent in the 2003 budget.
Reform the wage bargaining system through decentralisation. *Eliminate indexation clauses in wage negotiations*.	No action.
Allow more independence to schools, including setting incentives through wage differentiation.	The new law on Education Quality provides for more independence of school directors.
Reform university finance by raising tuition fees while providing more grants or subsidised loans to less well-off students.	A new reform of university education improves the system to choose professors and creates a new agency to evaluate teaching and research standards.
Improve incentives for R&D *and eliminate obstacles to the diffusion of new technology*.	
HOUSING MARKET	
Simplify the criteria and reduce the waiting period for obtaining building permits.	No action.
Implement tax neutrality between home ownership and rental housing by phasing out tax allowances for home ownership.	A new tax allowance for owners who put residential dwellings to rent has been introduced in the personal income tax.

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Table 14. **Recommendations for further structural reforms** (*cont.*)
Based on previous and current *Surveys* and actions taken

New (in *italics*) and previous recommendations	Actions taken since previous *Survey*
PRODUCT MARKETS	
Increase the resources of the competition authorities and consider consolidating decision processes by merging the Competition Tribunal and the Competition Service.	Resources for the Competition Tribunal and Service have been significantly raised, while more budgetary independence has been granted to the Competition Tribunal.
Ensure that regional competition tribunals do not add substantially to costs of competition policy. *Ensure that they do not favour local firms.*	A new law co-ordinates all competition tribunals, discharging the national tribunal from minor cases.
Envisage splitting of production and distribution companies in energy sectors if anti-competitive practices persist.	Choice of provider for small consumers has started in 2003, but price liberalisation has been postponed for this group. A new tariff system has been approved until 2010 that implies annual price increases below projected inflation, but above the price reductions achieved in recent years.
Study options to import electricity.	Agreements have been reached with France and Portugal to increase interconnections by 2005.
Allow competitors to build their own retail network in gas distribution before the current deadline.	No action.
Ensure that access tariffs to the local loop of Telefónica respond to long-term incremental cost. Increase powers of the sectoral regulator (CMT). *Withdraw prices for long distance national and international calls and* ADSL *connections from the price cap for Telefónica.*	The CMT depends now on the Ministry of Science and Technology. The price cap for Telefónica remains in the hands of the Ministry of Economy. A new and cost-based Subscriber Loop Offer has been approved by the regulator including a complete overhaul of price settings and procedures.
Define a framework conducive to investment in mobile telephony, by providing a stable framework to UMTS investment.	The radio-spectrum tax for UMTS was lowered in 2001 and fixed for the period 2002-06.
Autonomous communities should not restrain the establishment of new hypermarkets in their territories.	No action. The region of Madrid has approved a new law that liberalises opening hours further.
Further liberalise postal services.	The EC directive has been transposed into Spanish legislation going further in terms of the scope of liberalised services.
Liberalise the rail transport. Consider franchising in railways and urban transport.	A new rail model has been presented by the government, which includes liberalisation of the services (passenger and freight), beyond the standards of the EC legislation.
Lower public support to industry and concentrate on horizontal targets to prevent distortions to competition. Focus national support to agriculture more on restructuring incentives and promote transparency in the distribution of subsidies by the State and the regions.	No action.

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Table 14. **Recommendations for further structural reforms** (*cont.*)
Based on previous and current *Surveys* and actions taken

New (in *italics*) and previous recommendations	Actions taken since previous *Survey*
FINANCIAL MARKETS	
Ensure a level playing field in banking between commercial and savings banks, and promote ownership forms for savings banks that allow market-driven restructuring of the banking sector.	The new Financial Law clarifies the regime for savings banks to issue special securities (*"cuotas participativas"*) that are quoted in the stock market, though do not incorporate voting rights.
Implement the Comisión Aldama recommendations on corporate governance through legislation.	The Financial Law also introduces a wide range of reforms on accounting and corporate governance.
PUBLIC SECTOR	
Reform the pension system to make it actuarially neutral.	Incentives to postponing retirement have been strengthened by the personal income tax reform and by legal measures to establish a flexible retirement system.
Speed up implementation of hospital management reforms and adopt global budgets instead of line budgeting.	Only new hospitals have adopted the status of public foundations.
Extend and speed up the authorisation of generics. Introduce means-tested co-payment on pharmaceuticals for pensioners.	Consumption of generics rose to 6.7 per cent by May 2002.
Develop a new financing system for regions that promotes fiscal responsibility of autonomous communities. *Enhance co-ordination among regions and increase incentives to raise efficiency of public finance management.*	A new permanent financing system has been approved that improves fiscal responsibility of regional governments by increasing taxing powers and reducing revenue guarantees.
Adjust automatically every year the schedule and allowances of the personal income tax to inflation in order to avoid fiscal drag. Suppress the special tax system for small incorporated businesses (*sistema de módulos*).	The personal income tax reform reduces marginal rates, raises tax allowances for work and family circumstances, and improves incentives to work for the old, mothers of young children and for unemployed workers.

Source: OECD.

signed, but has not been followed by any concrete steps. Likewise, the government sponsored new talks in 2001 between the social partners to agree on a reform, but no agreement was reached.

Wage moderation has been prevalent during the 1990s, partly reflecting the social partners' support for joining EMU. Still, growth of unit labour costs between 1990 and 2001 has been systematically above the euro area average. In addition, the wage bargaining process has not helped in adapting wages to the imbalances in the labour market, with unemployment rates ranging from less than 6 per cent in Baleares and Navarra to more than 20 per cent in Andalusia and

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Extremadura. Even though negotiations at the sectoral and provincial level have been prevalent, which might suggest that wages could adapt to local conditions, wage increases during the 1990s have not varied much across regions, in particular failing to respond to differences in unemployment rates.[50] Labour mobility, which is very low, has not worked either to reduce differences in unemployment rates.

Recent efforts to reform the present system should continue. The draft reform elaborated by the government in 2001, which was not implemented, would be a good starting point. It provided for some areas of negotiation to be settled in national sectoral agreements, leaving wage increases, holidays or working time to be determined at the firm level. Since wage negotiations could be costly in small enterprises, which account for a large share of all firms (over 70 per cent), wage determination could continue to take place at the provincial level, as it does now, in order to reflect local labour market conditions. But in this case the compulsory application of agreed wages to all firms should be suppressed, since this is the element that eliminates wage dispersion across firms, and should be substituted by an *opt-in* system, that allows firms to adhere to the wage agreement if they wish to. In the absence of reform, national wage agreements like the one in 2002 are likely to prevail. In this pact, a recommendation was issued to negotiators at lower levels to take the official inflation target of inflation of 2 per cent as a reference for wage increases, resulting in lower agreed wages since the end of 2001. But wage drift has increased in 2002, partly due to the special clauses in the agreement (*cláusulas de revisión salarial*) that provide for *ex post* wage increases if the inflation outcome is above the reference value. Eliminating such clauses and using inflation projections for the whole euro area or core inflation as the inflation reference would help to reduce nominal wage rigidity in the face of negative price shocks. However, re-modelled national agreements are only a second best solution to a profound reform of the wage bargaining process.

A *recent reform has changed several aspects of unemployment benefits…*

An international comparison of unemployment insurance (UI) systems shows that the Spanish replacement rate of 70 per cent of the contribution base (60 per cent after six months) is close to the average (Table 15). Benefit duration depends on time worked and can reach up to 24 months, which is longer than in most OECD countries.[51] After the expiration of UI, or if the worker has not been employed previously during 12 months, unemployment assistance (UA) is in some cases provided at a flat rate, which is at a similar level to that in other countries, although extra benefits for unemployed with a family are lower than elsewhere.[52] The coverage of unemployment benefits is relatively low as compared to other OECD countries at just over 60 per cent of total registered unemployment. This is explained by the high incidence of young unemployed who do not have access to unemployment insurance (UI) if they have never worked, and by the high share of

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Table 15. **International comparison of unemployment benefits**
1999

	Unemployment Insurance (UI)			Unemployment Assistance (UA)	
	Payment rate % of previous gross earnings	Minimum benefit USD PPP/year	Duration Months	Payment rate % or USD PPP/year	Duration Months
Spain	**70-60 after 6 months**	**5 576**	**24**	**5 576**	**18**
Austria	56	1 571	10	92% of UI	No limit
Belgium	60-43 after 1 year	7 415	No limit	. .	. .
Canada	55	. .	11	. .	. .
Denmark	90	. .	60	. .	. .
Finland	90	5 119	25	5 119	No limit
France	75	8 214	60	4 479	No limit
Germany	60	. .	12	53% of previous net	No limit
Greece	40	. .	12	17% of UI	. .
Ireland	Flat rate	5 271	15	5 163	No limit
Italy	80	. .	6	. .	. .
Japan	80	. .	10	. .	. .
Netherlands	70	9 922	60	9 922	24
New Zealand	. .	. .	. .	5 698	No limit
Norway	62.4	. .	36	. .	. .
Portugal	65	5 714	30	4 571	24
Sweden	80	6 383	15	6 383	15
Switzerland	70	. .	7	. .	. .
United Kingdom	Flat rate	. .	6	4 084	No limit
United States	50	4 524	6	. .	. .

Source: OECD, *Benefits and Wages*: OECD *Indicators*, 2002.

workers with fixed-term contracts (over 30 per cent of all employees) who usually do not fulfil the minimum eligibility requirement. The coverage has nonetheless risen over the last two years, which is explained by cyclical reasons.

At the end of 2002 the government reformed several aspects of the unemployment benefit system, with the objective of enhancing incentives to search for work. The law somewhat modifies a previous decree-law issued in May 2002, which contained more radical measures but was heavily contested by the trade unions. The reform does not touch replacement rates and benefit duration, but it includes other measures that encourage a more active job search while unemployed. *First*, it defines eligibility conditions more clearly, resulting *de facto* in a stricter application. Specifically, after an initial period of 100 days in unemployment, the unemployed will have to accept to follow any ALMP measure proposed to him. An unemployed worker who rejects three suitable job offers by INEM will lose the unemployment benefits. A sanction of benefit suppression for three and six months is imposed on those who reject a first and a second offer, respectively.

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Previously, a single rejection led in theory to the loss of benefits, making it the strictest regime in the OECD, but in practice it was the worker who determined whether the job offer was adequate or not. The law now defines the suitability of an offer and it will be up to the INEM to enforce these criteria.

A *second* measure deals with the special non-contributive unemployment regime of seasonal agricultural workers in Andalusia and Extremadura (PER).[53] The system, which provides benefits of 75 per cent of the minimum wage during up to six months if workers have been employed at least for 35 days a year, will be replaced by a new subsidy, but only for new entrants, since current participants in the scheme will continue to receive benefits under the same conditions.[54] The scheme for new entrants is similar to the earlier one, but is conditional on participation in labour market programmes, and will be provided up to a maximum of six times, which in practice will result in a phasing out of the system.[55] At the same time, an unemployment insurance system for temporary agricultural workers based on contributions, that did not exist until now, will be implemented in the whole country.

A *third* aspect of the reform relates to the *salarios de tramitación* – wages paid by firms while waiting for courts to decide if a layoff is justified or unjustified, which carry different severance payments.[56] These are an additional barrier to layoffs, since they are paid by the enterprise on top of severance payments, although the State reimburses the firm for the wages paid in excess of two months if the judicial process is delayed beyond that period. In practice, these two months wages paid by the firm are an additional flat rate firing cost, independent of tenure. The new law allows the firm to pay the higher severance payment immediately and avoid paying wages during litigation, which was not possible under the previous regime. Given the high proportion of "expensive" dismissals – those that are considered as unjustified by the tribunals (90 per cent of all individual dismissals) – the new measure will probably encourage firms to avoid litigating with workers on the nature of the dismissals by disbursing higher severance payments from the start.

F*inally*, the law also includes other measures that improve employment incentives:

- Unemployed workers who become self-employed will be able to capitalise part of their unemployment benefit (up to 20 per cent of benefits rights) for initial investment and to use the remaining rights to pay social security contributions corresponding to their new activity.[57]
- It will be possible to accumulate unemployment benefits with salaried work (during a limited period) for those older than 52 and for those unemployed that are hired to temporarily substitute workers in training courses.
- Help will be provided for moving costs to those unemployed who accept a job in another region.

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… and is a step in the right direction…

The reform of the special non-contributive unemployment regime of seasonal agricultural workers in Andalusia and Extremadura is an overdue measure, since it reduces incentives to search for a job in more productive activities or upgrade skills. A slow phasing out of these subsidies should allow the reallocation of work to other activities in a gradual manner and to reduce the social impact of the reform. The new regulation on *salarios de tramitación* barely reduces the high employment protection for permanent workers, because firms may avoid paying those wages, but by doing so they will have to accept to pay higher severance payments, whereas in the present situation they have a small probability of paying less. As for the reform of the eligibility conditions of UI, it is a step in the right direction since it improves incentives to search for a job while unemployed. Empirical evidence for Spain shows that exit from unemployment is high after the job loss and in the weeks before the expiration of benefits (Bover *et al.*, 1996; Gonzalo, 2002), suggesting that those who do not find a job rapidly wait until the duration of benefits is finished. Other research across a large number of countries points in the same direction, showing that job search efforts do not seem to be closely linked to the replacement rate of UI, but that they are higher when benefit duration is shorter. By better linking benefits to the propensity of workers to accept job offers, the reform may contribute to reduce long-term unemployment and public spending on UI. Other countries apply similar rules, which in general seem to be effective to shorten the duration of unemployment.

… but needs to be better co-ordinated with active labour market policies

The effective application of the reform requires a well performing public employment service, able to provide job offers or adequate training to the unemployed. Public outlays on active and passive labour market policies in terms of GDP are not high in international comparison if the unemployment differential is taken into account. Funding is considerably lower than in some European countries, but higher than in Italy, the United Kingdom or the United States (Table 16). Spending on active labour market policies (ALMP) is also comparatively low (Figure 16). In addition, the allocation of funds among different ALMPs differs from that of other countries. Spending on training for workers and on subsidies to support new permanent contracts, which can carry important dead-weight costs, is relatively high, while funding of the employment service and for training of unemployed workers is very small.

A large number of OECD studies on ALMPs demonstrate that the amount spent on ALMPs does not have a clear impact on reducing unemployment, but that ALMPs are useful if they are very well targeted. In Spain, ALMPs do not seem to have a significant impact on the exit rate of unemployment. Results of a report (Sáez *et al.*, 2001) reveal that the probability of finding a job for participants in

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Table 16. **Public expenditure on labour market policies**

As a percentage of GDP, 2001[1]

	Total	Passive measures	Active measures					
			Total	*of which:*				
				Administration PES[2]	Training for unemployed	Training for workers	Subsidies	Other
Spain	**2.17**	**1.33**	**0.84**	**0.09**	**0.11**	**0.11**	**0.40**	**0.13**
Australia	1.43	0.98	0.46	0.20	0.02	0	0.11	0.13
Austria	1.60	1.07	0.53	0.14	0.18	0.02	0.11	0.08
Belgium	3.48	2.18	1.30	0.17	0.16	0.08	0.77	0.12
Canada	1.13	0.72	0.41	0.17	0.17	0	0.03	0.04
Denmark	4.56	3.00	1.56	0.12	0.67	0.18	0.17	0.42
Finland	2.96	2.02	0.94	0.12	0.26	0.03	0.29	0.24
France	2.96	1.65	1.31	0.18	0.22	0.03	0.37	0.51
Germany	3.13	1.92	1.21	0.23	0.34	0	0.25	0.39
Greece	0.93	0.47	0.46	0.06	0.12	0.07	0.08	0.13
Japan	0.86	0.55	0.31	0.20	0.03	0	0.08	0
Netherlands	3.44	1.86	1.58	0.26	0.24	0.07	0.38	0.63
New Zealand	1.96	1.41	0.55	0.11	0.16	0	0.09	0.19
Norway	1.23	0.44	0.79	0.12	0.06	0	0.01	0.60
Portugal	1.52	0.90	0.62	0.11	0.07	0.09	0.09	0.26
Sweden	2.28	1.19	1.09	0.23	0.30	0	0.24	0.32
United Kingdom	0.92	0.56	0.36	0.13	0.04	0.01	0.01	0.17
United States	0.45	0.30	0.15	0.04	0.04	0	0.01	0.06
Country mean	2.05	1.25	0.80	0.15	0.17	0.04	0.19	0.25

1. Or latest available data.
2. PES: Public employment service.

Source: OECD, OECD *Employment Outlook*, June 2002.

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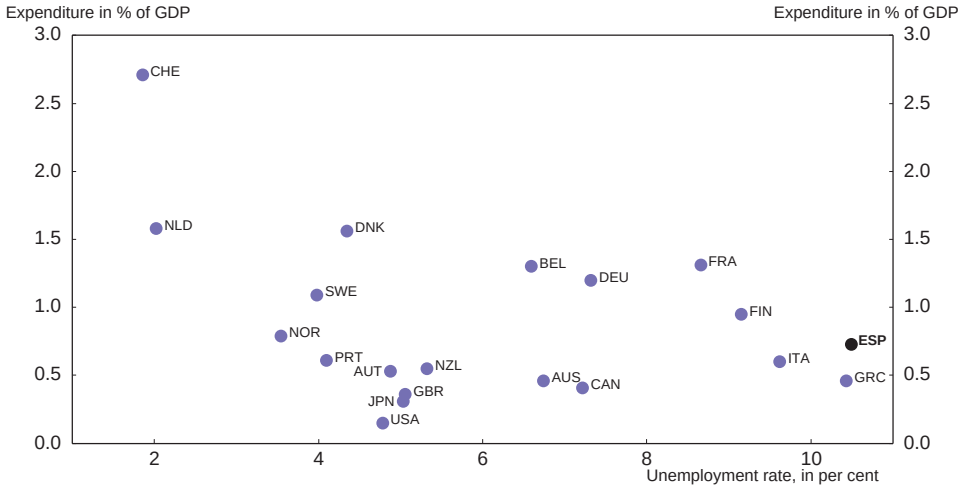
Figure 16. **Public expenditure on active labour market policies and unemployment rates**

2001



Source: OECD, *OECD Employment Outlook,* 2002.

these programmes improves in the following six months, but that their effect in the longer term is not statistically significant. In the INEM, personnel resources at the end of 2000 were 7 203 people, one for every 216 registered unemployed worker, though a large part of administrative resources are spent on registering contracts (15 million contracts per year, because of the large number of temporary contracts renewed every few weeks). Their counselling and job matching functions are undermanned, as shown by the fact that the INEM only intermediates in 17 per cent of job findings of unemployed workers. Management has been transferred to autonomous communities, and co-ordination across them is poor and still lacks a functioning information network across regions. Considering these shortcomings, ALMPs should be reformed, including a re-balancing of funds across different policies and an overhaul of the INEM, which should focus more on providing effective support to the unemployed. Regional public employment services could follow the example of Switzerland, where employment agencies receive their resources as a function of their performance, taking into account the socio-economic background of the area (OECD, 2002a). Independent evaluation of different policies should also be carried out and published on a regular basis, and the results should be used to better tailor ALMPs to different groups of workers.

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In the area of prevention of work accidents, where Spain has higher rates than the other EU countries, the government has approved several measures that include higher social security contributions for those firms with worse accident records, as recommended in the previous S*urvey*. The measures also include higher sanctions in case of firms' responsibility in work accidents and additional resources for work inspections.

Interactions across different labour market policies could be exploited further

Though structural unemployment has fallen, it still remains high at close to 11 per cent. Despite the labour market reforms of 1997 and 2000, which introduced a new type of permanent contract with lower firing costs (OECD, 2001a), severance payments are still the highest in Europe and much higher than in the countries with low unemployment (the United States or the United Kingdom, for example). As a consequence, the labour market continues to be segmented with two groups of workers co-existing, one with permanent contracts protected by high firing costs, and another consisting of unemployed and temporary workers that suffer from job insecurity and lower wages, creating horizontal inequalities. Human capital formation within the firm is also damaged by segmentation, since workers do not have a career perspective, while firms are encouraged to turn over their labour force in order to chain temporary contracts. However, more substantive reforms in this area, and in other key labour market aspects – like the wage bargaining system – have been difficult to introduce.

This situation should be tackled by taking into account interactions across different policies, so as to make reforms politically feasible. One strategy, already mentioned, would be the combination of improvements in ALMPs with other areas of the labour market reform like that of the wage bargaining system. Another possibility would be to reduce firing costs for permanent workers while implementing stricter controls for temporary contracts, which should be easy to achieve by using the INEM database and the social security registers to control abuses of contracts.[58] A third possibility would be to introduce non-linear severance payments, by lowering protection for those with long employment histories (who receive large lump-sum payments when they are fired), while introducing a small amount of protection for all permanent workers independent of time worked. A package of measures might be easier to introduce and could raise the acceptance by trade unions and employers' federations.

Education and R&D

Human capital formation is a key factor underpinning productivity growth and for achieving convergence in per capita GDP. Currently, the stock of human

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Table 17. **Educational attainment by age**
Percentage of population that has attained at least upper-secondary education or tertiary education by age group, 2001

Age	25-64	25-34	35-44	45-54	55-64
Upper-secondary					
Spain	40	57	45	29	17
OECD[1]	64	74	68	60	49
Tertiary					
Spain	17	24	18	13	8
OECD[1]	15	18	16	14	10

1. Unweighted country mean.
Source: OECD, *Education at a Glance* – OECD *Indicators*, 2002.

capital is fairly low, since the proportion of adults with upper-secondary education is only slightly more than half the OECD average, though the share of those holding university degrees is similar to the average (Table 17). Progress in the last 20 years has been remarkable, especially concerning access to tertiary education, where the student to population ratio at age 20 is amongst the highest in the OECD (38 per cent against an OECD average of 31 per cent). However, participation in upper-secondary education remains below that of most OECD countries (Table 18), while the dropout rate is also very high, at around 30 per cent. The quality of secondary education is also below the OECD average. Indeed, the 2000 OECD PISA report, which compares achievements of 15 year old students, ranks Spain well below the average on most scores (Table 19). At the university level, there is also a widespread perception that the increase in the number of graduates in the last two decades has not been accompanied by a parallel increase in the quality of education.[59] Moreover, vocational training has been underdeveloped for a long time, in part because it lacks reputation and is considered as a residual solution for students who drop out of secondary education.

The State is responsible for framework legislation and for ensuring homogeneity of curricula, while financing and managing education is the competence of the autonomous communities. Recently, the government has addressed problems in all parts of the education system with a new series of reforms: the *Law on Qualifications and Vocational Training* aims at homogenising vocational training programmes in schools with those of adult and work-based training, while the *Law on University Reform* and the *Law on Education Quality* aim at raising standards of university education and of compulsory education, respectively. The common goal of these reforms is to improve the quality of education services.

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Table 18. **Participation in education at typical ages**
Net enrolment rates, 2000

	In secondary education at age 16	In tertiary education at age 20
Spain	**90**	**38**
Australia	94	33
Austria	91	19
Belgium	99	46
Canada	93	37
Czech Republic	100	23
Denmark	91	11
Finland	95	31
France	96	42
Germany	97	15
Greece	94	56
Hungary	95	26
Iceland	90	14
Ireland	91	35
Italy	83	26
Japan	96	n.a.
Korea	96	57
Luxembourg	87	n.a.
Netherlands	107	32
New Zealand	90	33
Norway	94	28
Poland	92	33
Portugal	87	30
Sweden	89	23
Switzerland	90	14
United Kingdom	84	33
United States	87	38
Country mean	93	31

Source: OECD, *Education at a Glance* – OECD *Indicators*, 2002.

Pre-primary schooling is well developed

Education for children aged 3-6 is well developed, with enrolment rates above the OECD average.[60] This is important, since early education is considered a key factor in education achievements later in life (OECD, 2001d) and helps participation of women in the labour market and higher fertility, which are both very low in Spain. It also favours the linguistic and social integration of immigrants (Chapter IV). Education of the 3-6 year olds is free in publics schools, while in some private schools it is also financed by autonomous communities. The new *Law on Education Quality* extends free education for these ages to all schools, public or private. For children below three years, where financing is private and little subsidised by local governments, enrolment rates are much lower.[61] The new law provides lower requirements

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Table 19. **Student performance at age 15**

	Reading literacy scale				Mathematical literacy scale			
	Rank	Mean	10th percentile	90th percentile	Rank	Mean	10th percentile	90th percentile
Spain	**18**	**493**	**379**	**597**	**21**	**476**	**358**	**592**
Australia	4	528	394	656	5	533	418	647
Austria	10	507	383	621	11	515	392	631
Belgium	10	507	354	634	9	520	367	646
Canada	2	534	410	652	5	533	423	640
Czech Republic	19	492	368	610	17	498	372	623
Denmark	16	497	367	617	12	514	401	621
Finland	1	546	429	654	4	536	433	637
France	13	505	381	619	10	517	399	629
Germany	21	484	335	619	19	490	349	619
Greece	24	474	342	595	25	447	303	586
Hungary	22	480	354	598	20	488	360	615
Iceland	10	507	383	621	12	514	407	622
Ireland	5	527	401	641	15	503	394	606
Italy	20	487	368	601	23	457	338	570
Japan	8	522	407	625	1	557	440	662
Korea	6	525	433	608	2	547	438	650
Luxembourg	26	441	311	564	26	446	328	559
Mexico	27	422	311	535	27	387	281	496
New Zealand	3	529	382	661	3	537	405	659
Norway	13	505	364	631	16	499	379	613
Poland	23	479	343	603	22	470	335	599
Portugal	25	470	337	592	24	454	332	570
Sweden	9	516	392	630	14	510	386	626
Switzerland	17	494	355	621	7	529	398	653
United Kingdom	7	523	391	651	7	529	412	646
United States	15	504	363	636	18	493	361	620
OECD *total*		**499**	**363**	**622**		**498**	**358**	**628**
OECD *average*		**500**	**366**	**623**		**500**	**367**	**625**

Source: OECD, *Knowledge and Skills for Life: First Results from* PISA 2000.

for educational staff at this level,[62] which might help increasing supply. More public finance in the future would help to expand provision for these age groups, where externalities from education seem to be very high (OECD, 1999).

Primary and secondary education has been reformed…

The reform of primary and secondary education mainly concerns organisational issues, and does not modify financing, which is mostly public, nor does it provide additional funds for education. Expenditure on secondary education as a share of GDP is close to the OECD average, although spending per student is below average (Figure 17). Higher costs due to the reform will have to be paid by

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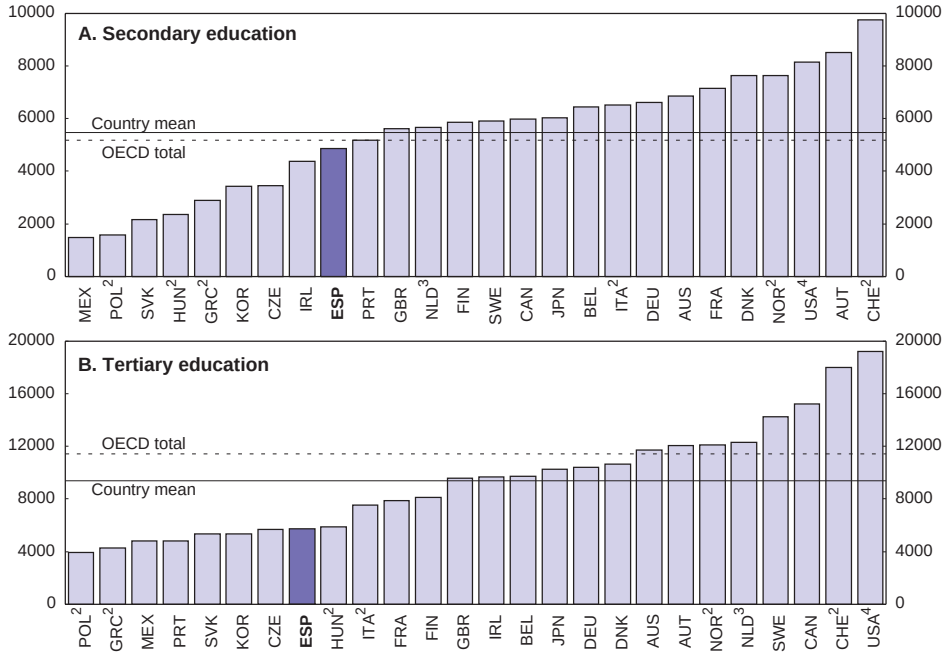
Figure 17. **Annual educational expenditure per student,[1] 1999**

Expenditure on public and private institutions, based on full-time equivalents



1. In US dollars converted using PPPs.
2. Public institutions only.
3. Public and government-dependent private institutions only.
4. Public and independent private institutions only.

Source: OECD, *Education at a Glance – OECD Indicators,* 2002.

central governments and all the regions. Compulsory education was raised from age 14 to 16 in 1990, and is free for parents. It also covers most students enrolled in private schools (24.5 per cent of all students).[63]

In primary education (age 6-12), the reform has increased the number of weekly hours devoted to core subjects, since the lack of basic skills is one of the main sources of failure later in secondary education. An exam will be implemented at the end of the period, with no curricular value and only for information purposes for the administration and the school. Reform has been deeper in secondary education (age 13-17), where a very high dropout rate accompanies low perceived abilities of students. The PISA report detected a low average level in all three core subjects (mathematics, reading comprehension and scientific knowledge). However, the

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dispersion of individual results around the average is not wide, which is a characteristic of those countries with better results, revealing that a high performance is compatible with a low dispersion of results (Finland, Korea, Japan). But, in Spain, low dispersion also implies a lack of outstanding students as compared to other countries. The reform addresses this problem with several measures. F*irst*, the separation of students into different subject paths is brought forward one year to age 14, with the aim to create a "technical" path for those who will probably go on to vocational training after compulsory education. These paths are voluntary and flexible, so that students are not locked in an early age. To reduce the dropout and increase the opportunities for all, "vocational initiation programs" are defined and introduced at age 16 into the formal system.[64] S*econd*, the possibility of course repetitions has been increased, while reinforcing measures will be taken for those students with special difficulties (with separate courses for some subjects). T*hird*, national tests will take place at the end of secondary school and will be required to obtain the degree.[65] F*ourth*, more autonomy will be given to schools. New steps are created in the professional career of teachers, and incentives will be given to those that show "special dedication" to the school or those who teach in an innovative way.

... but further increasing school autonomy would help to avoid failure and maintain equity

Some of these measures go in the right direction. More autonomy for schools is clearly positive since the main goal of the reform – raising quality – needs to be underpinned by new approaches, which can only be achieved by flexible management. Autonomy is also needed to adapt methods and resources to the needs of schools in areas with very different socio-economic backgrounds. Incentives for teachers – economic and professional – are also welcome, since wage differentiation of Spanish teachers is very low in international comparison. Interestingly, teachers' starting salaries in upper-secondary school are high – equivalent to 1.58 times per capita GDP, the highest ratio in the OECD, but the ratio of teachers' wages after 15 years of experience with respect to the starting salary is only 1.17, the lowest in the OECD after Norway and Turkey. This reveals that incentives for improvement are very poor. Allowing schools to manage their own budgets more freely would be useful to strengthen incentives. In this respect, New Zealand provides a good example, since individual school managers there are able to determine additional payments as a function of a wide range of criteria (management responsibilities, teaching students with special needs or outstanding teaching performance).[66] National tests are also a good measure to ensure equal treatment of students in different schools, and may be a powerful tool to obtain information on quality improvements. However, course repetitions are not an option in some of those countries that lead the PISA report, where schools apply all available means to avoid the failure of individual students, and manage to maintain a high performance with little dispersion of results. In the Spanish reform course repetition is considered as a way to promote effort and responsibility, but OECD

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experience shows that they may be used by some schools as an easy fix to solve problems associated with "bad" students and may not help to reduce the ratio of dropouts from secondary education. As for separation in paths, countries that separate students at early ages (Germany and Luxembourg) perform below the average in the PISA report, suggesting that these measures do not bring quality *per se*. However, separation of students in Spain is made at a relatively late age, while the flexibility to change between paths is a positive measure.

The reform of vocational training aims at building a bridge to life-long learning

Vocational training (*Formación Profesional*) for the young is less developed in Spain than in other OECD countries. Only 31.2 per cent of those in upper secondary studies follow vocational training, as compared to an average of 47 per cent in the OECD. After upper secondary studies, 41 per cent of students choose to follow advanced vocational training, against an average of 58 per cent in the European Union, despite the fact that employability of these students just after completion is higher than after graduation at university. The system suffers from a lack of social prestige, as it was considered as a secondary choice for those who could not make it into university. This perception is no longer correct, as now 25 per cent of those who finish advanced vocational school end up in tertiary education. The new law on vocational qualifications co-ordinates vocational training at school with other types of vocational training (for adults and firm-specific) by creating a national qualifications system (a major feature of which is a catalogue of 139 disciplines) common to all three types of training in a module system, favouring life-long learning.

Firm level training is subsidised by public funds (partly financed by the European Union), and is carried out by the FORCEM, an institution managed by trade unions and employers. Past episodes of misuse of funds (whereby firms devoted training funds to other uses) have prompted the government to enter the management of the FORCEM.[67] The annual financing and the number of people trained per year are large,[68] but the efficiency of these measures is unknown. Independent evaluation should address this problem, by comparing outcomes of participants in different courses with a control group of non-participants.

The university system has a large potential for efficiency gains…

The university system has developed rapidly in the last two decades. The process of transfer of competencies for university education to autonomous communities was recently completed. Regional governments have invested heavily in new universities, in part accompanying the arrival to adulthood of the baby-boom generation (which was born later in Spain than in other countries), but also because universities are considered as a "prestige" investment by regional authorities. As a result, every town with more than 50 000 inhabitants has at least one university. This allows many students to stay in their parents' home until late

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ages, since they are likely to find a university nearby, inducing a very low student mobility (Mora and García, 1999). Together with the relatively low level of fees, this makes for a relatively easy access to tertiary education. The level of current spending on universities is relatively low, however (Figure 17). The spending per student as a per cent of per capita GDP was only 30 per cent in 1998, well below the OECD average of 44 per cent.

The *Law on University Reform* includes two new elements that may affect the quality of university education. *First*, a new system to choose professors and researchers through a national competition will be introduced, substituting competitions at the departmental level which are more prone to manipulation in favour of insiders. This measure addresses the well-known problem of in-breeding in university departments, which translates into a high share of vacancies being filled with internal candidates. *Second*, the law creates a new national agency that will evaluate teaching and research activities of university departments. Both measures are undoubtedly positive. National competitions go in the right direction, since they will improve transparency of the selection process, although the competition tribunals will still be filled with the national experts in each area, which are usually a small group of people. Opening such tribunals to renowned international experts could improve transparency. As for the evaluation agency, it could help university authorities to detect potential problems, and students to choose the best universities. The choice for students has been widened from campuses in the own region of residence to all the national territory, which should reinforce the pressure on universities to provide quality education.

However, other structural problems remain. On the supply side, the public financing of universities does not provide incentives to improve quality. Public money received by universities is independent from their performance. This should be modified by linking funding to transparent criteria.[69] University staff are public servants and therefore it is very difficult to vary remuneration according to teaching or research records. In the long term, the ability to choose among universities could help to direct students to the best ones; but, in practice, low student mobility limits the scope for raising competitive pressure. Extended budgeting autonomy that includes staff salaries should be implemented to give university heads a lever to improve quality. On the demand side, the effective number of years spent at university goes well beyond the official duration of studies (Bricall *et al.*, 2000). Students can take each exam up to six times (though with increasing costs each time they repeat), while fees as a share of the actual cost (between 20 and 25 per cent), together with the fact that many students live with their parents, make the cost of studying relatively low. The opportunity cost is also small, since the probability of finding a job is low with youth unemployment rates at more than 25 per cent for the 16-24 year olds. Overall, long studies result in larger costs for the public sector, but do not result in a greater amount of human capital.

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... which could be seized by raising university fees

While private costs of education are low, private benefits over the lifetime are high, due to the higher wages received by people with university degrees, together with the lower probabilities of becoming unemployed. Average rates of return of investment in tertiary education are above returns on other assets (including the stock market), since the difference in the discounted stream of revenues between university graduates and those with upper secondary education is much larger than the discounted private cost of education (Box 2). This suggests that public subsidies to tertiary education are paying for benefits that accrue mostly to the individual, and that most students, in the absence of credit market

Box 2. **Private rates of return from non-compulsory education**

The private rate of return on non-compulsory education has been estimated for Spain following the methodology used in Blondal *et al.* (2002) for ten countries (Table 20). The private rate is the discount rate that equates future private economic benefits during lifetime of prolonging education to private costs. Benefits taken into account include higher future wages and lower unemployment probabilities (but not higher pensions), while costs include lost earnings for not working while studying and tuition fees (but not living expenses). Other benefits from education, like non-economic benefits and economic externalities, are excluded in these calculations.

As in other countries, the results show that investment in human capital yields much higher real returns than other forms of investment, including the stock market. Completing upper secondary education yields a real return of 8 per cent for men and nearly 12 per cent for women, the difference being mostly explained by a higher earnings gap for women between those who complete secondary education and those who do not. Returns of enrolling in tertiary education (*vis-à-vis* starting work after secondary education) are also very high at near 11 per cent and very close to those in other countries. Although these are average and not marginal rates of return, they reveal that incentives for an individual to enrol in non-compulsory education are likely to be very significant. Being higher than those of alternative investment forms, they point to some sort of credit market failure that would call for more provision of grants or subsidised loans for less well-off students. They also suggest that private gains from education are high, especially in tertiary education, giving scope for higher university fees. A simulation exercise shows that raising the share of private funding of expenditure in tertiary education from the present 24 per cent (which includes tuition fees in public and private universities) to 100 per cent would reduce private returns from 10.8 to 8.1 per cent for men, and from 11.4 to 7.6 per cent for women, leaving them still well above returns on other forms of investment.

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Table 20. **Private rates of return on education**

Per cent

	Upper-secondary education		Tertiary education	
	Men	Women	Men	Women
Spain (1998)[1]				
Return based on pre-tax earnings	12.0	14.1	11.7	13.2
Impact of:				
Taxes	–3.2	–2.1	–2.6	–3.4
Unemployment risk	–0.8	–0.2	3.1	3.9
Tuition fees and student support			–1.4	–2.3
Comprehensive rate	**8.0**	**11.7**	**10.8**	**11.4**
Comprehensive rate in other countries[2]				
United States (1999)	16.4	11.8	14.9	14.7
Japan (1999)	6.8	9.4	7.9	7.2
Germany (1998)	10.8	7.0	9.1	8.4
France (2000)	13.5	17.9	14.3	15.4
Italy (1998)	11.2	n.a.	7.5	n.a.
United Kingdom (2000)	15.1	n.a.	18.5	16.1
Canada (2000)	13.6	12.7	8.7	9.9
Denmark (1999)	11.3	10.5	11.5	11.1
Netherlands (1997)	7.9	8.4	12.1	12.5
Sweden (1999)	6.4	n.a.	11.4	10.8
Unweighted average	**11.3**	**11.1**	**11.6**	**11.8**

1. For Spain, data on earnings are taken from the *Encuesta de presupuestos familiares*, for 1998. Unemployment rates by age and education level are taken from the Labour Force Survey. Tuition fees are calculated as the proportion of private expenditures in education times total expenditure in education by education level, and come from OECD (2001d).
2. See Blondal *et al.* (2002) for methodology of calculation and data sources.

Source: OECD.

failure, would be ready to pay more for their education. Moreover, these subsidies are regressive, because they consist of a lump sum (in terms of low fees) paid independently from family revenues, and because they go to medium to high-income families or to students from less wealthy backgrounds who are likely to earn in the future more than the average worker.

Higher fees would liberate resources for more grants and subsidised loans or for other purposes. Greater funding of secondary education could be one alternative, since at this level of education the gap with other countries in terms of completion rates is still important; higher investment in vocational training is another option. Higher fees would also increase the marginal cost of repeating courses and would raise incentives for students to graduate early. They do not necessarily imply a reduction in the number of graduates. Although some authors have mentioned the existence of over-education in Spain (demonstrated by a large number of young graduates working in jobs for which they have excess skills), the calculation of the social benefits of university education for other countries, taking

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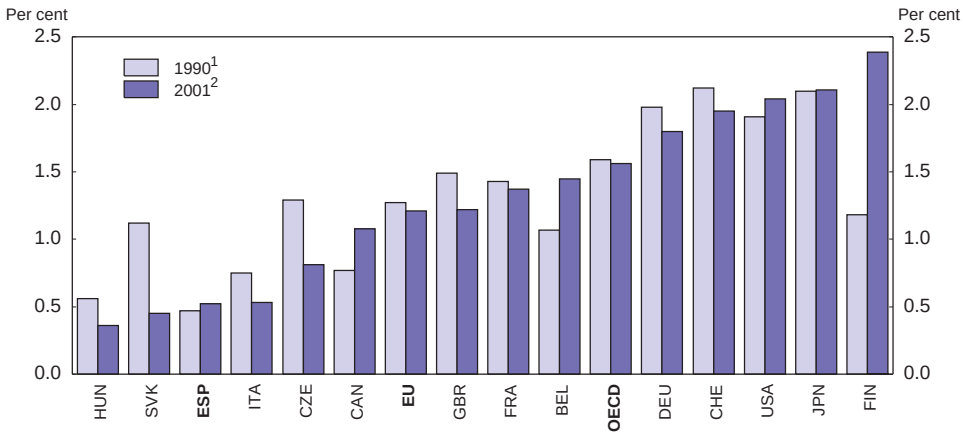
into account the direct cost of education and the opportunity cost of lost output, are still high at over 6 per cent for most countries (Blondal *et al.*, 2002). Moreover, other benefits not taken into account in these calculations (like externalities from having a well educated population, such as lower crime rates, or the fact that educated parents are the most important variable determining the success of children's education), make it advisable to avoid a reduction in the number of graduates. For those individuals desiring to enter university, but facing liquidity constraints, the resources freed by higher fees could be used to expand grant programmes for less well off students, or to subsidise loan programs, which are almost non-existent in Spain.[70]

Spending on research and development should be enhanced and better targeted

As the education system, production and, especially, the use of new technologies can have a considerable impact on productivity and growth, as highlighted by the OECD Growth Study (OECD, 2002b). Spanish businesses spend comparatively little on R&D (Figure 18), despite having by far the most generous tax credits for R&D outlays in the OECD.[71] R&D spending of public institutions is also comparatively low (0.4 per cent against an EU and OECD average of 0.6 per cent), though other indicators, like the number of researchers in universities and public institutes relative to the labour force, are above the OECD average, while

Figure 18. **Business R&D expenditure**
As a percentage of GDP



1. Or nearest available data.
2. Or latest available data.

Source: OECD, *OECD Science, Technology and Industry: Outlook 2002.*

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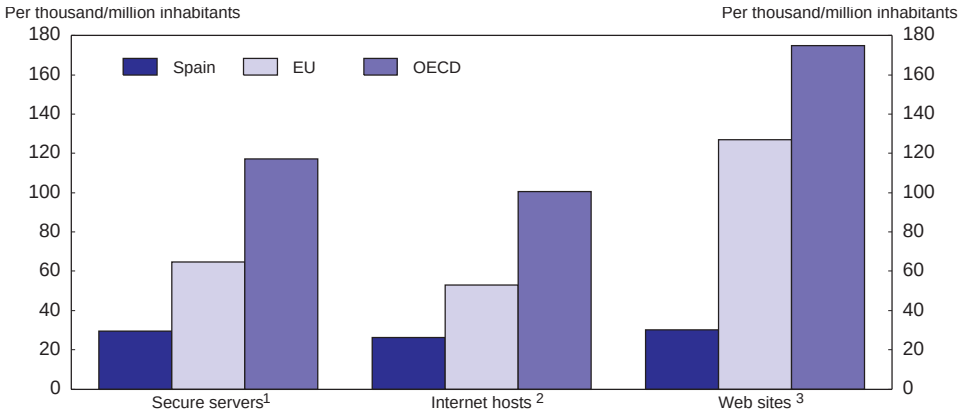
Figure 19. **Internet infrastructure**
July 2001



1. Secure servers per 1 million inhabitants.
2. Internet hosts per 1 000 inhabitants.
3. Web sites per 1 000 inhabitants.

Source: OECD, *OECD Information Technology Outlook: ICTs and the Information Economy,* 2002 ed.

the participation of private funding in public R&D projects is relatively high (OECD, 2001c). Utilisation of new technologies depends, among other factors, on the use of ICT, which can have a positive impact not only on new technology sectors, but also on improving productivity in traditional sectors like industry, health care and some types of services. In this respect, the internet penetration is rather limited (Figure 19), although the number of scientists and engineers working in businesses is rising rapidly.[72]

Given its present low levels, R&D spending and diffusion should be reinforced by inducing a more rapid incorporation of new technologies in businesses through appropriate policy settings. As noted above, labour market segmentation may slow the diffusion of new technologies. Moreover, greater competition in broadband telecommunications is important, as it is likely to raise the use of ICT more quickly. Public support to private R&D should also be rethought. The current very generous tax support could target incremental R&D spending rather than the level of R&D outlays, as is currently the case in the United States. The savings could then be used to channel support via other instruments like grants, which may engender less dead-weight losses (OECD, 2002b). However, the government should continually monitor the costs of such support against the would-be benefits.

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Housing market

Policy undermines the functioning of the housing market

Several features of the Spanish property market suggest that the sector is not functioning well. Property prices having risen more steeply in real terms since 1985 than in most other OECD countries, this being due in part to distortions affecting both housing demand and housing supply. Underlying demand for dwellings has been very dynamic since the mid-1980s for demographic reasons.[73] More recently, the growing housing demand is explained by the improvement of the general economic conditions which is reflected in a significant expansion of employment, by the fall in real mortgage interest rates and by the evolution of the stock market that has driven savings to other forms of investment. That trend, which is also underpinned by growing demand for second homes by non-residents, has been strengthened by tax incentives encouraging households to purchase their own homes.[74] The pressure exerted by demand has been only partially met because the increase in the supply of land has been limited, although the elasticity of land supply has been higher than in previous episodes of strong demand, while the building of new dwellings has been twice as high as that during previous expansions. However, in big cities land scarcity is due to limited supply, this being one of the main reasons for the rise in property prices: whereas the real cost of construction has fallen since 1980, the cost of land has been accounting for a growing share of house prices.[75] Both the large share of residential investment in GDP compared with other countries (Figure 20) and the share of real estate in households' wealth (Table 21) point to the existence of distortions in resource allocation. These distortions also affect the structure of the market, which is ownership dominated, whereas the supply of private and low-cost rental housing is much below the European average (Table 22). The poor return on rental property investment is due to the very restrictive regulations that governed the sector until 1985,[76] which resulted in a sharp drop in the supply of rental accommodation, particularly on the part of institutional investors, and in the worsening state of repair of this housing segment.[77] Legal problems to recover unpaid rents also explain the lack of attractiveness of this kind of investment for private owners. The inadequate supply of rental housing makes it difficult for population groups in an insecure situation or with modest incomes to find accommodation, and tends to make young adults stay with the family for longer than in neighbouring countries (Holdsworth and Irazoqui Solda, 2002).[78]

Housing policy has had a very strong influence on the way the market is organised. The main effect of that policy has been to encourage households to buy their own homes. Its implementation, which is shared by central government, the autonomous communities and the municipalities, is at present based on two main schemes. First, income tax relief is granted on the purchase of a main

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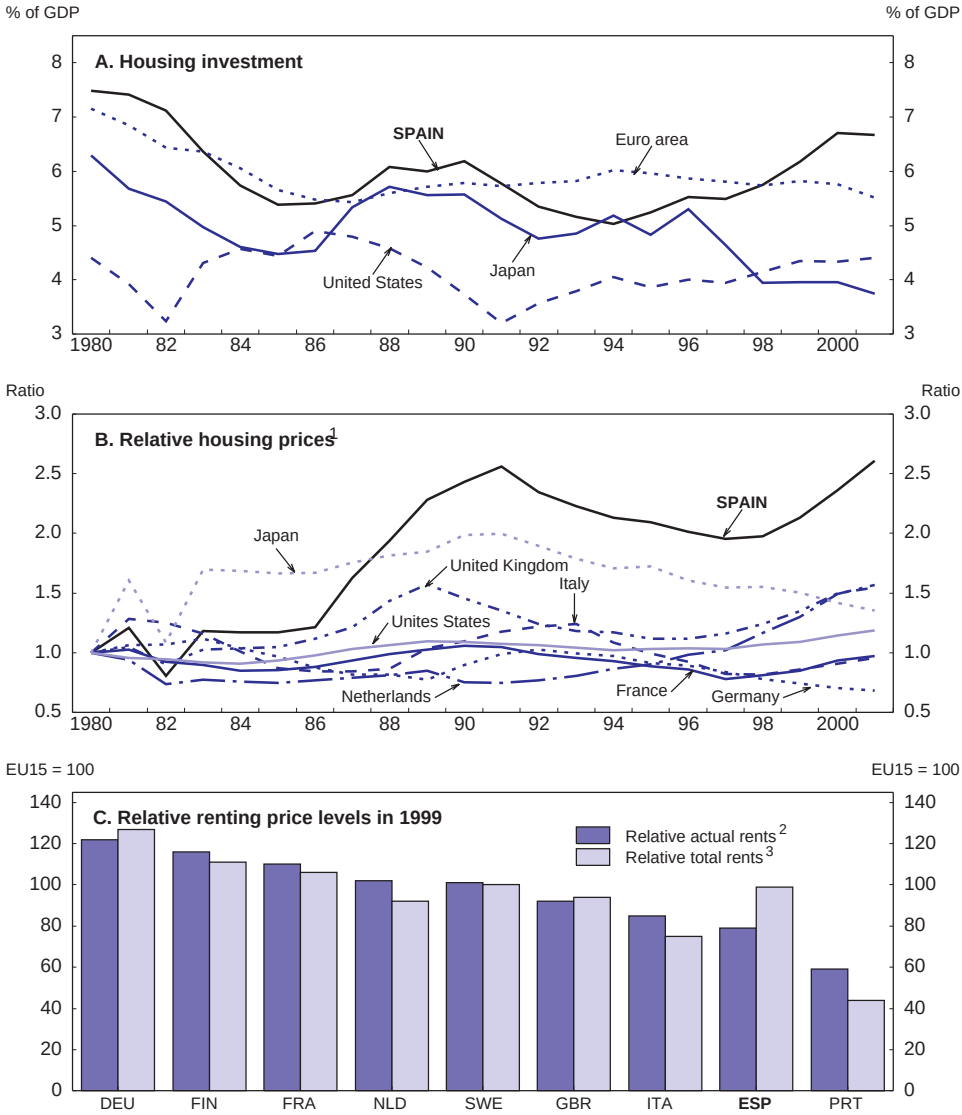
Figure 20. **Housing market trends**



1. Defined as the nominal housing price over the average production worker gross earnings.
2. Rents paid by tenants over private consumption.
3. Sum of actual rents paid by tenants and imputed rents by owner-occupiers over private consumption price level.

Source: BIS, EUROSTAT and OECD.

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Table 21. **Household assets**
Per cent of total household assets

	Housing assets					Other assets in 1998		
	1970	1980	1990	1995	1998	Equity	Other financial assets	Other tangible assets
Spain[1]	**. .**	**65**	**73**	**71**	**67**	**9**	**20**	**4**
United States	22	27	27	23	21	20	50	8
Japan	10	14	8	10	10	3	44	43
Germany	. .	. .	34	34	32	3	35	30
France[2]	34	44	43	42	40	3	47	9
Italy	36	40	37	35	31	17	39	13
United Kingdom	. .	40	44	33	34	12	47	7
Canada	21	22	23	22	21	17	39	23

1. 1980 data refer to 1984.
2. 1998 data refer to 1997.

Source: OECD *Financial Accounts of* OECD *countries*; United States, Federal Reserve, *Flow of Funds Accounts of the United States*, September 2000; Japan, Economic Planning Agency, *Annual Report on National Accounts*, 2000; Germany, Deutsche Bundesbank, *Ergebnisse der gesamtwirtschaftlichen Finanzierungsrechnung der Deutschen Bundesbank*; France, INSEE, 25 *ans de Comptes de Patrimoine* (1969-1993), and *Rapport sur les Comptes de la Nation*. (France's estimates for household equity holdings have been corrected to exclude non-quoted shares. Data presented are based on national authorities' estimates of the ratio of quoted to non-quoted shares in household portfolios); Italy, Banca d'Italia, *Supplementi al Bolletino Statistico* and unpublished estimates (the data on equity holdings in Italy include both quoted and unquoted shares); United Kingdom, Central Statistical Office, *United Kingdom National Accounts, Financial Statistics*; Canada, Statistics Canada, *National Balance Sheet Accounts*; Spain, *El balance nacional de la economía española* 1984-2000 by J.M. Naredo and O. Carpintero.

residence, without any means test (Chapter II). This reduction in tax, to which must be added a rather favourable capital gains tax regime, would appear generous by comparison with other savings products and in relation to the international average (OECD, 2000). However, the 1998 income tax reform reduced the generosity of the regime. Second, as part of a housing plan that is renewed every four years, assistance (accounting for 0.1 per cent of GDP in 2002) is granted either on a means-tested basis or to specific groups.[79] While this policy of assisting home ownership does make for a well-maintained housing stock, it also has the effect of increasing the share of real estate in households' wealth by reducing the real mortgage rate. It can, moreover, reinforce an imbalance in the economy in the event of an overheating housing market.

Being aware of these distortions, the authorities have sought to expand the supply of rental accommodation. Rental contracts were liberalised in 1985. Although rental leases extend for a minimum of five years since 1994, specific measures to promote low-cost rental investment were also included in the housing plans, but have not so far had the hoped-for effects. The liberalisation of rental leases in fact concerned only contracts dating from after 1985, with the result that a large proportion of rented accommodation still comes under the rules enforcing

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Table 22. **Tenure and availability of housing**

	1999				2000	2000	
	Per cent of the stock of main residence						Vacant dwellings in per cent of total stock
	Owner-occupied	Rented			Dwellings per 100 households	Dwellings per 1000 inhabitants	
		Total	Social	Private			
Spain (1960)	55	45	2	43			
Spain (1980)	79	21	2	19			
Spain	**86**	**14**	**2**	**12**	**147**	**485**	**14.8 (1991)**
Austria	55	45	23	22	115	465	4.6 (1990)
Belgium	63	37	7	30	110	455	3.9 (1990)
Denmark	55	45	19	26	104	470	3.9 (2000)
Finland	67	33	14	19	109	483	8.3 (1999)
France	54	46	17	29	121	494	6.9 (1999)
Germany (western part for housing tenure)	43	57	10	47	98	452	7.5 (1998)
Ireland	80	20	11	9	100	345	10.2 (1991)
Italy	78	22	6	16	121	434	21.3 (1991)
Netherlands	48	52	35	17	96	416	1.2 (1999)
Portugal	68	32	4	28	135	465	10.5 (1991)
Sweden	62	38	22	16	102	483	0.9 (2001)
United Kingdom	69	31	21	10	98	411	3.6 (2000)
EU average	61	39	18	21	112	452	

Source: Trilla (2001); Euroconstruct (2001); Housing Statistics in the European Union.

low rents and the indefinite extension of contracts.[80] As a result, there is a rental accommodation demand unsatisfied, mainly from medium and low income groups. Yet, from 2003 onwards, the supply of rental accommodation could be increased as a result of the liberalisation of those contracts that were regulated by the transitory regime introduced in 1994. Moreover, increases in rentals actually exposed to market forces are restricted by the limited finance available to low income groups, in the absence of government rent support, which exists in certain OECD countries. Where wealthier households are concerned, rent levels are also restrained by the fact that housing policy makes it possible to purchase accommodation on favourable terms. Average rent levels (excluding imputed rents) are lower in Spain than in the other European countries despite their rapid rise in recent years (Figure 20), which is explained to some extent by the still large proportion of rental contracts that remain under the pre-1985 regulation. Although this difference no doubt partly reflects the inferior quality of this housing segment, the rate of return on rental investment has remained low, and the available housing stock is still under-utilised, as the large proportion of unoccupied homes

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shows.[81] Other factors such as an asymmetric VAT treatment that favours construction for purchases rather than for renting, and the legal uncertainty faced by private owners when trying to recover unpaid rental income, reduce the supply of rental housing.

The trend rise in real property prices is both increasing owners' wealth and also encouraging people to buy their homes rather than rent. Despite the government's efforts in recent years, it has not proved possible to stop the rise, which is largely associated to the increase in the price of land for building. Central government has little leeway to change land management, which is under the responsibility of the autonomous communities and municipalities.[82] Nonetheless, the central government has reached an agreement with the autonomous communities in order to rationalise land management. This agreement is expected to be extended to the municipalities through further negotiations. The latter control the supply of building land and lay down town-planning rules whose excessive rigidity explains why supply is rationed. The rules, which vary from region to region, are the reason for the limited effect of the 1998 reform which stipulated that all land should *a priori* be considered to be developable, unless expressly decided otherwise.[83] It is in the interest of the municipalities to keep prices high by using arbitrary criteria to issue building permits. Landowners have to transfer part (10 per cent) of their land to the municipality if they sell, and real estate taxes are the local authorities' main source of tax revenue (nearly 66 per cent in 1999).

Distortions should be removed

All told, improving the functioning of the housing market requires on the demand side a re-balancing of incentives to rent and purchase houses and, on the supply side, a re-balancing between the return on rental investment and on alternative investments (Box 3). The authorities' policy of encouraging the supply of rental accommodation, such as the recent reform to reduce the taxation of net rental income by 50 per cent in conjunction with the ability given to municipalities to tax non-occupied homes more (Chapter II) tends to rebalance the market through renewed government intervention. While it does make for better utilisation of the available housing stock, this approach would seem to be less effective than phasing out home ownership assistance, which would restore incentives to rent and build new rental housing. Increasing the supply of such housing is very relevant to the effectiveness of the recent labour market reforms encouraging geographic mobility, and also to the better integration of the fast-growing immigrant population (Chapter IV). A reduction of the tax relief for future house buyers combined with rent support measures for households with modest means would be justified in view of the fall in financial costs resulting from the reduction in mortgage rates.[84] Such a reform has been introduced in other OECD countries in recent years, France being an example. It would also help to slow the trend rise in real

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Box 3. Analysis of rental supply and demand incentives

The existence of an adequate rental market depends on the incentives affecting household demand, *i.e.* relative rental costs compared with the costs of buying a home, and also on the supply of these housing services, *i.e.* the return on rental investment as compared with alternatives, *e.g.* financial investment. It is important to ensure a balance between these relative prices so that rental demand and supply match.

To show what role these incentives play, a small model has been built to calculate the effect on household saving of a decision to rent rather than buy one's home. The model, outlined in Annex IV, compares the amount of assets accumulated by a household depending on the housing solution it opts for, taking into account the prevailing tax liabilities and allowances, mortgage rates and rates of return on financial investments, the cost of renting as opposed to buying and the trend in inflation and in property prices.

In the case of a typical household buying a home by taking out a fixed-interest rate loan, the value of its wealth in a given year is calculated as the difference between the sales price of the property and the capital still to be repaid. The value of the asset is, moreover, increased by the cumulative amount of tax relief attaching to the purchase. In the case of a household renting a home, its wealth is calculated as its stock of saving, assuming that the household systematically invests on the financial market the surplus being equal to the difference between the rent paid and the loan repayments it would have had to make if it had bought. In this latter case, account is taken of the tax on the lessee's income from saving and the index-linking of rents.

The model highlights the size of the distortions

Assuming that inflation and real property prices rise at the same rate (2 per cent per year), buying one's home is a much better option than renting if the average rental price provides a return comparable to that on financial investments.* The assets built up by the buyer exceed those of the lessee by the third year and are on average 50 per cent greater on a discounted basis (Figure 21).

The higher the rate of inflation, the greater the incentive to buy because of the decline in the real cost of repaying the loan, whereas rents are index-linked. If the cost of the property increases because tax relief is capped, on the other hand, incentives are smaller.

Because of the trend rise in property prices and in tax allowances in favour of owners, a 40 per cent fall in rents is needed to achieve equilibrium between the incentives to rent and to buy one's home. However, that sort of fall in rents would imply a considerable decline in the return on rental investment compared with alternative forms of investment, and this would prevent the expansion of rental supply.

If expectations as to real property prices could be stabilised, the advantage of buying rather than renting accommodation would decline because of the narrowing of the difference between the value of the buyer's cumulative assets and

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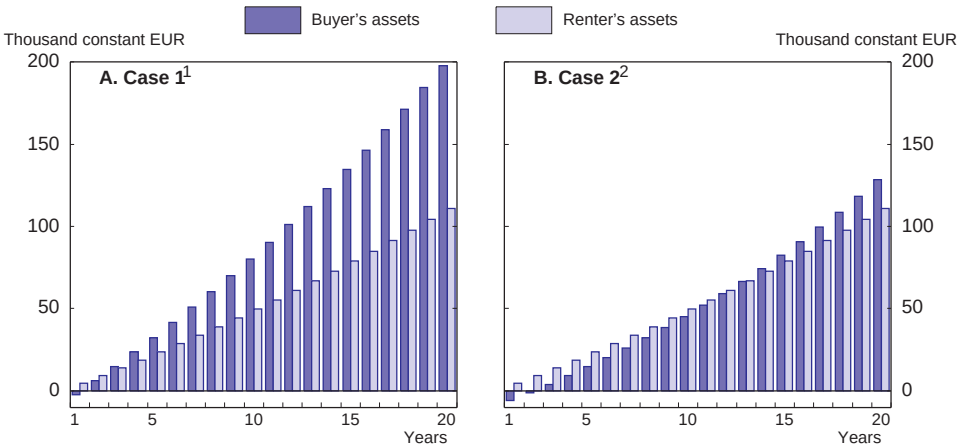
Box 3. **Analysis of rental supply and demand incentives** (*cont.*)

those of the lessee. In this case, the buyer's assets exceed the lessee's as of the sixth year and the average differential in favour of the buyer is in the region of 20 per cent.

Abolishing tax allowances encouraging the purchase of property, coupled with the stabilisation of expectations as regard real property prices, would balance the incentives to buy and to rent accommodation, while at the same time maintaining attractive returns on rental investment. In this case, the buyer's cumulative assets would not exceed those of the lessee until the fourteenth year and the differential in favour of buying would on average be eliminated over the period as a whole (Figure 21).

* Average annual rental costs, which are assumed to be between 5 and 5.5 per cent of the price of the property, give a return close to that on public debt. Rents of that order are similar to those observed on the market (San Martín, 2000).

Figure 21. **Wealth developments linked to housing purchase/rental**



1. Tax subsidies for the buyer and an annual appreciation of the property of 2% in real terms.
2. No tax subsidies for the buyer and no appreciation of the property in real terms.

Source: OECD.

property prices by easing demand pressures. The stabilisation of property prices that would be encouraged by more homogeneous, transparent and flexible management of the supply of land by the communities and municipalities would also seem important. In order to reform land management, the central government has

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already taken steps to reach an agreement between the different levels of government, and this could be facilitated by reforming the system of municipal financing in a way that increases their financial resources and makes them less dependent on property and real estate taxes.

Competition policy

More resources have been provided for the competition authorities

As recommended in previous *Surveys*, the resources of the competition authorities have been considerably enhanced between 2001 and 2003, including budgetary allocations for the Competition Tribunal (TDC) that more than doubled during this period. The Competition Service (SDC), which prepares cases for the TDC and depends on the Ministry of the Economy, has also seen its resources increased, and its personnel comprises 95 people (of which are 38 economists, lawyers and other professionals) while its administrative rank has been raised to one of a general directorate. The TDC has also been provided with more administrative autonomy, while 50 per cent of fees paid by firms for the review of mergers has been channelled into its budget.

After a 1999 ruling of the Constitutional Tribunal that attributes competition enforcement competence to autonomous communities in cases that affect only a region, a new law on the co-ordination of regional competition authorities has been approved. Two regions (Madrid and Cataluña) have already started to set up regional tribunals, while other communities have declared their intention to do so.[85] This division of enforcement responsibility should make it possible to discharge the national competition institutions from the need to investigate smaller cases, permitting them to devote more resources to other matters. Institutional mechanisms have also been set up to settle disputes that might arise among different regional institutions in the initial allocation of cases and to ensure uniform interpretation among different tribunals. The authorities should ensure that these mechanisms maintain a consistent national competition policy, and that regional tribunals do not become means for giving advantages to local firms.

Regional entry barriers to retail distribution should be scrapped

In retail distribution, which is a large sector (14 per cent of total employment), opening hours (90 hours per week) are not particularly restrictive as compared to other EU countries, and in practice they are higher because shops are allowed to open at least 13 Sundays per year (depending on the region, since regional governments can raise that number).[86] Small shops and drugstores are free from restrictions. However, since the 1996 Commerce Law was approved following a ruling of the Constitutional Tribunal, a second permit, in addition to the municipal permit, is needed from regional governments to open a new hypermarket. This has

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been used as an entry barrier, since many regional governments have explicitly blocked the issuance of new permits or have delayed concessions.[87] Other regions (Cataluña and Navarra) have approved new taxes applying only to hypermarkets or hard discount retailers. Such entry barriers should be scrapped, since they tend to protect local companies and undermine the functioning of this market, resulting in higher inflation and efficiency losses. In the pharmacy sector, where framework legislation is provided by the central government but regulation is approved by autonomous communities, some price competition has been introduced for para-pharmacy products, but retail margins are fixed for other products and entry barriers still exist in the form of minimum distance between pharmacies and minimum population per pharmacy. These barriers, which are reflected in high prices for the transfer of pharmacy licences in some areas, should be eliminated, while ensuring provision in areas with low population density through other means, such as subsidies.

Reforms in telecommunications are following technological innovation

Since liberalisation started in 1998, prices of fixed telephony have dropped by more than 50 per cent, and are close to the OECD average (Figure 22). Consumers also benefit from a wider choice of suppliers, with more than five operators for local, long distance and international calls and for direct access. Although new entrants are improving their position, the market share of the incumbent operator has remained high and has decreased less than in other countries after liberalisation.[88] Unbundling of the local loop was implemented in January 2001, but has only started to be used by competitors in the second half of 2002, when the sectoral regulator (CMT) imposed much stricter conditions on Telefónica. The CMT has also improved the Reference Interconnection Offer, which includes the terms of access to the incumbent's network: prices were reduced by 30 per cent in August 2001, and a new pricing system was implemented based on transmission capacity instead of time that has been heavily contested by Telefónica but should benefit its competitors by allowing them to diversify their offers to customers. Prices of Telefónica are still capped by the government, but will be fully liberalised by 2005. For 2003, and in the context of financial difficulties for telecommunications companies, the reduction approved has been much more moderate than in previous years,[89] as margins were already low for other competitors (which are price followers of the incumbent operator). As in other countries, broadband connections have developed rapidly during the past year, with asymmetric digital subscriber line (ADSL) services reaching more than one million customers in January 2003, which represents a high broadband penetration rate in terms of internet users (16 per cent), but is still relatively low with respect to population (3 per cent of households). Prices of ADSL are capped for Telefónica, and followed by other companies that have to negotiate wholesale agreements with the incumbent operator, who controls 78 per cent of the market. Commercial development of

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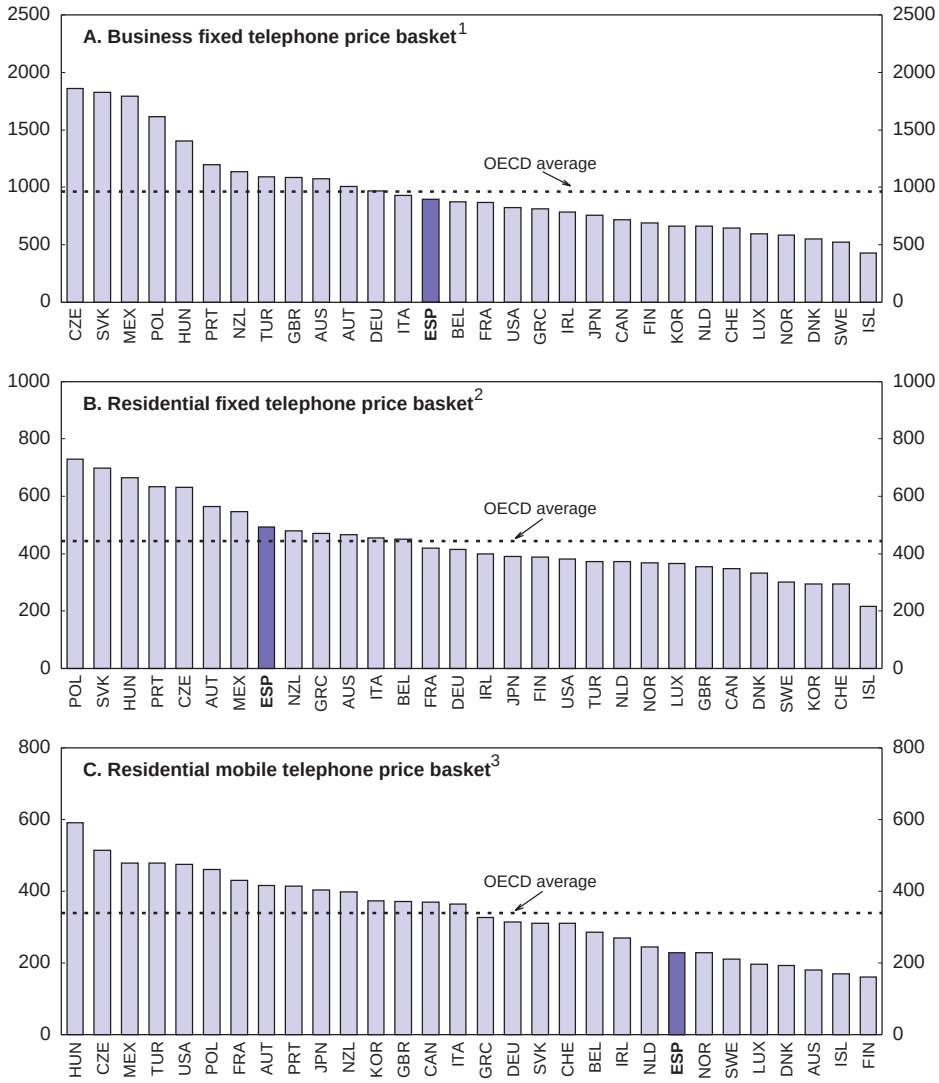
Figure 22. **Telecommunications prices**
In USD PPP, May 2002



1. Excluding VAT. Calls to mobile networks and international calls are excluded.
2. Including VAT. Calls to mobile networks and international calls are excluded.
3. Including VAT. The basket includes 50 national minutes and excludes international calls.

Source: OECD, *Communications Outlook* database.

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cable transmission has also increased strongly during 2002. Cable reaches 5.4 million households and firms, one third of the potential market, and is used by more than 800 000 customers. Given the still high market share of Telefónica in fixed telephony and to a lesser extent in data transmission, the development of cable has started to be a potential alternative for both types of services.[90]

In mobile telecommunications, the market is now mature (29 million users, out of a population of 40 million). Prices have dropped much more slowly in 2001 and 2002 than in previous years, and are close to the OECD average. In 2002 the regulator forced the two incumbent mobile operators (Telefónica and Vodafone) to cut their termination charges by 30 per cent. A similar cut was imposed to the third operator, Amena, some months later. As in other countries, universal mobile telephone systems (UMTS) investment and services have been delayed with respect to original deadlines, but they are expected to be developed during 2003 under pressure from the government.[91] The radio-spectrum tax paid by UMTS suppliers was lowered in 2001, and fixed for the period 2002-06, providing a more stable investment environment for companies.

The authorities should continue to regulate Telfónica's wholesale tariffs as long as the unbundling of the local loop is not extensively used, or until cable networks provide an effective alternative to the incumbent's network, but there is scope for reducing the coverage of the price cap so as not to include national or international long distance calls and ADSL retail services. The sectoral regulator should continue to scrutinise all segments of the market for anti-competitive practices.

The energy sectors are dominated by a small number of firms

In the energy sector, a number of measures were approved in June 2000 to accelerate liberalisation and strengthen competition, and to facilitate the entrance of new operators (OECD, 2001a). Currently, those activities, which are natural monopolies, are in private hands but regulated, with prices directly determined by the government (electricity and gas) or privately negotiated by market participants (oil sector). Recent measures have gone in the direction of developing the June 2000 decree:

- In *oil distribution*, ownership separation of the transport monopoly (CLH) is almost complete, and the three largest companies (Repsol YPF, Cepsa and British Petroleum) now own less than 50 per cent of CLH, while conditions of storage and transport contracts with distributors are now published by the sector regulator, improving transparency. In retail distribution, the temporary limits on the expansion of the two largest distributors (Repsol YPF and Cepsa), applied since 2000, have not resulted in lower market shares. The limits were defined in terms of distributors and not quantity sold, and both firms have been allowed to

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close small stations and open larger ones. The Competition Tribunal has also sanctioned them for imposing restrictive conditions in their long-term contracts with independent retailers. Following the June 2000 measures, several new petrol stations have been opened in hypermarkets pushing prices down somewhat. Prices, which were remarkably uniform across stations, have started to diverge at the end of 2002. However, the entrance of new competitors is still slow, and the number of stations per inhabitant remains far below the EU average.

– In *gas import and distribution*, the process of divestiture of the largest players in the company that manages large pipelines (ENAGAS) has been completed, and now the largest player (Gas Natural) owns less than 35 per cent of the company, which should ensure a fair access to the network of new market entrants. Other measures envisaged in June 2000, like the publication of the formula for calculating transport fees and the auction of part of the contract of Gas Natural for importing gas from Algeria have also been approved. In order to cope with the future demand of gas from electricity plants under construction, at least two new re-gasification plants are being built, while pipelines with France and North Africa are being expanded, which should allow for more competition. Retail distribution, *i.e.* the regulated activity of providing physical access to final customers, is still a local monopoly franchised to Gas Natural until 2005 to encourage new investment in the network, which is still underdeveloped. Moreover, fair access to the distribution network has been granted through a regulated system that establishes fees for its use. Supply of natural gas has been fully liberalised and at present, more than 35 suppliers are registered and may present competitive offers to all consumers.

– In *electricity markets*, an expansion of interconnections with France is planned, and should reach 11.6 per cent of projected demand by 2005, increasing potential competition from imports which are currently low at less than 6 per cent. Choice of provider has been extended to all consumers in 2003. As a result of the liberalisation, consumers will have access to liberalised prices, while the market regulator will continue to publish a reference tariff for those consumers who decide not to buy electricity in the market, which in practice will set a maximum price. This tariff should last until 2007 for large consumers and at least until 2010 for households. A new formula for determining this tariff has been approved at the end of 2002, which implies annual price increases of around 1.4 per cent, with an upper limit of 2 per cent, below projected inflation but above the price reductions achieved in recent years. Payment of stranded costs, including payments arising from the moratorium

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on nuclear constructions and subsidies to national coal production will continue to be included in the price calculation.

Liberalisation of the energy sectors has resulted in efficiency gains and important price reductions in recent years, but energy markets are still dominated by a small number of companies. Competition from new entrants has been limited so far, although the legal conditions for entry have improved and interconnections with foreign suppliers are being expanded. The highly concentrated nature of production and distribution of energy products has resulted in some instances of anti-competitive behaviour and a significant increase in profits in part of the energy sector (Figure 9). The authorities should continue to be vigilant and ensure that competition works. They should consider divestitures of assets for the main players or vertical ownership separation of production from distribution.

EU *aid to modernise the fishing fleet will be phased out by* 2005

Overfishing has been a serious problem in EU waters, with fish stocks outside safe biological limits in many fishing areas (ICES, 2002). The decline in fish stocks has also contributed to significant over-capacity of the fishing fleet, with the European Commission estimating there are almost twice the number of vessels needed for the capture of a sustainable harvest. In Spain the fishing sector represents 0.5 per cent of GDP, and directly accounts for 0.4 per cent of total employment. Although activity is spread along the coastline, production is more concentrated in areas such as Galicia and Andalusia, where per capita GDP is well below the national average. Being a very large consumer, Spain is also a net importer of fish. Fishing policy is mostly a responsibility of the European Union, which annually determines fishing quotas for different species and provides subsidies for the modernisation of the fleet. Despite a sharp reduction, Spain still has the second largest fishing fleet of the Union (after Denmark), though it is older on average. European subsidies to the Spanish fleet amounted to EUR 128 million in 2001, while the total budget for fishing (including European funds) was EUR 240 billion (about 8 per cent of the sector's value added). At the end of 2002, the European Commission launched a fishing policy reform, in the context of needed catch reductions to avoid over-fishing. The Commission proposed to stop providing help for construction, modernisation and export of vessels so as to halt the expansion of the sector. The final agreement postpones this measure, extending this aid until the end of 2004,[92] improves control measures against illegal fishing and reduces fishing quotas for 2003 though to a higher level than originally proposed. Most urgent quota reductions approved (for instance, for cod fishing in the North Atlantic) will not affect Spain, since until this year Spanish and Portuguese fishermen were not allowed to fish in the northern EU waters. From 2005, the elimination of investment subsidies for improving the fleet and the funding of alternative activities in fishing areas is a welcome

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measure that should help to reduce the least efficient part of the fleet, while aid will concentrate on developing alternative activities in affected regions.

Financial markets

A *new reform of financial markets has been approved*

A new Finance Law was approved in November 2002 that addresses a wide range of topics. Some of the changes adapt regulations to European legislation, to the increased use of ICT technologies in financial markets and to a more competitive environment after the introduction of the euro.[93] Other changes aim at improving the protection of small investors; for example, through the creation of a new Commissioner for the Defence of Investors that deals with abuses of financial institutions, including those in internet banking.[94] The law also includes some measures to improve access of small and medium-sized enterprises (SMEs) to financial markets, addressing an alleged problem of credit market failure. In particular, the scope of the securitization of mortgage-backed credits has been widened to include those not aimed at financing real estate; factoring of credits with the public sector will be permitted, which should favour SMEs since up to now they found it difficult to participate in public contracts due to the erratic payment patterns of some public administrations. The Financial Law also modifies the tax regime of venture capital entities. Venture capital was reformed in 1999, introducing tax advantages and putting venture capital firms under the supervision of the stock market regulator (CNMV). After a slow start, investment in venture capital rose steeply in 2000, but fell in 2001 following the flight from risk, and has reached cumulated assets equivalent to 0.18 per cent of GDP. This is still well below the level in other European countries,[95] though it is the only one, except for the United Kingdom, that has grown in 2002. The new tax regime extends tax advantages of venture capital investors to the period after their partial sale in the stock market.[96]

The Enron scandal spurred measures that aim to improve the transparency of financial market and auditing obligations of firms. Insider information and market manipulation issues are addressed by adopting the EU Directive on Market Abuse. Any operation of the dominant owner of a firm with other large investors will have to be disclosed. New norms are set for investment counsellors and financial analysts, like the obligation of communicating any potential conflict of interest and the ban on dealing in securities which they analyse. The new regulation on company auditing imposes compulsory rotation of external auditors every nine years (every seven years for large firms or for firms quoted in stock markets), determines sanctions for individual auditors and auditing firms and defines a new incompatibility regime. Extensive supervisory powers are given to the Spanish Institute for Accounting Standards and Auditing (ICAC), an independent body. The

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law does not address specific issues of corporate governance, which have been left to an independent Commission (*Comisión Aldama*) which has published a list of recommendations. Given the need to improve confidence in financial markets at the present juncture, it would be advisable that key recommendations of this commission be translated into legislative regulations rather than being left to voluntary codes of good practice.

The new law also introduces some regulatory changes for savings banks. These banks are defined as foundations, with no capital, and have been gaining market share after a process of regional consolidation and expansion outside their region of origin (OECD, 2001a). To avoid increased political interference, the voting rights of any public entity have been limited to 50 per cent, while the nomination of board representatives has been made irrevocable during their official mandate. Furthermore, in order to increase their own resources, the regulatory framework of special securities (*cuotas participativas*, whose remuneration is linked to profits though they do not incorporate voting rights) has been improved. These securities, which were previously allowed but were scarcely used due to restrictive issuance conditions, will provide more resources for further expansion and therefore could accentuate market share gains *vis-à-vis* commercial banks. Currently, savings banks may buy commercial banks but not the opposite (since savings banks have no equity). If this asymmetry creates an excessive competitive advantage for savings banks, it should be addressed by limiting their expansion through acquisitions or by allowing the reconstitutions of savings banks as share companies.[97]

Bank solvency remains high despite losses in Argentina

Bank profitability has been eroded by the squeeze of intermediation margins due to lower interest rates and the stock market crisis, and by losses due to the Argentinean crisis. After-tax net consolidated profits fell by 6.4 per cent in the first half of 2002, while the return on equity was 12.9 per cent in mid-2002 as compared to 13.4 per cent in 2001; however, this reduction of profits has been milder than that of other European banks (Bank of Spain, 2002c).[98] Credit risk levels have increased in part because of a rise in bad loans, many of them in Argentina, while market risk in the area is still important.[99] But solvency ratios for Latin American affiliates, which the Bank of Spain follows closely, are very high (at close to 20 per cent), while overall solvency for Spanish banks is well above the statutory BIS minimum of 8 per cent.[100] In the domestic market, the ratio of bad loans (1 per cent) is still very low, and credit risk has fallen since the end of 2000 due to more moderate credit growth. However, mortgage loans, which account for half of total credit and were growing at 18.5 per cent in the first half of 2002, should continue to be closely monitored by the Bank of Spain, since they are weighted as low risk assets in standard risk measures but would be damaged if a sharp fall in housing prices occurs.

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Since June 2000, the Bank of Spain has implemented a regulation that obliges banks to provision loans from the moment when a loan is granted, taking into account the average risk associated with that type of loan during the business cycle instead of the risk perceived when the loan is made. This implies that provisions increase when credit rises (expansions), instead of when risks increase (usually in downturns). Banks can use their own models for evaluating the risks by type of loans over the cycle, though a standard model has been developed by the Bank of Spain to be applied by those banks that do not have the technical capacity to develop their own model. The system is welcome, since it improves protection against banking crises, which usually appear when risks are re-assessed during downturns. It may also contribute, to some extent, to limiting excessive credit growth during expansions, since provisions have a cost that is now higher when the loan is granted.

The benefits from deepening structural reforms

The deepening of structural reforms in labour and product markets would bring important macroeconomic benefits. A reform of the wage bargaining framework, together with more competition in some protected sectors would contribute to closing the inflation gap with the euro area, which erodes competitiveness. Further progress in market liberalisation, including a reduction in the segmentation of the labour market, would underpin a more rapid diffusion of new technologies in the production process and invigorate productivity growth, which would also be favoured by improvements in R&D investment and in education outcomes. In addition, a more balanced housing market that would favour a better allocation of resources and greater geographic labour mobility would contribute to reducing structural unemployment.

The OECD medium-term baseline scenario, which extends short-term projections until 2008, already incorporates a structural improvement of the economy due to earlier reforms. This baseline scenario, which is similar to the Spanish Stability Programme, which extends to 2006, shows a slight budget surplus and GDP growth of 3 per cent between 2004 and 2008. In this scenario, a decline in the unemployment rate to 9½ per cent at the end of the period would be compatible with a reduction of inflation, which would stabilise at around 2 per cent in 2007-08. The further reduction of the natural rate of unemployment (NAIRU), together with a negative gap at the beginning of the projection period, would allow activity to be sustained above potential growth, which is on average 2.8 per cent even under the assumption of trend productivity growth not accelerating with respect to the recent past. The Spanish economy would continue to perform better than the euro area average, although less markedly due to some exceptional factors, like the reduction in real interest rates coming to an end. Between 2003 and 2008, the favourable growth differential with the euro area could be about ¾ percentage

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point, while the inflation differential would stabilise slightly above ½ per cent by 2007-08.

Two alternative scenarios have been simulated, using the OECD Interlink model. They illustrate, on the one hand, the potential positive effect of implementing the ambitious structural reforms suggested in this chapter and, on the other, the possible consequences of the lack of flexibility of labour and goods markets, and of excessive nominal rigidity of wages and prices, that would hinder the reduction of the inflation differential with the euro area. The main results of these simulations are presented in Figure 23 in terms of differences with respect to the performance of the euro area.[101] In the pessimistic scenario (unchanged inflation differential with the euro area), it is also assumed that the lower flexibility of the labour market would translate into a smaller reduction of the NAIRU than in the baseline scenario. Indeed, the structural unemployment rate, estimated at 11½ per cent in 2002, would not fall below 10¾ per cent by 2008, instead of 9½ per cent in the baseline scenario, which would reduce the growth potential to 2½ per cent between 2003 and 2008. In that case, the growth gap with the euro area would narrow considerably, to at most ¼ percentage point between 2003 and 2008, mainly reflecting a less favourable export performance resulting from competitiveness losses. Lower growth would also affect the labour market, while the public accounts would be worse, with a deficit of ¾ per cent of GDP by 2008.

Conversely, the scenario of intensified structural reforms assumes that the inflation differential with the euro area disappears by 2006. In addition, the positive effect of reforms on productivity and structural unemployment is assumed to improve potential GDP growth by ¼ points on average between 2003 and 2008 with respect to the baseline scenario, to reach 3¼ per cent at the end of the period. In this case, growth would be above potential from 2003 onwards without generating inflationary pressures. The favourable effect of disinflation on competitiveness and wage growth induced by efficiency gains would favour higher exports and internal demand growth despite the higher real interest rates with respect to the baseline scenario. Higher growth would improve the budget balance, which could reach a surplus of 1 per cent of GDP in 2008. In this scenario, the growth difference with the euro area could reach 1 percentage point.[102] If such a differential were to persist, the gap in per capita GDP with the European average (14½ per cent in 2002) would be closed by 2014, ten years earlier than in the baseline scenario.

Sustainable development

There is growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental, and social outcomes. This section looks at three issues at the

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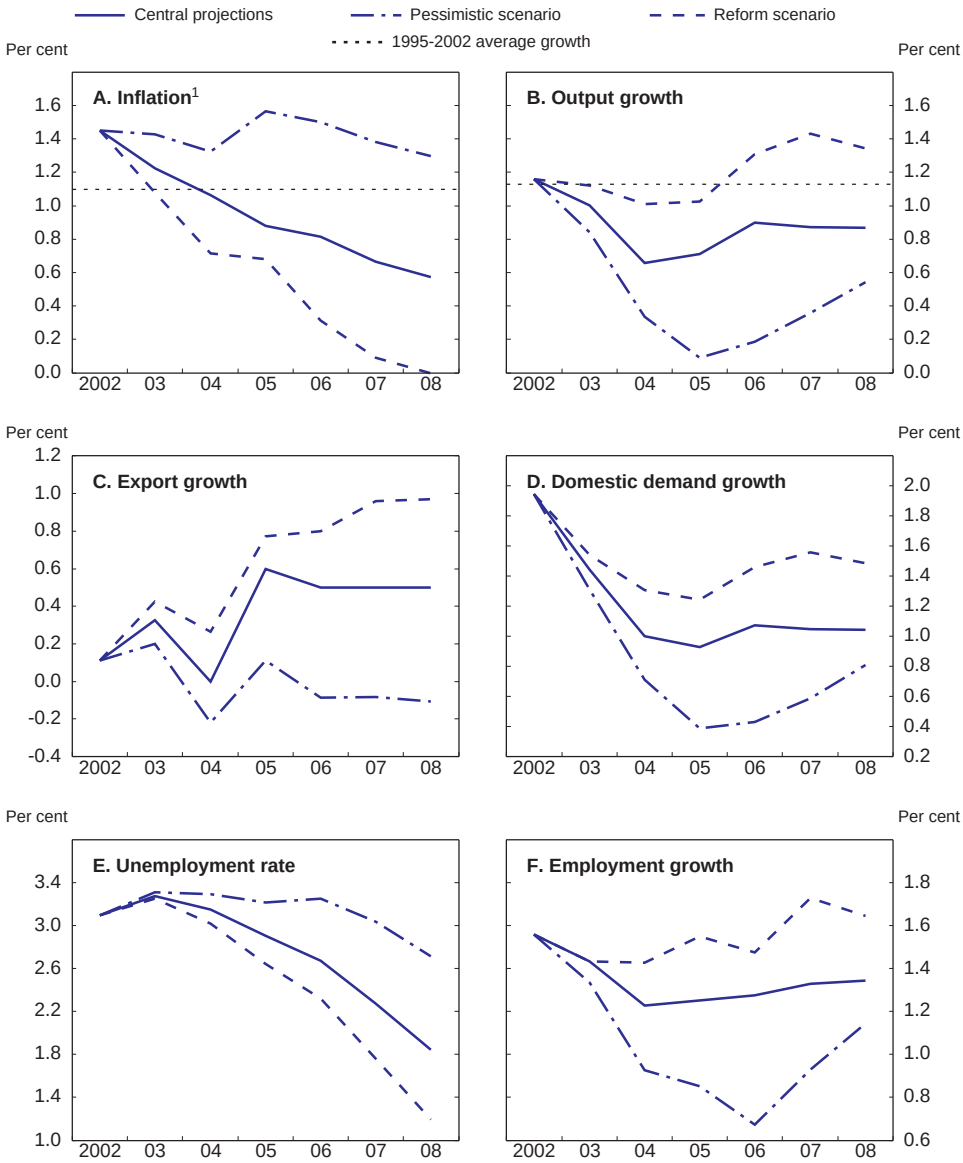
Figure 23. **Medium-term effects of structural reforms**
Performance differential with the euro area



1. Change of the private consumption deflator over the previous year.
Source: OECD.

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interface of the environmental and economic dimensions of sustainable development, drawing out the social implications when relevant, that are of particular importance for Spain. In each case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies that affect the issue. The section also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (Box 4). The section considers first Spanish policies in the realm of *climate change*. It then looks at *waste management* policies. Finally, the section examines developments in ensuring the *sustainable use of natural resources*.

Box 4. The integration of environmental concerns into government policy*

Sustainable development has risen in importance in Spain. One of the major motivations for environmental policy initiatives are the requirement to transpose EC directives into national law and Spanish international commitments, often reinforced at the EU level. The Ministry of the Environment takes the lead in drafting national policies for specific areas. These plans are either integrated into existing regional plans or form the basis for future regional legislation. The main institutional mechanisms for ensuring co-ordination are regular sectoral conferences, which consist of representatives from both central and regional government. Several specialised advisory bodies also provide advice on major legislation and plans. Furthermore, environmental impact assessments and cost-benefit analysis are often used for larger projects and policies.

Policy integration is being developed at present. Due to the ongoing evolution of the allocation of competencies and financing between levels of government, the new demands of sustainable development policies can create implementation difficulties. For example, while autonomous communities currently have major responsibility for environmental protection, when more than one autonomous community is involved central government intervention is required. Conflicts over proposed legislation between the regions and central government can result in policies being challenged in the constitutional court, leading to delays in implementation. Strategic environmental analysis will become a systematic requirement for major public plans in Infrastructures when the European Directive 2001/42 will be transposed, before July 2004. Regarding environmental impact assessment, it is required for major projects since 1986.

* The sections of this report dealing with climate change, waste management and the sustainable use of natural resources are inputs into the Organisation's follow up on Sustainable Development as mandated by the Ministerial Council in May 2001.

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Climate change

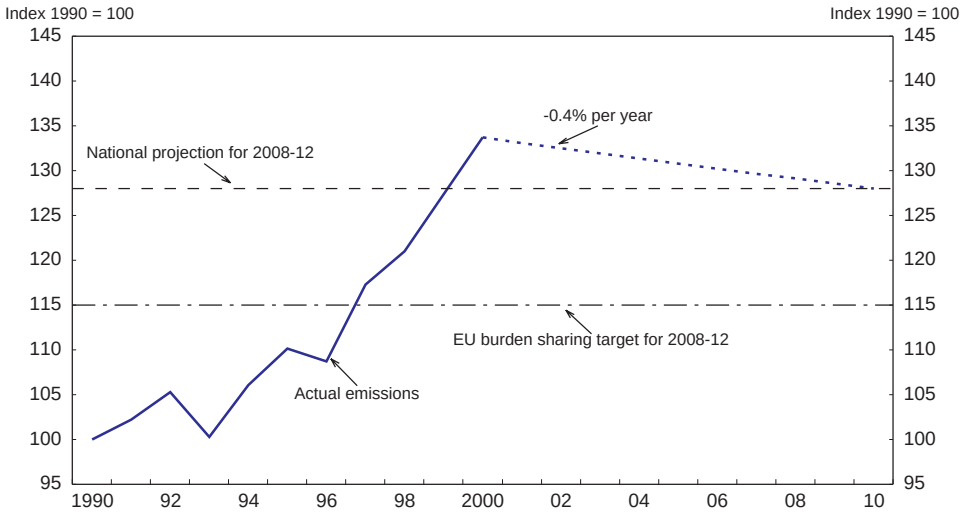
Main issues

Anthropogenic emissions of greenhouse gases (GHGs) are widely agreed to have been the principal cause of the increase in global average temperatures over the past century. With global emissions continuing to grow, climate change may impose significant costs on countries, including OECD and non-OECD countries. Spain has a target of limiting the growth of GHG emissions to 15 per cent above their 1990 level, as part of the EU's so-called "Burden Sharing Agreement".[103] However, rapid economic growth during the 1990s has contributed to raising emissions significantly above this limit. In consequence, the main issue for Spain is to push down emissions to its international commitments without imposing an excessive burden on the economy.

Performance

Emissions of GHGs in Spain have risen considerably faster during the 1990s than in almost all other OECD countries (Table A3, Annex I). By 2000, GHG emissions were more than 30 per cent above their 1990 level (Figure 24), with emissions per unit of GDP rising in contrast to the fall in virtually all other

Figure 24. **Greenhouse gas emissions**



Source: UNFCCC.

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Table 23. **Main indicators: climate change**

Indicators of greenhouse gas (GHG) emission intensity, gram of CO_2 equivalent per USD[1] of GDP, in 1995 prices

	Level of emissions, 1999				Average annual percentage change 1990-99			
	Total	CO_2 from electricity	CO_2 from transport	Other	Total	CO_2 from electricity	CO_2 from transport	Other
Spain	**537**	**127**	**130**	**280**	**0.41**	**1.12**	**1.28**	**–0.26**
France	416	32	103	280	–1.69	–2.04	0.16	–2.26
Germany	536	169	96	271	–4.00	–3.86	–0.57	–5.05
Greece	813	275	130	408	–0.24	0.07	0.74	–0.73
Italy	439	105	92	242	–1.05	–0.82	0.37	–1.64
Portugal	540	149	106	285	0.41	2.58	3.37	–1.39
United Kingdom	526	132	108	287	–3.66	–5.30	–1.38	–3.61
United States	792	278	196	318	–1.89	–0.60	–1.18	–3.28
OECD total	649	196	140	312	–1.80	–0.98	–0.38	–2.83
EU	506	120	103	283	–2.36	–2.60	–0.16	–2.95

1. National currencies converted to USD using purchasing power parities.

Source: Greenhouse gas emissions from national submissions to the United Nations Framework Convention on Climate Change (UNFCCC) and national publications; carbon dioxide emissions for electricity and transport from the International Energy Agency and GDP from OECD, National Accounts database.

OECD countries (Table 23). Despite the overall increases, there was some de-coupling of emissions from output trends in the manufacturing sector. However, the carbon intensity of electricity generation and road transportation remained broadly stable in the 1990s, thus failing to offset the impact of strong increases in electricity and vehicle use on emissions. In the residential sector, rising emission intensities compounded the effects of the strong rise in private consumption.

Policies

The rapid rise in carbon emissions partially reflects the absence of an integrated national strategy in this area. Nonetheless, a number of policies include carbon abatement among their objectives. Such policies include the promotion of renewable sources of energy and co-generation, and the improvement of energy efficiency as part of the national energy strategy (Ministry of Environment, 2002a). Renewable energy and co-generation was supported through obliging electricity companies to purchase electricity from renewable sources at a higher price, leading to a six-fold increase of the share of co-generation and renewables to 13 per cent of total electricity supply during the 1990s.[104] As these policies have also targeted other objectives, the support has been very uneven across different types of energy supplies, with abatement cost varying between EUR 97 and 1300 per ton of carbon abated.[105] Policies in the 1990s were also aimed at improving energy efficiency, including through the granting of subsidies

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and preferential loans specified under voluntary agreements.[106] However, although targets set in the Energy Savings and Efficiency Plan of 1992 were almost met in the industrial sector, little progress was recorded in energy savings from transport and buildings. Moreover, measures were not systematically assessed on the basis of cost effectiveness (IEA, 2001). In all these cases, a more cost-efficient policy would have been to use a uniform carbon tax or an emission trading scheme. In the particular case of renewable energy, market-based systems (such as green certificates) have been used in other countries to ensure that the most efficient use is made of subsidies. Nonetheless, the policies pursued during the 1990s appear to have slowed the growth of emissions of GHGs somewhat, as emissions had been projected in the early 1990s to rise 42 per cent by 2000 against the actual increase of 30 per cent.

Other policies have reduced, or at least restrained, the amount of emission abatement. There has been a moratorium on nuclear construction that was confirmed in 1994.[107] Moreover, it was not until 1996 that the main gas distribution company started expanding its domestic pipeline network and by 1999 there were still only 4 million natural gas customers. Moreover, hardly any electricity was generated from gas, though gas made inroads in the industrial sector amongst firms that produced power for their own use. Even so, only 9 per cent of electricity was generated from gas by 1999. Furthermore, natural gas imports were probably constrained by a law that imposed geographic diversification in natural gas imports, so that supplies from any one country were less than 60 per cent of not just overall Spanish imports but also the total imports of each individual gas distributor. The overall impact of these policies in the gas sector may have been one of the factors behind the failure of emission intensities to fall in the electricity sector, as occurred in the rest of the European Union.

Past policies in the coal sector have probably delayed progress towards abatement.[108] A number of restructuring plans have lowered coal output during the 1990s. However, each power station has a commitment to purchase a minimum quantity of domestically-produced coal, even if this quantity is set to fall by 28 per cent in the period 1998 to 2005 (IEA, 2001). Recent changes have meant that electricity generators no longer pay a high price for this coal but now pay the world market price with the government subsidising the coal producers in order to keep them competitive. Whether this system adversely affects carbon emissions depends on the choices that electricity producers would make if freed from these commitments. In any case, a continued quota system would work against any EU-wide cap and trade scheme for carbon emissions by preventing substitution.[109]

A national strategy for addressing climate change has been in preparation for over a decade, and is scheduled to be completed in 2003 but a number of policies have already been put in place. The plan should not include sectoral targets for domestic abatement. There is a danger that the marginal abatement costs

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between sectors could diverge, leading to meeting the overall abatement target at a higher cost than necessary.[110] Indeed, wide divergences in energy efficiency across sectors exist, implying that care is needed to ensure an equalisation of marginal abatement costs (Feijoó *et al.*, 2002). Without waiting for an overall abatement strategy, Spain aims at doubling the contribution of renewable energy to total energy supply to 12 per cent by 2010, in line with the EC White Paper on Renewable Energy.[111] With the liberalisation of the natural gas and electricity markets, natural gas is likely to make rapid inroads into the electricity generation market, especially if international pipeline and regasification capacity is expanded. Indeed, current plans envisage a further 75 per cent increase in natural gas's share in energy supply by 2010.[112] To this end, the government is implementing the Gas and Electricity Infrastructure Plan. Forestation is also being promoted in order to absorb GHGs, with the ancillary benefit of helping to combat soil erosion and deforestation (see below).

The Spanish government supports the use of international instruments to reduce the required abatement at home. These include the use of a tradable emission allowance scheme at the EU level that is likely to be opened to project-based flexible mechanisms allowed in the Kyoto Protocol. Indeed, trading of emission permits among those countries that ratify the Kyoto protocol in combination with the EU emission allowance trading scheme for companies may offer Spain the most cost-effective way of meeting its target. In addition, investments abroad that lead to abatements in GHG emissions – the so-called clean development and joint implementation projects – could also contribute to meeting targets. However, the international trading scheme will not come into place until 2008, though the EU emission allowance trading scheme will most likely start in 2005. Carbon taxation would also be a cost-efficient abatement policy and could focus on those sectors and installations not covered by the EC trading scheme.[113]

Conclusions

Spain has experienced one of the most rapid increases in GHG emissions in the OECD area in the 1990s. In part this was due to rapid economic growth, but it also reflected delay in formulating a climate change strategy and an initial resource endowment that left it with a very low ability to switch to less emission-intensive fuels in the electricity and residential sectors. However, the delay in overcoming this deficiency through the operation of market forces in the electricity and natural gas sectors, and the curtailment of the nuclear programme, has undoubtedly boosted emissions. The electricity and energy markets have been liberalised but there are still restrictions on natural gas imports and power stations are committed to purchasing a minimum quantity of domestic coal. Such policies need to be revised. The planned expansion of the natural gas network will help ensure a larger penetration of natural gas in power generation. The soon to

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be announced national climate strategy should put greater emphasis on using the flexible mechanisms allowed under the Kyoto protocol and the EU emission trading system as well as intensifying domestic efforts to meet targets at least cost. A green certificate market for supporting renewable energy production could help achieve the target of increasing this source of electricity generation in the most cost-efficient manner. The strategy should acknowledge that carbon taxation offers a way to ensure optimal investment in abatement and should be preferred to sectoral targets.

Waste management

Main issues

If badly managed, the disposal of waste can cause long-lasting damage to ecosystems and to human health. Waste treatment facilities may also involve serious disamenity costs that can destroy property values in addition to spoiling the environment. Waste management policy within the European Union has set increasingly demanding targets for individual member states in an attempt to increase significantly the recovery of waste and reduce as much as possible the environmental impacts of waste management. The main issues facing Spain are designing policies that comply with these requirements while achieving the targets at least cost to society.

Performance

The volume of municipal waste has grown very fast during the 1990s, both in absolute terms and relative to private consumption, partially due to high rates of economic growth (Table 24).[114] The majority of municipal waste has been disposed of in landfills (Table 25). At the beginning of the 1990s over one-third of land-filled waste went to uncontrolled sites, but this declined to under one-fifth by 1999. Nonetheless, there is still a substantial number of landfill sites that fail to meet Community legislation on waste management.[115] Spain has committed itself to close a large number of uncontrolled landfills, but still faces significant challenges in adapting existing sites to the requirements of the new Directive on the landfilling of waste. The share of recycling of municipal solid waste is below the EU average. After falling in the early 1990s, the share of biodegradable waste composted started to grow once again from the mid-1990s (Table 25). Finally, incineration accounts for a growing but still small share of municipal waste treatment. while the incineration of waste is increasingly coupled with energy recovery.

Policies

Waste management policy has changed markedly in recent years with the transposition of several EC directives into Spanish law, embodying the polluter-pays

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Table 24. **Performance indicators: waste**

	Municipal waste						Composting and recycling				
	Kg per capita		Annual growth		Level, 1 000 tonnes	Relative to private consumption, annual growth	Level			*Of which*: recycling Level	
	Year	Level	Period	Percentage	Latest year	Percentage	Year	1 000 tonnes	Percentage of total	1 000 tonnes	Percentage of total
Spain	**1999**	**464**	**1990-99**	**4.3**	**18 377**	**2.0**	**1999**	**4 056**[1]	**22**	873	5
France	1999	524	1993-99	1.7	30 744	0.0	1999	5 788	19	3 198	10
Germany	1998	549	1991-98	–1.5	44 995	–2.8	1998	. .	. .	16 318	41
Greece	1997	371	1990-97	3.8	3 900	1.8	1997	339	9	307	8
Italy	1999	497	1990-99	4.0	28 364	2.3	1997	4 416	17	1 874	7
Portugal	2000	437	1992-99	4.2	4 364	1.5	1999	418	10	190	4
United Kingdom	1999	558	1990-99	2.3	33 200	–0.2	1999	3 640	11	2 880	9
United States	1999	764	1990-99	1.3	208 520	–1.9	1999	57 961	28	46 068	22
OECD average		501		1.3	20 666	–1.2		4 269	22	3 731	17
EU average		526		1.9	13 056	0.0		2 438	22	2 349	18

1. Total recycling including composting and industrial glass and paper. If it only included separate collection of recyclable materials and composting, the figures would be 1 315 and 7, respectively.

Source: OECD and Eurostat.

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Table 25. **Treatment systems for waste in Spain**

	1990	1995	1999
Level (1990 = 100)	100.0	120.0	146.0
Municipal waste management treatment			
Uncontrolled landfill	26.2	17.4	12.8
Controlled landfill	48.1	62.9	58.8
Incineration with energy recovery	2.9	4.2	5.3
Incineration without energy recovery	1.9	0.4	0.3
Composting facilities	20.3	13.8	18.0
Separate collection of paper	0.0	0.0	2.0
Separate collection of glass	0.5	1.3	1.8
Separate collection of other materials	0.0	0.0	0.9
Total	100.0	100.0	100.0

Source: Ministry of the Environment.

principle.[116] The 2000 National Plan for Urban Waste and the Law on Packaging and Packaging Waste call for: a reduction of waste and packaging waste of 6 and 10 per cent respectively by 2006;[117] gradually expanding separate collection; increasing re-use, recycling, and energy recovery; eliminating all uncontrolled landfills by 2006; and improving the technical standards of landfills in operation.

The packaging waste law introduced the principle of extended producer responsibility (EPR) for such waste. Packagers are required to take responsibility for the collection and disposal of waste, including meeting targets for re-use and recycling. In practice, most packagers have opted to delegate this responsibility to third party not-for-profit organisations, with several organisations specialising in different types of packaging materials.[118] In comparison with other EPR schemes in Europe, the costs incurred by these organisations, reflected in their fees for different materials, are low (Table 26). In contrast to some countries, the EPR system only pays the extra costs that local authorities incur in the collection and treatment of recyclable waste, thereby offering economies of scale through integration with normal waste collection facilities of local authorities. Separate collection of packaging by local authorities has been expanding but it was not obligatory until recently. Indeed, the principal method for collecting recyclable glass and paper was voluntary bring systems. This method restricts the recycling effort to the quantity that consumers bring to collection points, so bearing a large part of the extra collection and transport costs on a voluntary basis.[119] Costs were also contained as municipal authorities can choose to use the EPR organisation, which contracts with recyclers on a competitive tender basis, or an alternative contractor if that is cheaper. As the recycling targets were realistic, the overall costs of recycling, with the exception of plastics, appeared to be competitive with incineration.

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Table 26. **Fees[1] for extended producer responsibility schemes in Europe**

		Spain	Austria	Belgium	France	Germany	Luxembourg	Portugal
	Kg	Fees in € per thousand containers						
Glass bottle (1 litre)	0.35	**2.40**	30.52	6.77	0.75	28.46	5.99	0.52
Tetrabrick (1 litre)	0.027	**2.25**	5.47	6.14	2.99[2]	25.28	5.69	0.27[2]
PET bottle (1 litre)	0.03	**3.53**	32.90	10.44	3.47	45.12	8.59	1.20
Aluminium can (33 cl)	0.014	**0.76**	6.92	2.40	0.45	13.65	2.00	0.52
Steel can (33 cl)	0.03	**0.93**	11.97	1.74	0.42	11.61	1.24	0.52
Cardboard box	1	**15.47**	202.76	37.68	74.09[3]	190.64	31.23	9.98
		Fees in € per tonne of material						
Glass bottle		**7**	87	19	2	81	17	1
Tetrabrick		**83**	203	227	111	936	211	10
PET bottle		**118**	1 097	348	116	1 504	286	40
Aluminium can		**54**	494	171	32	975	143	37
Steel can		**31**	399	58	14	387	41	17
Cardboard box		**15**	203	38	74	191	31	10

1. These fees are valid for the year 2000. They are exclusive of VAT.
2. A paper-cardboard tariff is applied (main material).
3. A 10 per cent rebate is awarded to cardboard packaging containing more than 50 per cent recycled material.

Source: European Commission, *European Packaging Waste Management Systems – Final Report* by ARGUS, February 2001.

By 2001, through the implementation of waste management policies, the targets set by the 1997 law were successfully met. Yet costs could rise quite substantially in the future. Since January 2001, separate collection of packaging is obligatory for municipalities with a population higher than 5000 inhabitants, that account for 85 per cent of total population. Also, the National Plan for Urban Waste sets demanding targets for 2006. In particular, the targets for plastic recycling, which can be particularly costly, are twice those required by the corresponding EC directive on packaging and packaging waste, which itself will be costly to achieve (Table 27). Efforts to increase the separate collection of different materials are already complicated by contamination of materials in collection containers.

Cost increases may result from more stringent requirements in EC directives to divert waste streams from landfills, which will entail considerable investment in separate collection and building of sorting, recycling, composting, and energy recovery plants. Planned investment in such facilities will amount to almost EUR 1.5 billion between 2000-06, accounting for 70 per cent of all public investment in the waste sector. Part of the remaining expenditure is necessary to ensure that landfills conform to environmental regulations that limit externalities. Indeed, the cost of improving landfills will have large impacts in many regions that rely heavily on landfill as the main disposal option. Once landfill sites are controlled, have methane recapture facilities in place and no longer pollute nearby

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Table 27. **Recovery and recycling rates and targets**

	Commission proposals	National urban waste targets	Actual 1999
Overall recovery target	60-75	>70	42
Overall recycling target	55-70	>50	38
Material specific targets			
Glass	60	75	38
Paper/board	55	75	54
Metals	50	90	24
Plastics	20	40	14

Source: European Commission, Ministry of the Environment.

water sources, there will be less justification for the extremely ambitious targets that have been set for increasing the amount of biodegradable material being composted instead of going to landfill. In such cases, the external benefits of composting such waste can be quite small (Eunomia Research and Consulting, 2002). They could also be offset by the increased costs of the required separate collection. Past costs for composting in some parts of Spain have been comparatively high in relation to other countries (Table 28). In this context, the government should undertake a cost-benefit analysis, taking into account the likely evolution of the costs of different waste treatment options, external benefits, and the value of compost, which can be used as soil improver in land suffering from severe erosion problems. Such an approach would also help to meet efficiently the targets set in the EC Directive on the Landfill of Waste.

Past neglect of the conditions under which landfills were operated has generated considerable soil contamination. Almost 5 000 cases of soil contamination were identified in an extensive assessment of the scale of the problem, many of which were the result of land-filling (Ferguson, 1999). Central government is paying half the costs for improving contaminated soils, as the costs of remedial action are prohibitively expensive for the autonomous communities on their own. Due to the high cost for each acre of land treated, funds are targeted at sites that threaten water supply or human health.[120]

Conclusions

In the field of waste management policy, Spain has made progress in a relatively short time and at a comparatively low cost. Nonetheless, a significant portion of landfill sites still does not comply with environmental regulations and should be rapidly brought into conformity. However, new targets to reduce the use of landfills will require more effort and may be more costly to achieve. In this light, policies should be designed to reach these targets in a least-cost manner. Indeed,

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Table 28. **Costs of composting in selected European countries**[1]

		Capacity, tonnes	Set-up costs, '000 euros	Operating costs, euros per tonne	Total financial costs, euros per tonne	Cost per household, euros
Spain	**Baix**	**4 000**	**6 000**	**45.0**	**220.9**	**110.4**
	Barcelona	**10 700**	**5 400**	**108.0**	**179.9**	**35.0**
	Montejurra	**10 000**	**4 000**	**222.0**	**279.0**	**121.3**
France	Gironde	36 000	175	20.0	20.7	37.2
	Niort	8 500	520	30.5	39.2	27.8
	SIVOM	6 000	3 300	37.0	115.3	58.6
Ireland	Cork	1 000	42	21.0	27.0	0.2
	Kerry	500	193	209.0	264.0	74.7
	Limerick	950	766	16.0	130.8	44.4
Italy	Cupello[2]	315	0	380.0	380.0	57.0
	Monza[2]	10 000	0	6.3	6.3	1.1
	Padova	16 500	2 200	400.0	419.0	67.4
Portugal	Amtres	250 000	58	17.5	17.5	29.2
	Lipor	30 000	5 400	8.5	34.1	41.0
United Kingdom	Castle Morpeth	5 000	225	20.5	26.9	6.6
	Wyecycle	250	18	89.0	99.3	24.8

1. Assuming that the set-up costs bear an opportunity cost of 7 per cent per annum.
2. Monza and Cupello use same equipment as normal collection and claim it is therefore costless.
Source: European Commission (2000), *Success Stories on Composting and Separate Collection*.

once the principal external costs of landfills have been internalised through environmental regulations, policy should remain conservative in setting targets for recycling. It would appear better to set a cap on the allowable packaging levies reflecting the marginal alternative methods of waste treatment and allow the amount of recycling to be determined by the market, rather than setting quantitative targets irrespective of costs. Significant progress has been made in identifying sites suffering from soil contamination. Due to the large costs involved in remedying contaminated soil, the government should continue to prioritise areas that present the greatest threat to health and the environment.

Sustainable use of natural resources

Main issues

Climatic and geographic conditions place considerable pressures on Spain's water and land resources. And, as in the other Mediterranean countries, water-intensive activities are relatively important. The main issues in the sustainable use of land-based natural resources is altering policies so as to reflect their cost, including environmental externalities in order to protect scarce water resources and prevent

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future land degradation. It is also important to ensure that sea-based resources are harvested in a sustainable fashion, an issue that is considered above in the discussion of fisheries in the product market section of this *Survey*.

Performance

The pressure on water resources in Spain is second only to Belgium in the OECD area (Table 29). In over 70 per cent of the country, abstractions of water are over 40 per cent of the flow of available water, while in the less dry area abstraction rates are below 25 per cent. Agriculture is the biggest end user of water, and the area under irrigation has almost doubled over the past four decades. The south and south-east (particularly, Murcia and Almería) are the areas experiencing the greatest stress on water resources. As these regions typically rely heavily on groundwater, local aquifers suffer from over-exploitation and saltwater intrusion is already a serious problem (Revenga *et al.*, 2000). During periods of drought, water supply has to be rationed for some users. Pressure on water resources is expected to intensify over the next 20 years with demand rising by over one-quarter (Ministry of Environment, 2002b). Soil erosion is also an important problem and has grown in severity over time, though some successes in containing erosion may have occurred more recently.

Table 29. **Main indicators for water withdrawal and land degradation**

	Water withdrawals		Irrigation coverage, per cent of farmland	Land degradation	
	Per cent of available resources	Relative to GDP (1980 = 100)		Per cent of total area	Per cent due to agriculture
Spain	**36.8**	**65**	**19**	**38**	**16**
France	23.9		10	9	0
Germany	24.4	69	3	31	67
Greece	12.1		36	48	5
Italy	32.2		24	28	12
Portugal	15.0		24	21	100
United Kingdom	14.6		2	19	100
United States	19.9	55	12	30	75

Source: OECD; International Commission on Irrigation and Drainage; Food and Agricultural Organisation.

Policies

The publication of the White Paper on Water in 1998 recognised the need for an improvement in water management and a better pricing mechanism in order to create the conditions for an efficient use of this critical resource. Following this analysis, several reforms have been adopted, although their practical implementation

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is still limited. All water supplies are publicly owned, with water basin authorities managing major infrastructure and determining water abstraction quantities. Water users hold rights for abstractions and are organised in user associations, which have the responsibility for distribution amongst individual irrigators, dispute settlement, and the collection of irrigation tariffs. The operation of this system is flawed in some respects:

- Unclear abstraction rights covering almost half the irrigated acreage have prevented the affected basin authorities charging at all for abstracted water (Garrido, 2001). This situation derives from the existence of private property rights before the passage of the 1985 Water Law. The authorities are working towards completing the clarification of the abstraction rights within the next five years.
- Water prices for all users have been below full cost recovery, because a share of capital costs (and in some cases even operational costs) were not covered (OECD, 1997b; Garrido, 2001).[121] However, with the adoption of the National Hydrologic Plan, the principle of full-cost recovery has been established.
- Non-payment of water charges was a serious problem, accounting for around 50 per cent of the value of (already subsidised) invoices (Ministry of Environment, 1998). However, payment discipline has recently improved with an invoice recovery rate ranging between 65 and 95 per cent.
- The cost of environmental externalities (such as remedial action to restore depleted aquifers) are estimated to be up to EUR 0.5 per cubic metre of water (Garrido, 2001) in contrast to prices for irrigation water that vary between EUR 0.03 and EUR 0.15 per cubic meter.[122]
- Incentives for water conservation in agriculture, the main user, are frequently absent due to irrigation pricing based on land area rather than consumption.[123]

In 1999, two policy innovations partially addressed some of the shortcomings in water management practices. The first innovation was to require existing irrigators to install metering equipment so that the price of water can be adjusted if past water use is above or below an estimated need, based on the characteristics of the land and crops grown (Garrido, 2001).[124] Such a needs-based system does not reflect effectively the value of the water to the individual farmer. It would be more efficient to adjust prices until there was no excess demand for water when quotas would no longer be necessary. However, existing quotas could be equated to property rights and some form of compensation would be needed and would have to be based on the current allocations.[125] The second innovation was the adoption of framework legislation for a secondary market for water. Such a system has the potential to reveal more accurately the opportunity costs of water use and hence increase water use efficiency. However, further supporting legislation

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is needed to ensure that this system contributes to efficient water use, through revealing the value of water in each location.

Having in mind Spain's rainfall pattern with unequal distribution among regions, an important component of national policy is the transfer of water between different water basins, thus balancing resource availability. This policy is also designed to preserve agriculture employment in areas where irrigation is important. One of the key initiatives is the transfer of water from the Ebro river in the north-east to the south and south-east over a distance of 850 km, at an estimated cost of over EUR 3 billion. The objective of the government is that the price of transferred water should bear the full cost of infrastructure investment and an environmental fee to cover externalities in the donor water basin. This seems an unlikely outcome. The National Hydrological Plan puts the average cost of the water transfer as EUR 0.32 per m^3 assuming that all the water transferred is used.[126] However, the cost is likely to be much higher in some areas due to the greater transport and construction costs. One study of water supply and demand in the southern area puts the cost of water from this project at EUR 0.45 per m^3 against the current price of EUR 0.03 per m^3 for irrigation water (Albiac-Murillo *et al.*, 2002). Should demand be less than anticipated, due to the difference between the price for transferred water (based on full cost recovery) and current tariffs for irrigation water, unit costs could potentially rise to the levels of desalination costs.[127] If instead, aquifer abstraction were to be limited to a sustainable rate, supply and demand could be balanced by raising prices to EUR 0.18 per m^3 (*op. cit.*), still much less than the cost of the Ebro scheme that has been approved by Parliament.

A second major component of national water policy is encouraging the modernisation of agricultural irrigation, in order to reduce the waste of the resource. At present around 40 per cent of the irrigated area uses efficient irrigation techniques, such as drip irrigation (Figure 25).[128] Losses from the distribution network are substantial at around one-fifth (INE, 2001). To expand more efficient irrigation and reduce losses, public and private funds will be used to modernise irrigation systems.[129] To this end, public investment between 1998 and 2008 will total EUR 3 billion, with additional private investment estimated at EUR 2 billion, for modernising irrigation systems covering almost 40 per cent of irrigated land.

Soil degradation is an important problem in Spain, principally affecting arid regions along the Mediterranean coast. Soil erosion is a potential problem in cases where vegetation is removed. In regions with scarce income alternatives various type of agricultural production has been expanded. This may have been a consequence of price incentives under past European agriculture policy (Beaufoy, 2000). Due to the erratic rainfall pattern in this region production has been intensified through drip irrigation. On these intensified plantations the vegetative cover has been removed in order to allow modern tilling practices not appropriate for the ecological conditions. These practices have in turn contributed to soil erosion. However

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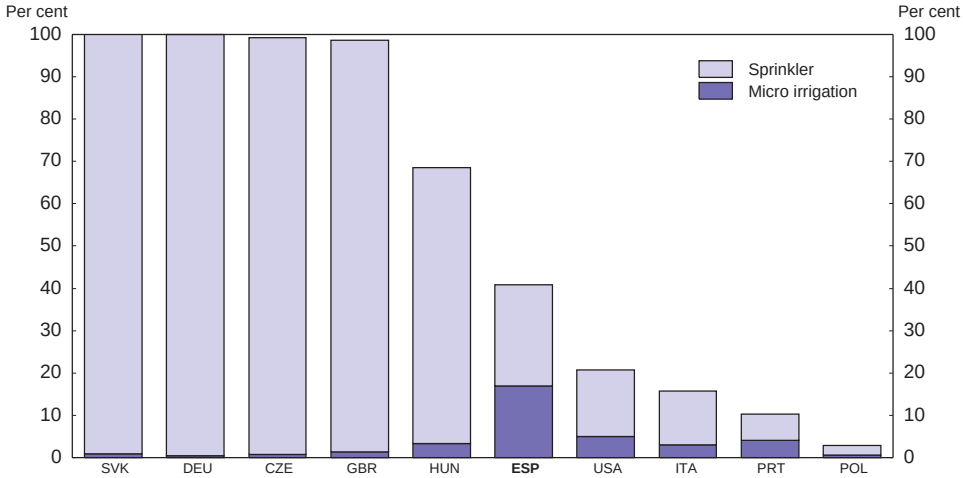
Figure 25. **Application of efficient irrigation methods**
As a percentage of total irrigated area



Source: ICID.

a significant part of the land affected by erosion is non-irrigated grassland and to a lesser extent areas of permanent crops (OECD, 2001e). Recent reforms have made production subsidies conditional on the development of good farming practices, which should contribute improving environmental outcomes. Further beneficial effects on the environment could probably be achieved by decoupling subsidies for these crops from production and moving completely to acreage payments. In addition, a reforestation programme, such as the 2002 National Forest Plan, could also address some soil erosion concerns and act as a sink for greenhouse gases. Improvements in the state of local aquifers would also help reduce soil degradation in some areas. In recent years, new policy measures have controlled the rate of abstraction from overexploited aquifers in some of the most affected areas. Moreover, in December 2002, further water legislation introduced new measures on the depth of water abstraction from aquifers. In addition, the central government is identifying areas affected by soil erosion and works with regional governments and private landowners.

Conclusions

Water and land management policies are important issues, and must be reinforced to face serious pressure. Water policy needs to strengthen the use of pricing to increase the efficiency of water use. As a start, payment discipline has to

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be enforced to give all users some incentive to husband resources. The overall quantity of water to be used should be determined with regard to the sustainability of aquifers and the minimum environmental flows. Water allocations based on a needs-based system does not reflect effectively the value of the water to the individual farmer. By adjusting water prices properly and further liberalising the nascent market in water trading by clarifying water rights, a more efficient use of water resources could be achieved. Getting the framework right for water management rapidly should complement the decision to invest in water transfer schemes to ensure this resource is used efficiently. Finally, soil erosion is a serious concern and while some successes in improving farming practices can help reduce the extent of the problem, government action will probably be needed to internalise the external costs of erosion. The government must, however, weigh carefully the various policies options promoted to ensure that erosion is limited in the most cost-effective manner.

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IV. The economic impact of migration

Introduction

For much of the 20th century, Spain experienced significant net emigration. But over the last decade, the net flow has reversed and Spain has become an immigration country. A number of factors, including its geographic position, historical links with Latin American countries, and a strong growth performance compared with most other EU countries, has made Spain an attractive recent destination for migrants. With the sharp rise in the inflow of immigrants, Spain is facing the challenge of defining an overall policy on immigration. And the challenges are numerous. These include most importantly decisions on the size of the immigrant inflow and the way immigrants should be selected in the case of non-EU immigrants, ways to cope with the still large number of illegal immigrants and means to integrate better the immigrants that stay.

This chapter lays out the interactions between immigration and economic performance and economic policies. After putting immigration flows and stocks into an international perspective and assessing the factors affecting them, the chapter outlines the evolution of migration policy and current arrangements. The economic aspects of immigration are assessed in terms of their interaction with the labour market, their impact on output and incomes, on public finances and on trade and the balance of payments. Policies for integrating immigrants are also reviewed, while the last section provides conclusions and recommendations.

Migration stocks and flows

From emigration to immigration

Immigration is a recent phenomenon, but has risen rapidly over the last decade. According to the census, 1.5 million foreign citizens[130] were living in Spain in 2001, four times more than in 1991, representing 3.6 per cent of the population. This is still a low proportion of the population compared with many other OECD countries (Figure 26), and there are still more Spaniards living outside Spain (2.2 million) than immigrants in Spain (Dolado, 2002). For more than a century, many Spaniards emigrated, initially to Spanish-speaking countries, and between

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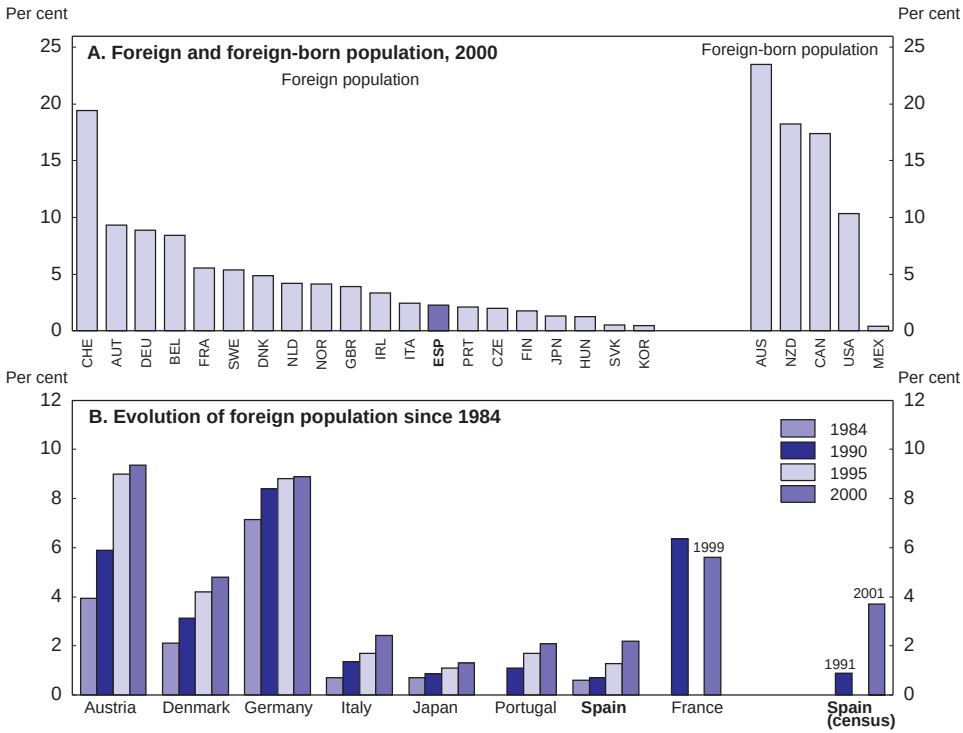
Figure 26. **Foreign population in selected OECD countries**
As a percentage of the total population



Source: SOPEMI 2002, based on population registries for Austria, Denmark, Germany, Italy and Japan; population census for France; resident permits for Portugal and Spain; and population census for Spain in 1991 and 2001.

the 1950s and 1970s to other western European countries, in particular France. In the second half of the 1970s, though, following the first oil price shock and rising unemployment in Europe, outflows started to decline, and Spanish emigrants started to return.

At the same time, stronger growth since the mid-1980s, combined with a more restrictive migration policy in some European countries, made Spain an attractive destination for migrants from countries with a lower income. In 1991, the migratory balance switched from negative to positive, and net immigration has grown continuously since then. This has been taking place in a context of rapidly decreasing Spanish fertility rates and an ageing population.[131] Indeed, while the natural growth rate of the population approached 0.9 per cent in 1970, at the end

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of the 1990s almost all the population growth was due to the rise in the foreign population.

There is no single complete and reliable data source to analyse migratory stocks and flows (Table 30), but the available statistics suggest that the increase in immigration has been particularly important since the second half of the 1990s.

Table 30. **Immigration statistics in Spain**

Statistic	Source	Period	Details	Comments
Foreign residents	Ministry of Interior	1991-2001 Some previous years (1980)	Age, sex, origin, destination	Problem: underestimates the number of EU citizens as they are not obliged to register in Spain. Some probably tend to keep their main residence in their country of origin. Probably provides a good estimate for the number of non-EU foreigners legally residing in Spain (non-signalled exit must be rare).
Residents registered in the "Padron"	Municipalities and INE	1996,1998-2000 (1st January)	Age, sex, origin, destination	Problem: no systematic trace of exit when a foreigner goes from one city to another. INE attempts to avoid double-counting. Not obvious that EU foreigners systematically register. Undocumented foreigners are allowed to register and probably do so to benefit from social services.
Population census and projection	INE, Census in 1991 and based on the Encuesta de Variaciones Residenciales for 1991-1997 and on the Padron for 1998- 2000.	1991-2001	Origin, age, etc. for census years; no detail for projected years	New Census in 2001 but complete results available only in 2003. The total number of foreigners from the Census has the advantage of being rather exhaustive. Problem: no detail on migrants' characteristics outside the census year.
Encuesta de Migraciones	INE, as part of the Encuesta de Población Activa	1980-2001	Age, sex, origin, activity status, profession, sector, education level	Problem: sample initially very restricted for foreigners; progressively extended (in 1987, 1992, 1995 and 2001) and methodology modified several times. Cannot be used over time. Better coverage now but still not completely representative (about 50 per cent of the foreigners in 1999). Covers only households (and not immigrants living in collective housing). Temporary immigrants are not surveyed either.

Source: OECD.

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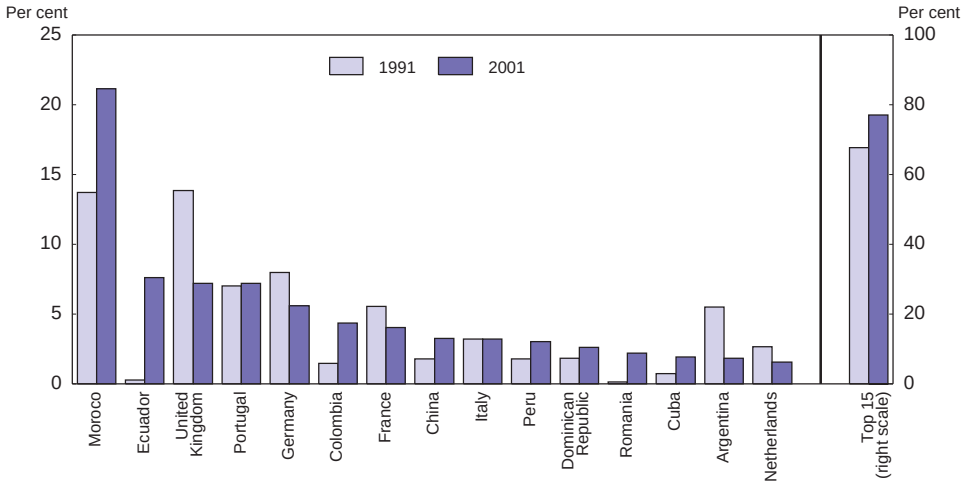
This is in contrast with most other EU countries, where net immigration flows increased during the 1980s, but declined in the 1990s, especially in the second half. The increase in the number of foreigners residing in Spain is important, but it is not exceptional in international comparison (Figure 26). However, since there were few immigrants in Spain a decade ago, the increase in immigration is perceived as being substantial.

Immigration from non-OECD countries, in particular Africa, has been the most important. While residents from the European Union represented about two thirds of the foreign population in 1980, their share shrank to only one third of the total in 2001. However, the latter figure is probably an underestimate, since many EU citizens do not declare their residence.[132] The number of African residents has quadrupled, led by Moroccans. Immigration from eastern European countries, though still low, has also been increasing very rapidly since the end of the 1990s, and that from Latin American countries has recently rebounded. Overall, the origin of immigrants is quite concentrated (Figure 27). In 2001, 82 per cent of the foreign residents were living in the six autonomous communities comprising the islands (Baleares and Canarias), the Mediterranean coast (Cataluña, Valencia, Andalusia) and in the capital (Madrid), where 64 per cent of the total population was located.

Figure 27. **Foreign population by nationality for the top 15 source countries**

As a percentage of total foreigners



Source: Dirección General de la Policía del Ministerio del Interior.

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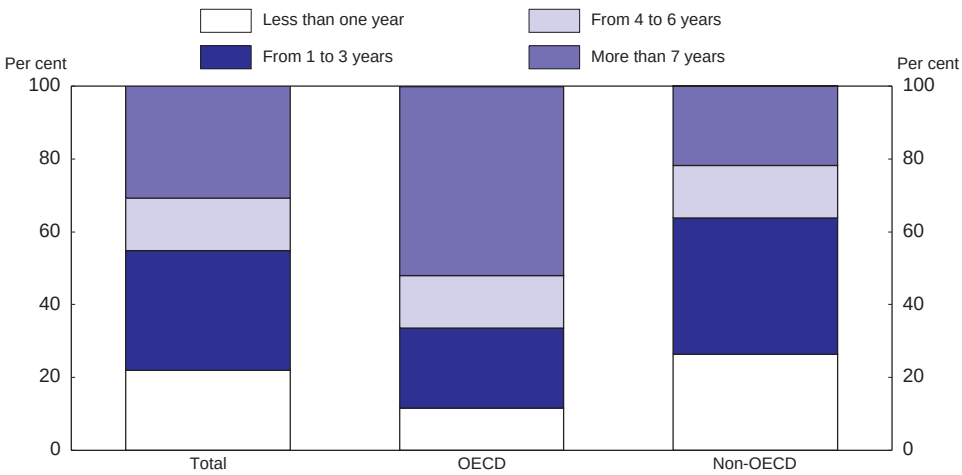
A large number of immigrants is entering and staying illegally, as revealed by the four legalisation processes that have taken place since 1990. Some immigrants cross the Strait of Gibraltar on fragile vessels ("pateras") landing in Andalusia or Canarias, with wreckages not uncommon. Although such events have received considerable attention, most immigrants enter Spain legally, with a visa or a residence permit, but a number of them stay illegally once these papers have expired. There is of course no reliable estimate of their number. The difference between the census data (1.5 million) and the number of officially-registered foreign residents (1.1 million) provides an indication (Figure 26), although it also probably includes EU citizens who have not declared their residence. The number of applicants for legalisation in 2000 and 2001 who were denied residency, that is about 186 000 at mid-April 2002 provides another indication; but some of the undocumented foreigners have not applied.[133]

The lack of data on exits from Spain, as well as the relative novelty of the immigration increase from developing countries, make it difficult to assess the extent to which migration is permanent or temporary. According to the only available source, in 2001, less than one third of foreigners had been in Spain for more than seven years, and they were mostly OECD citizens (Figure 28). About two-thirds of non-OECD foreigners had been in Spain for less than three years, but since their immigration is recent, it cannot be inferred that it is of a temporary nature.[134]

Figure 28. **Length of stay of foreigners**
2001



Source: INE, Encuesta de Migraciones.

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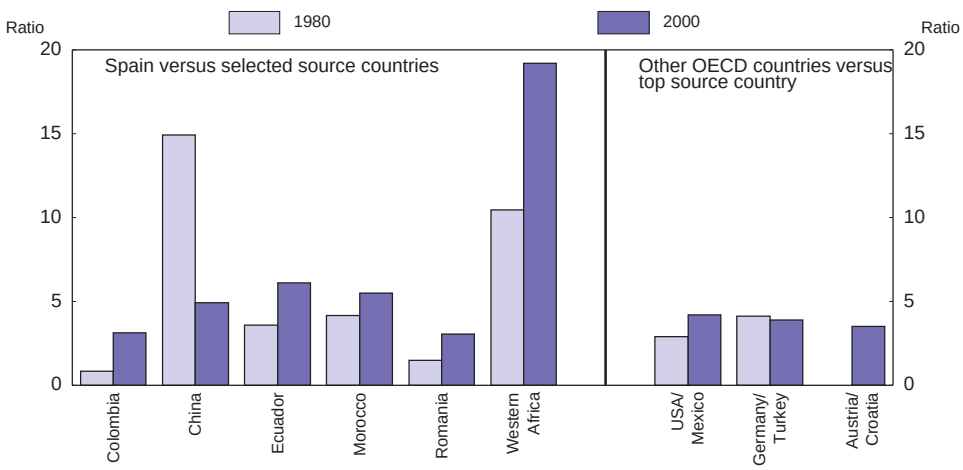
The factors inducing migration are going to persist

The economic incentive for people from developing countries to migrate, as highlighted by the difference in income per capita between Spain and important source countries, is quite large, and has been growing over the last two decades (Figure 29). In 2000, the Spanish income per capita was 5.5 times higher than in Morocco, against 4 times higher in 1980. The differential is even larger for Western African countries and has also widened against Latin American countries.[135] Lower travel costs and improved communications have been additional factors boosting migration, which make it very likely that the incentive for people living in developing countries to migrate to Spain will persist.

Structural factors in countries of origin, related to demography, economic and political reasons, are also at play. In the Maghreb, as well as in Ecuador and Colombia for example, fertility rates have started to decrease, but the cohorts of young people arriving at working age are still very important, a trend which is going to continue in the future (Figure 30), while in Western African countries fertility rates remain high. However, the increase in the stock of immigrants in Spain has been much larger than that of the young working-age population in source countries over the past decade (Figure 31), indicating that the role played by demographics *per se* in overall pressures is probably limited, and that economic and social factors are more important.

Figure 29. **Income differentials**[1]



1. Ratio of GDP per capita in current PPPs in Spain (or wealthy OECD countries) to that of source countries.
Source: World Bank, *World Development Indicators.*

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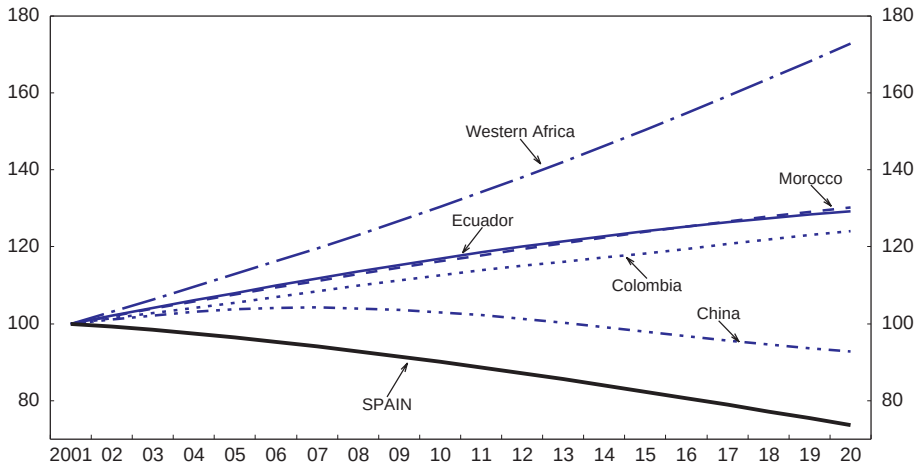
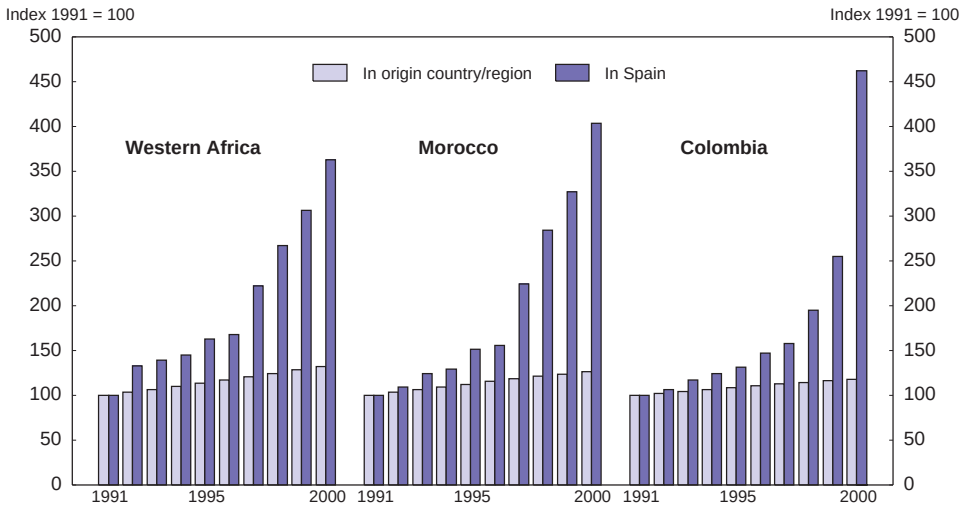
Figure 30. **Population projections for Spain and selected source countries**
Age 15-44, index 2001 = 100



Source: United Nations, *World Population Prospects.*

Figure 31. **Population of selected source countries/regions, aged 15-44**



Source: United Nations, *World Population Prospects.*

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Growth was slow in most of these countries, and particularly unstable in Northern Africa, and job markets have been unable to absorb these young cohorts, resulting in high unemployment. Focusing on Morocco, currently the main source country, about 30 per cent of people aged under 34 living in urban areas were unemployed in 2000.[136] The situation is no better in rural areas. The economy would have to grow between 6 to 8 per cent a year to absorb the people seeking work.[137] This figure is much higher than past performance, and not likely to materialise in the current environment. Unemployment rates have also risen over recent decades in Latin American and Eastern European countries. The absence of a favourable outlook concerning the economic and sometimes political environment often makes emigration one of the few options for any improvement in living standards. A survey conducted in 2001 showed for example that more than half of young Arabs wanted to emigrate to other countries, mostly to industrialised countries (UNDP, 2002).[138] These incentives to emigrate seem unlikely to diminish substantially in the near future and often seem to outweigh the risks, even to their lives, that emigrants incur.

While these factors make most OECD countries a desirable destination for migrants, a number of additional features favour Spain. In particular, Spain has experienced relatively strong growth. Migrants also know that they can find jobs for low-qualified workers, especially in labour-intensive agricultural activities – a sector having strongly developed since joining the European Union – and domestic services, even if undeclared. For Central and Latin Americans, the common language also makes Spain an easier destination. Among the policy factors, the various regularisation processes of undocumented immigrants in Spain combined with increasingly restrictive immigration policies in other EU countries have also made Spain a favoured destination.

Migration policy: still in the making

In line with the EU policy framework, Spain distinguishes two types of foreigners in its migration policy. Since the Treaty of Rome, EU citizens are free to enter, move and reside in another EU country, and there should be no discrimination based on nationality for labour market access. Since the Maastricht Treaty, EU citizens are not even required to hold a work permit to be able to work. They, and their family members, have access to the "Community regime", which provides a five-year resident card automatically. Foreigners from non-EU countries fall under the "General regime". Spain being a recent immigration country and the closest entry point to Europe from Africa, policies *vis-à-vis* non-EU foreigners have also been influenced by the EU-wide focus on border controls (Box 5).

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Box 5. **EU-wide immigration policies**

Visa matters, belonging to the first pillar of the Amsterdam Treaty, are part of community policy. Since 1999, two lists of countries have been established, whose nationals need or do not need a visa to enter the European Union or the Schengen area. Out of the top 15 immigration source countries (Figure 27), Argentineans are exempted from a visa requirement for periods up to three months; Ecuadorians used to be exempted too, but a visa requirement has been ratified by the European Parliament in January 2003.

Immigration issues as such, on the other hand, belong to the third pillar (Justice and Home Affairs), where policies remain under the sovereignty of national governments. A number of joint actions have been agreed regarding asylum seekers and illegal immigration, aiming at the harmonisation of practices, notably at the June 2002 Seville summit closing the Spanish presidency of the EU Council, but little action has been yet taken. Regarding conditions of residence and employment, the initiatives have failed to achieve any harmonisation of national practices. The rules on family reunification are very flexible, for example, allowing substantial national variations. Directives to regulate immigration, such as labour immigration and family reunification had been proposed by the Commission but were not adopted.

Past policies

The first law regulating the immigration of non-EU nationals was passed in 1985, a few months before Spain joined the European Communities.[139] It was relatively restrictive regarding work and residence permits – no permanent permit existed, and the requirements for the renewal of temporary permits were rather strict – and immigrants' rights were limited.[140] Despite the regularisation process of 1986 following the introduction of the law, a stock of undocumented immigrant workers built up and in 1991, at the same time as a visa requirement was introduced for Moroccans in preparation of Spain's joining of the Schengen area, another regularisation process took place, whereby more than 100 000 foreigners were provided with papers (Table 31). After that, to complement the general regime that deals with individual immigration demands of all types (for work, family reunification, etc.), a system of quotas for work permits was established in 1993 in an attempt to better manage inflows of non-EU workers. Work permits were to be issued every year to fill those jobs that had not been filled by Spanish nationals. In practice though, instead of managing inflows through the Spanish embassies in source countries, the quota system functioned more as a regularisation mechanism for undocumented workers already present in Spain, with 20 000 to 40 000 work permits issued almost every year between 1994 and 1999.

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Table 31. **Regularisation processes**

	Number of demands	Number accepted
1985-86	44 000	23 000
1991	148 000	110 000
1996	25 128	21 294
2000[1]	247 600	163 900
2001	350 100	216 300

1. It should be added 57 616 number of demands, and 36 013 accepted, as a consequence of a re-examination process.

Source: OECD based on Moreno Fuentes (2000), SOPEMI *report for Spain* 2000, and *Balance* 2000, Delegación del Gobierno para la Extranjería y la Immigración, Ministerio del Interior.

Table 32. **Annual quota for work permits in the 1990s**

	Permits demanded	Permits granted
1993	–	5 220
1994	37 093	19 953
1995	37 206	25 604
1997	67 522	24 615
1998	65 221	27 904
1999	97 033	39 879

Source: SOPEMI *Report for Spain*, 2000.

Despite these initiatives, another regularisation process was implemented in 1996 for foreigners who had previously had papers but had not been able to renew them (Table 31). The gap between work permits demanded and issued under the quota also grew over time (Table 32), indicating that the number of undocumented foreigners had built up in parallel.

Regarding the integration of immigrants, the decentralised institutional arrangements in place, together with the multi-dimensional nature of the issues at stake, resulted in some lack of co-ordination among the different authorities and gaps in the coverage of the basic needs of immigrants. In the case of health care, for example, immigrants contributing to social security had access, but long-term unemployed, the undocumented and those having only a residence permit did not.

In search for a new policy framework

In 2000, a new Immigration Act was adopted, and after two massive regularisation processes in 2000 and 2001 (in which about 380 000 immigrants obtained

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papers for a one-year period), the government is setting up a new policy framework. The declared objective is to better gear immigration to labour market needs, and to reduce the number of undocumented immigrants. It is intended that future labour immigration relies exclusively on a system of yearly quotas for short-term work permits (the "contingente") for foreigners living outside Spain, while measures to contain illegal immigration have been implemented. It is still early to evaluate the results of the new policy measures, but the available information points to some limits of the approach.

The new Immigration Act and the GRECO *plan*

The current Immigration Act (*Ley de extranjería*) was adopted in December 2000.[141] Compared with the previous legislation, it makes civil and social rights of immigrants residing legally in Spain equal to those of nationals, notably for access to welfare services and education, and establishes the possibility to vote at the local level. On the other hand, it draws a sharper distinction between documented and undocumented foreigners, restricting certain rights of undocumented foreigners and reinforcing the possibilities for expulsion.[142] However, if registered in the "Padrón" (the municipal residence register), undocumented foreigners are entitled to free education for their children (from 3 to 18), access to grants for education and health care.

The institutional setting has also changed. Immigration issues, which were previously in large part the responsibility of the Labour Ministry, have been brought under the supervision of the Interior Ministry, illustrating the importance attached to border and police aspects of immigration. The newly created Delegation of the Government for Foreigners and Immigration is now in charge of formulating policy and co-ordinating and directing actions in this field. Since March 2001, all migration policy measures are grouped in a programme called GRECO, being implemented over 2001-04, under four themes: global and co-ordinated design of immigration in the EU context, integration of foreign residents, admission regulation and inflow control, and management of refugees.

Current tools for managing entry, residence and work

Non-EU foreigners must obtain a residence permit (initially only temporary) to stay in Spain. The right to work is generally distinct from the right to reside, although a work permit is granted automatically to permanent residents. Moreover, a recent reform will grant foreigners with Spanish parents with an automatic work permit. As in most other European countries, and contrary to Australia, Canada and the United States, there is no permanent immigration policy (*i.e.* a foreigner cannot enter in Spain with a permanent visa). Except for asylum seekers who obtain refugee status, permanent residence permits are granted to non-EU foreigners only after five years of uninterrupted residence, resulting from right-recognition over

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time to immigrants that were initially seen as temporary immigrants. The number of refugee visas is very small (700 in 2000), and Spain has a relatively low recognition rate compared with other OECD countries.

As in the previous legislation, the system of work permits included in the 2000 Immigration Act is rather complex, including more than ten types of permits. It is composed of two parts, the general regime, which deals with individual applications, and a new quota system. In January 2002, the government announced that the quota system would be the main source of work permits. Outside this system individuals could apply for a first work permit only in specific cases.[143]

Work permits are grouped under two broad categories, wage earners and self-employed, the latter representing only about 5 per cent of the total in 2000 (Table 33).[144] The first permit that a non-EU foreigner can obtain is for a maximum duration of one year, and can (and often does) include restrictions on the sector

Table 33. **Work permits**

Type of permits	Duration	Characteristics	Number granted in	
			2000	Jan.-June 2001
For wage earners				
A	Maximum of one year. Can be extended if the activity requires	Temporary work for foreigners living outside Spain. Can be limited to a specific sector or region.	1 143	1 509
T	Maximum of 9 months	Seasonal or temporary work for foreigners living outside Spain. Can be limited to a specific sector or region. Employer guarantees transport (including return) and housing.	. .	. .
B initial	Maximum of one year	Can be limited to a specific sector and/or region.	37 472	33 247
B renewed	2 years	After B initial. No limitation.	5 529	2 336
C	2 years	After B renewed. No limitation.	30 406	27 320
F	5 years maximum	For cross-border workers.	795	517
Permanent			32 004	7 033
For self-employed				
D initial	One year	Can be limited to specific activity or region.	964	448
D renewed	2 years	After D initial. No limitation.	264	206
E	2 years	After D renewed. No limitation.	1 155	943
F	5 years maximum	For cross-border workers.	107	8
Permanent			3 614	840

Source: OECD based on the 2000 Immigration Act and Ministry of Labour and Social Affairs.

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and/or region in which it can be used (for example, agricultural work in Andalusia). When a non-EU foreigner asks for a work permit under the general regime for the first time, the law stipulates that the decision must take into account the overall employment situation. Public employment services have to certify either that the job offer has been published and not been filled or that there are no job seekers with the required profile. The process seems to be rather lengthy, and enterprises have been asking for more flexibility, in particular in the way the checks are carried out. The permit can be renewed if the foreigner still has a work contract or if he receives unemployment benefits (for a duration equal to the period of rights in the latter case). Refusals of work permits (as well as visas) do not have to be justified anymore under the new law.

The new quota system included in the 2000 Immigration Act (Box 6) aims at allowing a more active management of immigration inflows based on domestic labour market needs. In contrast with the previous quota system, it is composed of generic job offers which can be filled exclusively by foreigners residing outside Spain, with preference given to countries which have signed bilateral agreements on immigration with Spain. It also aims at improving the likelihood that temporary foreign workers will return to their country of origin.

Although it is too early to make a definitive judgement, the results of the implementation of the quota system in 2002 are not encouraging. 13 600 of the 32 000 job offers agreed upon in December 2001 were actually filled, including 3 100 "stable" jobs, largely by Polish and Romanian workers. The recent regularisation processes may partly explain why entrepreneurs have not followed through with concrete offers, as they may have been covered nationally. However, the system also seems to suffer from deficiencies. A criticism made by the social partners and non-governmental organisations (NGOs) relates to the length and administrative cost of the process. It takes about four months between the moment when entrepreneurs submit their requests and the date when the quota is effectively allocated amongst the employer organisations. Employer organisations underline in particular the necessity to simplify the administrative process. The lack of established rules to determine the quota as well as to allocate it amongst employers and provinces is another common criticism. As to the selection process, employers have underlined the relatively high costs associated with travelling abroad to select workers, while trade unions have emphasised the lack of established rules as well as the insufficient means and experience of embassies.

To overcome these problems, the government has modified the rules. The main changes agreed, published in January 2003, are the following: *i*) firms wanting to hire more than five foreign workers (and not only business organisations) can also make generic job offers; *ii*) jobs can also be offered to a worker who has previously had a work permit in Spain and has returned to the country of origin, but he can only return to his earlier employer; *iii*) if a province cannot provide

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Box 6. The new quota system for work permits

The quota system ("*Contingente*") distinguishes two types of work permits: temporary permits, for jobs with a maximum duration of one year, and with responsibility for the employer to provide decent housing and to organise the travel of the worker from the country of origin and back; and "stable" permits, for non-temporary jobs, of one year's duration.

The process of determining the size of the quota starts at the provincial level, with business organisations sending their requests for foreign workers (about 80 000 "stable" and 31 000 temporary jobs in 2002) to the Provincial Executive Commissions of the INEM, which, after evaluating the employment situation in the province, transmit a proposal to the Labour Ministry (about 33 000 "stable" and 29 000 temporary places in total). Taking into account the national employment situation, unemployment among foreigners, and, in 2002, the number of foreign workers regularised, the Ministry of Labour makes a new proposal. Finally, after consultation with the inter-ministerial commission for foreigners, business organisations, trade unions, and the superior council for migration policy, the government approves a quota (10 900 "stable" and 21 100 temporary places for 2002).

The allocation of the quota among business organisations in the various provinces is then based on the requests of the business organisations and set in collaboration with the provincial commissions of the INEM. Business organisations subsequently present generic job offers by type of occupation to the Labour Ministry, which transmit them to Spanish embassies and consulates, which then recruit the workers, possibly with the assistance of employers. Foreigners recruited then receive the work permits. Preference is given to countries with bilateral agreements with Spain; in 2001 and 2002, six countries (Morocco, Dominican Republic, Colombia, Ecuador, Romania and Poland) signed such agreements. Aiming to prevent illegal immigration and economic exploitation of undocumented foreigners, they include a general definition of the selection process of foreign workers (communication to the embassy of the offers defined in the quota, selection of candidates by a commission with possible participation of employer organisations, evaluation of living conditions), a guarantee that foreign workers obtain the same rights and working conditions as Spanish workers, specific provisions for temporary workers and voluntary repatriation assistance programmes.

the jobs that it has been assigned, the government can transfer them to other provinces; *iv*) administrative steps have been simplified; and *v*) the role of trade unions in the allocation of quotas has been increased. For 2003, a preliminary quota of 13 700 temporary and 10 600 "stable" job offers were agreed upon. However, it is not clear how far these changes will resolve the problems cited above.

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Dealing with irregular immigration

Police measures aimed at better controlling the entry of immigrants are high on the current policy agenda. The government has reinforced border controls, including along the coast, through a surveillance system. Spain has also been encouraging the Moroccan government to intensify controls on its side. The recent proposal by the European Commission to invest EUR 40 million to organise joint EU-Morocco patrols to control illegal immigration in the Strait of Gibraltar including from sub-Saharan Africans, goes in the same direction. Policing efforts seem to have started to bite in the Strait, but part of the flow seems to have been diverted to the Canaries instead. Increasing police activity against illegal immigration networks is also part of the GRECO programme. The government has recently announced its intention to increase the legal sanctions against people smugglers.

The two massive regularisation processes conducted in 2000 and 2001 have shown that a large number of foreigners were indeed staying in Spain without documents. More than 650 000 applications were received, out of which, by mid-April 2002, about 440 000 had been given a one year residency permit (and often a work permit), and 210 000 (or 32 per cent) had been filed or rejected.[145] At a minimum, undocumented foreigners involved in these regularisation processes were about 1.4 times as numerous as foreigners officially residing in Spain at end-1999 when the first process started.[146] These processes have also shown that people entering illegally in Spain are far from being the majority of the "stock" of undocumented foreigners: they represented about 20 per cent of the applicants in the first process, and Ecuadorians – who did not need a visa to come to Spain at that time – were the most important nationality in the 2001 process.

In this context, and given that most of the immigrants come to Spain to work, focusing on dealing with undeclared employment is an efficient way to reduce irregular immigration, as it influences incentives to immigrate. Reinforcing labour inspections is part of the GRECO programme, and the government is also planning to increase the sanctions for hiring undocumented workers. However, the means dedicated to labour inspection services seem to be relatively small; a report for the professional association of labour inspectors indicated that they were a maximum of 700, and had poor working conditions.[147] Besides, the aim of reducing undeclared employment sometimes conflicts with that of preserving activities relying on undeclared work. Indeed, part of the hiring of undeclared workers seems to result from the impossibility to hire them legally, given the difficulty to obtain a permit, as emphasised by employer organisations.

The government is also increasing the number of expulsions of undocumented foreigners.[148] Agreements for repatriation have been negotiated with Algeria, Ghana, Guinea Bissau, Morocco, Nigeria and a number of other countries.[149] However, beyond the human issues that are involved, expulsions are difficult to implement on a large scale. It is often impossible to identify the nationality of

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undocumented foreigners, and thousands of sub-Saharan Africans cannot be expelled because most African governments do not recognise them as their nationals. Transport and escort costs are also high. Hence, a number of foreigners arrested when entering illegally, often in the islands, are transferred to special centres in the peninsula, but released soon after. The problem is even larger when considering the number of undocumented foreigners already living in Spain. Looking just at the regularisation processes, a minimum of 90 000 persons have seen their applications rejected and are therefore staying in Spain without documents, and the overall number is certainly larger.[150] As noted above, the government had eliminated the possibility for undocumented workers to regularise their situation by closing the general regime in 2002, but it has been partially re-opened for 2003.[151] In addition, the partial re-opening of the general regime is allowing the regularisation of some immigrants living in Spain.

Economic aspects of immigration

An important part of the overall economic gains from immigration is probably appropriated by the migrants themselves. However, the economic effects of immigration on the native population are playing an important role in shaping the attitudes towards immigration and thereby the policies to influence it. While migration may also have external economic effects in the countries of origin, analysis of these effects is beyond the scope of this chapter. The chapter is not dealing with the economic impacts of emigration either. While these were probably important in the past, very little analysis is available on the subject, and the impacts are probably lower nowadays.

Labour market outcomes

Greasing the wheels of the labour market?

Given the still limited share of foreigners in the labour force, immigration has probably had relatively little effect so far on overall labour market performance. However, its impact may be important in some sectors. This section analyses the contribution of immigrants to labour market performance, notably how they affect wages and employment. It also assesses if immigration is playing a role in reducing some of the problems discussed in the previous chapter, namely the segmentation of the labour market and the regional disparities in unemployment.

- Wages, employment and labour conditions

Immigration of people of working age implies an increase in the labour supply and might thus be expected to affect wages and/or unemployment of natives. The size of this effect ought to depend on the extent to which immigrants complement or substitute for natives in the labour market, on the labour market

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situation, and on other factors, especially the possibly associated capital flows. There is little empirical analysis specifically on Spain. The only study covering the early 1990s by Dolado *et al.* (1996) concluded positive – albeit weak – effects of immigration on the labour market outcomes of native workers. However, studies linking wages and employment to migration densities suffer from an endogeneity problem, since migration is to some extent a function of labour market conditions – people are more likely to move where labour demand is relatively high – making it difficult to identify an independent effect of migration.[152] Besides, in the case of Spain, the importance of undeclared foreign workers make the results of studies based on official work permit statistics unreliable.[153] Nevertheless, the information available allows one to distinguish three main types of immigration flows that have different characteristics and thus probably a different impact on the labour market.

The first group consists of the OECD (mostly EU) foreigners, who are on average better educated than the Spanish average worker and generally occupy skilled jobs (Table 34). Working largely in multinationals or in tourism-related activities, they often occupy labour market niches, linked to their characteristics as

Table 34. **Education level and type of employment: foreigners *versus* natives**
Per cent of the total population of each category, 2001

	Foreigners		Spanish
	OECD	Non-OECD	
Population of more than 16 years by level of education			
Illiterate and primary school uncompleted	3.4	11.4	15.5
Complete primary school	11.6	19.8	26.0
Secondary school	40.3	43.4	39.9
Professional and technical education	11.8	8.3	5.6
University education and professional specialisation	33.0	17.1	13.0
Total (in thousands)	168.6	365.9	33 688.6
Employment by sector and type of occupation			
Managers	20.8	4.5	7.7
Technicians and scientific and intellectual professions	21.5	3.8	12.1
Technicians and support personnel	13.3	4.2	10.3
Administrative employees	7.2	3.6	9.7
Employees in restaurants, personal services and sales	12.0	18.3	14.0
Qualified workers in agriculture and fishing	1.2	2.5	4.2
Craftsmen and qualified workers in mining, manufacturing industry and construction	11.7	16.3	17.6
Machinery operators	3.6	5.0	10.4
Unqualified workers	8.8	41.8	13.5
Army	. .	0.0	0.6
Total (in thousands)	90.9	242.6	15 555.2

Source: INE and EPA.

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foreigners (language ability, cultural proximity for services to foreign residents). Their jobs are often linked to the internationalisation of the Spanish economy – including investment inflows which create jobs – and to the presence of other EU foreigners (tourists or residents), and in any case are located in sectors of employment that might be considerably smaller in their absence.

Second, a group of the highly educated non-OECD immigrants occupy high-skilled jobs, mostly in the commerce and professional service branches (Carrasco, 1999). Coming often from Chile or Argentina, they include a large share of self-employed and have often been residing in Spain for a relatively long time. The impact of that group on the Spanish labour market is not clear, but it seems likely that they contribute to Spanish economic development and growth.

The third group includes the majority of non-OECD immigrants, who, by contrast, are more concentrated in unskilled jobs.[154] Information on that group of workers is taken from a number of surveys conducted in the second half of the 1990s, which allow to pinpoint their labour market characteristics.[155] In part, they occupy jobs that are unattractive for Spaniards, due to harsh working conditions and low pay. This is the case for work in greenhouse farming, some construction jobs and domestic services, where virtually no Spaniards are working. In these jobs, immigrants seem to be complementary to the Spanish labour force, although competition among themselves is probably important. This is probably less so in other jobs or sectors, such as hotels and restaurants, external domestic services, tradesmen in the construction sector and qualified agricultural workers, where both immigrants and Spaniards are working. The situation varies across sectors (or jobs) and ethnic group, but this group generally faces precarious conditions in the Spanish labour market: *i*) they are often not declared;[156] *ii*) they mostly work on temporary contracts; and *iii*) they tend to work longer hours, sometimes paid below the contractual rate.[157] Hence, they are generally contributing to the "temporary work" segment of the Spanish labour market, which is very important compared with other OECD countries.

Undeclared work is not specific to immigrants. It is obviously difficult to estimate the importance of undeclared activities in the Spanish economy, and the various estimates available are not reliable, but there is evidence that it is relatively more important in Southern European countries than in Central and Northern Europe. A traditional explanation for the existence of undeclared activity is that the costs of declaring are too high, either due to labour market regulations or to a high tax wedge. The level of the minimum wage is relatively low in Spain compared with other EU countries,[158] and generally not considered as deterring employment of low-skilled people, and the sectors in which immigrants are working have rather loose labour regulations.[159] The tax wedge, on the other hand, although not high on average compared with other European countries, is relatively high for low-wage workers (OECD, 2000) so that the incentive not to declare

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may be higher in their case. The relatively low probability of being caught obviously also plays a role. Agriculture, construction and hotels and restaurants, which are particularly large in the Spanish economy, are mainly composed of small enterprises which are more difficult to control, typically resort more frequently to undeclared employment, and even more so for domestic services. The development of outsourcing is also making undeclared employment easier (OECD, 2000). This is all the more the case in Spain as labour inspections seem to be relatively weak in international comparison.

Immigrants are probably more likely in undeclared employment than Spaniards because they have to work to survive. The relative difficulty to get a permit or renew it also probably plays a role. In fact, the huge increase in recent years in the number of foreign workers affiliated to the social security system following the two latest regularisation processes – non-EU immigrants represented almost 4 per cent of the affiliated workers in June 2002 compared with only about 1.5 per cent at end-1999 – shows that providing papers to immigrants contributes to reduce undeclared employment (Table 35).

Table 35. **Workers affiliated with the social security system**
In thousands

	December 1999	December 2000	December 2001	June 2002
Non-EU foreigners	214.4	314.1	448.9	606.3
EU foreigners	120.6	140.5	158.1	184.8
Spanish	14 243.3	14 781.6	15 141.7	15 496.0

Source: Ministry of Labour and Social Affairs, *Estadísticas Laborales y de Asuntos Sociales*.

The third group of immigrants allows the maintenance, or in some cases the development, of activities based on low-skilled labour. Although in many cases they seem to complement Spanish workers, the presence of this flexible and easily exploitable workforce is probably exerting some downward pressure on wages and labour conditions of the low-skilled Spanish workers in the same sectors. In their absence the sectors they work in would either have to provide better pay and working conditions to attract workers or would not exist.

- Mobility

One feature of the Spanish labour market is the persistence of large differences in unemployment rates across regions. Immigrants are more mobile than natives, as they have less attachment to a particular region. They could therefore be expected to reduce regional disparities in unemployment resulting primarily from the low mobility of native workers (Chapter III). In practice, however, immigrants go

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where there is employment for them, and it is sometimes in regions where the unemployment rate is low, such as in Madrid and Barcelona, and sometimes in regions where it is high, like in Andalusia, reflecting the strong segmentation of the labour market. In fact, immigrants and natives by and large respond to a different set of incentives, as the former (at least the low-skilled ones) are in a much weaker position and do not have much choice. Thus immigrants have a minor role in evening out regional unemployment differences as they boost both employment and labour supply without much influence on the labour market situation of the natives.

Integration of immigrants into the labour market

The way in which immigrants integrate into the labour market very much depends on which group of immigrants is concerned. No study exists on the adaptation of EU foreigners in the Spanish labour market, but there does not seem to be any major problem. They have a lower unemployment rate than Spanish nationals according to the labour force survey (EPA) or a slightly higher level according to registered unemployment statistics (Table 36). Even less is known about the second group, the non-EU highly-skilled workers, but, when they have papers, they probably do not face major problems. As in other European countries, highly-skilled foreigners may sometimes face barriers due to lack of recognition of diplomas.

Table 36. **Unemployment rates of foreigners and natives**
In per cent, 2001

	EPA[1]	INEM[2]
Total	10.5	10.1
Spanish	10.4	10.0
OECD foreigners	8.4	10.8
Non-OECD foreigners	15.8	12.8
Total foreigners	13.9	12.2
EU	8.3	11.0
Other Europe	13.7	8.9
Latin America	15.2	11.5
Rest of the world	17.1	14.3
Africa	n.a.	16.5
North America	n.a.	8.3
Asia	n.a.	5.9
Others	n.a.	14.6

1. The EPA unemployment rate of a specific group is defined as the level of unemployment of that group over the labour force of the same group.
2. The INEM unemployment rate is defined as the registered unemployment over the workers affiliated to the social security system.

Source: EPA and OECD calculations based on *Anuario estadístico de extranjería* 2001 and social security data.

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The situation is different for the non-EU low-skilled workers, especially those who are employed without a work permit. The recent character of this immigration makes it difficult to assess how they integrate into the labour market, as time plays an important role in integration. Carrasco (1999) notes that the proportion of documented in total non-EU immigrants increases with the length of stay, as well as their wages (except for Africans). Non-EU immigrants have generally access to a limited number of sectors and seem to have difficulties escaping the temporary jobs at the low end of the labour market. Qualification and experience generally seem to be of relatively little relevance in their work position and evolution.[160] Sectoral and regional limitations included in some of the permits (which represented 35 and 47 per cent of the work permits granted in 2000 and the first half of 2001, respectively) may partly explain that situation. The difficulty immigrants had faced in renewing their work permit may also have played a role, as it often leads to spells of undeclared work until the next regularisation process. The short duration of the permits does not help in that respect. Unemployment is higher than among Spanish workers, women being more exposed than men (although the gender differential is smaller than for Spanish nationals), and Africans being by far the most exposed ethnic group (Table 36).[161]

Output, productivity, growth

Immigration raises output, as it increases both supply and demand, but how it affects output per capita is less obvious. In a first instance, immigration increases labour supply. Most immigrants come in their working age, and foreigners from non-EU countries have higher participation rates than Spanish nationals (79 per cent in 2001, against 59 per cent for EU foreigners and 68 per cent for the whole country).[162] Indeed, they have had a clearly substantial positive effect on the employment rate, foreign workers (mostly from non-EU countries) having contributed about 27 per cent to the increase in employment between end-1999 and mid-2002.[163]

Apart from this direct effect, immigration facilitates female participation in the labour market, through employment in domestic services. The total female participation rate, at 52 per cent in 2001, remains below the OECD average (59 per cent), but it has increased by about 10 percentage points over the past decade. The substantial increase in female education likely explains a large part of this increase, but the possibility to resort to foreign women to take care of children and parents has probably played a part too.[164] Indeed, institutional services to take care of young children and elderly parents are relatively little developed in Spain compared with other OECD countries; only 5 per cent of children aged less than 3 were covered by formal childcare arrangements at the end of the 1990s[165] and, by European standards, a relatively large proportion of Spanish women were providing elderly care, thus reducing participation (Spiess and Schneider, 2002).

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While this situation may lead to lower fertility instead of not participating in the labour market,[166] having a domestic employee is often the only way to combine family life and work.

The effect of immigration on output and growth also depends on the capital that immigrants may bring along or may be associated with. While most of those coming from developing countries do not have any capital, German or British pensioners coming to Spain to live in a sunnier place spend their retirement income. They not only consume goods and services, but also have an impact on demand through the acquisition of real estate, which has been growing by about 30 per cent per year over the last eight years (Bank of Spain, 2002a). As noted above, professionals working for multinationals and in the tourism sector, are "complementary" to these investment flows, although they do not own the capital. These investment flows raise the stock of capital in the Spanish economy.

How immigrants affect productivity depends on qualification, or at least on the type of jobs they occupy. While the highly-skilled Europeans certainly contribute positively, non-EU immigrants mainly occupy low-productivity jobs, and the increase in such immigration flows may have contributed to the slowdown in overall productivity growth since 1995.[167] However, a simple calculation based on the (rough) assumption that non-EU workers are less productive than their Spanish and EU counterparts, suggests that this effect is limited: assuming productivity levels of non-EU workers equivalent to 75 or 50 per cent of the Spanish workers would have reduced labour productivity growth by 0.1 and 0.2 percentage point per year, respectively, between 1995 and mid-2002.[168] Another angle of analysis, based on a simple growth accounting exercise, helps in evaluating how the shift in employment shares across sectors has affected productivity growth, and the role played by immigration of non-EU foreigners in this shift. This calculation suggests that the changes in employment shares have slowed down annual productivity growth by about 0.2 percentage point between 1995 and the second quarter of 2002, whereas the total slowdown amounted to 0.6 percentage point per year on average. Most of the sectoral shift is linked to immigration (Annex V), but the size of the effect is limited in any case. These estimates do not take into account other possible positive effects of immigrants on the labour productivity growth of other workers, as is generally thought to be the case when investment flows and immigration are complementary and when low-skilled labour is complementary to high-skilled labour.

Besides the overall macroeconomic effect, immigration is also having an effect on regional development. In some rural areas, the development of horticultural activities, which rely heavily on immigrant labour, has had a noticeable effect on overall activity and income. It has allowed some regions

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to catch-up *vis-à-vis* the national average in terms of income per capita, while others have widened their lead.[169] This is also the case in tourism areas with important immigration, such as Las Palmas.

Fiscal effects

The effect of migrants on the public finances will depend on their characteristics, behaviour, rights and obligations. If the typical migrant were similar to the average Spaniard, the effect of migration on per-capita budget values would be limited to lowering per-capita public debt. However, immigrants do differ from Spanish nationals. Highly-skilled workers from EU and non-EU countries are likely to pay relatively high income taxes, while they probably require less per capita spending and have limited pension rights. EU pensioners (who represent about one third of EU residents)[170] may have an effect through income taxes when they declare their residency; however, although precise information is not available, as mentioned above, it seems to be the case that a number of them do not declare residency in Spain. They are nevertheless contributing through indirect taxes. As to the lower skilled non-EU foreigners, lower income levels imply lower income tax payments, but their higher participation rate may imply more payment of social security contributions (when declared). On the spending side, they share the features of the EU workers and also often have only limited access to unemployment benefits because of too short contribution periods to the UI system resulting from their temporary work contracts. These features seem to be beneficial for the public accounts in the short run, but it is less clear in the longer run, which is important in the context of ageing.

Research on the fiscal impact of immigrants is scarce. Aparicio and Tornos (2000) provide the only estimate on the short-term impact of non-EU immigrants on the state budget (Table 37). Their contribution is clearly positive, as the revenues that immigrants contribute are more than twice the expenditure they generate.[171] The sums concerned were still very low in 1998, but they must have grown over the last years. In particular, social security contributions have boomed with the increase in social security affiliation of foreigners following the regularisation processes. Indeed, the legal status is instrumental in determining the fiscal impact. Immigrants who have no residence or work permit and/or work undeclared (they are not included in the above estimates), are likely to induce a net fiscal loss. By definition, they do not pay taxes or social security contributions, while, with the Immigration Act, they are entitled to receive health benefits, education for their children and other types of social assistance, when they are registered in the municipal "P*adrón*".

Welfare dependency of immigrants has been a subject of real or imagined concern in a number of European countries, but the Spanish situation seems quite different. While non-EU foreigners experience higher unemployment than Spanish

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Table 37. **Fiscal impact of non-EU immigrants**
1998

	% of GDP	% of public expenditure
State expenditure generated by immigrants		
Immigrant and refugee affairs	0.00	0.01
Public order and safety	0.02	0.05
Health	0.09	0.23
Education	0.05	0.13
Unemployment	0.02	0.04
Total	0.17	0.46
State revenue contributed by immigrants		
Income tax on individuals	0.06	0.15
Property tax	0.01	0.01
Special tax	0.00	0.01
Value added tax	0.00	0.01
Tax on economic activity	0.00	0.00
Total tax	0.07	0.19
Social security contributions	0.31	0.86
Total contributions by immigrants	0.38	1.04
Balance	0.21	0.59

Source: Calculations based on Aparicio and Tornos (2000).

nationals, they receive unemployment benefits less often: in December 2001, 77 per cent of the immigrants registered as unemployed received no benefits, against 45 per cent for the whole unemployed population (and 61 per cent for the EU foreigners). The temporary nature of their jobs combined with their frequent passage into an irregular situation explains why non-EU immigrants are *de facto* often excluded from unemployment benefits. Compared with the other European countries, the level of social transfers other than those related to health and pensions is low. Minimum incomes exist at the autonomous community level, but they are generally rather low and cover only a relatively limited number of households;[172] they are sometimes subject to being Spanish, and require a residence of between 3 to 5 years, thus often excluding immigrants. Family benefits are very low, as is the extent of other social services, *e.g.* for housing.[173]

In the context of increased decentralisation, some concern has been raised that the concentration of immigrants in a number of areas may put some strain on the public finances of the autonomous communities or local governments, raising an equity issue between regions. Spending on programmes specifically designed for immigrants is low,[174] but both health and education spending is now mainly concentrated at the autonomous community level (Chapter II and III). However, such concerns should not be exaggerated: while there

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are additional expenditure, increased immigration also generates additional tax revenues at the community level. Some concern could remain in the case of undocumented foreigners (which pay VAT but not direct taxes), and for the low-income documented foreigners. Yet, the problem raised is not specific to immigration, but more generally relates to differential population dynamics (and possibly differential dynamics in the social composition of the population) across regions and how it can be compensated for in the Spanish decentralised system. The new rules for allocating revenues to the autonomous communities provide for some adjustment across regions. Nevertheless, only in the case of a significant shock to the population size (when the annual growth rate in the number of pupils or the number of persons benefiting from health benefits deviates by more than three percentage points from the national average) is there compensation.

Another widely debated issue is that of the long-term fiscal impact of immigration, in particular its potential contribution to mitigating the effects of population ageing. The demographic changes currently taking place in Spain imply that ageing is going to occur later but more intensively than in other OECD countries, raising an important problem for the long-run sustainability of public finances. Immigration, by immediately increasing the working-age population and employment helps to improve the financing of the pay-as-you-go pension system. The relief provided is transitory, however, since in the longer-run immigrants retire too. Indeed, projections based on different assumptions on immigration flows find that the increase in public spending on pensions at the 2050 horizon remains high by international comparison (Box 7).

These projections assume that immigrants and natives are similar. In practice, this may not be the case on a number of counts. Immigrants can contribute to temporarily raising fertility rates, because the females of the first generation have higher fertility rates. When looking at public expenditure at large, immigrants arrive at working-age, implying that they contribute to the pension system but have not benefited from education expenditure.[175] These types of effects may imply that immigration alleviates public expenditure pressures, but they are only transitory. Besides, immigration may also require specific public expenditure, such as for example for the education of immigrants' children. More generally, immigration in itself does not change the fact that the parameters underlying the calculation of old-age pensions are generous in Spain. Currently pensioners receive, on a discounted basis more than they contribute. As shown in the previous OECD S*urvey*, the internal rate of return that equalises the discounted value of contributions paid with that of pensions received amounts to about 4 per cent for the general pension scheme (OECD, 2001a). It reaches even between 6½ and 9 per cent in the case of the agricultural and domestic employee regime, sectors where a large proportion of immigrants works. These implicit yields are much higher than the potential growth of the economy over the coming decades,

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Box 7. Projections of public pension expenditure

According to the joint projections by the OECD and the Economic Policy Committee of the EU, the old-age dependency ratio in Spain is expected to rise by 36 percentage points and public expenditure on pensions by 8 percentage points of GDP by 2050 (against 3 to 4 percentage points in other OECD countries) (OECD, 2001a). These projections assume net immigration of 60 000 persons per year from 2010 onwards, and do not incorporate the most recent immigration developments. Recent immigration inflows have been much larger than expected and the latest INE demographic projections embody an equilibrium inflow of 160 000 foreigners per year,[1] which would damp the increase in the dependency ratio by about 6 percentage points, bringing it to 56 per cent by 2050. Using the revised INE demographic projections, Jimeno (2002b) estimates that public pension expenditure should increase by about 7 percentage points of GDP between 2000 and 2050, which is still higher than in most other OECD countries.[2] In addition, such immigration levels would imply a strong and continuous growth in the foreign population, probably unprecedented in other OECD countries, which would result in the foreign or foreign-born population representing 21 per cent of the total population by 2050, which is quite high by international standard.[3]

1. The inflow of foreigners decreases from 360 000 in 2000 to 160 000 in 2005, keeping constant this figure up to 2050.
2. On the basis of simulations carried out in OECD (2001a), a permanent increase of immigration to 150 000 people a year would limit the rise in pension expenditure to 5½ percentage points of GDP between 2000 and 2050, which would still be one of the strongest increases in the OECD.
3. The recent reform relaxing citizenship laws for foreigners with Spanish parents could imply that a significant part of this foreign population might acquire Spanish citizenship and would no longer be counted as immigrants.

highlighting the absence of neutrality from an actuarial point of view.[176] All in all, these elements suggest that the extent to which immigration can alleviate the fiscal consequences of ageing is at best limited, and possibly negative in the long run if the system is not reformed.

Balance of payments: remittances and trade

Immigration may affect the balance of payments in a number of ways. Remittances of foreign workers to their country of origin are still low compared with those of Spanish emigrants to Spain (Figure 32), but they have grown significantly since 1998.[177] In the future however, given the expected increase in immigration

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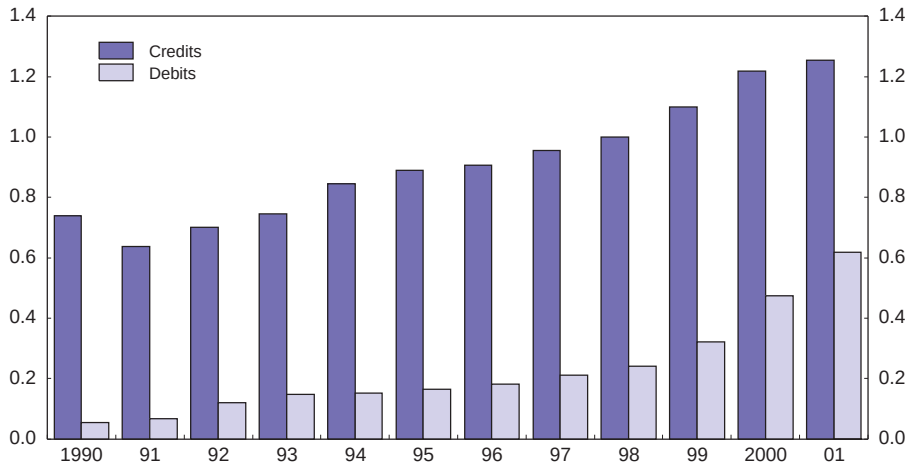
Figure 32. **Workers' remittances**
Per cent of total compensation of employees



Source: Bank of Spain, *Boletín Estadístico* and OECD.

flows from non-EU countries, remittances are likely to become increasingly important. The links between their country of origin and the host country that immigrants create may also generate future trade. This is not the case at the moment, however. Trade flows with the EU countries have in fact preceded migration of EU people, while immigration of non-EU foreigners is still very recent, and largely reflects already existing cultural and economic (including trade) links (*e.g.* with Latin American countries).

Theory suggests that movements of goods and labour can to some extent be substitutes; trade barriers lower the return of the abundant factor (labour in the source country) and increase the return of the scarce factor (labour in the host country), thus increasing the incentives to migrate on both counts (Faini, 2001). This link seems relevant in the case of immigrants working in the traded goods sector in Spain, and in particular agriculture. Sectors like fruits, vegetables and olive oil are still benefiting from protection, in particular through important trade barriers (Box 8).[178] Significant additional support is provided to water-intensive agricultural activities through water charges, which are often set below cost, and not related to the quantities consumed (Chapter III). These policy settings create a demand for cheap labour in Spain, and drive part of the migration flows.[179]

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Box 8. EU agricultural market access for South Mediterranean countries*

The European Union has signed a number of association agreements with countries on the other side of the Mediterranean, providing them with preferential market access. The advantages provided are nevertheless limited for agricultural products. The most important "Mediterranean" agricultural sectors are olive oil and fruits and vegetables. The first is subject to high specific tariffs (ranging from EUR 1 194 to EUR 1 737 per ton) and the second to *ad valorem* tariff from 0 to 21 per cent; duties are generally higher for some "sensitive" products and during periods of peak EU production. Besides, additional duties can be applied if certain "entry prices" are not respected, which in fact amounts to imposing a minimum import price. Quotas are also applied on some products, some of which are binding (*e.g.* tomatoes for Morocco and potatoes and oranges for Egypt in 1998 and 1999). Tunisia (more precisely the state-trading organisation that is the exclusive exporter of olive oil in Tunisia) also benefits from a zero tariff on exports up to an amount equivalent to 2 per cent of EU production; this tariff quota is regularly filled.

* The information provided in this Box is mainly based on CIHEAM (2001).

Integrating immigrants

The integration of immigrants encompasses several dimensions. Integration in the labour market, as discussed above, is the first, and it implies first acquiring a legal status. But social and cultural aspects matter too, as they condition social cohesion and growth potential, and policy may play a role in this respect. The new Immigration Act was conceived very much on the basis of the willingness to make progress in that area. Integration is not an issue for EU citizens and not so much for non-EU foreigners who occupy skilled jobs either, but, as seen above, foreigners coming from developing countries often occupy jobs at the lower end of the Spanish labour market, and are at risk of being marginalised. The general "integration philosophy" of the Immigration Act is to allow legally resident immigrants access to the same social services as Spanish citizens, as opposed to developing policies specifically targeted at immigrants. Immigrants, whether legally resident or not, are allowed access to compulsory education and health care as well as (for legal residents only) to social transfers, including housing assistance. There may be some problems in practice however, and, in some cases, this approach seems to be insufficient (because immigrants face specific problems and/or are over-represented in the low-income groups). A tension also persists

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between the integration objectives and the willingness to restrict immigration as far as possible to temporary immigration.

The responsibility for immigrants' integration policies is shared among the various administrative levels. Central government has unique responsibility on immigration matters, but regions are responsible for education, health, social assistance and housing policies. Local administrations also have the responsibility to provide social services. Co-ordination among the different competent administrations is organised through bodies like the Permanent Observatory of Immigration, the High Council of Policy of Immigration, or the Forum for the Social Integration of the Immigrants. At the central level, the Instituto de Migraciones y Servicios Sociales (IMSERSO), co-ordinates a number of activities through conventions and funding, including for NGOs, which supply a significant part of the assistance to immigrants, reflecting the rather limited government involvement in that area in the past. Overall, the IMSERSO spent EUR 34 million on programmes for immigrants and refugees in 2001, *i.e.* 0.03 per cent of overall central government outlays. At the regional level, Madrid and Andalusia have recently developed their first pluri-annual immigration plans, with yearly spending of between 0.3 and 0.5 per cent of their total budgets. In Madrid, about three-quarters of the financing is devoted to the new social centres for immigrants, managed by NGOs. In contrast with the general integration philosophy, these centres provide specific services to immigrants only, such as help for family reunification or return to the origin country, language courses, emergency housing, judicial assistance, and translation and interpretation. It is too early to evaluate the activity of these centres, but they seem to have a relatively limited coverage of the immigrant population.

Housing is probably one of the main problems for immigrants' integration. It is difficult for immigrants to find a place to rent, even when they can afford it. Many live in slums or shanties, or in sheds in rural areas, often without drinkable water, and sometimes no electricity and/or sewage, and, in rural areas, often outside the town on land that is not for construction.[180] Rent abuse exists, and owners may require very high guarantees.[181] A number of factors explain that situation. First, as highlighted in Chapter III, there is a general lack of supply of housing for rent, which affects particularly the groups of the population who cannot afford or do not have the credit-worthiness to buy a house, including immigrants. Social housing is also very little developed.[182] Second, immigrants often face discrimination from the owners, due to their ethnic origin. Regions are in charge of housing policies. In Andalusia, a plan for promoting private and public investment in rental housing for temporary workers with high mobility has been jointly adopted by the central and regional governments, but little is done to support the rehabilitation of existing housing (Defensor del Pueblo Andaluz, 2001). In Madrid, Navarra and Murcia, programmes providing guarantees for rental housing for immigrants are being developed, as a way to overcome the discrimination barrier, but on a

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very small scale. Finally, municipalities generally do not promote immigrants' access to housing. This situation highlights the need to reform housing and land policies along the lines suggested in the previous chapter, so as to allow the price of land to fall and make investment in rental housing profitable. Additional measures would also be needed, both in terms of developing access to housing for low-income households in general, and to overcome the specific barriers faced by immigrants due to discrimination, for example through guarantee programmes.

Language difficulties are also an important barrier to integration for some groups of immigrants. It is obviously not the case for Spanish-speaking immigrants, and apparently not either for foreigners from OECD countries, whose jobs often involve the practice of their mother tongue. But it is for the other groups, in particular Africans. There is no general strategy for teaching Spanish to adult immigrants and very little is known about the language ability of immigrants from non-Spanish speaking countries. Spanish courses are also proposed at the level of the autonomous community, competent in education matters. In addition, there are initiatives in some locations, either by NGOs or local governments, but not systematically, and the coverage seems to be low. The language problem is even larger in the regions where immigrants have to learn a second language, especially in Catalonia, which has a large number of non-EU immigrants.

Evidence from older immigration countries shows that language skills are important for labour market integration and for social integration at large. Language proficiency is found to have a significant impact on both employment probability and earnings in the United Kingdom and the United States for example (Dustmann *et al.*, 2000). Hence, a more active approach would be needed in this area, well targeted on immigrants' needs and encouraging their participation, as some groups of immigrants seem reluctant to learn Spanish. While part of the immigration may remain temporary, the experience of other countries shows that even if not intended, a large part of immigration becomes permanent, implying that there may be important gains associated with such an investment.

Regarding children, the awareness of the risk of marginalisation has led every autonomous community to develop integration plans for children of immigrants and provide financing for compensatory education. Children of undocumented immigrants face specific difficulties: they often lack the basic material, like textbooks, as their parents have precarious jobs and incomes, and frequent changes in location also explain their relatively poor performance, for example in reading (Figure 33). As in other countries, the concentration of children of immigrants in specific locations and schools is also challenging the Spanish educational system. The fact that compensatory education is provided only in schools attended by a large number of immigrant children is probably accentuating the

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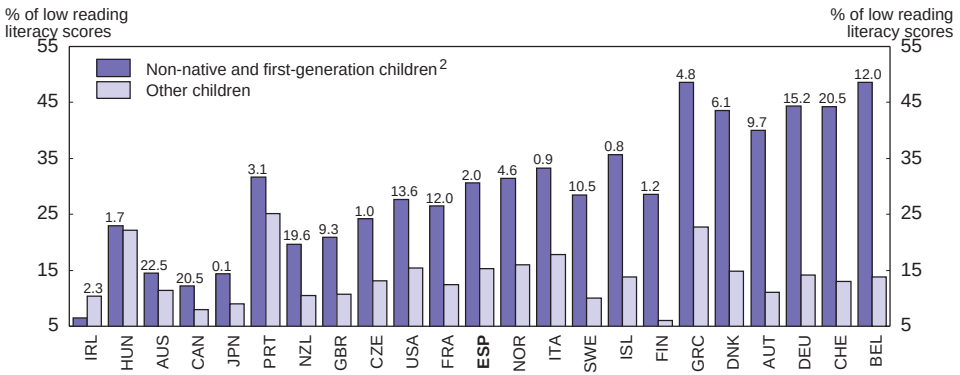
Figure 33. **Disadvantage in reading of 15 year-old children**[1]
2000



1. Countries are classified according to the difference in performance of "non-native and first- generation children" and other children.
2. The share of non-native and first-generation children – *i.e.* those whose parents were not born in the country – in the total is indicated above the bars of each country.

Source: OECD, *Knowledge ans Skills for Life – First Results from PISA 2000.*

problem (Cuesta Azofra *et al.*, 2000). A recent report of the "Defensor del Pueblo" indicates that 17 per cent of the schools in which more than 30 per cent of pupils are children of immigrants do not have any specific plans and means to attend their education needs.[183]

Conclusions and recommendations

Immigration on a large scale in Spain dates only from the mid-1990s, and migration policy has changed recently to respond to the challenge raised. The aim of the reform of the policy framework, which is still not fully implemented, is to better control immigration flows, in particular to reduce illegal immigration, while attempting to better match legal immigration flows with labour demand.

In large part, immigration results from the combination of the strong economic development of Spain and poor economic performance in source countries. Today, despite the strong increase in recent years, immigration remains relatively low and has therefore had rather limited macroeconomic impacts. However, the increase in immigration is likely to continue in the future and may become more important. In this context, it is important to assess the various economic impacts of immigration, notably how it affects some of the labour market problems, growth, and the sustainability of public finances in the long run.

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Immigration has had a positive effect on economic growth. With the exception of EU pensioners, whose presence is overall positive for the economy, most foreigners come to Spain to work. With the inflow of highly-skilled workers (mostly from the Union), which are often accompanying capital inflows, Spain is reaping the benefits of its closer integration with the European Union and, more generally, of the increased internationalisation of its economy. Non-EU foreigners, on the other hand, by often occupying jobs at the bottom of the qualification ladder that Spaniards can afford to refuse, have contributed to sustaining or developing the activity of a number of sectors relying on low-skilled labour. They have helped in raising the employment rate both directly through their high participation rates, and indirectly by facilitating female participation through domestic services. The overall effect on labour productivity is less clear. The development of low-productivity sectors through immigrants' work has somewhat moderated the growth of overall labour productivity, but this does not imply that natives' productivity and standards of living have been negatively affected; instead they have probably been improved, notably due to the increase in female participation. Besides, non-EU immigrants' productivity is probably much higher than if they had stayed in their countries, implying important net gains for themselves, especially when they are legally resident.

On the labour market, immigration has largely contributed to the strong growth in employment registered over the recent period – particularly after the regularisation processes – probably moderating inflation pressures at the same time. To the extent that non-EU immigrants are largely complementary to Spanish workers, they probably affect the unemployment of natives rather little. Immigration seems to have played a limited role in reducing regional differences in unemployment linked to the low mobility of the Spanish labour force. Most non-EU immigrants work on temporary contracts, contributing to the already high share of such contracts, highlighting the inflexibility of the "mainstream" labour market.

As to the effect on public finance, immigration will mitigate the expected increase in the dependency ratio, but the relief is transitory because immigrants will also retire. Given the high implicit yields embedded in the current pension system (especially in some of the regimes where immigrants are affiliated), increased immigration will only contribute to postpone the fiscal consequences of ageing, and reforms of the pension system remain necessary. In the short-run, however, immigrants are not dependent on the welfare system and, when legal residents, actually make a positive contribution to public finances. Although their unemployment rate is higher than that of natives, the combination of entitlements and the short-term nature of their jobs and work permits imply that they receive unemployment benefits much less often. Abolishing the sectoral and geographic limitations on work permits may also help to reduce the unemployment rate of immigrants.

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Migration policy faces a difficult task. Reducing illegal immigration is an important policy target, and the number of new work permits issued to non-EU persons (other than in the regularisation exercises) seems to have been much smaller than the estimated number of undocumented immigrants in the country. But the demand for labour in certain sectors, and hence the incentive to immigration, is strong; tightening controls on immigration enough to enforce the formal restrictions would affect the expansion of these sectors. Where this expansion is partly due to sectoral support, as in the case of agriculture through trade protection at the EU level and undercharging for water, this may be less of a concern.

Since illegal migration occurs because of the relative ease with which migrants can find undeclared employment, measures to reduce undeclared employment (by increasing labour inspections) also act to reduce illegal immigration. The government is reforming the new quota system to improve its functioning and make it more flexible. For any given level of quotas, the government should take into account the need to align the resources devoted both to border controls and to labour inspection (and taking into account the relative efficiency of the two sets of measures) if illegal entry is not to continue to undermine the formal immigration targets.

A difficulty encountered in analysing immigration flows and policies is the relative lack of publicly available statistics. Improving their collection and diffusion would lead to better policy analysis on this highly sensitive topic. The effort already provided to extend the coverage of the foreign population in the labour force survey could be intensified, to draw a more precise picture of how they affect the labour market, and allow some analysis through time, currently precluded by the recurrent changes in methodology and coverage. There are still few statistics publicly available concerning migration policy. Publishing the number of residence permits granted every year, for example, would be useful, as well as continuing the publication of the number of work permits granted, which was interrupted in 2002.

Integration of immigrants is important for both economic and social reasons. The new Immigration Act clarifies the rights of immigrants in terms of access to health and education. The government targets its integration efforts on legally resident immigrants to avoid providing incentives for illegal immigration. Housing seems to be an obstacle, which reinforces the necessity to reform housing policies and to develop the supply of rental housing, along the lines suggested in Chapter III. Developing guarantee schemes at the regional or local level may also be a tool to overcome discrimination. There is also room for a more active policy in developing Spanish language courses for immigrants, and encouraging participation in them, in particular as they may improve their ability to better integrate in the labour market. There is also probably much to gain from reinforcing the policy to improve the performance of children of immigrants at school.

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Notes

1. As in the rest of the euro area, the introduction of the fiduciary euro resulted in a fall in demand for notes during the course of 2001, households being compelled to clear their cash reserves. That saving, put at 2½ per cent of GDP by the BBVA (2002a), seems to have been partly spent on consumer durables and housing in late 2001. This may have stimulated consumption by between ¼ and ½ percentage point in 2001, which would partly explain why it grew more slowly in early 2002 (Laborda, 2002).
2. According to Farré and Raymond (2002), the wealth effect on household behaviour is significant and of the same order of magnitude in Spain as in the other main OECD countries: a rise in total wealth by one euro results in a fall in the saving ratio by nearly 4 cents. These authors claim that the bulk of the reduction (by 4 percentage points) in the household saving ratio between 1995 and 2000 was due to the increase in wealth (over 40 per cent in real terms), itself stemming from the rise in the relative prices of financial assets and real estate. Since 2000, the continuing rise in house prices has offset the stock market fall.
3. The number of dwellings has increased on average by nearly 500 000 over the last four years, whereas the number of new households has risen by only 200 000 (Bank of Spain, 2002a).
4. According to Tinsa Real Estate Appraisals, approximately one-third of real estate purchases in autumn 2002 were for investment purposes. This investment, which takes the form of purchases of second homes or of property for children, is one of households' favourite ways of saving to protect themselves against the risk of inflation.
5. The proportion of households with both members working rose between 1992 and 2000 by 12 percentage points to 43 per cent, though this remains well below the European average (60 per cent). Measured against household income, the effort required to finance an average mortgage loan remains fairly moderate from an historical point of view, despite the recent rise in real estate prices (Bank of Spain, 2002a). A discussion of the risks of a speculative bubble is also provided by Balmaseda *et al.* (2002).
6. In the Balearic Islands, moreover, the introduction of a special green tax on tourism may have had an effect.
7. Prices of hotel and tourism services have climbed by an annual average of 9½ per cent since 1997, which is 6½ per cent more than in France and 4½ per cent more than in Italy.
8. For example, lower wage costs in countries such as the Slovak Republic prompted the car manufacturer Seat to transfer part of its production to that country during autumn 2002. According to Muñoz (2002a), foreign firms have been cutting back on their net investment in the industrial sector in Spain, though not in services. This

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author argues, however, that this process has more to do with the restructuring by multinationals throughout Europe than with Spain losing its comparative advantage in terms of wage costs.

9. Temporary employment accounted for only 13.2 per cent of total employment in the EU in 2001. The reduced use made of temporary contracts in the private sector in Spain was partly offset by an increase in the public sector.

10. The capital-labour ratio has fallen in recent years and the share of salaries in value added, which went from 52.2 per cent in 1993 to 49.8 per cent in 1996, rose slightly to 50.1 per cent in 2001.

11. At end-2003, the National Statistical Institute (INE) will present the revised ESA 95 national accounts figures, calculated for the base year 2000. There are a number of indicators, such as the buoyancy of tax receipts and the steeper-than-expected population rise (revealed by the 2001 census) resulting from increased immigration, which suggest that GDP might be revised upwards.

12. Between 1995 and 1999 (the period for which detailed data are available), 70 per cent of jobs (measured in full-time equivalent) were created in sectors where productivity was lower than the national average. Over 50 per cent of jobs were created in sectors where apparent labour productivity was less than 75 per cent of the national average. See also Estrada and López-Salido (2001) on this subject.

13. According to a study carried out by the Bank of Spain (Estrada *et al.*, 2001a) for the period 1996-99, nearly three-quarters of Spain's productivity gap with the EU is due to a different production structure, whereas only a quarter of the gap derives from a lower level of sectoral productivity.

14. The statistical changes that the INE introduced in the labour force survey with effect from the first quarter of 2002 brought some major revisions in labour market statistics. As reported in the previous survey (OECD, 2001a), there have been three main changes: *i*) a revision of the labour force to take account of new immigration figures; *ii*) an adjustment in the system of reweighting the sample deriving from the labour force survey; *iii*) the way unemployment is calculated has been made consistent with Eurostat norms. For 2001, these changes meant revising employment upwards by 8 per cent and the labour force by 5 per cent, while the unemployment rate was reduced by 2½ percentage points.

15. According to the INE, the total cost of work, per worker and per month, has risen by some 5 per cent on average across the whole economy since early 2001. This is 1½ percentage points higher than in the euro area for which Eurostat publishes homogeneous statistics. In the case of Spain, the sharp rise in labour costs partly reflects a bigger increase in non-wage costs than in the other countries, this being due to the raising of the ceiling on social security contributions for the least skilled workers, which is now the same for all job categories. The rise in non-wage costs may also be partly attributable to the suspension of the subsidies, which certain contracts had hitherto benefited from.

16. Rogers (2001), for example, draws attention to the existence of a price convergence phenomenon in the case of Spain. The empirical findings presented in this study depend, however, on the specification of the equation estimated.

17. As indicated in the previous OECD survey (OECD, 2001a), for example, it is hardly likely that price convergence between Spain and the euro area is the result of rising wages in the service sector due to a catch-up in industrial productivity, as described by Balassa and Samuelson. Productivity growth has been lower in Spain than in the

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other countries in recent years and the increase in prices of services is closely linked with the trend in profit margins in the latter sector.

18. According to the Spanish Government, the effect of the introduction of the euro on inflation in 2002 may have been slightly more marked than in the other countries because of the special difficulty of converting pesetas into euros.

19. Taylor rules typically relate short-term interest rates to the output gap and inflation developments relative to inflation target. Such rules, estimated on the basis of underlying inflation for the euro area but used for Spain, suggest that short-term interest rates should be around 5¼ per cent if a 1½ inflation target is assumed and around 4½ per cent in the case of a 2 per cent target. The three-month interest rate averaged only 2.85 per cent in early January 2003.

20. As in 2001, activity in 2002 turned out lower than forecast. The 2001 budget was initially based on an assumption of real GDP growth of 3.6 per cent, and of 2.9 per cent for the 2002 budget. In 2001, growth finally reached 2.7 per cent and was around 2 per cent in 2002.

21. In 2002, with inflation overshooting the initial target by nearly 2 percentage points, pensions expenditure again had to be increased by a total of EUR 2.1 billion, or 0.3 per cent of GDP, which is spread evenly over the 2002 and 2003 budgets.

22. Monitoring of the budget out-turn was also difficult in 2002 because of the ongoing process of decentralisation and the lack of information at the regional level. Discussions concerning the quarterly publication of government (including regional) accounts are under way with a view to improving the monitoring of the budget outturn. A new Eurostat regulation will effectively be in force as from 2003 whereby all levels of government will be obliged to report quarterly budget outturns. One of the obstacles to the implementation of this regulation is the difficulty of applying the ESA 95 methodology to lower levels of government.

23. The agreement in the public service also provides for a 3.2 per cent increase in 2004, of which 0.5 per cent represents a deferred wage increase in the form of a pension fund set up for civil servants.

24. Over the past 15 years, the number of civil servants has risen by 800 000, which is 3 per cent per year (Caixa, 2002). This sharp increase is attributable in particular to decentralisation and the development of the autonomous communities, where over 80 per cent of these jobs were created, whereas the number of central government public servants has declined. Despite having risen so rapidly, the proportion of government jobs, at 14.5 per cent of total employment, was not high in international comparison (OECD, 2001a). Recently, the restrictions on recruitment to the civil service have resulted in an appreciable increase in temporary jobs, especially at the regional and local level. The share of temporary employment in the public sector, for example, rose from 16.6 to 22 per cent between 1997 and 2002, whereas it fell from 38.8 to 34 per cent in the private sector.

25. Sometimes information on general government is presented with reference to the forecast end-2002 outturn, which is useful, but is usually made in comparison with the initial 2002 Budget, which is inadequate.

26. The projections are based on the following assumptions: oil prices are assumed to stay at USD 31 per barrel in the first half of 2003 before falling to USD 25 in the second half and remain there until the end of the projection period; exchange rates are fixed at their level of 7 February 2003 (1 EUR = 1.08 USD), implying an appreciation of the euro *vis-à-vis* the US dollar of 20 per cent since the first half of 2002.

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27. The basket also includes the tax on electricity and on car registration and devolved taxes (on wealth, inheritance and donations, gambling and stamp duty).
28. The previous income tax reform lowered the top rate from 56 to 48 per cent, and the minimum rate from 20 to 18 per cent.
29. Revenues from mutual funds are considered as capital gains by the 1999 law, not capital income, so that they are taxed at a flat 15 per cent rate (instead of the marginal rate), if generated in more than one year.
30. Tax exemption for rental income, though good for the development of the market of rental housing and labour mobility will also subsidise the numerous holiday rentals.
31. Those businesses whose activity falls outside the geographic application of the tax (like exporters and importers) will not be allowed to pay tax under this regime.
32. After the 1999 reform, the tax brackets were modified in 2000 but not in 2001 and 2002.
33. According to the 1998 law, local authorities can only resort to fees if a direct link can be established with the cost of the service to be financed. According to Martinez (2000), the municipalities' lack of information in this area partly explains why little use is made of this source of finance.
34. Spending efficiency will be enhanced by management improvements. A recent example is the extension to the whole of Spain of a control system developed in Andalusia for prescriptions of anti-inflammatory drugs. More generally, the indicators could gauge the budgetary management of the regions and the central government. The publication of such information in other federal countries, coupled with the existence of local referenda, represents a powerful incentive to take account of the aspirations of the population and keep tighter control over public finance (OECD, 2002a).
35. Hospital reform has also been slowed by the completion of the decentralisation of health spending to the 10 communities that did not as yet have that responsibility in 2002. In late 2000, the government had suspended the reform process, pending the transfers of responsibility.
36. A 1998 Court of Auditors study on hospital activity showed that some hospitals were paying up to twice as much as others for the same drug. According to the body that managed non-decentralised health services up until 2002 (INSALUD), the rules governing tenders for procurement contracts relating to advertising and competition, as laid down by law, were not applied in half of these contracts.
37. Despite the budgetary restrictions, the number of doctors has increased since the mid-1990s, suggesting wage restraint.
38. According to official estimates, these measures had the effect of slowing the growth of public spending on prescription drugs by 7 percentage points between 1999 and 2002. However, over 40 per cent of this reduction was due to the narrowing of margins on drugs, the effect of which is temporary. Consumption of generic products rose to 3.4 per cent of pharmacy spending in value terms and was put at 6.7 per cent in real terms in May 2002.
39. In 2003, minimum pensions will be raised by 5½ per cent, *i.e.* 1½ per cent more than the inflation-linked adjustment, whilst survivors' pensions will be increased by some 8 per cent. The budgetary cost of these measures amounts to EUR 335 million.
40. The bulk of the 28-page report on Spain's national strategy by the Ministry for Social Affairs concerning the future of the pension system deals with the period 2000-15, whereas the pressures liable to emerge after that period are discussed in only three lines. An annex to the report contains an extrapolation of pension expenditure projections up

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to 2040, *i.e.* 10 years before the demographic pressures peak. Also, the projected 3¾ per cent of GDP increase in pension payments between 2020 and 2040 seems very optimistic. The assumptions underlying this projection are, first, a decrease in the number of pensions for the population aged 65 and over – despite the rise in female labour force participation and the possibilities of combining retirement and survivors' pensions without means-testing – and, second, a decrease in pensions relative to productivity (and real wages), contrary to the trend during recent years. The Stability Programme 2002-06 (Ministry of Economy, 2002) includes also unofficial projections similar to those of the Ministry of Social Affairs until 2040. These projections are extended until 2050 and project an increase of public pension spending by 4½ per cent of GDP between 2020 and 2050.

41. The INE will publish at the end of 2003 new demographic projections based on the results of the 2001 census.

42. The uncertainties, which chiefly concern economic and demographic developments, are both positive and negative, and their implications are frequently subjected to sensitivity analyses (OECD, 2001a).

43. Since 1980 pensions have been growing on average by 1½ percentage points more than wages. Were this trend to continue, there would also be problems of income distribution, especially since, according to the authorities, the elderly are less subject to poverty than labour force participants and the unemployed in particular (Ministry of Labour and Social Affairs, 2002).

44. It would seem reasonable for persons who will be retiring at age 65 in 2015-20 to know at least by age 55, how their pensions will be calculated. This means that the reform has to be adopted between 2005 and 2010, *i.e.* during the next parliamentary term.

45. The INE will present at the end of 2003 revised national accounts, which will take into account, among other things, the new census for 2001.

46. The Bank of Spain periodically publishes several indicators of convergence with the European Union. For those on the stock of public and private capital, see Bank of Spain (2002b).

47. For an exhaustive account of such externalities, see OECD (1999) that follows Calmfors (1993).

48. However, the empirical evidence in favour of this hypothesis is not conclusive (OECD, 1997a).

49. For a background for these attempts, see Bentolila and Jimeno (2002).

50. Wage increases have been more moderate at the enterprise level than at the sectoral (provincial or national) level, although this is partly due to the fact that they are usually signed in larger firms, where wage levels are higher to start with, and workers receive higher non-wage benefits (Fina *et al.*, 2001).

51. Out of 18 OECD countries, it is longer only in the Netherlands, Denmark, France and Norway.

52. Countries like the United States and the United Kingdom do not have unemployment assistance after unemployment benefits have expired, while Australia and New Zealand do not have a UI system.

53. Out of 800 000 temporary workers in agriculture, 600 000 live in Andalusia and Extremadura.

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54. Abuses of the system were reported in the past. In particular, some enterprises issued false certificates of days worked, while some municipalities did the same in public works. This type of abuse seems to have almost disappeared in recent years.
55. The May 2002 decree suppressed the PER completely for new entrants, *i.e.*, for those who have not received the subsidy during the last three years. The new, temporary subsidy for new entrants in the PER system will be limited to those who have been resident in the area for at least ten years, which will exclude most immigrant workers.
56. Individual dismissals may be "justified" by the law depending on different criteria, including economic reasons. In practice, most dismissals are declared "unjustified" by courts, carrying higher severance payments.
57. Those unemployed workers who take up co-operative work and handicapped workers who set up as self employed were already able to capitalise their benefits as initial investment and will now be able to use part of their benefits to pay social security contributions.
58. Such a measure should probably be accompanied by a simplification of the different types of temporary contracts. There are too many, which makes contract management difficult.
59. Quality of university education is very difficult to pin down and measure, though the literature on this subject points to several weaknesses. The Bricall *et al.* (2000) report, written by a group of experts prior to the most recent reform, points out that the structure of university institutions is somewhat old fashioned and teaching is focused to knowledge transmission rather than learning processes. Mora *et al.* (2000) stress that high unemployment rates among the young with university degrees reflects, among other factors, the lack of preparation of Spanish graduates to changing professions and skilled requirements, as the education system is too focused on professional education. Mora and García (1999) show that the cost per student of university education is relatively low because the number of services offered other than lectures is also limited.
60. 100 per cent of children aged 5 go to school, 98 per cent at age 4, and 75 per cent at age 3. Only France, Italy and Belgium have higher rates.
61. Enrolment rates are 1.3 per cent for under one-year-olds, 5.7 per cent for those aged 1 and 13.7 per cent for those aged 2.
62. In particular, schooling at this stage will be considered "assistential" as well as educational, implying that not all pedagogic staff will be required to hold university degrees.
63. Most private schools are financed by the State through special agreements (*conciertos*) covering current expenses. Investment in private education is paid by the school, which can receive extra finance from students through fees. Private schools that receive public funds are obliged to follow official guidelines in their curricula and the same admission criteria apply as in public schools.
64. That is an another flexible way to obtain the compulsory secondary education certificate and provides a link with the other post-compulsory formal education paths.
65. The new national tests (*Prueba General de Bachillerato*) will substitute a previous exam that was required only for those who wanted to access university.
66. See OECD (2002c).
67. The FORCEM has been renamed after the entry of the government in its management, as the *Tripartite Foundation for Training for Employment*.

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68. The number of demands for training covered 4.7 million workers in 2000; about 80 per cent were approved and implemented.

69. Good indicators of university performance are not easy to find. One possible good indicator is a low drop-out rate (although, in this case, provisions would have to be made so as to ensure that universities do no minimise drop-out rates by accepting only the best students).

70. Spending on grants is 11 per cent of total public expenditure in tertiary education (against 12 per cent in the OECD), covering mostly fees, and in some cases some help for books and transport. The fact that grants do not cover the actual cost of living is also a factor behind the low mobility of students.

71. The OECD index of tax advantages for R&D expenditures reveals that, for each unit spent by large firms on R&D, one third is recovered in lower tax payments. This is the highest ratio out of 24 OECD countries reviewed, and almost double that of the second country ranked, Canada.

72. Annual employment growth for scientists and engineers was 8 per cent between 1995 and 1999, much higher than average employment growth and one of the highest in the OECD.

73. The rise in the birth rate between 1960 and 1975 increased households' residential investment owing to the higher rate of acquisition of housing by those aged 22 to 32 years (Eguía and Echevarría, 2001), *i.e.* at the time of getting married (Namkee, 2001).

74. House purchases by non-residents have increased sharply since 1993. The pressure exerted by foreign demand in tourist areas is also causing property price disparities to widen across the country. Up to 1991, tax incentives also encouraged Spanish residents to buy second homes.

75. Land prices have more or less doubled in Madrid since 1997 (Bank of Spain, 2002a).

76. These restrictions on the rental housing market were removed by the 1985 reform.

77. Leases dating from before 1985 could be extended indefinitely. Rents relating to contracts signed before 1964 were frozen, whereas those signed between 1964 and 1985 could be linked to inflation.

78. A study of the Economic and Social Council (CES, 2002) indicates that the proportion of youngsters between 20 and 34 that have left their parents' home has decreased from 44 to 33 per cent between 1977 and 2001. This phenomenon, which reflects longer time spent at studying and problems of unemployment and unstable employment for the young, seems to be more intense than in other countries, and has probably contributed to reducing fertility rates.

79. The object of this assistance is to facilitate the purchase of a home at reduced prices by means of loans at preferential rates and subsidies.

80. In Barcelona, for example, some 50 per cent of rental accommodation comes under rules imposing low rents and the renewal of contracts until the death of the tenant or the tenant's spouse (or until their children are 25 years old) (Trilla and Carreras, 2001).

81. According to Pareja and San Martín (2002), the rate of return on rental investment has been lower than that on public debt since 1984, although the gap has narrowed since 1998.

82. The competence of the municipalities and autonomous communities as regards land management was confirmed by the decision of the Constitutional Court to rescind the

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measures taken by central government in June 2000 with the objective of restricting the communities' discretionary powers as regards issuing building permits.

83. Since the Act of 1998, an urban plot of land, *i.e.* land complying with certain technical characteristics, has been considered *a priori* to be developable, unless a decision to the contrary is taken and is properly argued. Under the communities' town planning rules, however, other conditions have been imposed which *de facto* prevent owners from exercising their right to build until they have obtained formal permission from the municipal or regional authorities. The process for disposing of land for construction are long (three years on average, sometimes much more), complex and uncertain.

84. Income from principal homes occupied by their owners (imputed rents) ceased to be taxed in 1999.

85. Comunidad Valenciana, Andalusia and the Basque Country.

86. In practice, most regional governments stick to the national minimum of 13 Sundays. The autonomous community of Madrid allows the largest number of Sunday openings, namely 21.

87. The association of hypermarkets claims that about 50 new shopping centres are on the waiting list for opening permits.

88. Telefónica's share of access lines was 95 per cent at the end of 2001, 82 per cent in the national long distance market (share of minutes) and 83 per cent in international traffic minutes.

89. The approved price cut was 2 per cent, as it was foreseen in the original price cap system, although the connection fee has been taken out of the price cap system and has been increased by 8 per cent, ending the tariff re-balancing process. Simultaneously, the regulator adopted some other measures to foster competition, like a new single invoicing mechanism for pre-assigned customers, and reinforced rules for public procurement of telecommunication services.

90. Telefónica lowered its investment in cable networks in 2000 when it focused its broadband strategy towards ADSL technologies. The main company in the cable business is Auna, which also owns Retevisión (the second fixed telephony operator) and Amena (the third mobile operator).

91. The government has pressed operators to accelerate its development, and could withdraw the deposits paid by the operators when the licences were given, since investments have been delayed. The government has recently liberated cable and radio-telephony operators from their investment agreements, but not UMTS operators.

92. Modernisation measures will only apply to older vessels (5 or more years old) and only for improving security, not efficiency of catches; construction will be linked to somewhat stricter decommissioning rules for old vessels; and exports to third countries in joint venture agreements will only be possible with those countries that have signed a fishing agreements with the European Union.

93. For instance, the legislation has allowed the recent merger of the clearing and settlement systems of all secondary markets, which has taken place as a result of this initiative. Although a *de facto* monopoly system, the law leaves the door open for other competing systems.

94. New regulations on the take over bid (OPA) regime in stock markets are also being prepared, following some episodes of sales of controlling stakes which were just at the limit that obliges the launch of an OPA, although they will depend on the corresponding European directive on the issue.

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95. Venture capital investment is equivalent to 0.25 per cent in Italy, 0.38 per cent of GDP in France, 0.48 per cent in the Netherlands and 0.85 per cent of GDP in the United Kingdom.

96. To protect the initial stability of stock values, stock market regulations oblige the main owner of a firm to keep a participation for some time when it is publicly offered. Up to now, this resulted in the loss of any tax advantages for the venture capital entity, which prevented them from offering the participated firms in the stock markets. The new regime addresses this problem by extending tax advantages for two years after admission to trading on a regulated market.

97. In fact, saving banks have acquired very few and very small commercial banks. On the other hand saving banks reinforce competition in the market. The possibility of privatising saving banks is not on the politic agenda.

98. In a comparison by the Bank of Spain for the period June 2001 – June 2002, profits of Spanish banks fell by 10 per cent, whereas those of French banks fell by 20 per cent and those of German and Italian banks by more than 35 per cent.

99. The two largest Spanish banks, BBVA and Santander, invested heavily in Latin America in past years. Consolidated foreign claims of Spanish institutions in the area (which exclude loans of affiliates of Spanish banks in domestic currencies, that are less prone to devaluation risks) amounted to USD 48 billion in December 2001. Of these, 32 per cent originated in Argentina, 30 per cent in Mexico, 10 per cent in Brazil and 9 per cent in Chile. Claims on Argentina fell strongly during the first half of 2002 due to devaluation and to the partial reduction of positions in the country by Spanish banks.

100. In part because the Bank of Spain imposes stricter solvency ratios for Spanish banks.

101. These two alternative scenarios assume the same exchange and nominal interest rates as the baseline scenario. Furthermore, public spending is maintained stable in real terms with respect to the reference scenario.

102. Potential growth in the euro area, which is estimated at about 2 per cent between 2005 and 2008, tends to decline slightly due to the negative effect of ageing.

103. Council Decision (2002/358/EC) of 25 April 2002 concerning the approval, on behalf of the European Community, of the Kyoto Protocol to the United Nations Framework Convention on Climate Change and the joint fulfilment of commitments thereunder.

104. Some co-generation facilities are ineligible for support as they fail to meet efficiency criteria.

105. In 2001, for example, the premium per kilowatt hour for renewable energy varied from EUR 0.03 for electricity generated from biomass to EUR 0.36 for small-scale solar energy electricity production (National Commission for Energy, 2002). Energy recovery from the incineration of waste also receives a feed-in premium. From 2007 this system will end support for larger co-generation facilities.

106. Support was granted for energy saving projects, the installation of co-generation capacity, and promoting the use of natural gas and renewable energy sources.

107. Some expansion of existing plants has been allowed.

108. Not only are emissions of carbon dioxide in this sector the highest per kilowatt hour of electricity generated, but the externalities due to other conventional pollutants are also substantially higher than for other major electricity generating sectors (CIEMAT, 1997). For example, the sulphur content of domestically produced coal is high (IEA, 2001).

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109. The domestic coal industry is regionally concentrated and so appropriate, temporary assistance might be required for these regions. However, concerns over the security of supply of coal are less important (IEA, 2001).

110. For example, Capros and Mantzos (2000) estimate that marginal abatement costs of a sectoral approach with identical reduction targets would rise to EUR 348 per tonne of carbon from EUR 103 in the case of an optimal allocation of abatement efforts in Spain alone.

111. Almost three-quarters of the financing is expected to come from the European Community.

112. The expansion of gas powered electricity generation will also allow a further downsizing of support to the coal industry.

113. For emitters not participating in the EU trading scheme, carbon taxation could be usefully applied, if the implicit tax on carbon embedded in fuel taxation does not exceed the price of allowances. In particular, carbon taxation on the fuels used by small-scale combustion plants that will fall outside the EC emission trading scheme could help reduce the overall cost to the economy of achieving greenhouse gas targets.

114. Waste statistics in Spain have recently been reviewed. Reporting methodologies across autonomous communities have differed substantially over time making the determination of trends extremely difficult.

115. In particular, Directives 75/442/EEC on waste and 99/31/EC on the landfilling of waste. For this reason the European Commission has opened infringement proceedings against Spain on the grounds of several actions of illegal dumping with the view of bringing the Kingdom of Spain to the European Court of Justice.

116. N*ormativa e instrumentos de gestion en materia de residuos.*

117. In reducing the amount of waste generated the government has required larger firms to prepare waste reduction strategies. In the case of packaging waste, the reduction is expressed in terms of packaging weight relative to the weight of the product.

118. These include Ecoembes (for general packaging waste), Ecovidrio (glass), and Sigre (pharmaceutical packaging). If companies do not join such a system, they are required to implement a deposit-refund scheme. Industrial packaging must be sent to an authorised operator for disposal.

119. The packaging law (following the EC directive) required that between 25 and 45 per cent of all packaging must be recycled, but that recycling of an individual packaging material need only reach 15 per cent.

120. Annual outlays have been above EUR 10 million in recent years, with remedial action taking place at 275 contaminated sites.

121. The EC Water Framework Directive will require full-cost recovery by 2011.

122. These prices refer to those charged in the Jucar and Almeria counties (Albiac-Murillo *et al.*, 2002).

123. Agricultural support policies may also be indirectly contributing to demand for irrigation water. See also Chapter IV.

124. These changes were introduced in the 1999 amendments to the Water Act. New irrigators would pay half the capital costs for new infrastructure and be required to cover operation and maintenance costs.

125. Such a policy would be equivalent to grand-fathering existing rights and allowing trading.

126. The national analysis uses a 50 year timeframe for the cost analysis with a real discount rate of 4 per cent.

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127. Desalination costs vary considerably according to the size of plant and the amount of electricity that can be sold. A range of cost of between EUR 0.3 and 0.4 per m^3 is suggested by Uche *et al.* (2001).

128. Almost a third of the land under irrigation is based on systems over a century old (Ministry of Agriculture, Fishery and Food, 2002).

129. Public investment is limited to existing and uncompleted irrigation systems, though new irrigation schemes of a "social nature" may also receive funding.

130. Foreigners referred to in this chapter are foreign citizens, as Spain bases its statistics on nationality rather than place of birth.

131. For an analysis of ageing in Spain see OECD (2001a).

132. Up to 2003 EU citizens were obliged in principle to obtain a residence card. As from March 2003 this obligation no longer applies to those working or studying in Spain, following the implementation of the 28 July 2000 Marseille Declaration, although EU citizens must apply for a number of identification of foreigner (N.I.E), which is necessary for bank operation, etc.

133. The total number of undocumented foreigners is uncertain. In June 2001, Enrique Fernández-Miranda, the Delegate of the Government for Immigration, declared that Spain had about 1.2 million immigrants on its territory (*El Mundo*, 10 June 2001); the number of foreign residents officially-registered by the Ministry of the Interior on 31 December 2000 was around 896 000 and 1.1 million at end 2001.

134. There are no statistics on the return of foreigners to their country of origin or to any other country.

135. Besides, income distribution is generally wider in these countries, and has often worsened since the 1980s, which makes the incentives to migrate all the greater for low-income people.

136. This figure is in addition probably underestimated, since, as in most developing countries, informal activity is high and under-employment too.

137. See OECD (2002d).

138. The countries covered in this survey are the 22 members of the League of Arab States, which includes North African countries and Arab Middle-Eastern countries, with a population of 280 million in 2000.

139. Up to 1985, Spanish legislation characterised immigration-related issues as matters of "public order" (and therefore the exclusive responsibility of the Ministry of Interior) and issues related to settlement of foreign nationals in Spain were very weakly regulated. For a review of Spanish migration policy until 1999, see Moreno Fuentes (2000).

140. Rights to meeting and association were limited, although these aspects were softened on a number of occasions by the courts; family reunification rights were not recognised.

141. The Immigration Act is the result of two laws. The "L*ey de extranjeria* 2000/4" was adopted in December 1999, after two years of negotiation between all political parties in a parliamentary commission, and put a strong focus on integration issues. It was reformed a few months later by the "L*ey de extranjeria* 2000/8", that put more emphasis on the control of immigration flows. One of the main points of disagreement was the introduction of a permanent possibility for undocumented foreigners to obtain a temporary residence permit if they could prove that they had stayed two years in Spain and had sufficient means of subsistence, which was denounced as providing incentives for immigrants to come illegally. The new Act extends the period of stay to five years.

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142. Undocumented foreigners do not have the rights of meeting, association, unionisation and strike. Being in Spain without authorisation is a serious offence that can be punished by expulsion instead of a fine. Documented foreigners committing offences that can be punished by one to six years of jail can also be expelled.

143. The decision was published in early January, at the same time as the quota was announced, to restrict individual applications to specific categories of people, such as those for which looking at the national employment situation is not required (*e.g.* family members of foreign residents, students, Chilean and Peruvian nationals due to preferential international agreements).

144. The publication of the number of work permits granted has been recently interrupted, and figures are not available for the whole year of 2001.

145. The persons whose application was rejected (whose number is probably inferior to 210 000 because some may have applied in the two subsequent processes) were not expelled and are likely to remain on the territory.

146. This is under the assumption that all the persons whose application was rejected in the 2000 process did apply in the 2001 process.

147. See *El País*, 14 October 2001.

148. The government has proposed a 2003 budget for expulsions of EUR 8 million, almost three times higher than in 2002.

149. France, Italy, Portugal, Bulgaria, Poland, Slovakia, Estonia, Letonia, Lithuania and Rumania.

150. 90 000 is a minimum as it is the number of persons whose application was rejected in the 2001 process. 70 000 applications were also rejected in the 2000 process, but part of those rejected in the first process have certainly participated in the second. Those who did not apply, those who entered after the processes, and those who fell into illegality because their permit expired and could not be renewed since then due to lack of a work contract should also be added.

151. The government has also announced its intention to eliminate the permanent mechanism of regularisation "*por arraigo*" (*i.e.* for settlement reasons) included in the Immigration Act, which is allowing foreigners able to prove that they have been living in Spain for five years to obtain papers.

152. Besides, migration decisions of natives may also respond to labour immigration (Brücker *et al.*, 2001). For a critical analysis of this type of studies suggesting that immigration effects on the labour market may be underestimated see Borjas (2002).

153. Dolado *et al.* (1996) consider the number of work permits granted from 1990 to 1992. In 1991 and 1992 they are dominated by the regularisation process, so that it is not the effects of immigration on the labour market *per se* that are captured, but rather the effects of the legalisation of immigrants.

154. According to the labour force survey, non-OECD immigrants as a whole are better educated than Spanish nationals. Although the share of the second and third groups of workers referred to here cannot be exactly established (due to lack of data), the third group (*i.e.* low-skill jobs) is more important than the second (high-skilled jobs). To some extent, the fact that non-OECD workers are on average better educated than Spanish nationals may reflect under-utilisation of immigrants' human capital. However, this may also reflect the non-accuracy of the figures: as noted above, the labour force survey is not completely representative for foreigners, especially for non-OECD immi-

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grants, who are more likely to live in collective housing (not covered) and have temporary visas.

155. See Colectivo IOÉ (1998) for the construction sector, Colectivo IOÉ (1999) for hotels and restaurants, and Carrasco (1999). Mendoza (1997) provides information on African employment in agriculture and construction, and Martínez Veiga (1999) for employment in domestic services, agriculture and construction.

156. 24 per cent of the immigrants surveyed by Carrasco (1999) were without contracts; in the construction sector, 62 per cent of the Poles and 19 per cent of the Moroccans surveyed by Colectivo IOÉ (1998) had no contracts, against 1.5 per cent of the Spaniards; in restaurants, 42 per cent of the Peruvians, 23 per cent of the Dominicans and Chinese, 12 per cent of the Moroccans and 6 per cent of the Philipinos and the Spanish had no contract.

157. In the construction sector, 52 per cent of the Moroccans were working more than 40 hours a week, against 39 per cent of Spanish; overtime hours are more often paid at the normal rate for Moroccans than for Spanish and Polish workers (Colectivo IOÉ, 1998). Foreigners also tend to work longer hours in hotels and restaurants; more than two thirds of the Dominicans and Filipinos were not paid for extra-hours, against one third for Spanish and Moroccan workers (Colectivo IOÉ, 1999).

158. At end-1997, it was one of the lowest in the European Union in PPP terms (OECD, 1998).

159. For example, domestic employees can be fired without notice and do not have access to unemployment benefits.

160. Comparing the situation of young natives and young non-EU immigrants both in Belgium and Spain at labour market entry based on the EU Labour Force Survey, Kalter and Kogan (2002) find that while differences in education levels explain the disadvantage that non-EU youth face compared with natives, it is not the case in Spain.

161. It is also likely that there is more variability in their unemployment rate than for natives, *i.e.* that they play the role of cushioning the variations in activity. It is however not possible to verify this based on the labour force survey since data for foreigners are published only annually.

162. EU foreigners, on the other hand, include a large number of pensioners.

163. Employment data used here is the affiliation with the social security system, available for foreigners starting in December 1999.

164. About 55 000 foreigners were registered under the domestic employee social security regime in June 2002. It is a sector where undeclared employment is common, so that overall domestic employment is higher.

165. See OECD (2001f).

166. See Álvarez-Llorente (2002) and Muñoz (2002b).

167. The important increase in temporary jobs for natives may also have played a role. Unemployment dropped from about 18 per cent in 1995 to about 11 per cent in 2002, with new employment largely situated at the low-skilled, low-productivity end.

168. Productivity here is measured as GDP divided by the number of workers affiliated with the social security system. The number of non-EU foreigners affiliated with the social security system is not available for 1995, so that the number of existing work permits was used instead, after checking that they were quite similar to the number of affiliations in the period where both are available.

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169. In Almeria, income per capita was equivalent to 82 per cent of the national average in 1999, against 73 per cent in 1991. In Huelva, the Rioja and Castellon, income per capita was equivalent to 73, 119 and 117 per cent of the national average, respectively, against 69, 110 and 108 in 1991.

170. According to the 2000 Census, one out of three EU residents were aged more than 55.

171. Revenues from the VAT on consumption are not included.

172. According to de Ayala Cañón *et al.* (2001), 70 000 households or 180 000 individuals (0.5 per cent of total population) were benefiting from a minimum income.

173. According to the OECD database on social transfers, in 1998, the benefits provided to compensate for low income represented 0.07 per cent of GDP, family benefits and services to families 0.4 per cent of GDP, and housing benefits 0.07 per cent of GDP (excluding income tax relief granted for purchasing a main residence).

174. Aparicio and Tornos (2000) estimated that they were inferior to 2.8 per cent of the expenditure they estimated at the central level in 1998. They may have somewhat increased recently.

175. See Collado *et al.* (2002).

176. As shown by Samuelson (1958), in the steady-state the system's internal rate of return must be less than or equal to the real rate of output growth in order to keep the pension regime in equilibrium. Given the effect of immigration and under the assumption of a stable annual growth of 1¾ per cent of productivity, real output could rise on average by 1¾ per cent a year over the 2000-50 period, a rate much below the estimated level of the internal rate of return of the pension system.

177. Moroccan and Peruvian immigrants surveyed in the Madrid area by Aparicio and Tornos (2000) were sending on average one to two months of their salaries in remittances to their origin country in 1999.

178. The fruit and vegetable sector, which employs many immigrants, is not benefiting from direct subsidies from the CAP, but benefit from trade protection; other sectors including the olive oil sector enjoy both trade protection and production subsidies. The regime of EU subsidies to olive oil was significantly modified in November 1998. The main modification entailed replacing the intervention price system by a private storage aid, abolishing aid to consumption and introducing an overall cash ceiling on the amount of production subsidies (these changes are not definitive; they have been extended up to 2004/05). This ceiling is distributed across countries according to the national guaranteed production level. In the event that national production exceeds the quota, the subsidy per tonne is reduced proportionately.

179. The demand from Latin American countries for reduced protection of EU markets, in particular for agricultural products, was one of the most contentious points in the last Ibero-American summit in November 2002. In March 2002, the Spanish Secretary of State for Foreign Affairs defended (during a meeting of entrepreneurs from Spain and Maghreb countries on investment opportunities in the Maghreb) the view that free trade between the EU and Maghreb, including in agricultural products, should be sought by 2010 so as to help the development of these countries and slow immigration flows. The president of Catalonia also supported this view (see *Noticias Infoagro*, 22/03/2002). For an analysis of the impact of liberalisation of Euro-Mediterranean trade in agricultural products, see Lorca Corrons (2002).

180. According to a report by the Defensor del Pueblo Andaluz (2001), in western Almeria, at least 65 per cent of the immigrants did not live in decent housing. More than 50 per

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cent had no access to drinkable water, and 25 per cent to electricity, and most had no sewage.

181. For example, in Madrid, rents can vary between EUR 150 and 240 per month for a bed in shared rooms. The immigrants are also often asked to provide a bank guarantee or are required to pay a number of monthly rents in advance, while the norm is one month (*El País*, 9 April 2002).

182. According to Trilla (2001), social housing accounts for 2 per cent of the housing stock, against 18 per cent on average in the EU.

183. See *El País*, 23 January 2003.

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Glossary of acronyms

ADSL	Asymmetric digital subscriber line
ALMP	Active labour market policy
BBVA	*Banco Bilbao Vizcaya Argentaria*
BIS	Bank for International Settlements
CAP	Common agricultural policy
CES	*Consejo Económico y Social* (Economic and social council)
CLH	Transportation monopoly
CMT	*Comisión del Mercado de las Telecomunicaciones* (telecommunications sector regulator)
CPI	Consumer price index
EC	European Commission
ECB	European Central Bank
EMU	European Monetary Union
EPA	Labour force survey
ESA	European System of Integrated Economic Accounts
EU	European Union
EUR	Euro
Eurostat	European Union Statistical Institute
FAO	Food and Agricultural Organisation
FORCEM	Foundation for the training of workers
GDP	Gross domestic product
GHG	Greenhouse gas
GRECO	*Programa Global de Regulación y Coordinación de la Extranjería y la Inmigración en Espãna* (Global Programme to Regulate and Co-ordinate Foreign Residents' Affairs and Immigration in Spain)
IAE	*Impuesto de actividades económicas* (tax on business activities)
IEA	International Energy Agency
IBI	*Impuesto sobre bienes inmuebles* (real estate tax)
ICAC	*Instituto de Contabilidad y Auditoría de Cuentas* (Institute for Accounting and Auditing)
ICES	International Council for the Exploration of the Sea
ICID	International Commission on Irrigation and Drainage
ICT	Information and Communication Technology
IMF	International Monetary Fund
IMSERSO	*Instituto de Inmigración y Servicios Sociales* (national agency for migration and social services)
INE	*Instituto Nacional de Estadística* (National Statistics Institute)
INEM	*Instituto Nacional de Empleo* (National Agency for Employment)

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INSALUD	*Instituto Nacional de la Salud* (National Health Institute)
NAIRU	Non-accelerating inflation rate of unemployment
NGO	Non-governmental organisation
OPA	*Oferta pública de adquisición* (take-over bid)
PER	*Plan de Empleo Rural* (special unemployment and investment regime for seasonal agricultural workers in Andalusia and Extremadura)
PES	Public employment service
PISA	Programme for International Student Assessment
PPP	Purchasing power parity
R&D	Research and Development
SCP	Stability and Convergence Programmes
SDC	*Servicio de Defensa de la Competencia* (Competition Service)
SME	Small and medium-sized enterprise
TDC	*Tribunal de Defensa de la Competencia* (Competition Tribunal)
UA	Unemployment assistance
UI	Unemployment insurance
UMTS	Universal Mobile Telephone Systems (third generation mobile telephone systems)
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
USD	United States dollar
VAT	Value added tax

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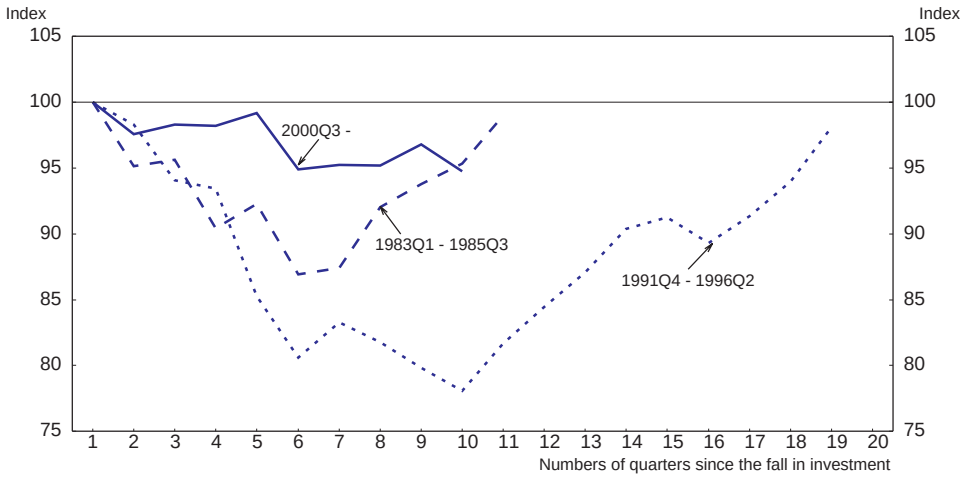
Annex I
Supporting material

Figure A1. **Investment cycles**[1]



1. Investment in machinery and equipment.
Source: OECD.

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Table A1. **Balance of payments**

Million euros

	1998	1999	2000	2001	2002 Q1	2002 Q2	2002 Q3
Trade balance	–18 480	–28 585	–37 778	–35 265	–25 136	–32 247	–36 650
(As a % of GDP)	–3.5	–5.1	–6.2	–5.4	–3.7	–4.7	–5.2
Non-factor services (excluding tourism)	–2 689	–3 726	–3 539	–2 833	–3 133	–1 494	–1 743
Tourism	22 315	25 250	27 782	29 971	20 377	29 124	37 506
Net investment income	–6 751	–8 904	–8 985	–10 655	–12 390	–10 387	–16 566
Net current transfers	3 006	2 853	1 528	1 836	8 937	–2 294	5 099
Current balance	–2 598	–13 112	–20 992	–16 947	–11 344	–17 299	–12 355
(As a % of GDP)	–0.5	–2.3	–3.4	–2.6	–1.7	–2.5	–1.8
Capital balance	5 680	6 552	5 181	5 556	6 967	8 959	7 652
Financial balance (net change)[1]	–105	15 800	21 300	18 827	22 644	14 185	15 797
Assets (net change)	67 162	84 367	139 732	66 740	61 296	120 635	52 390
Spanish investment abroad							
Direct	17 002	39 501	59 344	31 072	10 520	22 633	19 086
Portfolio and financial derivatives	41 521	43 816	63 025	49 185	46 339	55 105	14 913
Other investment[2] and reserve assets	8 639	1 051	17 363	–13 517	4 438	42 897	18 391
Liabilities (net change)	67 057	100 167	161 032	85 567	83 940	134 820	68 186
Foreign investment in Spain							
Direct[3]	10 592	14 791	40 728	24 340	27 776	15 826	11 829
Portfolio[4]	15 400	42 688	63 644	30 838	31 176	34 830	41 186
Other investment[2]	41 065	42 688	56 659	30 389	24 989	84 164	15 172
Errors and omissions (net)	–2 977	–4 682	–5 488	–7 436	–18 267	–5 845	–11 094
Memorandum items:							
Terms of trade, goods and services (% change)	0.0	–0.3	–2.2	2.1	–0.7	1.6	–0.2
As a % of GDP							
Foreign direct investment in Spain	2.0	2.6	6.7	3.7	4.1	2.3	1.7
Direct investment abroad	3.2	7.0	9.7	4.8	1.6	3.3	2.7
Net direct investment	1.2	4.4	3.1	1.0	–2.6	1.0	1.0

1. Changes in financial assets and liabilities are both net of repayments. Financial derivatives have been included in the change in financial assets although they are obtained as the balance of assets less liabilities.
2. Mainly loans, deposits and repo operations.
3. Does not include direct investment in listed shares but includes portfolio investment in non-listed shares.
4. Includes direct investment in listed shares but does not include portfolio investment in non-listed shares.

Source: Bank of Spain and OECD.

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Table A2. **Inflation decomposition: comparison between Spain and the euro area**

Year-on-year percentage change

	Average 1994-99	2000	2001	2002[1]	2002 S2
A. Spain					
GDP deflator[2]	3.3	3.5	4.2	4.4	4.7
Labour cost[3]	2.6	3.7	4.2	4.3	4.3
Wages	3.3	3.7	4.1	4.3	4.5
Productivity	1.1	0.7	0.3	0.6	0.5
Employee ratio[4]	0.4	0.7	0.4	0.5	0.4
Net indirect taxes	6.9	3.9	1.9	2.3	3.1
Profits	3.4	3.1	4.8	5.1	5.5
Real wages	–0.1	0.6	0.8	0.7	0.7
B. Euro area					
GDP deflator[2]	2.0	1.3	2.4	2.3	2.2
Labour cost[3]	1.1	1.5	2.8	2.5	2.2
Wages	2.2	2.4	2.6	2.8	3.1
Productivity	1.4	1.2	0.0	0.4	0.9
Employee ratio[4]	0.2	0.4	0.3	0.2	0.0
Net indirect taxes	4.6	0.3	0.4	2.8	3.1
Profits	2.6	1.4	2.4	1.8	2.0
Real wages	0.0	0.2	0.2	0.7	0.9
C. Difference					
GDP deflator[2]	1.3	2.2	1.8	2.1	2.4
Labour cost[3]	1.5	2.2	1.3	1.7	2.1
Wages	1.0	1.4	1.5	1.5	1.4
Productivity	–0.2	–0.5	0.3	0.2	–0.4
Employee ratio[4]	0.2	0.3	0.1	0.4	0.3
Net indirect taxes	2.2	3.6	1.5	–0.5	–0.0
Profits	0.8	1.7	2.4	3.3	3.5
Real wages	–0.1	0.3	0.6	0.0	–0.2

1. OECD estimates.
2. The breakdown of inflation (measured by the GDP deflator) is obtained using the following accounting identity:

$$\frac{\text{GDP}}{\text{GDP volume}} = \frac{\text{Total compensation of employees}}{\text{GDP volume}} + \frac{\text{Net indirect taxes}}{\text{GDP volume}} + \frac{\text{Profits}}{\text{GDP volume}}$$

The first term represents the total labour cost in the economy, the second term represents the costs linked to net indirect taxes and the final term (defined as a residual of the identity) represents the contributions of profits and other revenues.

3. Wages less productivity plus the ratio of employees/total employment.
4. Ratio of employees to total employment.

Source: OECD calculations based on national accounts data.

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Table A3. **Greenhouse gas emissions and sectoral indicators**

Annual average percentage change, 1990-99[1]

	Total level in 1999[2]	Total emissions	CO_2 emissions				Indicators per unit of GDP	
			Per kilowatt hour of electricity	Manufacturing, per unit of industrial production	Residential, per unit of private consumption	Road transport, per vehicle	Electricity use	Industrial output
Spain	**371**	**2.9**	**0.1**	**–1.5**	**1.9**	**–0.2**	**1.0**	**–0.4**
France	549	–0.1	–2.8	–1.5	–1.2	0.1	0.8	–0.2
Germany	994	–2.3	–2.2	–2.7	–5.0	–2.6	–1.7	–0.9
Greece	124	1.9	–1.7	–1.6	8.0	–1.1	1.8	–1.4
Italy	540	0.4	–1.6	–1.8	0.4	0.4	0.8	–0.4
Portugal	86	3.1	0.6	1.2	1.9	–0.4	2.0	–1.2
United Kingdom	647	–1.5	–4.7	–1.7	–0.7	–0.2	–0.7	–1.6
United States	6 830	1.2	0.2	–5.3	–2.3	1.1	–0.8	1.2
OECD[3]	14 257	0.5	–0.7	–3.1	–1.6	0.1	–0.3	0.1
OECD excluding USA[3]	7 428	0.0	–1.8	–2.0	–1.1	–0.6	0.0	–0.6
European Union	4 048	–0.4	–2.4	–2.2	–2.0	–0.6	–0.2	–0.5

1. 1990-98 for road transport CO_2 emissions. 1991-99 for German CO_2 emissions.
2. In million tonnes CO_2 equivalent.
3. The OECD total covers 27 countries.

Source: Greenhouse gas national submissions to the United Nations Framework Convention on Climate Change (UNFCCC), national sources and UNFCCC; carbon dioxide data from the International Energy Agency; industrial production and private consumption from OECD.

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Annex II

The decentralisation and financing reform of autonomous communities and local authorities

With the transfer of health service provision to several autonomous communities in 2002, the decentralisation of the biggest spending items was complete, making Spain one of the most decentralised OECD countries (Table 10). The autonomous communities, created in 1979, now manage one-third of consolidated public expenditure and employ more than twice as many civil servants as central government (excluding social security). Health spending apart, the regions' main responsibilities concern the social services (excluding contributory pensions), active labour market policy, education, culture and some public infrastructure. This further decentralisation was accompanied by a reform of the financing method which had been in force since 1997 in 15 of the 17 communities subject to the common scheme.[1] The new system is designed, like its predecessor, to increase the regions' self-sufficiency and fiscal responsibility. It does, however, make good the main shortcomings of the previous system and its validity is not limited to five years, in contrast with the system in force up until now. It thus broadens the field of regional tax receipts and makes them more stable. Previously they were excessively dependent on fluctuations in personal income tax. More importantly, and in conjunction with the increased fiscal responsibility of the regions, the reform extended the communities' tax powers and abolishes the State guarantee which ensured that their main tax resource increased by at least as much as GDP.

The regional financing relies now mainly of fiscal revenues complemented by unconditional transfers. This system also includes a set of conditional transfers reinforcing the solidarity between communities. In order to set up this new financing system, the authorities began by calculating the needs of the communities as a whole, making a distinction between three types of expenditure: general purposes (including education), health and social services. The breakdown between the regions was determined on the basis of their demographic and geographic diversity and differences in per capita income in the base year (1999). In the case of medical services, minimum spending standards were set on the basis of the 1999 level and the future growth of the State's tax revenue. To finance the spending, which is determined as explained above, the regions now have tax revenues based on a broad basket of taxes including 35 per cent of VAT, one-third of the personal income tax and 40 per cent of the excise duties on alcoholic beverages, tobacco and fuel oils collected within their territory. The basket also includes taxes on electricity, on certain forms of transport and "retroceded" taxes (on wealth, inheritance and donations, gambling and stamp duties).[2] Total tax receipts account for some 65 per cent of the regions' resources, as against 29 per cent under the previous system. The State guarantee, ensuring a minimum increase in these receipts equal to that of nominal GDP, is now only temporary (up to 2004) and partial (applying only to additional resources linked to health-related responsibilities). On top of these resources, which are not earmarked, are transfers which safeguard solidarity between the

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regions. These transfers are partly unconditional (F*ondo de* S*uficiencia*) and are designed to offset the discrepancy (positive or negative) between each Community's spending responsibilities and its fiscal revenues.[3] The regions also receive conditional transfers so that they can guarantee a minimum provision of health and education services throughout the territory (A*signaciones de nivelación*) and correct regional economic imbalances (F*ondo de compensación interterritorial*). The share of these conditional transfers has been reduced to less than 2 per cent of total receipts, compared to about 45 per cent under the previous system.

The reform also extends the communities' powers with regard to most of the taxes they collect. This is true in particular of personal income tax, with the regions now free to amend the scales (as long as they do not touch the number of brackets and the tax remains equally progressive) and adjust the scale of housing-related tax allowances. This is also the case of wealth and inheritance taxes, with the Communities having total control over tax rates and extensive powers with regard to tax allowances. On the other hand, VAT and excise duties on tobacco, mineral oils and alcohol totally escape the Communities' powers because of the norms set by the European Union in this area and the inherent difficulties that different VAT rates inside a country would create. Regions can, however, adjust fuel oil taxes through the regional part of the Tax on Retail Sales of Certain Mineral Oils to finance environmental and public health programmes, as the Community of Madrid recently decided. They are also more closely involved in tax management, collecting certain taxes directly (special taxes on fuel oils and transport) and taking part in the management of those they do not collect but which have an impact on their resources.

As mentioned in the main text, by the way it has been organised, decentralisation to the regions has the potential to improve the incentives for greater efficiency in public finance management, while maintaining sound fiscal control through the simultaneous application of the fiscal stability law. However, the efficiency gains would be strengthened by a widening of the use of benchmarking and a public management more oriented towards the outputs or outcomes and with greater co-ordination among regions. It is also important to recognise that, despite the planned equalisation mechanisms, it is difficult to predict the impact on trends in wealth differentials between regions with the introduction of extensive tax competition with a tax-sharing system. All in all, in view of the complexity of the new system and the uncertainty surrounding its actual use, a mechanism for assessing the reform ought to be programmed within a few years' time in order to correct any possible defects in its application.

Following decentralisation to the Communities, a reform of the local authority financing system was also launched. This is warranted for two main reasons: first, the need, in the wake of the adoption of the fiscal stability law, to increase the municipalities' fiscal responsibility, as was done in the case of the communities; second, the need to remedy certain shortcomings in the local taxation system. The three main local taxes, which affect real estate, business activity and motor vehicles, have rigid tax bases, which often depend on administrative decisions that make them costly to manage. Moreover, the tax on business activities (IAE) affects companies without taking account of their income or wealth, poses equity problems and affects incentives to recruit and invest.[4] To remedy these difficulties, a reform has been adopted which will be implemented progressively in 2003-04. It abolishes the IAE for about 95 per cent of taxpayers by exempting small firms and individual carrying out professional or economic activities (with total net turnover of less than EUR 1 million) and those recently set up. Local taxation, over which the municipalities will have more extensive powers, is also being modernised: the incentives that are unfavourable towards employment for firms still liable to the IAE are to be abolished; in fact, the reduction in the IAE on small firms is partly offset by an increase in this tax which is now levied on the incomes of large and medium-sized companies; the municipalities will have more leeway to differentiate between the property tax applying to firms and to households, reduce that imposed on large families and

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increase that applying to vacant housing; by applying tax exemptions, they will also be able to encourage the production and use of the least polluting forms of renewable energy. Lastly, local authorities' resources, which will be at least equal to those currently at their disposal, in the case of municipalities with fewer than 75 000 inhabitants, will be supplemented by the transfer of part of State taxes (income tax, VAT and excise duties) and additional transfers linked to the rise in central government tax revenues.[5]

Apart from reforming local authority financing, discussions have also started on reforming their responsibilities (*Pacto local*). The share of municipality spending in total public expenditure, which has remained stable at between 13 and 15 per cent for 20 years, is smaller than in other OECD countries, including federal countries, which has prompted calls for these authorities' powers to be stepped up (Table 10). It is important to realise, however, that there are advantages and drawbacks to extending their powers. While perhaps desirable, increasing local powers involves identifying those areas which might generate efficiency gains without impairing budgetary control. An international comparison shows that local authorities in Spain account for a share of education, health and welfare spending that is smaller than the OECD average (Table A4). However, a detailed assessment of both the benefits and also the costs involved in any transfer of responsibility is necessary because of the possible losses of scale economies or the existence of geographic externalities (spillover effects). Local authorities in Spain are numerous (8 000) and small in size (85 per cent of municipalities have fewer than 5 000 inhabitants). Without a high degree of inter-communal co-operation, many cannot take on the responsibility of providing services with high fixed costs or requiring specialised staff. Also, decentralising certain services such as health care and social services could cause problems relating to co-ordination with other policies such as hospital management or active labour market policy which will no doubt continue to be regionally administered for scale economies reasons. Decentralising basic education provision to the local level would no doubt cause fewer difficulties, and it would enable municipalities to better take into account the specific needs and preferences of the population and would encourage innovation in educational methods. In view of the centralised management

Table A4. **Local governments' expenditure by function**

In per cent of total local governments' expenditure

	Education	Health	Social security and welfare	Housing and community amenities	Total
Spain (1997)	**4.2**	**2.6**	**4.9**	**22.7**	**34.5**
Canada (2000)	41.6	1.2	7.3	5.5	55.6
Denmark (2000)	12.3	16.1	57.5	0.9	86.8
France (1993)	19.6	2.3	17.7	24.1	63.7
Germany (1996)	13.0	14.5	24.6	15.3	67.4
Ireland (1997)	11.3	45.5	5.2	14.9	76.8
Netherlands (1997)[1]	17.9	2.6	22.6	20.0	63.1
Switzerland (1999)	22.8	18.3	14.7	8.4	64.2
United Kingdom (1998)	28.7	0.0	32.5	5.4	66.7
United States (1999)	43.9	8.7	7.6	2.3	62.5
Non-weighted average	23.5	12.1	21.1	10.7	67.4

1. Provincial and local governments.

Source: IMF, *Government Finance Statistics Yearbook*, 2001.

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of curricula in Spain, decentralising basic education would not discourage labour mobility.[6] It could, however, give rise to equity problems between wealthy and less wealthy authorities, which would need to be resolved. Decentralising this competence, which is currently covered by the Communities, would also necessitate an equivalent transfer of resources in order to guarantee the local authorities' financial self-sufficiency.

Notes

1. The last two regions, the Basque Country and Navarra, are covered by a specific financing system (OECD, 2000) which the reform has barely altered. The system was extended indefinitely.
2. The communities may also raise taxes and user charges on regional utilities.
3. The amount of transfers received or disbursed by the communities was set for the base year 1999. The increase in these transfers in future years will be linked to the total tax receipts of central government (the case of transfers received by one region) or regional government (if the transfers are paid by the region).
4. The IAE is the second most important municipality tax, representing 20 per cent of local tax receipts (0.3 per cent of GDP) and affecting all companies, whether or not they are making profits. It is based on fixed elements such as the number of employees, the size of the municipality and the surface area of the premises. Its parameters vary according to the type of activity and their location (including in the same town). It depends little on economic activity (Poveda and Sánchez, 2002).
5. Small municipalities will be financed by their own tax receipts and by State transfers allocated on the basis of their population size, tax effort and taxation possibilities.
6. There will be *a priori* no obstacle to the geographic mobility of people because their children can continue the same curriculum in another region.

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Annex III

Main measures of the personal income tax reform

- The number of tax brackets is reduced from 6 to 5. The top marginal rate is set now at 45 per cent (previously 48 per cent), and the lowest rate at 15 per cent (previously 18 per cent).
- The general tax allowance for net labour income, which is based on a non-linear formula, is raised by 16.67 per cent for those who earn less than EUR 8 200 per year, and by 6.6 per cent for those who earn more than EUR 13 000.
- To encourage participation of older workers, the general income allowance will be doubled for those older than 65 who continue working.
- In order to foster mobility of unemployed workers, the general net labour income allowance will be doubled during two years for those unemployed who accept a job in a different municipality.
- Mothers who work will receive a tax rebate of EUR 100 per month during three years from the moment a new child is born or adopted. This rebate can also be received in advance in monthly payments.
- The flat rate for capital gains is reduced from 18 to 15 per cent, in line with the reduction of the lowest marginal rate.
- The deduction for labour income, income from economic activities and income on capital held for more than two years is raised from 30 to 40 per cent.
- To foster competition among mutual funds, capital gains from such investments will not be taxed when changing funds.
- The ceiling of tax-exempted contributions to private pension funds is raised by 11 per cent (to EUR 8 000 per year).
- Contributions to a new type of insurance contract (*Planes de previsión asegurados*) that provide pension income at a guaranteed interest rate will also be tax-exempted up to EUR 8 000 per year.
- To favour the supply of rental housing, a new tax allowance of 50 per cent of net rental income for owners who offer residential dwellings for rent is introduced. There will be deductions for all types of expenses (including 3 per cent for depreciation).
- The special regime for small self-employed and unincorporated businesses (*sistema de módulos*), which is based on special parameters rather than on profits, is subject to the respect of two new conditions. Businesses whose purchases of intermediate goods are above the threshold of 300 000 euros per year, or those whose activity falls outside the geographic application of the tax (exporters and importers), will not benefit from this special regime.

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- Family allowances are raised, and extended to new family situations. Deductions for children range from EUR 1 400 per year for the first child to EUR 2 300 for the fourth, while the additional reduction for children younger than 3 is four folded (to EUR 1 200). The deduction for old-age dependants is raised by 33 per cent if they are older than 65, and a new reduction of EUR 1 000 is introduced for social care, without the need of justifying the expenses. Tax allowances for the disabled are also raised.
- Several measures are introduced to facilitate tax filing and the calculation of withholding tax.

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Annex IV

Rental supply and demand incentives

The existence of an adequate rental market depends on the incentives affecting household demand, *i.e.* relative rental costs compared with the costs of buying a home, and also on the supply of these housing services, *i.e.* the return on rental investment as compared with alternatives, *e.g.* financial investment. To show what role these incentives play in the case of Spain, a small model has been built to calculate the effect of a decision to rent rather than buy one's home on household wealth. The model compares the amount of assets accumulated by a household depending on the housing solution it opts for.

The model developed for this analysis relies on the following assumptions:*

- *Characteristics of the house purchased*: the price of the house considered amounts to EUR 96 162, corresponding to a medium-category house. In addition, real property prices are assumed to rise by 2 per cent per year, which is close to its trend since 1990.
- *House purchase conditions*: it is assumed that the buyer borrows the total amount of capital required to acquire the house, including the costs and taxes associated with the transaction, which are estimated at 9 per cent of the house price. To simplify the analysis, the buyer is supposed to take out a fixed-interest rate loan, with a nominal rate of 5½ per cent or 3½ per cent in real terms.
- *Rental conditions*: initially, the level of the rent is set such that the rate of return on rental investment is the same as for alternative forms of investment, *e.g.* financial investment, with a return of 5 per cent (3 per cent in real terms). The annual rent level is hence set at EUR 4 800 the first year. It is indexed on inflation, which is supposed to rise by an annual rate of 2 per cent during the following years.

In the case of a household buying a home, the value of its wealth in a given year is calculated as the difference between the sales price of the property and the capital still to be repaid. This estimated wealth takes into account the cost related to anticipated repayment of the capital borrowed (estimated at 2 per cent of its value). It also includes the cumulative amount of tax relief attached to the purchase. These aids, which are granted without any means test, with a ceiling of EUR 1 803, decrease after two years, falling to EUR 1 578 from the third year onwards. In the case considered, the tax relief attached to house purchased represents 97 per cent of the ceilings.

In the case of a household renting a home, its wealth is calculated as its stock of saving, assuming that the household systematically invests on the financial market, the surplus being equal to the difference between the rent paid and the costs if it had bought. These costs include loan repayments but also the property tax (IBI, based on a cadastral value representing half of the actual value of the house and an average tax rate of 1 per cent). The nominal rate of return

* A comparable analysis to the one carried out here is presented in San Martín (2000), with similar conclusions.

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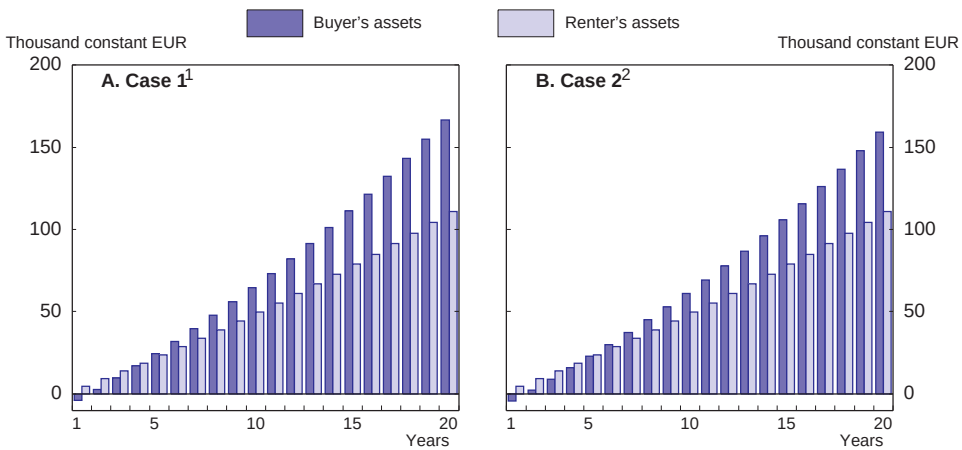
of this saving is assumed to be 5 per cent, while account is taken of the tax on the lessee's income from saving (15 per cent) and the index-linking of rents.

As indicated in the main text, buying one's home is a much better option than renting if the average rental price provides a return comparable to that on financial investments. The assets built up by the buyer exceed those of the lessee by the third year and are on average 50 per cent greater on a discounted basis (with an actualisation rate of 3 per cent, Figure 21). The higher the rate of inflation, the greater the incentive to buy because of the decline in the real cost of repaying the loan, whereas rents are index-linked. With a 3 per cent inflation rate, the average difference between the buyer's assets and the lessee's reaches 70 per cent on a discounted basis. Moreover, if the real housing prices increase by 1 percentage point to 3 per cent a year, the wealth gap between the buyer and the lessee rise by 10 percentage points to 60 per cent. On the other hand, if the cost of the property increases, incentives to buy are smaller because tax relief is capped.

The lack of incentives for rental demand is likely to induce a reduction in rental prices. With a trend rise in property prices and tax allowances in favour of owners, a 40 per cent fall in rents is needed to achieve equilibrium between the incentives to rent and to buy one's home implying a stronger rental demand. However, that sort of fall in rents would imply a considerable decline in the return on rental investment compared with alternative forms of investment, and this would prevent the expansion of rental supply. Recent tax incentives granted to lessees included in the 2003 personal income tax reform aims at strengthening the profitability of rental house investments, which is limited by relatively low rents.

If expectations as to real property prices could be stabilised, the advantage of buying rather than renting accommodation would decline because of the narrowing of the difference between the value of the buyer's cumulative assets and those of the lessee. In this case, the buyer's assets exceed the lessee's as of the sixth year and the average differential in favour of the buyer is close to 20 per cent (Figure A2). Eliminating the tax allowances encouraging

Figure A2. **Wealth developments linked to housing purchase/rental**



1. No tax subsidies for the buyer and an annual appreciation of the property of 2% in real terms.
2. Tax subsidies for the buyer and no appreciation of the property in real terms.

Source: OECD.

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the purchase of property, would also reduce the incentives in favour of home purchasing. The buyer's assets exceed the lessee's as of the fourth year and the average differential in favour of the buyer is close to 30 per cent (Figure A2).

Abolishing tax allowances encouraging the purchase of property, coupled with the stabilisation of expectations as regards real property prices, would balance the incentives to buy and to rent accommodation, while at the same time maintaining attractive returns on rental investment. In this case, the buyer's cumulative assets would not exceed those of the lessee until the fourteenth year and the differential in favour of buying would on average be eliminated over the period as a whole (Figure 21).

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Annex V

The effect of employment shifts on labour productivity growth and the contribution of immigration of non-EU foreigners

Methodology

The growth accounting exercise is based on the following equation, where Y is GDP, L total employment, and L_i employment in sector i:

$$(Y/L)^t = \sum_i (Y_i / L_i)^t * (L_i / L)^t$$

or $P^t = \sum_i P_i^t * S_i^t$,

where P is productivity and S_i the share of sector i in total employment.

The change in productivity over n periods can be decomposed as:

$$\begin{aligned} P^t - P^{t-n} &= \sum_i P_i^t * S_i^t - \sum_i P_i^{t-n} * S_i^{t-n} \\ &= \sum_i \left(P_i^t - P_i^{t-n}\right) * S_i^{t-n} + \sum_i P_i^t * S_i^t - \sum_i P_i^t * S_i^{t-n} \\ &= \sum_i \left(P_i^t - P_i^{t-n}\right) * S_i^{t-n} + \sum_i P_i^t * \left(S_i^t - S_i^{t-n}\right) \end{aligned}$$

The first term of the last line represents the contribution of individual sectors' labour productivity growth to total labour productivity (the "within" effect), and the second term measures the contribution of shifts in employment shares among sectors with differing productivity levels. The effect of immigration can then be isolated by calculating what would have been the changes in employment shares with no immigration.

Data and assumptions

The period considered runs from the first quarter of 1995 to the second quarter of 2002. GDP at constant 1995 prices and employment from the national accounts were used to calculate the level of productivity per full-time equivalent job.

The sectors considered are the following: agriculture, energy, industry, construction, low productivity services where immigrants often work (trade and repair, hotels and restaurants, domestic services) and other services.

Non-EU foreign employment is taken from the work permit statistics for 1995 and from the social security registers for 2002 (checking that the two sources did not significantly differ

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in periods where both are available, *i.e.* 1999 and 2000). From these data, the change in sectoral shares attributable to immigration was calculated, and applied to the national account employment figures. Table A5 displays some of the data and the results.

Shifts in sectoral employment are estimated to have slowed down labour productivity growth by about 0.23 percentage points per year. If there had been no immigration between 1995 and 2002, the shift effect would have been negligible, *i.e.* annual labour productivity growth would have declined by only 0.02 percentage point.

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Table A5. **The effect of employment shifts on labour productivity growth and the contribution of immigration**

1995 – Q2 2002

		Total	Agriculture	Energy	Industry	Construction	Services with immigration	Other services
Productivity level in 1995	(1)	8.4	8.4	32.3	8.6	7.0	6.6	9.8
Change in productivity level 1995-2002	(2)	0.4	0.9	16.6	0.3	–0.3	0.2	0.4
Employment share in 1995	(3)	100	8.0	1.1	18.5	9.4	24.6	38.4
Change in employment share 1995-2002	(4)	0	–1.9	–0.3	–0.3	1.8	0.9	–0.3
Change in employment share without immigration	(5)	0	–2.1	–0.1	–0.1	1.8	0.6	–0.2
Within effect	= (3) * (2)	0.49	0.07	0.18	0.05	–0.03	0.05	0.17
Shift effect	= (1 + 2) * (4)	–0.13	–0.11	–0.15	–0.02	0.12	0.06	–0.03
Shift effect without immigration	= (1 + 2) * (5)	–0.01	–0.12	–0.03	–0.01	0.12	0.04	–0.02
Shift effect in annual percentage point of labour productivity growth		–0.23						
Shift effect without immigration in annual percentage point of labour productivity growth		–0.02						

Source: OECD calculations based on national accounts, social security registry and work permits to foreigners.

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Annex VI

Calendar of main economic events

2001

May

The ECB cuts interest rates by 0.25 points to 4.5 per cent.

July

The University Law is approved by the government.

August

The ECB cuts interest rates by 0.25 points to 4.25 per cent.

September

After the September 11 attacks in New York, the ECB lowers its key interest rate by 0.5 points to 3.75 per cent.

October

Gas liberalisation continues with the partial sale of the natural gas contract with Algeria's Sonatrach to competitors of the incumbent operator, Gas Natural.

November

The ECB lowers its key interest rate by 0.5 points to 3.25 per cent.

December

The government issues a decree allowing for retirement after age 65 and introducing some incentives to postpone retirement.

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2002

January

The 2002 budget enters into force, accompanied by a new financial regime for autonomous communities. The tax powers of the regions are widened, while the transfer of competencies on health policy is finished.

March

The EU summit of Barcelona approves, among other measures, investment in cross-frontier interconnections of the electricity network.

April

The CMT imposes a reduction of the lease price of the local loop on Telefónica.

May

A decree is issued that reforms the unemployment insurance system. It includes a clarification of eligibility conditions for receiving benefits, a phasing out of the special unemployment subsidies for temporary agricultural workers in Andalusia and Extremadura and the suppression of the payment of wages between a lay-off and the tribunal decision on its adequacy.

The two Satellite TV companies, S*ogecable* and V*ía* D*igital*, announce their intention to merge.

July

The prime minister appoints a new government.

The Law on Education Quality is approved by the government.

Publication of the new census, which puts the Spanish population at 40.8 million, 2 million more than in the previous census of 1991.

September

The government announces a package of measures for improving competition in telecommunications, including a modification of the price cap of Telefónica.

October

Some elements of the unemployment insurance reform of March are modified, softening the eligibility conditions for benefits.

The P*restige* oil tanker sinks off the Galician coast.

November

The Financial Law is approved by Parliament, affecting a wide range of issues such as accountancy and transparency rules for firms and regulation of savings banks.

A reform of the financing system of municipalities is launched by the government, including the suppression of the tax for business activities (IAE) for small firms.

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The government issues, for approval, a draft law on co-ordination and quality of the health system.

December

The ECB lowers its key interest rate by 0.5 points to 2.75 per cent.

The EU Economic Council approves a reform of the Common Fisheries Policy.

The budget for 2003 is approved, including a reform of the personal income tax which reduces marginal rates, increases some tax allowances and introduces incentives to seek work for some groups of workers.

A new method for calculating electricity tariffs is approved by the government, which should provide a stable framework until 2010.

2003

January

The final stage of the liberalisation of electricity and gas markets enters into force, with choice of provider for all consumers.

The Budget Stability Law enters into force.

The reform of the special regime for agricultural workers of March is complemented with transition benefits during six years for those who cannot enter the system.

A national wage agreement is reached for 2003 between the social partners, recommending wage increases of 3.1 per cent.

March

A new law for "Health Co-ordination and Quality" was approved by the Parliament.

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